Prometric Physical Therapy Examination

Review & Study Guide

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PART II

QUIZ BANK

Preface

One of requirements to be licensed to practice physiotherapy in the states of Gulf Cooperation Council is to pass the prometric physical therapy examination which is not an easy as this exam often requires candidates to combine basic physical therapy knowledge with clinical experience in order to interpret, evaluate, or solve problems that may occur in clinical situations. In order to protect the public, Prometric use examination results as the main resource to determine whether a candidate has demonstrated minimal standards necessary for safe and effective practice.

So, it was our mission and it was a great mission to make an inclusive,wide and accurate source to help anyone who seek to work at these states or to be licensed to pass this exam easily.

Since fifteen years we had started collecting all exams for most of states in humble book, updating them every year until 2022 and the idea to make a great step has been established by managing this humble book to be a trusty source for any PT to completely depend on it. The title of our book **"PPTE"** an acronym for **"Prometric Physical Therapy Examination"** from its mission that we wish to do to the fullest.

Our book consists of two parts one as brief study guide, which is selectively collected from trusty sources in physiotherapy in orthopedics, neurology, cardiovascular, pulmonary, pediatric, therapeutic exercises and therapeutic modalities, etc.. the other part is the quiz bank which consists of prometric exams questions parted in pattern that follow the first part topics, almost all questions are answered, some of them have rationale for the true questions by an elite of professors and teaching staff in notable universities in ARE.

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Latest PPTE Quiz + VIP Notes

Physical Therapy Specialist Licensure Examination
Pre & Post Exam Guidelines

Pre-Exam Guidelines

General Rules

What are Licensure Examinations?
The Licensure Examinations are assessments to ensure that the public will not be harmed by the incompetence of healthcare practitioners. It assesses the ability to apply knowledge, concepts, and principles that constitute the basis of safe and effective health care.

What is the Physical Therapy Specialist Licensure Examination?
The Physical Therapy Specialist Licensure Examination is an exam that assesses your readiness to practice. It is a three-hour MCQ examination. It consists of 150 MCQs which may include up to 10 pilot questions. These questions have four options from which the candidate will choose one answer.
The examination shall contain recall questions that test knowledge and questions with scenarios that test other skills (interpretation, analysis, decision making, reasoning and problem solving).

What is a test blueprint, and what is its purpose?
A test blueprint is a document that reflects the content of your specialty licensure examination. The blueprint is the plan used for "building" the exam. The purpose of the blueprint is to ensure including questions related to what you are expected to know.
Note: to know more about Physical Therapy Specialist Licensure Exam Blueprint and References, (See Appendix A & B).

Application and Eligibility

Eligibility for the Classification Examination?
To apply for the Classification Examination, Please follow the most updated information available on SCFHS website.

Physical Therapy Specialist Licensure Examination
Pre & Post Exam Guidelines

Exam Day

Instructions for examination day:

• You will be continuously monitored by video, physical walk-throughs and the observation window during your test. All testing sessions are video and audio recorded.

• You must bring valid (unexpired) and acceptable ID(s) (Saudi ID, Resident ID, or Passport) and exam schedule printed out to the examination hall, and present it at the registration desk.

• Any clothing or jewelry items allowed to be worn in the test room must remain on your person at all times. Removed clothing or jewelry items must be stored in your locker.

• You may not leave the examination hall before thirty minutes have elapsed and always accompanied by an invigilator if you wish to return.

• You must conduct yourself in a civil manner at all times when on the premises of the testing center. Exhibiting abusive behavior towards the Test Center Administrator (TCA), or any other staff member of the test center, may result in legal prosecution.

• To protect the privacy of all testers, the TCA can neither confirm nor deny if any particular individual is present or scheduled at the test center.

• Repeated or lengthy departures from the test room for unscheduled breaks will be reported by the TCA.

• You must return all materials issued to you by the TCA at the end of your test.

• You are required to sign out on the test center roster each time you leave the test room. You must also sign back in and show your ID to the TCA in order to be re-admitted to the test room.

Physical Therapy Specialist Licensure Examination
Pre & Post Exam Guidelines

Prohibitions
Before the examination:

• Seeking, providing, and/or obtaining unauthorized access to examination materials, providing false information or making false statements on or in connection with application forms, scheduling permits, or other exam-related documents.

• Applying for an examination for which you are not eligible.

• Communicating or attempting to communicate about specific test items, cases, answers, and/or exam results with an examiner, potential examiner, or formal or informal test developers at any time before, during, or after an examination.

During the examination:

• Taking an examination for which you are not eligible

• Taking an examination for someone or engaging someone to take an examination for you giving, receiving, or obtaining unauthorized assistance during the examination or attempting to do so

• Making notes of any kind while in the secure areas of the test center, except on the writing materials provided at the test center for this purpose

• Failing to adhere to any exam policy, procedure, or rule, including instructions of TCA.

• Verbal or physical harassment of test center staff or other examination staff, or other disruptive or unprofessional behavior during the registration, scheduling, or examination process

• Possessing any unauthorized materials, including photographic equipment, communication or recording devices, and cell phones, in the secure testing areas

• Any other electronic communication device, not herein mentioned, are prohibited in the examination hall irrespective if they are turned off, and no provision will be made to store them

• Communicating or attempting to communicate about specific test items, cases, and/or answers with another examinee, or formal or informal test preparation group at any time before, during, or after an examination.

After the examination:

• Altering or misrepresenting examination scores.

• Any reproduction by any means, including, but not limited to, reconstruction through memorization, and/or dissemination of copyrighted examination materials by any means, including the internet.

• Communicating or attempting to communicate about specific test items, cases, and/or answers with another examinee, potential examinee, or formal or informal test preparation group at any time before, during, or after an examination.

• Failure to cooperate fully in any investigation of a violation of the SCFHS rules.

Post-Exam Guidelines

General Rules

When and Where Can I Find My Score?
A detailed report will be provided upon completing your test.

After Getting Your Passing Score:
Congratulations. Passing the licensing test is one of the requirements for registration. Now you may complete your registration at SCFHS by fulfilling all the requirements.

After Getting Your Failing Score:
We regret that you have not pass. We recognize that you have spent much time and exerted efforts to prepare for the exam. Thus, you must not lose hope and start over. Consider your failure as a beneficial experience that enables you to know more about your weakness and strength points and how so strengthen them and be determined to make it on the next chance, if Allah will. Be sure

that you are not the first one who fail in this world as many other people have experienced the same and they have made their failure a strong motivation to succeed.

If you would like to request a rescoring:

Read carefully the SCFHS rescoring Process. Then , fill the Rescoring Request Form and submit it to Tawasul During two weeks from announcing the result.

If you would like to request a formal review:

Read carefully the Policy and Procedures for the Formal Review of SCFHS Examinations. Types of Requests may include: disruption during examination, faulty examination equipment, or other similar issues related to examination process. Then , fill the Formal Review Request and submit it to Tawasul. Please make sure your Request is submitted during two weeks from exam date.

Appendix A: Physical Therapy Specialist Licensure Examination Blueprint

Evaluation Item	Content	Relative Percentage	Passing Score
(150 MCQs, 3 hours)	**Topics:**		
	1. Medical Knowledge	14%	55%
	2. PT Knowledge	24%	
	3. Orthopedics	24%	
	4. Neurology and Pediatrics	19%	
	5. Cardiopulmonary and Geriatrics	19%	
	Total	**100%**	

Note:

1. Blueprint distributions of the examination may differ up to ±3% in each category.
2. See the SCFHS website for the most up-to-date information.

Appendix B: References

1. Muscle: Testing and Function with Posture and Pain. 5th Edition, Florence Peterson, 2005.

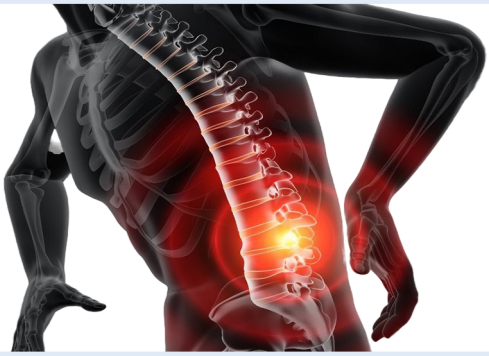
2. Clayton`s Electro Therapy, Theory and Practice, Angela Forster, 9th Edition 1999.

3. Tidy`s Physiotherapy, 13th Edition.

4. Physical Rehabilitation, Assessment and Treatment, 3rd Edition, Susan B.Q., Sullivan, 1994.

5. Tide`s Physical Therapy, Ann Thomason, 12th Edition.

Note: This list is intended for use as a study aid only. SCFHS does not intend the list to imply endorsement of these specific references, nor are the exam questions necessarily taken from these sources.





Chapter

1.A Anatomy

Question 1

What is this position called :

1. half kneeling
2. **prone kneeling**
3. sitting kneeling



Correct Answer: 2
rationale:
prone kneeling known as" quadruped position" when all four limbs are on the ground, typically down on the hands and knees. in which hands rested on the ground , back parallel to ground and vertical to both arms and thigh .
Note for the other answers
half kneeling : placing one knee down directly under the hip and the other foot should be in line with the knee,
sitting kneeling : Sitting on the knees is a sitting style where your knees are bent and your legs are folded under you

Question 2

measurement of aerobic capacity

1. Max heart rate
2. **O2 consumption**

Correct Answer: 2
rationale:
Your aerobic capacity is the maximal amount of oxygen your body can consume during maximal intensity exercise. And measured with a VO2 max test, (O2 consumption)

Question 3

Which of the following is used as a measure to estimate aerobic capacity in a normal person?

1. Maximum heart rate
2. Maximum cardiac output
3. Maximum exercise tolerance
4. **Maximum rate of oxygen consumption**

Correct Answer: 4
rationale:
Your aerobic capacity is the maximal amount of oxygen your body can consume during maximal intensity exercise. And measured with a VO2 max test

Question 4

One of the following does not use phosphotognase of the body:

1. **weight lifting**
2. jogging
3. running a marathon
4. diving



Correct Answer: 1
rationale:
weight lifting is anaerobic exe.which does not use phosphotognase .The biochemistry of anaerobic exercise involves a process called glycolysis, in which glucose is converted to adenosine triphosphate (ATP), which is the primary source of energy for cellular reactions
All other answers is aerobic exercises (jogging . running a marathon and diving)

Question 5

One of the following is not an aerobic exercise:

1. **weight lifting**
2. jogging
3. running a marathon
4. diving

Correct Answer: 1
rationale:
weight lifting is not an aerobic exercise
All other answers is aerobic exercises (jogging . running a marathon)

Question 6

One of the following is not depends on oxygen of the body

1. **weight lifting**
2. jogging
3. running a marathon
4. diving

Correct Answer: 1
rationale:
weight lifting is not depends on oxygen as weight lifting is anaerobic exe
All other answers is aerobic exercises that depend on oxygen

Question 7

Which of the following features is about Pacinian corpuscles?

1. **Rapidly adapting**
2. Change in muscle length
3. Change in muscle tension
4. Detection of acceleration and deceleration

Correct Answer: 1
rationale:
Pacinian corpuscles are rapidly adapting (phasic) receptors that detect gross pressure changes and vibrations in the skin
All other answers is incorrect because :
Change in muscle length is a function of ***muscle spindles***
change in muscle tension is a function of ***Golgi tendon organ***
dynamic mechanoreceptor whose brief, high-velocity discharges signal joint acceleration and deceleration with both active and passive joint movements.

Question 8

One of the following is not depends on oxygen of the body:

1. aerobic
2. **anaerobic**

Correct Answer: 2
rationale: Anaerobic exercise is not depends on oxygen

Question 9

What type of exercise where energy needs are not provided by oxygen?

1. Aerobic
2. Isotonic
3. Isometric
4. **Anaerobic**

Correct Answer: 4
rationale: Anaerobic exercise is not depends on oxygen

Question 10

Which of the following is the most helpful to diagnose a myocardial infarct?

1. **Troponin**
2. Aminotransferase
3. Gluco phosphokinase
4. Creatine phosphokinase

Correct Answer: 1
rationale: The main use of testing troponin levels is to help rapidly diagnose acute myocardial infarction (MI), or heart attack. Troponin is a protein very specific for the heart muscle, and when a heart attack occurs, troponin levels in the blood begin to rise within 2 to 4 hours of onset, and continue to be elevated for about 2 weeks.

Question 11

Which of the following is the main benefit of adaptation in aerobic exercise?

1. Decreases mental stress
2. Decreases hypercholesterolemia
3. Enhances neuromuscular coordination
4. **Increases efficiency of cardiovascular system**

Correct Answer: 4
rationale: training adaptations serve to redesign muscle and lead to an improved capacity for oxygen exchange between capillary and tissue and to an improved control of metabolism within the muscle fibers leading to Increases efficiency of cardiovascular system
all other answers may be a result of exercise but the MAIN benefit of adaptation is to Increases efficiency of cardiovascular system



Question 12

A resting pulse rate of 40-60 beats per minute would unlikely to be found in:

1. elderly individuals with arteriosclerotic heart disease.
2. patients taking digitalis.
3. **children under five years of age.**
4. marathon runners.

Correct Answer: 3
rationale:
resting pulse rate of 40-60 beats per minute cant be found in children under five years of age because their normal resting heart rate is high ranging from 70 bpm to 100 bpm.
Note :. people with high physical fitness, such as long-distance runners, can have a heart rate as low as the 40s when at rest without any problem. it is not necessarily a bad thing

Question 13

During the opening of a patient's mouth, a palpable and audible click is discovered in the left temporomandibular joint. The physician informs the therapist that the patient has anteriorly dislocated disk. This click most likely signifies

1. **the condyle is sliding anterior to obtain normal relationship with the disk.**
2. the condyle is sliding posterior to obtain normal relationship with the disk.
3. the condyle is sliding anterior and losing normal relationship with the disk
4. the condyle is sliding posterior and losing normal relationship with the disk.

Correct Answer: 1
rationale:
the initial click is created by the condyle slipping back into the correct position under the disk with opening of the mouth.
In this disorder, the condyle is resting posterior to the disk before jaw opening. With closing, the click is caused by the condyle slipping away from the disk.

Question 14

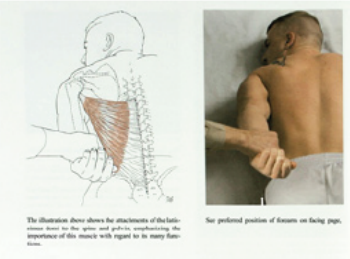
A patient is referred to physical therapy with a history of tempomandibular joint pain. The therapist notices that the patient is having difficulty closing his mouth against minimal resistance. With this information, which of the following muscles would not be a target for strengthening exercise to correct this deficit

1. Medial pterygoid muscle
2. Temporalis
3. Masseter
4. **Lateral pterygoid muscle**

Correct Answer: 4
rationale:
In this case the therapist must focus on muscle that close the mouse and Lateral pterygoid muscle open jaw and protrude the mandibular and move the mandibular from side to side so it would not be a target for strengthening exercise to correct this deficit
All other answers considered the a target for strengthening exercise to correct this deficit because they lead to elevation of the mandible (close the jaw)

Question 15

A patient is in prone position with his head rotated to the left side. The left upper extremity is placed at his side and fully internally rotated. The left shoulder is then shrugged toward the chin. The therapist then grasps the mid shaft of the patient's left forearm. The patient is then instructed to "try to reach your feet using just your left arm." This movement is resisted by the therapist. The test is assessing the strength of what muscle?



1. Upper trapezius
2. Posterior deltoid
3. **Latissimus dorsi**
4. Triceps brachialis

Correct Answer: 3
rationale: training adaptations serve to redesign muscle and lead to an improved capacity for oxygen exchange between capillary and tissue and to an improved control of metabolism within the muscle fibers leading to Increases efficiency of cardiovascular system
all other answers may be a result of exercise but the MAIN benefit of adaptation is to Increases efficiency of cardiovascular system

Question 16

The following muscles are medial rotators of the arm

1. **Medialpectoralis major, latissimus dorsi and subscapularis**
2. pectoralis minor, pectoralis major and teres major
3. teres major, teres minor and pectoralis major
4. serratus anterior, pectoralis minor and teres major.

Correct Answer: 1
rationale:
medial rotators of the arm subscapularis, latissimus dorsi, teres major, pectoralis major, anterior fibers of deltoid.

Question 17

A physiotherapist receives a referral for a patient. The referring physician has outlined all clinical findings in the referral, one of which is "ptosis".What does this term mean?

1. **Drooped eyelid**
2. Disruption of sensation causing abnormal sensation
3. Inability to perform purposeful voluntary movements
4. Involuntary repetitive movement characterized by twisting

Correct Answer: 1
rationale:
Ptosis is a condition where the upper eyelid droops. It is also called blepharoptosis, or upper eyelid ptosis.
Terms of other anwers :
Paresthesias : .Disruption of sensation causing abnormal sensation
Apraxia: Inability to perform purposeful voluntary movements
Dystonia.: Involuntary repetitive movement characterized by twisting

Question 18

A therapist assessing radial deviation range of motion at the wrist. The correct position of the goniometer should be as follows: the proximal arm aligned with the forearm and the distal arm aligned with the third metacarpal. What should be used as the axis point?

1. Lunate
2. Scaphoid
3. **Capitate**
4. Triquetrum



Correct Answer: 3
rationale:
Capitate is in the middle of hand and should be used as the axis point

Question 19

A 24-year-old basketball player fractured his left hand. Which carpal bone is most likely fractured?

1. Lunate
2. Hamate
3. Pisiform
4. **Scaphoid**

Correct Answer: 4
rationale:
Scaphoid fractures are by far the most common of the carpal fractures, and account for 10 percent of all hand fractures and about 55 percent of all carpal fractures

Question 20

The most appropriate position to strength sternocleidomastoid is

1. supine
2. **sitting**
3. prone
4. standing

Correct Answer: 2
rationale: sitting is the best position for strength sternocleidomastoid

Question 21

which muscle close the mouth :

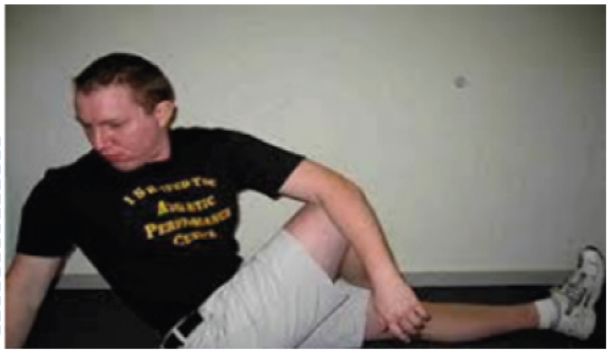
1. **masseter**
2. lateral ptyrgoid
3. platysma

Correct Answer: 1
rationale: Elevation of the Jaw (close the mouth): Produced by the masseter, temporalis and medial pterygoid.
Depression of the Jaw (open the mouth): Produced by the lateral pterygoid

Question 22

Which of the following group of hip muscles is being stretched (see image)?

1. Flexors and abductors
2. Extensors and adductors
3. Extensors and internal rotators
4. **Extensors and external rotators**



Correct Answer: 4
rationale: Sitting Rotation Stretch position ,is a position to stretch Piriformis . buttocks, as well as at your side . Main muscles worked: Piriformis, external oblique rotators, internal oblique rotators

Question 23

which of the following muscles compress the checks

1. **Buccinator**
2. corregator
3. frontalis
4. Depressor angulii

Correct Answer: 1
rationale:
Buccinator ,This muscle is sometimes referred to as an accessory muscle of mastication due to its role in compressing the cheeks inwards against molars, thus, aiding in chewing and swallowing

Question 24

which of the following elevate the jaw:

1. **masseter**
2. levator lipii
3. mentalis
4. orbicularis oris

Correct Answer: 1
rationale:A-Masseter is one of muscles of mastication which elevate the jaw .

Question 25

muscle that moves angle of mouth inferior and back

1. **Platysma**
2. Mentalis
3. Masseter

Correct Answer: 1
rationale:
platysma , moves angel of the mouse inferior and back (it is attached to the base of the mandible and angle of the mouth)

Question 26

What is the function of the muscles in the image (see image)?

1. Flexion of the hip
2. Extension of the hip
3. **Adduction of the hip**
4. External rotation of the hip



Correct Answer: 3
rationale:
Scaphoid fractures are by far the most common of the carpal fractures, and account for 10 percent of all hand fractures and about 55 percent of all carpal fractures

Question 27

Which of the following muscles is important while someone adducts, extends, and medially rotates the arm?

1. Teres minor
2. **Latissimus dorsi**
3. Pectoralis major
4. Serratus anterior

Correct Answer: 2
rationale:
Latissimus dorsi function as Arm internal rotator , Arm adductor, Arm extensor ; Assists in respiration

Question 28

A 35 - year - old man with right facial palsy presents to the Physical Therapy clinic. During the examination of the facial muscles, the patient is unable to close his mouth. Which of the following muscles is most likely damaged?

1. Frontalis
2. Masseter
3. Temporalis
4. **Orbicularis oris**

Correct Answer: 4
rationale:
the orbicularis oris one of muscles of facial expression which is supplied by facial nerve and can close the mouth.

Question 29

which muscle Raise the skin of the chin.:

1. buccinators
2. levator libii
3. **mentalis**
4. platysma

Correct Answer: 2 **rationale:** mentalis we use it for shaving our faces

Question 30

which muscle Close and protrude the lips, as in whistling?

1. masseter
2. mentalis
3. **orbicularis oris**
4. depressor libii

Correct Answer: 3
rationale:
The main action caused by the orbicularis oris muscle is the pursing or puckering of the lips. These actions of the lips allow you to kiss someone, play a wind instrument, and whistle your favorite song

Question 31

which muscle close the eyes ?

1. buccinators
2. orbicularis oris
3. **orbicularis occuli**
4. masseter

Correct Answer: 3
rationale:
The orbicularis oculi : it surround the orbit and has 3 parts (orbital part (tight closure of the eye), palpebral part (light closure of the eye) and lacrimal part

Question 32

A 15-year-old boy with left facial palsy receives treatment in the Physical Therapy Clinic. During the assessment, the physical therapist notices the patient was unable to close the eyelids. Which of the following muscles is most likely affected in this case?

1. Risorius
2. Frontalis
3. Masseter
4. **Orbicularis oculi**

Correct Answer: 4
rationale: The orbicularis oculi is a muscle in the face that closes the eyelids. And is supplied by the facial nerve So any defict in this muscle makes the patient unable to close the eyelid

Question 33

which of the following Draws the eyebrow down ward and inward, with vertical wrinkles

1. buccinators
2. **corrugator**
3. frontalis
4. depressor angulii

Correct Answer: 2
rationale: Corrugator a muscle whose contraction causes wrinkling of the brow. As it draw the two eyebrows together medially and inferiorly, thereby producing vertical wrinkles over the glabella and an expression of frowning.

Question 34

What are the three muscles attached to the first cuneiform bone?

1. **The anterior tibialis, the posterior tibialis, and the fibularis (peroneus) longus**
2. The extensor digitorum, the flexor hallucis, and the fibularis (peroneus) longus
3. The anterior tibialis, the peroneus brevis, and the fibularis (peroneus) longus
4. The flexor carpi ulnaris, the flexor pollicis brevis, and the flexor pollicis longu

Correct Answer: 1
rationale:
this table to illustrate Muscle attachments to first cuneiform bone (Medial cuneiform)

Muscle	Direction	Attachment
anterior Tibialis	Insertion	Medial cuneiform
Fibularis (peroneus) longus	Insertion	Medial cuneiform
posterior Tibialis	Insertion	Medial cuneiform

Question 35

which of following muscle doesn't closes the mouth:

1. masseter
2. ptrygoidus medialis
3. temporalis
4. **suprahoid ms**

Correct Answer: 4
rationale:
The suprahyoid muscles are a group of muscles located superior to the hyoid bone of the neck. They all act to elevate the hyoid bone and depress the mandible helping in opening of mouth

Question 36

A 55-year-old woman with elbow dysfunction is referred to physical therapy. The physical therapist assessed the patient and found a weak brachioradialis muscle. Which of the following exercises is recommended to strengthen this muscle?

1. Elbow flexion with the forearm rm in a pronated position
2. Elbow flexion with the forearm in a supinated position
3. Elbow extension with the forearm in a pronated position
4. **Elbow flexion with the forearm in a midway (neutral) position**

Correct Answer: 4
rationale:
Because the action of brachioradialis is flexion of forearm from neutral position

Question 37

The muscle used for smiling:-

1. procerus
2. **Zygomaticus**
3. Platysma
4. risorius

Correct Answer: 2
rationale:
The zygomaticus major muscle is a muscle that controls facial expression, drawing the mouth's angle upward and outward produces the expression of smiling

Question 38

Which muscle draw eyebrow to upward making horizontal (true)wrinkling:-

1. **frontalis (occipto-frontalis)**
2. corrugator
3. orbicularis oris
4. lateral pterygoid

Correct Answer: 1
rationale:
The function of occipitofrontalis muscle is to elevate the eyebrows and wrinkle the forehead skin with its frontal part



Question 39

The circular band of muscles that surround the eyes are the _____ oculi, while the circle muscle that surrounds the mouth is the _____oris. (the same answer for both blanks

1. **Orbicularis**
2. Annualus
3. Circularis
4. Muscularis

Correct Answer: 1
rationale:
The orbicularis oculi is a muscle in the face that closes the eyelids , the orbicularis oris muscle controls movements of the mouth and lips

Question 40

Origin of ms : posterior aspect of head of fibula and medial aspect of tibia ?

1. **soleus**
2. Tibialis anterior
3. Tibialis posterior

Correct Answer: 1
rationale:
Soleus muscle originate from soleal line, medial border of tibia, head of fibula, posterior border of fibula

Question 41

Which muscle originates from the posterior fibula and mid of tibia ?

1. soleus
2. Tibialis anterior
3. **Tibialis posterior**

Correct Answer: 3
rationale:
Tibialis posterior muscle originate from posterior surface of tibia, posterior surface of fibula and interosseous membrane

Question 42

from supine when there is shortening of this muscle the arm raise above table and not down ,the MS is

1. biceps
2. brachioradialis
3. brachialis
4. **pectoralis major**

Correct Answer: 4
rationale:
pectoralis Major because it`s adductor muscle and the others flexors of the elbow.

Question 43

from supine when there is shortening of this muscle the arm raise above table and not down ,the MS is

1. biceps
2. brachioradialis
3. brachialis
4. **pectoralis major**

Correct Answer: 4
rationale:
pectoralis Major because it`s adductor muscle and the others flexors of the elbow.

Question 44

PT is evaluating patient who is unable to open the door using supination ; so he expects weakness of

1. brachioradialis
2. **biceps brachii**
3. anconeus
4. flexor carpi ulnaris

Correct Answer: 2
rationale:
biceps brachii make a flexion with supination

Question 45

physical therapist is evaluating elbow flexion from neutral position ; which muscle should be evaluated ?

1. biceps brachii
2. brachialis
3. **brachioradialis**
4. anconeous

Correct Answer: 3
rationale:
Brachioradialis can do Flexion on Mid prone position



Question 46

A 26-year-old patient with shortened biceps brachii muscle after casting the elbow for few weeks. The physical therapist decided to stretch this muscle. Which of the following positions of the upper extremity will maximally elongate the biceps brachii?

1. **Shoulder extension, elbow extension, and forearm pronation**
2. Shoulder extension, elbow extension, and forearm supination
3. Shoulder external rotation, elbow extension, and forearm pronation
4. Shoulder external rotation, elbow extension, and forearm supination

Correct Answer: 1
rationale:
Shoulder extension, elbow extension, and forearm pronation are the opposition of biceps action so it cause biceps brachii stretch

Question 47

Which of the following muscles is an abductor of the shoulder joint and responsible for the first 30° of shoulder abduction

1. Teres minor
2. Teres major
3. Subscapularis
4. **Supraspinatus**

Correct Answer: 4
rationale:
supraspinatus muscle functions to initiate abduction and depress the humeral head against the upward pull of the deltoid
electromyography (EMG) studies have demonstrated a high degree of supraspinatus activity during the initial 30° of abduction.

Question 48

Sterno-clavicular joint is enforced by except :

1. sterno-clavicular lig
2. intra-articular disc
3. intra-clavicular lig
4. **acromio-clavicular lig**

Correct Answer: 4
rationale: Acromioclavicular ligament support the acromioclavicular joint superiorly

Question 49

Which nerve supplies the muscle in the picture

1. **Axillary**
2. Supascapular
3. Long thoracic
4. Musculocutaneous

Correct Answer: 1
rationale:
Axillary nerve supplies the teres minor and deltoid muscle, abductors of the arm at the shoulder.

Question 50

A patient presents to an outpatient physical therapy clinic with a severed ulnar nerve in the upper extremity. What muscle is still active and largely responsible for the obvious hyperextension at the metacarpophalangeal (MCP) joints of the involved hand?

1. Dorsal interossei
2. palmar interossei
3. **Extensor carpi radialis brevis**
4. Extensor digitorum

Correct Answer: 3
rationale:
the extensor digitorum communis extends medial four digits at the metacarpophalangeal joints and secondarily at the interphalangeal joints and supplied by posterior interosseus nerve (deep radial nerve

Question 51

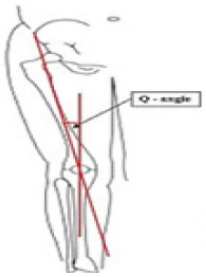
A physical therapist is fabricating a splint for a patient who received four metacarpophalangeal joint replacements. The surgical joint replacement was necessary because of severe rheumatoid arthritis. Which of the following is the correct placement of the metacarpophalangeal joints in the splint?

1. Full flexion and slight radial pull
2. Full flexion and slight ulnar pull.
3. **Full extension and slight radial pull**
4. Full extension and slight ulnar pull.

Correct Answer: 3
rationale:
the extensor digitorum communis extends medial four digits at the metacarpophalangeal joints and secondarily at the interphalangeal joints and supplied by posterior interosseus nerve (deep radial nerve

Question 52

if a straight line across the middle of the patella and a straight line going down through the center of the tibial tuberosity and another straight line going upward toward the anterior superior iliac spine is drawn, then an angle can be obtained (see image).Which of the following is the angle that will be obtained

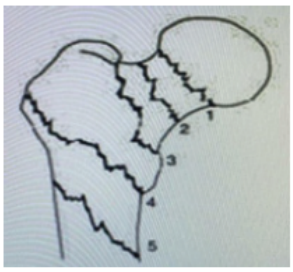


1. A
2. L
3. K
4. **Q**

Correct Answer: 4
rationale:
Q angle, which is also known as quadriceps angle, is defined as the angle formed between the quadriceps muscles and the patella tendon

Question 53

Hip fractures are classified according to their location (see image). Which of the following is the name given to the fracture site labeled as "4" in the figure?



1. Subcapital
2. Basicervical
3. Subtrochanteric
4. **. Intertrochanteric**

Correct Answer: 4
rationale:
Intertrochanteric fractures are defined as extracapsular fractures of the proximal femur that occur between the greater and lesser trochanter

Question 54

Femoral neck fractures are classified according to its location (see image). Which of the following is the name given to the fracture site labelled as "1"?



1. **Subcapital**
2. Basicervical
3. Transcervical
4. Subtrochanteric

Correct Answer: 1
rationale:
Subcapital fracture is the most common type of intracapsular neck of femur fracture. The fracture line extends through the junction of the head and neck of femur

Question 55

A therapist is evaluating a patient's ability to perform flexion, medial rotation, and abduction of the hip. Which of the following muscles is the therapist evaluating?

1. Sartorius
2. Hamstring
3. **Tensor fascia lata**
4. Gluteus maximus

Correct Answer: 3
rationale:
Subcapital fracture is the most common type of intracapsular neck of femur fracture. The fracture line extends through the junction of the head and neck of femur

Question 56

Which of the following is the term for the leg posture deformity while standing (see image)?



1. Coxa varum
2. Genu varum
3. Genu valgum
4. **Genu recurvatum**

Correct Answer: 4
rationale:
Genu recurvatum is operationally defined as knee extension greater than 5 degrees. Individuals who exhibit genu recurvatum may experience knee pain, display an extension gait pattern, and have poor proprioceptive control of terminal knee extension

Question 57

A therapist is evaluating a patient's ability to perform flexion, medial rotation, and abduction of the hip. Which of the following muscles is the therapist evaluating?

1. Sartorius
2. Hamstring
3. **Tensor fascia lata**
4. Gluteus maximus

Correct Answer: 3
rationale:
Tensor Fascia lata Actions: Hip - flexion, medial rotation, abduction and knee - lateral rotation, Torso – stabilization
Sartorius Actions: At the hip joint, it is a flexor, abductor and lateral rotator. At the knee joint, it is also a flexors
Gluteus Medius Action: abduction of the hip; preventing adduction of the hip. Medial rotation of thigh.



Question 58

A 51-year-old man comes to the Physical Therapy Clinic for assessment of right hip pain (lateral side) that started 2 weeks ago. Pain worsens when the physical therapist tests a muscle that abducts, flexes, internally rotates the hip.
Which of the following muscles is the most likely cause of this pain?

1. Iliopsoas
2. Psoas minor
3. Psoas major
4. **Tensor fascia latae**

Correct Answer: 4
rationale:
Tensor fasciae latae trigger points and tightness can cause pain in the hip joint area and antero-lateral portion of the thigh which can extend as far as the knee. This pain can feel like deep hip pain which can be further aggravated by walking or laying down on the affected side

Question 59

What muscle make lateral rotation of Humerus :

1. Pectoral major
2. **Teres minor**
3. Teres major
4. Rotator cuff

Correct Answer: 2
rationale:
Teres minor is lateral rotation of the humerus and stabilization of the humerus

Question 60

Which muscle not related to rotator cuff:

1. teres minor
2. **teres major**
3. supraspinatus
4. infraspinatus.

Correct Answer: 2
rationale:
Rotator cuff muscles are Supraspinatus, infraspinatus , teres minor, subscapularis

Question 61

What is the muscle originating from transverse process of C7 and T1 and inserts into medial border of scapula.

1. evator scapulae
2. latissimus dorsi
3. rhomboid major
4. **rhomboid minor**

Correct Answer: 4
rationale:
rhomboid minor is a cylindrical muscle that originates at the ligamentum nuchae and C7 and T1 vertebra. It inserts at the scapula's medial border near the base of the spine of the scapula

Question 62

A 28 - year - old patient is being examined for excessive subtalar pronation. Which of the following are other possible associated motions or postures expected in this patient?

1. **Tibial, femoral, and pelvic internal rotation,**
2. Tibial, femoral, and pelvic external rotation,
3. Tibial and femoral external rotation with external pelvic rotation.
4. Tibial and femoral internal rotation with external pelvic rotation.

Correct Answer: 1
rationale:
Excessive pronation has been linked to numerous lower extremity injuries, including medial tibial stress syndrome, stress fractures , excessive pronation induces an increased compensatory internal rotation

Question 63

which muscle act in Med rotation, adduction and extension of shoulder joint?

1. pectoralis minor
2. serratus anterior
3. teres minor
4. **latissmus dorsi**

Correct Answer: 4
rationale:
Latissmus Dorsai make a medial rotation with Extension and Adduction of the shoulder

Question 64

Which of the following is not part of the triangular fibrocartilage complex of the wrist?

1. Dorsal radio-ulnar ligament
2. Ulnar collateral ligament
3. **Radial collateral ligament**
4. Ulnar articular cartilage

Correct Answer: 3
rationale:
Redial Collateral ligament is triangular fibrocartilage complex of the wrist which is consist of ulnar collateral ligament, ulnar articular cartilage & dorsal radio-ulnar ligament

Question 65



To strengthen this muscle, you should

1. **Maintain dorsiflexion and walking some steps on heel**
2. Walking on toes
3. Sit at edge of plinth, rest foot on ground and fully dorsiflexion the foot

Correct Answer: 1
rationale:
Maintain dorsiflexion and walking some steps on heel as tibialis anterior (TA) is the strongest dorsiflexor of the foot

Question 66

Which of the following is not part of the triangular fibrocartilage complex of the wrist?

1. flexor Carpi ulnaris
2. **flexor Carpi radialis**
3. flexor digitorum profundus
4. biceps brachii

Correct Answer: 2
rationale:
the main function of FCR is providing flexion of the wrist and assisting in abduction of the hand and wrist

Question 67

physiotherapist asks patient to flex the wrist and adduct it ;

1. **flexor Carpi ulnaris**
2. flexor Carpi radialis
3. flexor digitorum profundus
4. biceps brachii

Correct Answer: 1
rationale:
flexor carpi ulnaris action is flexion with adduction

Question 68

Shortness of whitch muscle cause limitation in shoulder abduction and lateral rotation

1. **pectoralis major true**
2. serratus ant
3. sub scapularis
4. teres major

Correct Answer: 1
rationale:
pectoralis major action is adduction with internal rotation so its shortness limit the opposite action

Question 69

All of the following muscles affected in tennis elbow except

1. Ext carpi radials brevis
2. Ext carpi ulnaris
3. Ext digitorum
4. **flex carpi ulnaris**

Correct Answer: 4
rationale:
exact tendon most commonly involved in tennis elbow connects to a muscle called the extensor carpi radialis brevis. , then extensor carpi ulnaris and extensor digitorum

Question 70

All of the following are adductors of hip joint except

1. Gracilis
2. Pectineus
3. **Sartorius**

Correct Answer: 3
rationale:
as we described aboveprimary hip adductors are the pectineus, adductor longus, gracilis, adductor brevis, and adductor magnus

Question 71

When you pushing automobile with shoulder flexion, elbow extended the main muscle of action is

1. **serratus anterior**
2. biceps
3. pectorals major
4. triceps

Correct Answer: 1
rationale:
serratus Anterior action is stabilize the shoulder and protract the scapula that`s mean stabilize the arm in push anything



Question 72

Injury to long thoracic nerve leading to weakness of

1. pectorals major
2. **serratus ant**
3. sub scapularis
4. teres major

Correct Answer: 2
rationale:
Serratus anterior → Long Thoracic nerve
Pectoralis major → Lateral and medial pectoral nerve
Subscapularis and teres major → Upper & lower subscapular nerve

Question 73

winging of scapula due to weakness of

1. pectorals major
2. **serratus ant**
3. sub scapularis
4. teres major

Correct Answer: 2
rationale:
scapular winging is almost always caused by damage to one of three nerves that control muscles in your arms, back, and neck: the long thoracic nerve, which controls the serratus anterior muscle. the dorsal scapular nerve, which controls the rhomboid muscles. the spinal accessory nerve, which controls the trapezius

Question 74

Winging of the scapula could be due to damage to this nerve:

1. **long thoracic nerve**
2. thoracodorsal nerve
3. lower subscapular nerve
4. upper subscapular nerve.

Correct Answer: 2
rationale:
Scapular winging occur due to injury of the long thoracic nerve, which controls the serratus anterior muscle

Question 75

First bone to ossify in foetal (embryo) life is

1. femur.
2. tibia.
3. **clavicle.**
4. sternum

Correct Answer: 3
rationale:
clavicle is the first bone to ossify in the devel- oping embryo. At approximately 5 weeks of gesta- tion, primary ossification of the clavicle proceeds by membranous ossification of two centers with no prior cartilaginous anlages

Question 76

which part not included in posterior neck triangle

1. clavicle
2. trapezuis
3. strenoclidomastoid
4. **strenum**

Correct Answer: 4
rationale:
these borders include the trapezius muscle posteriorly, the sternocleidomastoid muscle anteriorly, and the middle one-third of the clavicle inferiorly

Question 77

which is not related to the shoulder triangle

1. Clavicle
2. **Sternum**
3. Humurars

Correct Answer: 2
rationale:
shoulder is made up of three bones: the scapula (shoulder blade), clavicle (collarbone) and humerus (upper arm bone).

Question 78

Which of the following is not included in shoulder complex?

1. clavicle
2. scapula
3. humerus
4. **sternum**.

Correct Answer: 4
rationale:
shoulder is made up of three bones: the scapula (shoulder blade), clavicle (collarbone) and humerus (upper arm bone).

Question 79

Protraction and retraction of shoulder occurs at

1. glenohumeral joint
2. **scapulothoracic interfance**
3. acromioclavicular joint

Correct Answer: 2
rationale:
protraction and retraction of the scapula happens around a vertical axis passing through the lateral end of clavicle

Question 80

What is the action of this muscle shape ?

1. hip adduction
2. **hip extenstion**
3. back extension



Correct Answer: 2
rationale:
primary hip extensors include the gluteus maximus,

Question 81

The main muscle responsible for climbing stairs and coming from reclined position, bringing knee to chest

1. rectus femoris
2. **iliopsoas**
3. quadratus lumborum
4. pectineus

Correct Answer: 2
rationale:
iliopsoas is the prime mover of hip flexion, and is the strongest of the hip flexors (others are rectus femoris, sartorius, and tensor fasciae latae). The iliopsoas is important for standing, walking, and running

Question 82

Weakness of which the following muscle interferes stair climbing, walking up an incline, and getting up from the inclining position?

1. soleus
2. hamstring
3. quadriceps femoris
4. **iliopsoas.**

Correct Answer: 2
rationale:
The iliopsoas is important for standing, walking, and running

Question 83

While on supine and bending your knees towards the chest which muscle is doing that?

1. **Iliopsoas muscle**
2. quadriceps muscles
3. quadratus lumborum
4. pectineus.

Correct Answer: 1
rationale:
Because iliopsoas attaches the knee to chest

Question 84

Which of the following muscles unlock the knee?

1. vastus medialis obliqus
2. **popliteus muscle**
3. gastrocnemius muscle
4. sartorius muscle

Correct Answer: 2
rationale:
to 'unlock' the knee, the popliteus muscle needs to contract, causing flexion and lateral rotation of the femur on the tibia, thereby giving the muscle the term - 'key' to the locked knee

Question 85

Muscle response for getting up form sitting and claiming stair and walking:

1. gluteus maximus
2. quadriceps
3. Tensor fascia Lata.

Correct Answer: 2
rationale:
stair climbing activates your core muscles in your legs, such as your hamstrings, quads, calves

Question 86

A physical therapist is evaluating a female distance runner who complains of intermittent medial ankle pain. In static standing, the therapist palpates excessive lateral deviation of the head of the talus. From this information, in what position is the subtalar joint during palpation

1. Supination
2. Pronation
3. Neutral
4. Unable to determine from the information given

Correct Answer: 2
rationale:
subtalar joint is located within the rearfoot This joint consists of the articulation between the facets on the inferior surface of the talus and the matching facets on the superior surface of the calcaneus

Question 87

Best muscle to strength for ascending stair and initiating stand up from sitting:

1. gluteus Medius
2. **quadriceps**
3. hamstring
4. soleus.

Correct Answer: 2
rationale:
You use your quads to perform a variety of movements, including kicking, running, jumping and walking



Question 88

All the following is true about ligaments except

1. high collagen content
2. laxity lead to hyper mobility
3. connect on bone to another
4. **usually heals fast and need not support**

Correct Answer: 4
rationale:
ligament healing is often slow and incomplete. Joint laxity caused by ligament injury improves slowly over a period of six week to a year

Question 89

A physical therapist should place the knee in which of the following positions to palpate the lateral collateral ligament (LCL)?

1. Knee at 60 degree of flexion and the hip externally rotated.
2. Knee at 20 degree of flexion and the hip at neutral.
3. **Knee at 9o degree of flexion and hip externally rotated.**
4. Knee at 0 degree and the hip at neutral.

Correct Answer: 3
rationale:
patient position : Legs crossed with ankle resting on opposite knee (90° knee flexion, hip abduction and external rotation) In this position the iliotibial band relaxes and makes the LCL easier to isolate

Question 90

supraspinatus muscle palpation?

1. **slight abduction with internal rotation**
2. abduction with external rotation

Correct Answer: 1
rationale:
slight abduction with internal rotation , So what you do is you have the person put the hand behind their back

Question 91

Palpation for piriformis

1. From ASIS to greater trochanter
2. **From PSIS to greater trochanter**
3. From coccyx and iliac crest
4. From sacrum to greater trochanter

Correct Answer: 2
rationale:
piriformis muscle is palpated 1 to 2 cm below the middle third of a line drawn between the posterior superior iliac spine and the upper boundary of the greater trochanter

Question 92

which of the following you cannot palpate while examining shoulder

1. 2nd rib
2. Sterno clavicular joint
3. **1st rib**

Correct Answer: 3
rationale:
first rib is difficult to palpate. That's because it lies both below and behind the clavicle, and also because there's a thick layer of muscle in front of it

Question 93

The shoulder work with 3 joints except:

1. glenohumeral joint
2. acromioclavicular (AC) joint
3. **costosternal joint.**
4. sternoclavicular

Correct Answer: 3
rationale:
joints in the shoulder allow it to move: the acromioclavicular joint, where the highest point of the scapula (acromion) meets the clavicle, and the glenohumeral joint and sternoclavicular joint

Question 94

Absorption of calcium in the intestine depends on which of the following vitamins?

1. A
2. B
3. C
4. **D**

Correct Answer: 4
rationale: The intestinal absorption of calcium and phosphorus has received considerable attention in recent years. The evidence has clearly indicated that calcium is absorbed by two processes: active transport and diffusion.
Vitamin D appears to affect both processes, and has a significant effect at the brush border of the intestinal cell. Vitamin D also affects the release of calcium from the intestinal cell to the lamina propria

Question 95

All of the fallowing in the shoulder except :

1. sternoclavicular joint
2. acromioclavicular joint
3. glenohumeral joint
4. **costosternal joint.**

Correct Answer: 4
rationale:
Costosternal joint is synovial plane joints of the costal cartilages of the true ribs with the sternum



Question 96

Which three muscles attach to the medial cuneiform bone

1. **The anterior tibialis, the posterior tibialis, and the fibularis (peroneus) longus**
2. The extensor digitorum, the flexor hallucis, and the fibularis (peroneus) longus
3. The anterior tibialis, the peroneus brevis, and the fibularis (peroneus) longus
4. The flexor carpi ulnaris, the flexor pollicis brevis, and the flexor pollicis longu

Correct Answer: 1
rationale: first cuneiform bone = medial cuneiform bone
anterior tibialis Insertion :Medial cuneiform bone, base of metatarsal bone 1
posterior tibialis Insertion :Tuberosity of navicular bone, all cuneiform bones, cuboid bone, bases of metatarsal bones 2-4
fibularis (peroneus) longus Insertion : Medial cuneiform bone, metatarsal bone 1
insretion of the other muscles
The extensor digitorum :Extensor expansions of digits 2-5
the flexor hallucis : Base of distal phalanx of great toe
the peroneus brevis : Tuberosity of the 5th metatarsal bone
The flexor carpi ulnaris, the flexor pollicis brevis, and the flexor pollicis longu all are muscles in upper limb

Question 97

Which of the following is correct of the female pelvis when comparing it with the male pelvis?

1. **the distance between the female ischial spines is greater**
2. the angle of the female pubic arch is smaller
3. the female pelvis as a whole is deeper, and the bones are heavier and thicker
4. the female iliac bones are less flared

Correct Answer: 1
rationale:
2 and 4 isn't correct because the adult male pelvis is narrower and less flared, exhibiting an oval or heart-shaped pelvic inlet, and the angle of the pubic arch is less than 90 degrees. The adult female pelvis is usually broader and exhibits a round pelvic inlet, and the angle of the pubic arch is greater than 90 degrees.
3 isn't correct because The male pelvis bone is heavier, taller, and much thicker compared to a female pelvis that is thinner and denser.

Question 98

The most important ligament for supporting the weight of the scapula when the person is standing is the:

1. **coracoclavicular**
2. glenohumeral
3. coracohumeral
4. transverse humeral.

Correct Answer: 1
rationale:
function of the coracoclavicular ligament is to allow complex shoulder movement without separation of the scapula from the clavicle and supporting scapular weight

Question 99

Which of the following articulate with the middle cuneiform?

1. **Navicular**
2. Talus
3. First metatarsal
4. Cuboid

Correct Answer: 1
rationale:
second cuneiform = Intermediate cuneiform has an articulation with the navicular proximally and with the base of the second metatarsal and the medial cuneiform distally.

Question 100

Which of the following is the additional caloric intake per day, needed for an exercising pregnant patient?

1. **300**
2. 400
3. 500
4. 600

Correct Answer: 1
rationale:
the total energy cost of pregnancy is about 85,000 kca If averaged over 280 days, the cumulative energy cost of pregnancy is about 300 kcal per day

Question 101

the name of muscle cell membrane is

1. **sarcomer**
2. actin
3. myosin
4. no name

Correct Answer: 1
rationale:
The cell membrane of a muscle cell is known as the sarcolemma and the cytoplasm is called sarcoplasm *Sarcomer = sarcolemmaay

Question 102

Which of the following didn't pass through acromio-humeral?

1. supraspinatus
2. tendon of long head of biceps
3. infraspinatus
4. **superior portion of glenohumeral capsule**

Correct Answer: 4
rationale:
acromiohumeral distance (ACHD) is defined as the shortest distance between the inferior cortex of the acromion and the top of the humeral head and it doesn't contain superior portion of glenohumeral capsulem



Question 103

The name of muscle cell is:

1. **myocyte**
2. actin
3. myosin
4. no name

Correct Answer: 1
rationale:
A muscle cell, known technically as a myocyte, is a specialized animal cell which can shorten its length using a series of motor proteins specially arranged within the cell. While several associated proteins help, actin and myosin form thick and thin filaments which slide past each other to contract small units of a muscle cell.

Question 104

Which don't present in dermis ?

1. **tactile**
2. blood vessels

Correct Answer: 1
rationale:
Tactile tissue isn`t present in dermis they are primarily located in glabrous skin just beneath the epidermis within the dermal papillae.

Question 105

These thin filaments are a major component of the contractile apparatus of the skeletal muscle:

1. collagen
2. titin
3. **actin**
4. myosin

Correct Answer: 3
rationale: the striated appearance of skeletal muscle fibers is due to the organisation of two contractile proteins: actin (thin filament) and myosin (thick filament).

Question 106

All of the following contents of dermis except:

1. sweating gland
2. **tactile corpuscles or tissue**
3. blood vessels.

Correct Answer: 2
rationale:
The dermis contains hair roots, sebaceous glands, sweat glands, nerves, and blood vessels. The tactile corpuscle, They are distributed on various areas of the skin, but concentrated in areas especially sensitive to light touch, such as the fingers, lips and male prepuce More specifically, they are primarily located in glabrous skin just beneath the epidermis within the dermal papillae.

Question 107

Anterior horn cell in gray matter not include

1. ventral root
2. alpha motor neuron
3. gamma motor neuron
4. **sensory neuron**

Correct Answer: 4
rationale:
Anterior horn cell contains the cell bodies of motor neurons not sensory that affect the skeletal muscles

Question 108

all is true about Anterior horn cell ,except

1. Origination of Alfa motor neuron
2. Its in ventral root
3. **Origination of afferent sensory neuron**

Correct Answer: 3
rationale:
Anterior horn cell isn't Origination of afferent sensory neuron as it concerned with the innervation of skeletal muscle (efferent)

Question 109

Brain consist of

1. two part
2. **Three parts**
3. Four part
4. Multipart

Correct Answer: 2
rationale:
Brain consists of cerebrum, cerebellum, and brain stem)

Question 110

Acromegaly is due to malfunction of :

1. **Pituitary gland**
2. Thyroid gland
3. thymus

Correct Answer: 1
rationale:
Acromegaly is due to malfunction of Growth Hermon, a syndrome that results when the anterior pituitary gland produces excess growth hormone (GH) after epiphyseal plate closure at puberty



Question 111

Which gland regulate the growth of bones:

1. thyroid gland
2. **Anterior pituitary gland**
3. Adrenal Glands

Correct Answer: 2
rationale:
Anterior pituitary gland is responsible for release of growth hormone GH. Bone growth is under the influence of growth hormone

Question 112

increase growth of bone :

1. **ant lobe of pituitary gland**
2. thyroid

Correct Answer: 1
rationale:
Anterior pituitary gland is responsible for release of (GH) ,
Bone growth is under the influence of growth hormone from the anterior pituitary gland and sex hormones from the ovaries and testes

Question 113

What portion of the *adult* knee meniscus is vascularized

1. **Outer edges**
2. Inner edges
3. The entire meniscus is vascular
4. The entire meniscus is avascular

Correct Answer: 1
rationale:
During the first year of life the meniscus contains blood vessels throughout its body but when the weight bearing starts the vascularity and the circulatory network diminish and only 25-33% area remain vascular by the capillaries of the capsule and synovial membrane. The vascularity diminishes so much that in 4th decade of life only the periphery is vascular whereas the center of the menisci is avascular. The center portion is completely dependent upon the synovial fluid diffusion for nutrition. The central avascular portion of menisci either does not heal completely or heal at all after injury

Question 114

Hyaline cartilage innervated by

1. blood vessels
2. **synovial fluid**
3. epiphyseal growth

Correct Answer: 2
rationale:
Hyaline Cartilage is not innervated and therefore relies on diffusion to obtain nutrients by synovial fluid

Question 115

All is true about medial meniscus except:

1. **more injury than lateral meniscus**
2. absorb of about 90% of load on knee joint
3. injury from hyperextension

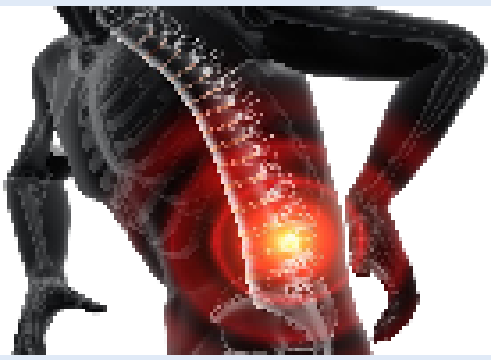
Correct Answer: 1
rationale:
1 is true because Injury to the medial meniscus is about 5 times more common than injury to the lateral meniscus.
2 is true because Knee loading can increase up to 550 % body weight during sudden balance loss, showing the importance of the interaction of the muscles. It is well known that the medial compartment takes significantly more load than the lateral one,
3 isn't true because Damage to the meniscus is due to rotational forces directed to a flexed knee

Question 116

Sensory end organs are absent from the:

1. muscle
2. periosteum
3. **synovial membrane**
4. articular cartilage
5. joint capsule

Correct Answer: 3
rationale:
Hyaline Cartilage is not innervated and therefore relies on diffusion to obtain nutrients by synovial fluid





Chapter

1.A Biomechanics



Question 1

Patient walking on slippery surface we teach him to do:

1- COG keeping close to the body

2- Wide base of support

3- Walking fast

Correct Answer: 1

rationale:

The closer the body's center of gravity is to its base of support, the more stable it is.

Question 2

The therapist is treating a 52-year-old woman after right total hip replacement. The patient complains of being self-conscious about a limp. She carries a heavy briefcase to and from work every day. The therapist notes a Trendelenburg gait during ambulation on level surfaces. What advice can the therapist give the patient to minimize gait deviation?

1- Carry the briefcase in the right hand.

2- Carry the briefcase in the left hand.

3- The patient should not carry a briefcase at all.

4- It does not matter in which hand the briefcase is carried.

Correct Answer: 1

rationale:

Trendelenburg gait is also seen after hip replacement surgery and femoral fixation with intramedullary nail. In patients with hip replacement, trendelenburg gait ensues due to the surgical dissection of the gluteus medius muscle during surgery to expose the hip joint; thus the dysfunction in the abductor muscles. at patient with a positive Trendelenburg sign the pelvis drops on one side " contralateral " , the trunk lurches to the other sides " ipsilateral ", in an effort to maintain balance so she can Carry the briefcase in the right hand to help in maintaining balance and decrease deviation .

Question 3

patient ask to go upstairs with crutch

1- Sound limb go up 1

2- Affected limb go up 1

Correct Answer: 1

rationale:

same answer as we described above in Q (23)

Question 4

A 36-year-old woman comes to the Physical Therapy Clinic with right heel pain for more than 3 months. The physical therapist decides to do a gait analysis to this patient. Which of the following is the most appropriate step length?

1- Distance between 2 consecutive foot flat.

2- Distance between 2 consecutive heel strikes.

3- Distance between 2 consecutive foot flat by the same leg.

4- Distance between 2 consecutive heel strikes by the same leg.

Correct Answer: 2
rationale:
The most basic spatial descriptors of gait include the length of a stride and the length of a step. Stride length is the distance between two successive heel contacts of the same foot and Step length, in contrast, is the distance between successive heel contacts of the two different feet. Comparing right with left step lengths can help to evaluate the symmetry of gait between the lower extremities.

Question 5

A 23 - year - old man presents to the clinic with pain in his right knee before 10 days ago. During treatment, the physical therapist observed a difficult knee joint flexion during motion at heel strike. What is the degree of knee flexion during normal gait?
1- 10
2- 20
3- 30
4- 40

Correct Answer: 2
rationale:
At heel contact the knee is flexed approximately 5 degrees, and it continues to flex an additional 10 to 15 degrees during the initial 15% of the gait cycle. This slight knee flexion, controlled by eccentric action of the quadriceps, serves the purpose of shock absorption and weight acceptance as body weight is progressively transferred to this lower extremity.

Question 6

A 30 - year - old man presents to the clinic with right foot pain 2 weeks ago and starts treatment. The physical therapist conducts a gait analysis to detect any gait abnormalities. What is the percentage of the swing phase in the normal gait cycle?
1- 20
2- 40
3- 60
4- 80

Correct Answer: 2
rationale:
At normal walking speed, the stance phase occupies approximately 60% of the gait cycle, and the swing phase occupies the remaining 40%.

Question 7

elderly person 65 year of age during his gait:
1- swing phase more than 40%
2- swing phase less than 40%
3- swing phase

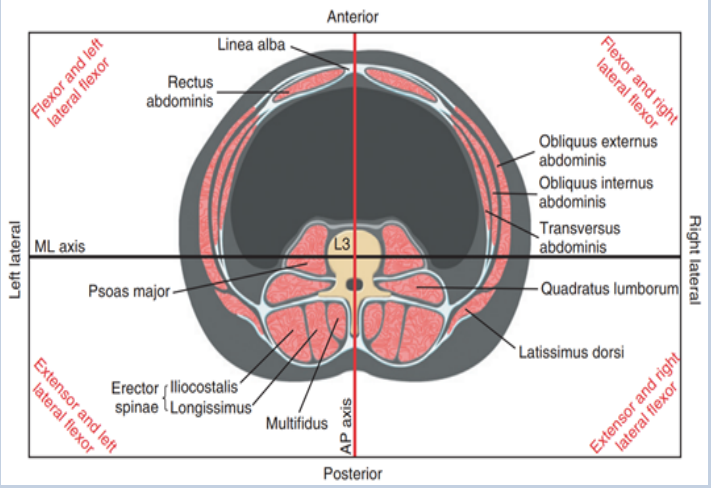
Correct Answer: 2
rationale:
Ageing is marked with cognitive decline, reduction in joint motion with reduction in acuity for auditory, vestibular, visual and somatosensory system results in increase in prevalence of gait disorders among older adults such as (the stance and gait base is widened, the swing phase will be limited or reduced) Because of the need for more stability due to weakness in muscles .

Question 8

What motion takes place in the lumbar spine with right lower extremity single limb support during the gait cycle?

1- Left lateral flexion
2- Right lateral flexion
3- Extension
4- Flexion

Correct Answer: 2
rationale:
Activation of both erector spinae (before heel contact to about 20% and from 45% to 70% of the gait cycle) and rectus abdominis (between 20% and 40% and again at between 70% and 90% of the gait cycle) and both muscles share function of ipsilateral lateral flexion as shown in fig below :



Question 9

While examining a patient's gait, the physical therapist observes muscle weakness in right side hip abductors and also shows lateral deviation of trunk as compensatory reaction. What is the compensatory gait the patient used?

1- Vaulting
2- Crouched
3- Scissoring
4- Trendelenburg

Correct Answer: 4
rationale:
Trendelenburg gait as we described above in Q2

Question 10

A physical therapist is ordered to provide gait training for an 18-year-old girl who received a partial medial meniscectomy of the right knee one day earlier. The patient was independent in ambulation without an assistive device before surgery and has no cognitive deficits. The patient's weight bearing status is currently partial weight bearing on the involved lower extremity. Which of the following is the most appropriate assistive device and gait pattern?

1- Crutches, three-point gait pattern.
2- Standard walker, three-point gait pattern.
3- Standard walker, four-point gait pattern.
4- Crutches, swing-to gait pattern.

Correct Answer: 1
rationale:
Partial meniscectomy patients usually need to use two crutches for only a few days. You should remember to put the involved foot flat on the ground, even when lightly weight bearing. Increase the weight on the foot as tolerated. You can advance to one crutch for the next few days and then a cane if needed. Most patients can be full weight bearing by the end of the first week. Note that two crutches with uninvolved limb considered 3 point gait.

Question 11

A physical therapist is evaluating a patient with muscular dystrophy. The patient seems to "waddle" when she walks. She rolls the right hip forward when advancing the right lower extremity and the left hip forward wheadvancing the left lower extremity. Which of the following gait patterns is the patient demonstrating?

1- Gluteus maximus gait
2- Dystrophic gait
3- Arthrogenic gait
4- Antalgic gait

Correct Answer: 2
rationale:
Waddling Gait occurs when there is extreme muscular weakness in the thigh and hip muscles as commonly found in pseudohypertrophic muscular paralysis and muscular atrophy or dystrophy. In this gait, the shoulders are thrown back, the lower section of the spine is lordotic, the pelvis is tilted greatly, and when walking, the subject swings from side and his hip drops with each step.

Question 12

A physical therapist is treating a 24-year-old woman with lumbar muscle spasm. Part of the patient's treatment involves receiving instruction on correct sleeping positions. Which of the following would be most comfortable?

1- Supine with no pillows under the head or extremities.
2- Prone with a pillow under the head only.
3- Sidelying with a pillow between flexed knees.

Correct Answer: 3
rationale:
Lower back pain makes it hard to get comfortable enough to fall asleep or may provoke night-time awakenings when pain surges. The best sleeping position for lower back pain (lumbar muscle spasm) is on your side with partial bending in knees , Keeping the knees bent helps balance the body and reduces pressure on the lumbar spine. and it sometimes helpful to put a small pillow between knees to make this position more comfortable.

Question 13

While gait training a patient following a cerebral vascular accident, you observe the knee on the affected side going into recurvatum during stance phase. The most likely cause of this deviation can be attributed to:

1- Severe spasticity of the hamstrings or weakness of the gastrocnemius- soleus.
2- weakness or severe spasticity of the quadriceps.
3- Weakness of the gastrocnemius-soleus or spasticity of the pretibial muscles.
4- Weakness of both the gastrocnemius-soleus and pretibial muscles.

Correct Answer: 2
rationale:
Genu recurvatum (GR), a gait pattern commonly seen in hemiplegic patients, is defined as full extension or hyperextension of the knee in the stance phase , Knee hyperextension can be caused by several causes. These include muscle weakness, especially of the muscles in the top of the thigh (quadriceps) .

TABLE 15.9 Gait Deviations at the Knee Secondary to Specific Knee Impairments

Observed Gait Deviation at the Knee	Likely Impairment	Selected Pathologic Precursors	Mechanical Rationale and/or Associated Compensations
Genu recurvatum during stance	Knee extensor weakness	Poliomyelitis	Secondary to progressive stretching of the posterior capsule of the knee.

Question 14

Max Knee flexion occur in any phase from this:

1- mid stance
2- toe off
3- mid swing
4- heel strike

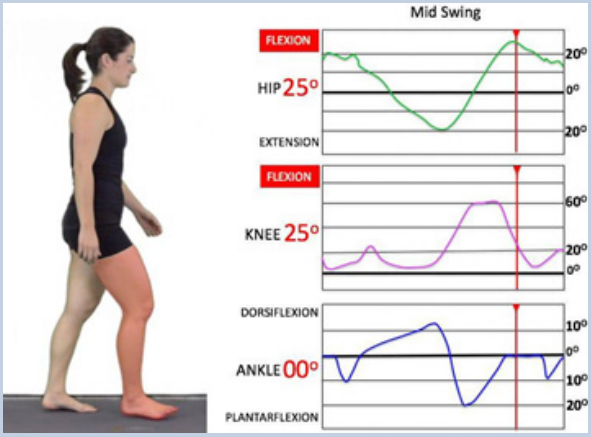
Correct Answer: 3
rationale:
Maximum knee flexion of approximately 60 degrees is assumed by the beginning of mid swing (75% of the gait cycle).
NOTE
5° flexion -> initial contact
20° flexion -> 15% of gait
knee approches near full extension -> (30% to 40%) of gait cycle
35° flexion -> toe off (60% of gait)
max flexion (60°) -> mid swing (75% of gait)

Question 15

A 43-year-old woman with walking dysfunction and inability to clear her foot from the ground during swing phase. The physical therapist assessed her gait and found the following in mid-swing: 25-degree hip flexion, 25-degree knee flexion, and 25-degree ankle planterflexion. Where is the expected dysfunctional site that needs intervention?

1- Hip
2- Knee
3- Ankle
4- Spine and pelvic

Correct Answer: 3
rationale:
25 hip flexion -> normal in mid swing
(Maximum flexion to 30 degrees is reached by continued iliacus action)
25 knee flexion -> normal in end of mid swing
(Relaxation of the flexor muscles allows the knee to extend passively and By the end of mid swing, knee flexion (30 degrees) equals that of the hip, and the tibia is vertical)
25 ankle plantarflexion -> abnormal in mid swing
(During the swing phase the ankle is again dorsiflexed to a neutral position to allow the toes to clear the ground).

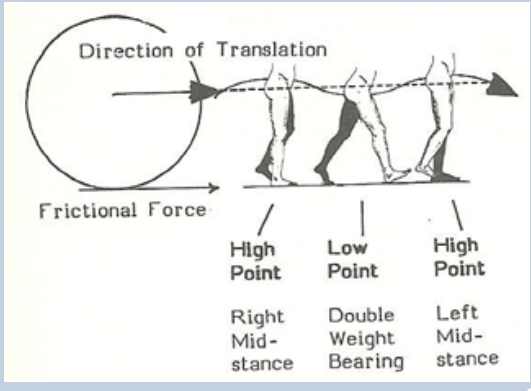


Question 16

C.O.G during locomotion observation can change the gait due to:

1- c.o.g go upward and oblique to stance phase.
2- downward during double limb support.
3- side to side oscillation.
4- move forward and backward.

Correct Answer: 4
rationale:
the most notable displacement of the body during gait is in the forward direction. Superimposed on this forward displacement, however, are two sinusoidal patterns of movement that correspond to the movement of the CoM in the vertical (upward during stance and downward during double limb support) and side-to-side directions.

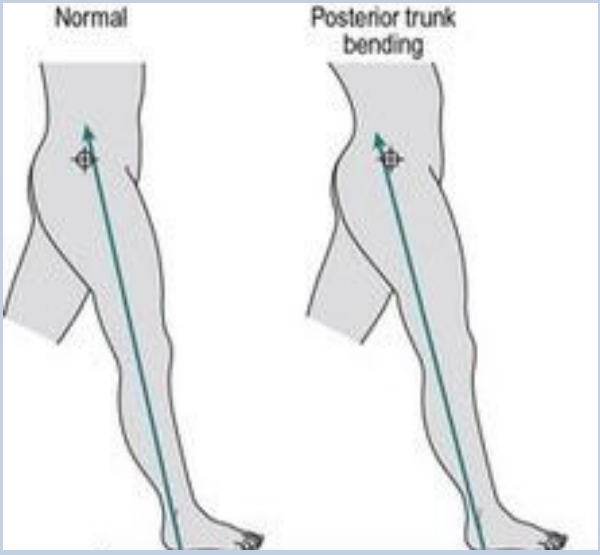


Question 17

During evaluation of a patient, the therapist observes significant posterior trunk lean at initial contact (heel strike). Which of the following is the most likely muscle that the therapist needs to focus on during the exercise session in order to minimize this gait deviation?

1- Gluteus medius
2- Gluteus Maximus
3- Quadriceps
4- Hamstrings

Correct Answer: 2
rationale:
Posterior trunk lean is done to compensate for ineffective (weak) hip extensors. The line of the ground reaction force early in the stance phase normally passes in front of the hip joint . This produces an external moment which attempts to flex the trunk forward on the thigh and is opposed by contraction of the hip extensors, particularly the gluteus maximus. Should these muscles be weak or paralysed, the subject may compensate by moving the trunk backwards at this time, bringing the line of action of the external force behind the axis of the hip joint



Question 18

A 53-year-old patient complains of difficulty in walking. Manual muscle testing reveals weak (grade 3) abductors of the left hip joint. Which of the following findings will be observed during the stance phase of the affected leg when the patient is asked to walk?

1- Lifting of the pelvis on the left side.
2- Lifting of the pelvis on the right side.
3- Dropping of the pelvis on the left side.
4- Dropping of the pelvis on the right side.

Correct Answer: 4
rationale:
The gluteus medius and minimus, the two primary hip abductors, are most active during the first 40% of the gait cycle, especially during single-limb support. The primary function of the abductors is to control the slight lowering of the contralateral pelvis on the side of the swing limb.

Question 19

Patient uses crutches and move the crutches forward and move his body to the crutches the gait is: OR patient move crutches first then move legs to them:

1- swing through
2- swing to
c- 2 gait point.

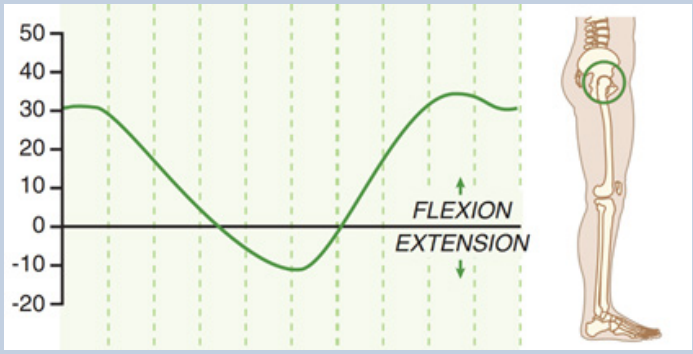
Correct Answer: 2
rationale:
the swing to gait is where the patient will move both crutches forward then they will hint swing or move both legs forward and place them at the placement of where the crutches are located.

Question 20

Which is not correct about hip during gait?

1- max hip flexion occur in terminal swing.
2- Slight hip extension before ic.
3- Hip extension occur during double limb support.

Correct Answer: 1
rationale:
Maximum flexion (slightly more than 30 degrees) is achieved by mid swing and maintained in terminal swing just before heel contact. Note that at heel contact the hip has already started to extend in preparation for weight acceptance.

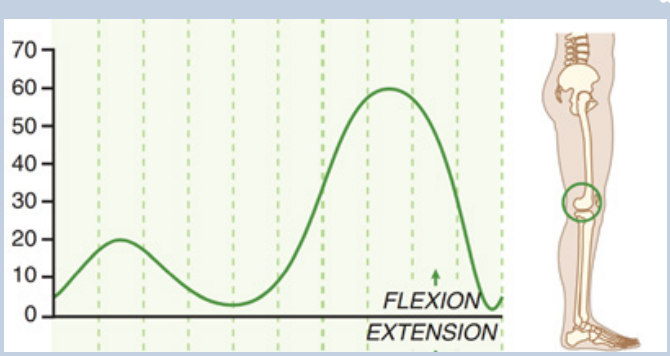


Question 21

When evaluating knee in gait cycle it maximum flexion is during?

1- initial contact
2- feet flat
3- mid swing
4- terminal swing

Correct Answer: 3
rationale:
Maximum knee flexion of approximately 60 degrees is assumed by the beginning of mid swing (75% of the gait cycle).

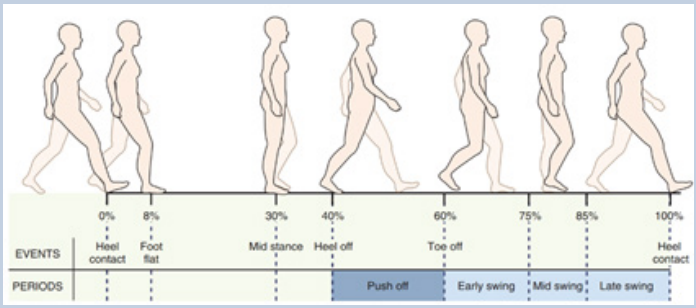


Question 22

what is not a part of stance phase?

1-deceleration
2-midstance
3-intial contact
4-preswing

Correct Answer: 1
rationale:
subdivision of stance phase -> " 1) Initial Contact – 2) Loading Response – 3) Midstance – 4) Terminal Stance – 5) Pre swing "



Question 23

which statement correct during training with cane up and down stairs

1- unaffected leg put first during upstairs.
2- unaffected leg put down during up stairs.
3- affected leg put first in down stairs.

Correct Answer: 1
rationale:
in ascending stairs , Going up the stairs lead with your uninvolved leg and followed by involved leg and in descending stairs , Going down the stairs lead with your involved leg and followed by uninvolved leg . For example, if right lower extremity affected , during ascending Patient leads with left leg on step and Right leg and cane follows and during descending patient Lead with cane and right leg on step below and Left leg follows.

Question 24

Patient walk using crutches , patient first make two elbow crutches move forward then , make two lower limbs advance?

1- swing two
2- swing through
3- 2 point gate
4- 4 point gate

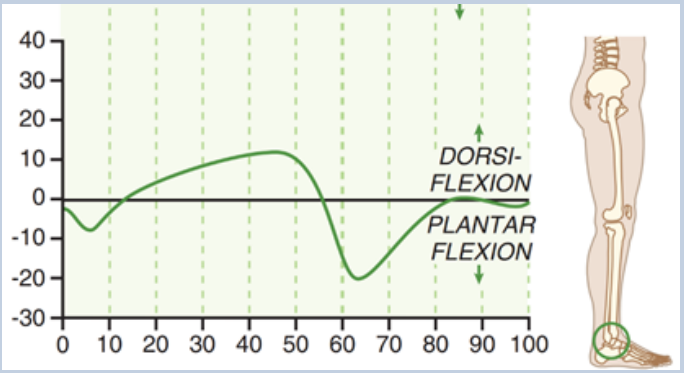
Correct Answer: 2
rationale:
the swing through gait is where th patient will move both crutches forward then the will move both legs forward and they will swing it pass the crutches placement.

Question 25

Which of the following is the most appropriate position of the ankle during the first phase of the gait cycle?

1- Neutral position
2- 5-degree dorsiflexion
3- 5-degree plantarflexion
4-10-degree plantarflexion

Correct Answer: 3
rationale:
At the ankle, heel contact (first phase of gait) occurs with the talocrural joint in a slightly plantar flexed position (between 0 and 5 degrees)



Question 26

All of the following in the gait cycle except:

1- Swing phase
2- Mid swing
3- Initial contact
4- Heal flat

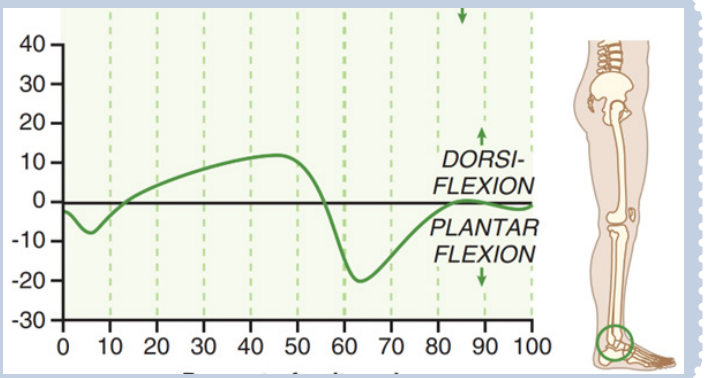
Correct Answer: 4
rationale:
Phases of the Gait Cycle -> " 1) Initial Contact – 2) Loading Response – 3) Midstance – 4) Terminal Stance – 5) Pre swing – 6) Initial Swing – 7) Mid Swing – 8) Late Swing "

Question 27

In the terminal swing phase of gait, what muscles of the foot and ankle are active?

1- Extensor digitorum longus
2- Gastrocnemius
3- Tibialis posterior
4- B and C

Correct Answer: 1
rationale:
During the swing phase the ankle is again dorsiflexed to a neutral position to allow the toes to clear the ground
Answer (a) -> ankle dorsiflexor
Answer (b and c) -> ankle plantar flexors





Question 28

Which of the following types of gait training are most likely described as both crutches moved forward together, both lower limbs then move towards the crutches?

1- 2-point
2- 4-point
3- Swing to
4- Swing through

Correct Answer: 3
rationale:
the swing to gait is where the patient will move both crutches forward then they will hint swing or move both legs forward and place them at the placement of where the crutches are located.

Question 29

Evaluating the gait of a patient with left hemiplegia, you note toe drag during mid swing on the left. The least likely cause of this deviation would be:

1- inadequate concentric activity of the ankle dorsiflexors.
2- excessive extensor synergy.
3- knee and ankle joint pain.
4- decreased proprioception.

Correct Answer: 3
rationale:
Hemiplegic gait, This gait pattern is described as unilateral weakness on the affected side with weakness in flexion (excessive extensor synergy) and dorsiflexion (inadequate concentric activity of ankle dorsiflexors). This causes the person to swing the paraplegic leg outwards and in a circular motion in order to bring the leg forward.

Question 30

Which muscle bilateral weakness patient use bilateral crutches

1- Gluteus medius
2- Gluteus maximus
3- Quadriceps
4- Hamstrings

Correct Answer: 2
rationale:
The Gluteus Maximus (GM) muscle is the largest and most powerful in the human body. It plays an important role in optimal functioning of the human movement system as well as athletic performance , It is an important muscle for activities of daily living and allow us to be able to maintain an upright erect posture. so bilateral weakness of gluteus maximus will need bilateral crutches because it will affect greatly the normal mechanics of gait cycle.

Question 31

If there is weakness of right gluteus medius dropping pelvis of occurs at

1- right pelvic drop at stance phase.
2- left pelvic drop at stance phase.
3- right pelvic drop at swing phase.
4- left pelvic drop at swing phase.

Correct Answer: 4
rationale:
The gluteus medius and minimus, the two primary hip abductors, are most active during the first 40% of the gait cycle, especially during single-limb support. The primary function of the abductors is to control the slight lowering of the contralateral pelvis on the side of the swing limb.

Question 32

Quadriceps function in swing phase:

1- accelerated knee flexion.
2- prevent excessive knee flexion.

Correct Answer: 2
rationale:
the quadriceps is active in the very late stage of the swing phase in preparation for heel contact . The major burst of activity, however, occurs shortly after heel contact. The function of the quadriceps at this time is to control the knee flexion that takes place in the first 10% of the gait cycle. Eccentric activation serves to cushion the rate of weight acceptance on the lower extremity (i.e., shock absorption) and to prevent excessive knee flexion.

Question 33

When comparing the gait cycle of young adults to the gait cycle of older adults, what would a therapist expect to find?

1- The younger population has a shorter step length.
2- The younger population has a shorter stride length.
3-The younger population has a shorter period of double support.
4- The younger population has a decrease in speed of ambulation.

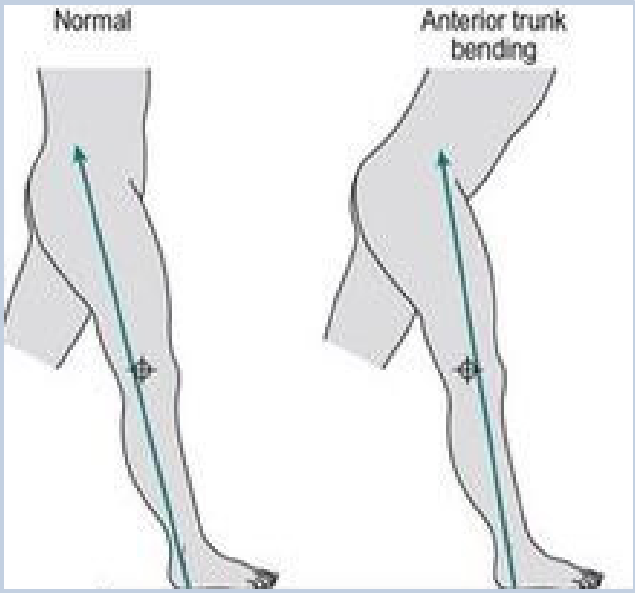
Correct Answer: 3
rationale:
Ageing is marked with cognitive decline, reduction in joint motion with reduction in acuity for auditory, vestibular, visual and somatosensory system results in increase in prevalence of gait disorders among older adults such as (the stance and gait base is widened, bipedal floor contact is prolonged " increase time spent in double stance phase " , step length becomes shorter, the feet are lifted less high during the swing phase, walking becomes slower " decrease the speed of ambulation " and the posture becomes stooped.

Question 34

A physical therapist begins gait training for a patient with bilateral knee flexion contractures at 30o at a long-term care facility. The therapist knows that the patient will have a forward trunk lean during gait because:

1- The patient's line of gravity is anterior to the hip.
2- The patient's line of gravity is anterior to the knee.
3- The patient's line of gravity is anterior to the ankle.
4- A and C.

Correct Answer: 2
rationale:
Anterior trunk lean is done to compensate for an inadequacy of the knee extensors , the line of action of the ground reaction force vector normally passes behind the axis of the knee joint and generates an external moment which attempts to flex it . anterior trunk bending is used to move the centre of gravity of the body forwards, which results in the line of force passing in front of the axis of the knee, producing an external extension (or hyperextension) moment .



Question 35

Which of these joints is classified as a biaxial diarthrosis?

1- the metacarpophalangeal joint
2- the elbow joint
3- the hip joint
4- the pubic symphysis

Correct Answer: 1
rationale:
A freely mobile joint is classified as a diarthrosis. These types of joints include all synovial joints of the body, which provide the majority of body movements. These joints are divided into three categories, based on the number of axes of motion provided by each (uniaxial – biaxial – multiaxial)
A biaxial joint allows for motions within two planes. An example of a biaxial joint is a metacarpophalangeal joint (knuckle joint) of the hand. The joint allows for movement along one axis to produce bending or straightening of the finger, and movement along a second axis, which allows for spreading of the fingers away from each other and bringing them together.

Question 36

At what point in the gait cycle is the center of gravity the lowest?

1- Double support.
2- Terminal swing.
3- Deceleration.
4- Midstance.

Correct Answer: 1
rationale:
In the vertical direction, the CoM oscillates up and down to describe two full sine waves per gait cycle , Minimum height of the CoM occurs at the midpoint of both periods of double-limb support (5% and 55% of the gait cycle).

Question 37

patient during walking raise his hip to clear toes from ground due to:

1- paralysis of dorsiflexors
2- paralysis of planter flexors
3- paralysis of quadriceps

Correct Answer: 1
rationale:
A person with foot drop is unable to raise their foot at the ankle , the muscles affected are the dorsi flexors (paralysis of ankle dorsi flexors) . a person suffering from foot drop may have to excessively lift their hip to compensate for the inability to clear the foot and toes from ground .

Question 38

An outpatient physical therapist is gait training a patient recently discharged from the hospital. The inpatient therapist's notes describe a decrease in left stride length due to pain with weight bearing on the right lower extremity. The outpatient therapist knows that the patient's gait deviation is:

1- An abnormally short distance from the left heel strike and the successive right heel strike.
2- An abnormally short amount of time between the left heel strike and the successive right heel strike.
3- An abnormally short amount of time in stance phase on the left lower extremity.
4- An abnormally short distance between the left heel strike and the successive left heel strike.

Correct Answer: 4
rationale:
A stride (synonymous with a gait cycle) is the sequence of events taking place between successive heel contacts of the same foot.



Question 39

use of can in contralateral hand will:

1- not affected gait pattern.
2- reduce energy expenditure in normal person.
3- decrease the floor reaction force on the opposite foot.
4- not affected floor reaction force on the opposite foot.

Correct Answer: 3
rationale:
A cane held in the contralateral hand reduces joint reactive forces through the affected hip up to 50% by reducing abductor muscle pull with significant reduction in peak vertical ground reaction force on the symptomatic limb (opposite foot).

Question 40

As a result of a gait analysis, a therapist has determined that the patient ambulates with excessive foot pronation. This deviation would not occur as a result of:

1- compensated forefoot varus
2- internal tibial rotation
3- a weak tibialis posterior
4- excessive ankle dorsiflexion

Correct Answer: 4
rationale:
commonly accepted compensation for a mild lack of ankle dorsiflexion is increased foot pronation. With or without toeing-out, excessive foot pronation can lead to greater stresses applied to the supporting soft-tissue structures of the plantar aspect of the foot.

Question 41

joint play R O M which occurs not named with:

1- necessary to completion of joints ROM
2- Passively occur
3- under voluntary control

Correct Answer: 3
rationale:
By nature, joint play which is essential for the proper functioning of normal, painless active and passive range of movements is an involuntary movement that is inherent to the musculoskeletal system and cannot be introduced by voluntary muscles. It provides roll, glide, distraction, and spin combinations for joint motion and occurs in the shape of the joint surfaces.

Question 42

Normal ROM of hip joint:

1- extension 35 --- flexion 125
2- extension 10 ---- flexion 125
3- extension 0 -----flexion 125

Correct Answer: 2
rationale:
1-Hip flexion range of motion (ROM) for adults is 120 degrees according to the American Academy of Orthopedic Surgeons (AAOS) . The hip flexion with the knee flexed will permit a range of motion of approximately 115° to 125°.
2- Beyond the midline the normal extension of the hip is of 10° to 15°

Question 43

When the knee is at its maximal amount of flexion during the gait cycle, which of the following muscles are active concentrically?
1- Hamstrings
2- Gluteus maximus
3- Gastrocnemius
4- All of the above.

Correct Answer: 1
rationale:
Maximum knee flexion mean that knee flexors are the concentric group of muscles and knee extensors are the eccentric group of muscles and main knee flexors are hamstrings.

Question 44

Patient complaining of knee pain, the therapist decides to test the available movement in the joint, what is this type of joint play called?
1- Manipulation
2- Mobilization
3- Arthrokinematics test
4- Osteokinematics fest

Correct Answer: 3
rationale:
arthrokinematics is the general term for the specific movements of joint surfaces . Normal joint surface movement is necessary to ensure long-term joint integrity and joint surface movement are sometimes called joint play motion or component motions .

Question 45

The number of joint and bones in the foot
1- 23 joint and 25 bones
2- 33 joint and 26 bones
3- 23 joints and 17 bones

Correct Answer: 2
rationale:
The foot is a complex mechanical structure of the human body composed of 33 joints, 26 bones, and more than a hundred muscles, tendons, and ligaments that all work together to bear weight, allow for locomotion, and transmit force.



Question 46

Knee joint ROM:
1- flexion 110 – extension 0
2- flexion 130 – extension 0
3- flexion 140 – extension 0

Correct Answer: 3
rationale:
The knee is a hinge joint and primarily only moves in one plane of movement, flexion and extension. A straight leg is at 0 degrees. A flexed knee is at about 140 degrees. It is a common misconception that 0-140 degrees is a normal range of motion, but most people have some hyperextension, too, taking their ROM into negative numbers.

- normal ROM is usually minus 5 to 143 degrees in women
- normal ROM is usually minus 6 to 140 degrees in men.

Question 47

elbow joint:
1- uniaxial
2-biaxial
3- three axial
d-multi-axial

Correct Answer: 1
rationale:
uniaxial joint only allows for a motion in a single plane (around a single axis). The elbow joint, which only allows for bending or straightening, is an example of a uniaxial joint.

Question 48

which is synovial joint:
1- TMJ joint
2- Symphysis pubis

Correct Answer: 1
rationale:
1-TMJ joints is a synovial joint is formed by the articulation of the mandible and the temporal bone of the cranium. the articular surfaces of the bones never come into contact with each other – they are separated by an articular disk. This disk splits the joint into two synovial joint cavities, each lined by a synovial membrane. The articular surface of the bones are covered by fibrocartilage, not hyaline cartilage.
2- symphysis pubis isn`t synovial joint it is a secondary cartilaginous joint.

Question 49

all of the following are synovial joints except:

1- Elbow joint
2- Knee joint
3- Hip joint
4-Symphysis pubis

Correct Answer: 4
rationale:
The pubic symphysis is a secondary cartilaginous joint (a joint made of hyaline cartilage and fibrocartilage) located between the left and right pubic bones near the midline of the body.

Question 50

Spyndolythesis:

1- vertbera above vertebra dislocat posterior.
2- vertbera below vertebra dislocat posterior.
3- vertbera above vertebra dislocat anterior.
4- vertbera belowe vertebra dislocat anterior.

Correct Answer: 3
rationale:
Spondylolisthesis is where one of the bones in your spine, called a vertebra, slips forward (anterior). It can be painful, but there are treatments that can help. It may happen anywhere along the spine but is most common in the lower back.

Question 51

Brodie abscess in osteomyelitis what stage?

1- Acute
2- Subacute
3- Chronic

Correct Answer: 2
rationale:
Brodie abscess is referred to as the most common subacute form of osteomyelitis.

Question 52

Osteomyelitis abscess spread:

1- Acute stage
2- Sub-acute stage
3- Chronic stage

Correct Answer: 2
rationale:
Brodies abscess is a suppurative collection of necrosis that is encapsulated by granulation tissue in the setting of subacute osteomyelitis.



Question 53

The joint between adjacent vertebrae that includes an in vertebral disc is classified as which type of joint?

1- multiaxial
2- synarthrosis
3- amphiarthrosis
4- diarthrosis

Correct Answer: 3
rationale:
An amphiarthrosis is a joint that has limited mobility. An example of this type of joint is the cartilaginous joint that unites the bodies of adjacent vertebrae. Filling the gap between the vertebrae is a thick pad of fibrocartilage called an intervertebral disc. Each intervertebral disc strongly unites the vertebrae but still allows for a limited amount of movement between them. However, the small movements available between adjacent vertebrae can sum together along the length of the vertebral column to provide for large ranges of body movements.

Question 54

Which of these joints is classified as a synarthrosis?

1- the epiphyseal plate
2- the shoulder joint
3 the pubic symphysis
4- an in vertebral disc

Correct Answer: 1
rationale:
primary cartilaginous joints, also known as synchondroses, only involve hyaline cartilage. These joints may be slightly mobile (amphiarthroses) or immobile (synarthrosis). A temporary form of synarthrosis joint called an epiphyseal (growth) plate, is one where the cartilage is converted into bone before adult life. Such joints are found between the epiphyses and diaphysis of long bones, between the occipital and the sphenoid bones, and during the early years of life, between the petrous portion of the temporal and the jugular process of the occipital bone. The epiphyseal plate is a hyaline cartilage plate in the metaphysis at each end of a long bone.

Question 55

Synovial joints _____.

1- are joints where the bones articulate with each other within a fluid filled joint cavity
2- may be functionally classified as amphiarthroses.
3- may be functionally classified as synarthroses.
4- are joints where the bones are anchored to each other by fibrocartilage.

Correct Answer: 1
rationale:
Answers b, c and d describe cartilaginous joint.

Question 56

Which of the following is not an example of a synarthrodial joint in the body?

1- Coronal suture
2- The fibrous joint between the shaft of the tibia and fibula.
3- Symphysis pubis
4- Metacarpophalangeal

Correct Answer: 4
rationale:
1-Metacarpophalangeal joint is a condyloid joint is classified as a biaxial diarthrosis .
2- Synarthrodial joint is a fibrous joint which mean A union of two bones by fibrous tissue such that there is no joint cavity and almost no motion possible; the types of fibrous joints are sutures, syndesmoses, and gomphoses. Synonym(s): immovable joint.

Question 57

Myositis ossificans most affected:

1- wrist Joint
2- Elbow Joint
3- Shoulder Joint
4- Knee Joint

Correct Answer: 2
rationale:
The most common sites affected are the hip, anterior thigh (quadriceps) and anterior arm (brachialis) . Hip and elbow the common site but if the question mentions the hip chose the hip is the most common.

Question 58

Treatment of myositis ossificans?

1- Active and active assisted movement
2- massage
3- strengthening ex
4- stretching ex

Correct Answer: 1
rationale:
Rehabilitation of patients with myositis should include restoration of ROM, strength and proprioception by Active and active assisted movement. Heat, continuous ultrasound, massage, stretching, and strengthening exercise should be avoided initially to prevent further bleeding in the area that may promote the progression to MO. Protective padding should be worn to prevent further injury to the area.



(*VIP*)

Question 59

Sudeck's atrophy which is not true:

1- sever burning pain
2- sympathetic hyperactivity
3- trophic changes occur
4- more perspiration

Correct Answer: 2
rationale:
Complex regional pain syndrome (CRPS), also known as Sudeck atrophy, is a condition that can affect the extremities in a wide clinical spectrum. (CRPS) is a broad term describing excess and prolonged pain and inflammation that follows an injury to an arm or leg. CRPS has acute (recent, short-term) and chronic (lasting greater than six months) forms. CRPS used to be known as reflex sympathetic dystrophy (RSD) and causalgia. People with CRPS have changing combinations of spontaneous pain or excess pain that is much greater than normal following something as mild as a touch. Other symptoms include changes in skin color, temperature, and/or swelling on the arm or leg below the site of injury. Although CRPS improves over time, eventually going away in most people, the rare severe or prolonged cases are profoundly disabling. Most CRPS illnesses are caused by improper function of the peripheral C-fiber nerve fibers that carry pain messages to the brain. Their excess firing also triggers inflammation designed to promote healing and rest after injury.

Question 60

Osteomyelitis is:

1- infective inflammation of the bone due infective bacteria entered into bone.
2- Decreased bone density and increased porosity.
3- due to repetitive trauma.
4- Vitamin D deficiency.

Correct Answer: 1
rationale:
Osteomyelitis is an infection in a bone. Infections can reach a bone by traveling through the bloodstream or spreading from nearby tissue. Infections can also begin in the bone itself if an injury exposes the bone to germs. most cases of osteomyelitis are caused by staphylococcus bacteria, types of germs commonly found on the skin or in the nose of even healthy individuals.

Question 61

A 27-year-old man presents to the Physical Therapy Department with pain, redness and fever on the leg. He has been diagnosed with osteomyelitis (see report). X-ray: Brodies abscess. Which is the following is the most likely form of osteomyelitis?

1- Acute
2- Chronic
3- Subacute
4- Tubercular

Correct Answer: 3
rationale:
Brodies abscess is a suppurative collection of necrosis that is encapsulated by granulation tissue in the setting of subacute osteomyelitis.

Question 62

An 11-year-old patient with post-fracture stiffness of the elbow has been referred to a therapist for mobilizing exercise. Which of the following risks make the physician strongly recommends against forced passive movements?

1- Ulnar nerve injury
2- Myositis ossificans
3- Brachial artery injury
4- Refracturing the bone

Correct Answer: 2
rationale:
In treatment of MO Contusions and strains are vulnerable to additional trauma during the first couple of weeks.so physician Avoid forced passive movements, heavy activity, massage, and forceful stretching.

Question 63

32 - year - old patient requires a physical therapy assessment to start the postoperative rehabilitation program. A surgeon performs partial osteotomy of the right shaft of the femur to correct its alignment. Which of the following describes the procedure of osteotomy?

1- Internal fixation of a bone.
2- removal of the periosteum.
3- Operative sectioning of a bone.
4- Surgical closure of an epiphysis.

Correct Answer: 3
rationale:
An osteotomy is a surgical procedure that involves cutting bone (and sometimes adding bone tissue) to reshape or realign your bones. Femoral osteotomy This surgery involves cutting and aligning your upper thigh bone (femur) to help restore hip function.

Question 64

most common site of estrogenic sarcoma is:

1- Femur, upper end
2- Femur, lower end
3- Tibia, upper end
4- Tibia, lower end



Correct Answer: 2
rationale:
Osteosarcoma is a bone tumor and can occur in any bone, usually in the extremities of long bones near metaphyseal growth plates. The most common site is Femur (42%, 75% of which are in the distal femur (lower end).

Question 65

Infection of microorganism lead to:
1- osteomyelitis
2- Rocket
3- Equinovarus

Correct Answer: 1
rationale:
Osteomyelitis is an infection in a bone. Infections can reach a bone by traveling through the bloodstream or spreading from nearby tissue.
Infections can also begin in the bone itself if an injury exposes the bone to germs. Most cases of osteomyelitis are caused by staphylococcus bacteria, types of germs commonly found on the skin or in the nose of even healthy individuals.

Question 66

Which of the following scientific terms is most likely used to describe an involuntary movement response to the stimulus that serves to maintain normal alignment of the head and body in space?
1- Balance
2- Motor program
3- Postural control
4- Righting reaction

Correct Answer: 3
rationale:
Postural control is the ability to maintain our body in space achieving both goals of stability and orientation.

Question 67

All of the following is clinical features in fracture neck of femur except
1- pain in hip region
2- abduction of leg
3- shortness of the limb
4- lateral rotation of the leg

Correct Answer: 2
rationale:
the clinical feature of Neck femur fracture is adduction with lateral rotation in the leg.

Question 68

A 27-year-old man presents to the Physical Therapy Department with pain and swelling (see report). X-ray: Burst fracture.
Which of the following is the most likely affected area?
1- Talus
2- Scaphoid
3- Vertebral
4- Femoral head

Correct Answer: 3
rationale:
Burst fractures are a type of compression fracture related to high-energy axial loading spinal trauma that results in disruption of a vertebral body endplate and the posterior vertebral body cortex.

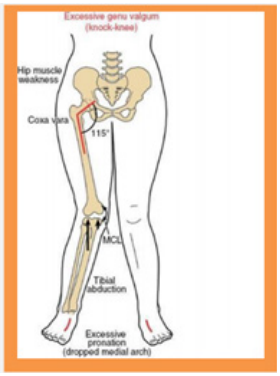
Question 69

The angle between Neck of femur and Shaft of femur :
1- 90 Degree
2- 120 Degree
3-150 Degree
4-170 Degree

Correct Answer: 2
rationale:
The femoral neck-shaft angle (NSA) or caput-collum-diaphyseal (CCD) angle is one of the most frequently applied measurements to assess hip morphology, in particular, the relation of the femoral shaft to the femoral head-neck axis. A known normal range of the caput-collum-diaphyseal (CCD) angle is generally considered 125°-135° 3, with a global mean of 126.4° and standard deviations are approximately 5.6° measured with a 360° goniometer in anatomical studies .Widely considered cut-off values are the following :
coxa Valga: >140 coxa Vara: <120°

Question 70

this angle in the picture means:
1- genovalgum
2- coxavalgum
3- coxavara
4- genurecrvatum



Correct Answer: 3
rationale:
Coxa Vara describes a deformity of the hip where the femoral neck-shaft angle is decreased, usually defined as less than 120 degrees.



Question 71

What is the approximate angle between the femoral shaft and neck in a healthy person?

1- 90°
2- 125°
3- 150°
4- 170°

Correct Answer: 2
rationale:
The femoral neck-shaft angle (NSA) or caput-collum-diaphyseal (CCD) angle is one of the most frequently applied measurements to assess hip morphology, in particular, the relation of the femoral shaft to the femoral head-neck axis. A known normal range of the caput-collum-diaphyseal (CCD) angle is generally considered 125°-135° 3, with a global mean of 126.4° and standard deviations are approximately 5.6° measured with a 360° goniometer in anatomical studies.

Question 72

which statement not correct about fracture neck of femur?

1- its fatal
2- need arthoplasty
3- lead to avascular necrosis
4- occur mostly in children

Correct Answer: 4
rationale:
Femoral neck fractures are a subset of proximal femoral fractures. The femoral neck is the weakest part of the femur. Since disruption of blood supply to the femoral head is dependent on the type of fracture and causes significant morbidity, the diagnosis and classification of these fractures is important. There are three types:
1-subcapital: femoral head/neck junction.
2- transcervical: midportion of femoral neck.
3- basicervical: base of femoral neck. Though anyone can fracture their femoral neck, it's considerably more common in elderly adults who have poor bone density. More than 90 percent of these fractures occur in people older than 50. They are more common in women.

Question 73

A 67-year-old patient with lower back pain referred to the Physical Therapy Clinic for assessment and treatment. The therapist decided to start with the 1st performance of the pelvic tilt exercise. Which of the following positions was performed?

1- Sitting
2- Standing
3- Lying supine on a firm surface
4- Lying supine on a soft surface

Correct Answer: 3
rationale:
The pelvic tilt exercise is performed with the patient supine on a firm surface; the hips are flexed to 45°, knees flexed to 90°; the patient is to then tilt the pelvis posteriorly, flattening the lumbar spine without raising their buttocks off the examining table or floor.

Question 74

which fracture is treated by traction?

1- femur fracture
2- ankle fracture
3- knee fracture

Correct Answer: 1
rationale:
The main goal of a traction splint is to provide stability to a femur fracture. When people suffer a severe femur fracture, such as that in an auto accident, there are major complications which might develop. Some of these include:

- •The development of a deep vein thrombosis
- •An acute hemorrhage which can lead to a life-threatening amount of blood loss
- •Severe pain
- •Nerve damage

The job of traction is to try and prevent this from happening. In a traction splint, the bone fragments are stabilized. This prevents the fragments from moving around, possibly damaging important structures in the area. Furthermore, even though traction might be painful at first, the fracture is, ultimately, stabilize. This will make the individual more comfortable.

Question 75

Right convex Scoliosis we find:

1- Lateral trunk flexion & concave side to RT.
2- Lateral trunk flexion &convex side to RT.
3- Lateral trunk flexion &convex side to LT.
4- Lateral trunk flexion & shorten RT musculature.

Correct Answer: 2
rationale:
Scoliosis is defined as an abnormal lateral curvature of the spine. It is quite common in young individuals and is often idiopathic and asymptomatic. Each curve of a scoliosis can be described in terms of the direction of convexity as levoscoliosis: curvature towards the left dextroscoliosis: curvature towards the right. It gives convex curvature to RT (Right Thoracic) as in the figure





Question 76

Which of the following is a possible cause of anterior pelvic tilting?

1- Weak lateral trunk muscles.
2- Tightened tensor fascia lata.
3- Tight hamstring.
4- Severe weakness of abdominal muscle.

Correct Answer: 4
rationale:
In addition to hip flexor tightening, anterior pelvic tilt can also result from weak abdominal muscles . Abdominal muscles help support a healthy pelvic tilt. If they are not strong enough to keep the pelvis in this alignment, it could easily tilt forward.

Question 77

incorrect about poor posture:

1- need surgery and physiotherapy.
2- spinal and frozen shoulder problems.
3-pain and affect strength of muscles.

Correct Answer: 1
rationale:
Absolutely, poor posture can be improved and corrected through physical therapy except some extreme cases of posture may need more invasive interventions, such as surgery. B and C are true because people have poor posture, you may experience symptoms like: Back pain -Muscle fatigue - Body aches – Migraines - Gait issues -Rounded shoulder and frozen shoulder.

Question 78

A physical therapist is speaking to a group of receptionists about correct posture. Which of the following is incorrect information?

1- Position computer monitors at eye level.
2- Position seats so that the feet are flat on the floor while sitting.
3- Position keyboards so that wrists are approximately in 20o extension.
4- Take frequent stretching breaks.

Correct Answer: 3
rationale:
1-Wrist Extension is a Bad posture is one of the most common problems typing posture, this occurs when your keyboard is higher than your wrists, requiring you to hyper extend your wrists to press the keys. And this may lead to wrist strain.
2- The ideal typing posture is such that the keyboard is below your elbow height when seated, with your wrists straight and not planted on the surface of the desk. Maintain a straight, neutral wrist as you type.

Question 79

The ability of the posture to modify external environment and preserve alignment of one body part to another?

1- Postural control
2- Strength
3- Balance

Correct Answer: 1
rationale:
Postural control is a term used to describe the way our central nervous system (CNS) regulates sensory information from other systems in order to produce adequate motor output to maintain a controlled, upright posture. The visual, vestibular, and somatosensory systems are the main sensory systems involved in postural control and balance.
Postural orientation and equilibrium are two main functional goals of postural control.

Question 80

which is incorrect about plump line of body in standing:

1- pass posterior to ear lobe.
2- pass through shoulder.
3- pass Posterior to greater trochanter.
4- pass anterior to ankle.

Correct Answer: 1
rationale:
It passes through the lobe of the ear, through the shoulder joint, posterior to the hip joints, and anterior to the knee and ankle.

Question 81

While assessing the standing posture of a patient, the therapist notes that a spinous process in the thoracic region is shifted laterally. The therapist estimates that T2 is the involved vertebra because he or she notes that it is at the approximate level of the:

1- Inferior angle of the scapula.
2- Superior angle of the scapula.
3- Spine of the scapula.
4- Xiphoid process of the sternum.

Correct Answer: 2
rationale:
- Superior angle of the scapula → at level of T2
- Spine of the scapula → at level of T3
- Inferior angle of the scapula → at level of T7



Question 82

Shoulder abduction and adduction movement occur in

1- sagittal plane
2- scapular plane
3- coronal plane (frontal plane)
4- thoracic plane

Correct Answer: 3
rationale:
Frontal Plane (coronal plane): divides the body into front and back. Movements in this plane are sideways movements, called abduction and adduction.
In GH joint (shoulder joint) Abduction and adduction are traditionally defined as rotation of the humerus in the near frontal plane around an axis oriented in the near anterior-posterior direction.

Question 83

In which of the following axis and plane on the human body, flexion and extension motion will occur?

1- Around the sagittal axis and in the coronal plane.
2- Around the coronal axis and in the sagittal plane.
3- Around the coronal axis and in the coronal plane.
4- Around the longitudinal axis and in the transverse plane.

Correct Answer: 2
rationale:
-Sagittal Plane: passes through the body front to back, so dividing it into left and right.
-Frontal axis (coronal axis) - passes horizontally from left to right and is formed by the intersection of the frontal and transverse planes.
-Movements in sagittal plane around coronal axis are the up and down movements of flexion and extension.

Question 84

Human body flexion and extension which plane occurs?

1- axial to transverse plane.
2- sagittal plane to coronal axis.

Correct Answer: 2
rationale:
-Sagittal Plane: passes through the body front to back, so dividing it into left and right.
-Frontal axis (coronal axis) - passes horizontally from left to right and is formed by the intersection of the frontal and transverse planes.
-Movements in sagittal plane around coronal axis are the up and down movements of flexion and extension.

Question 85

Internal rotation and external rotation on which plane can performed:

1- Longitudinal plane around horizontal axis.

2- Axial plane around vertical axis.

Correct Answer: 2

rationale:

Transverse Plane: or axial plane (also called the horizontal plane) This plane divides the body into top and bottom. Movements in this plane are rotational in nature, such as internal and external rotation, pronation and supination.

internal and external rotation at the joint is defined as an axial rotation in the horizontal plane occurs around vertical or longitudinal axis.

Question 86

A 22-year-old patient with weak shoulder flexor. The physical therapist decided to give him some exercises to strengthen the shoulder flexor muscle in standing. Which of the following planes of motion is the most appropriate to do these exercises?

1- Frontal

2- Sagittal

3- Coronal

4- Transverse

Correct Answer: 2

rationale:

The sagittal plane divides the body vertically into equal right and left halves. This plane is also referred to as the midsagittal plane because it is on the midline of the body. Movements within the sagittal plane are flexion and extension.

Question 87

Plane divides the body vertically?

1- frontal axis

2- transverse

Correct Answer: 1

rationale:

The coronal plane divides the body vertically into equal front (anterior) and back (posterior) portions. The coronal plane, (also referred to as the frontal plane) is always perpendicular to the sagittal plane. Movements within this plane are abduction and adduction (side-to-side movements of the arms and legs) and lateral flexion (side-to-side movements of the head, neck, and torso).



Question 88

In horizontal axis which divided into caudal and cranial

1- sagittal
2- coronal
3- axial
4- transverse

Correct Answer: 4
rationale:
The transverse plane, also known as an axial plane or cross-section, divides the body into cranial and caudal (head and tail) portions.

Question 89

What plane divides the body is two halves (a right side and a left side):

1- coronal
2- sagittal
3- transverse
4- horizontal

Correct Answer: 2
rationale:
The sagittal plane divides the body vertically into equal right and left halves. This plane is also referred to as the midsagittal plane because it is on the midline of the body.

Question 90

Which of the following is the basic body plane that is vertical, extends from side to side, divides the body into anterior, and posterior portions?

1- Sternal
2- Sagittal
3- Coronal
4- Transverse

Correct Answer: 3
rationale:
The Coronal Plane (Frontal Plane) is a vertical plane running from side to side; divides the body or any of its parts into anterior and posterior portions.

Question 91

Acute osteomyelitis is most commonly caused by:

1- staphylococcus aureus
2- actinomyces bovis
3- nocardia asteroids
4- borrelia Vicenti

Question 92

A physical therapist is beginning a gait evaluation. During heel strike to foot flat on the right lower extremity, which of the following does not normally occur?

1- The left side of the pelvis initiates movement in the direction of travel.
2- The right femur medially rotates.
3- The left side of the thorax initiates movement in the direction of travel.
4- The right tibia medially rotates.

Question 93

A 45 years male presented with an expansile lesion in the center of femoral metaphysis. The lesion shows endosteal scalloping & punctuates calcifications. Most likely diagnosis is:

1- Osteosarcoma
2- Chondrosarcoma
3- Simple bone cyst
4- Fibrous dysplasia

Question 94

in clinical examination you must do all of the following except:

1- observe the patient examine both affected unaffected.
2- review patient sheet.
3- focus only on areas of patient`s complains.
4- respect the sensitizer's of patients.

Question 95

which is not true about sudeck's atrophy?

1- osteoporosis of bones of hand.
2- fully curable by taking sympathetic nerve blocker.

Question 96

To prevent maximal compressive forces being placed on the patella, a therapist should minimize placing the patient:

1- prone and flexing the knee to 30 degrees.
2- in a sitting position with the knee flexed to 90 degrees.
3- supine and flexing both the hip and knee to 110 degrees.
4- prone and flexing the knee to 110 with the hip extended.

Question 97

If the line of gravity is posterior to the hip joint in standing, on what does the body first rely on to keep the trunk from moving into excessive lumbar extension?

1- Iliopsoas muscle activity.
2- Abdominal muscle activity.
3- Anterior pelvic ligaments and the hip joint capsule.
4- Posterior pelvic ligaments and the hip joint capsule.



Question 98

A physical therapist is performing a functional capacity evaluation on a patient with a L4-L5 herniated disc. Part of the evaluation consists of performing floor to waist lifts using 30 pounds as resistance. During the first trial, the physical therapist notices that patient exhibits decreased anterior pelvic tilt. What should the physical therapist do during the second trial?

1- Correct the deviation verbally before the lift.
2- Correct the deviation with manual contact during the lift.
3- Correct the deviation both verbally and manually during the lift.
4- None of the above.

Question 99

abduction to Adduction in which plan occurs

1- From sagittal plan to coronal plan.
2- From coronal plan to sagittal plan.
3- From longitudinal to transverse.

Question 100

patient has scoliosis c on right thoracolumbar we found:

1- lat flexion of thoracolumbar on right convex.
2- lat flexion and left convex.
3- lat flex and concave on left side.

Orthopedic

1- A patient present to physical therapy with a long-standing diagnosis of bilateral pes planus. The therapist has given the patient custom-fit orthotics. After using the orthotics for 1 week, the patient complains of pain along the first metatarsal. The therapist decides to use joint mobilization techniques to decrease the patient's pain. In which direction should the therapist mobilize the first metatarsal?
a-Inferiorly
b-Superiorly
c-Laterally
d-A and C

Rationale:
Superior glide of the proximal phalanx on the first metatarsal head. Before performing this manipulation, the examiner performs deep-tissue massage along the entire flexor hallucis brevis muscle. This is followed with a series of hold-relax stretches in which the patient lightly presses against the examiner's thumb (A). Upon releasing the pressure, the examiner dorsiflexes the joint a few degrees, gently pushing the first metatarsophalangeal joint into a more dorsiflexed position (B). This process is repeated until bony resistance is noted. At this point, the examiner performs a series of gentle mobilizations in which the distal aspect of the first metatarsal head is driven inferiorly (C), while the proximal phalanx is traction

2-A patient has limited motion in supination and calcaneal inversion at the subtalar joint, using manual techniques, the accessory motion of the calcaneus that needs to be emphasized in order to increase the motions that are limited would be: -
a-anterior glide
b- posterior glide
c- medial glide
d- lateral glide

Rationale:
As the lateral force that imparted to the calcaneus is an upward movement with the thumb on the calcaneus. This is called subtalar lateral glide for inversion.

3- In which of the following situations should the therapist be most concerned about the complications resulting from grade IV joint mobilization techniques?
a- A 37-year-old man with a Colles' fracture suffered 10 weeks ago
b- A 23-year-old woman with a boxer's fracture suffered 10 weeks ago.

c- A 34-year-old man with a scaphoid fracture suffered 12 weeks ago
d- A 53-year-old man with a Bennett's fracture suffered 12 weeks ago

Rationale:
As A Colles fracture is a type of fracture of the distal forearm in which the broken end of the radius is bent backwards. Which its symptoms may include pain, swelling, deformity and bruising. And its complication may include damage to the median nerve.
And the grade 4 joint mobilization is a small amplitude rhythmic oscillating mobilization at end range of movement so be carful to prevent the median nerve damage.

4- Following cast immobilization for a now healed supracondylar fracture of the humerus, a patient's elbow lacks mobility. To increase elbow range of motion, joint mobilization in the maximum loose-packed position should be performed at:
A-full extension
B-90 degrees of flexion
C-70 degrees of flexion
D-30 degrees of flexion

Rationale:
As the humeroulnar articulation consists of the trochlea is convex and it articulates with the olecranon fossa which is concave, so its maximum loose pack position for mobilization is elbow flexion 70 degrees and forearm supination 10 degrees.

5- A therapist is mobilizing patient's right shoulder. The movement taking place at the joint capsule is not completely to end range. It is a large amplitude movement from near the beginning of available range to near the end of available range. What grade of mobilization according to Maitland, is being performed?
a-Grade I
b- Grade II
c-Grade III
d-Grade IV

Rationale:
Maitland Joint Mobilization Grading Scale:
Grade I- Small amplitude rhythmic oscillating mobilization in early range of movement
Grade II- Large amplitude rhythmic oscillating mobilization in midrange of movement.

Grade III- Large amplitude rhythmic oscillating mobilization to point of limitation in range of
movement Grade IV- Small amplitude rhythmic oscillating mobilization at end range of
movement
Grade V- (Thrust Manipulation) - Small amplitude, quick thrust at end range of movement.

6-A 67-year-old woman presents to an outpatient facility with a diagnosis of right adhesive capsulitis. The therapist plans to focus mostly on gaining abduction range of motion. In which direction should the therapist mobilize the shoulder to gain abduction range of motion?
A. Posteriorly
B. Anteriorly
C. Inferiorly

Rationale:
As the direction of shoulder mobilization
Inferiorly to gain more abduction
posteriorly to gain more flexion,
Anteriorly to gain more extension

7- You have been treating a 61 year-old women over a period of four months for adhesive capsulitis which had resulted I a "frozen shoulder". Her recovery has been good, however she still complains that she cannot reach the upper shelves in her kitchen cupboards and closets. To help her achieve his goal, joint mobilization should focus on :-
a-inferior glide
b-posterior glide
c- anterior glide
d-grade II oscillations

Rationale:
As the direction of shoulder mobilization
Inferiorly to gain more abduction
posteriorly to gain more flexion,
Anteriorly to gain more extension

8- A 67-year-old woman presents to an outpatient facility with a diagnosis of right adhesive capsulitis. The therapist plans to focus mostly on gaining abduction range of motion. In which direction should the therapist mobilize the shoulder to gain abduction range of motion?
a-Posteriorly
b-Anteriorly

c-Inferiorly
d-Superiorly

Rationale:
As the direction of shoulder mobilization
- Inferiorly to gain more abduction
- posteriorly to gain more flexion,
- Anteriorly to gain more extension

9- A physician notes a vertebral fracture in the X-ray of a patient involved in a car accident. The fractured vertebra has a bifid spinous process. Which of the following vertebrae is the most likely to be involved?
a- Fourth lumbar vertebra
b- Fifth cervical vertebra
c- Twelfth thoracic vertebra
d- First sacral vertebra.

Rationale:
The spinous process of a typical cervical vertebra is short and bifid posteriorly. It is bifid because it develops from two separate secondary centers of ossification. This morphology is unique to cervical spinous processes.

10- Which of the following muscles lowest work when a therapist asks the patient to rise onthe toes and push the body weight directly upward
A. Soleus
B. Peroneus longus
C. Extensor halluces longus
D. Extensor digitorum longus

Rationale:
As regarding to the main action of these muscles
Soleus for planter flexion
Peroneus longus for eversion but helps planter flexion as an accessory motion
Extensor halluces longus for big toe extension which need for this position
Extensor digitorum longus for extend the lateral four toes at metatarsophalangeal joint which need for this position

11- 27-year-old woman is referred to a physical therapy clinic with a diagnosis of torticollis. The right sternocleidomastoid is involved. What is the most likely position of the patient's cervical spine?
a-Right lateral cervical flexion and left cervical rotation
b-Right cervical rotation and right lateral cervical flexion
c-Left cervical rotation and left lateral cervical flexion
d-Left lateral cervical flexion and right cervical rotation

Rationale:
As torticollis also known as wry neck is a condition in which the neck is twisted and causes the head to rotate and tilt at an odd angle which is the action of Sternocleidomastoid.

12-A 50-year-old woman has some difficulty sitting and rising from a chair. The physical therapist decided to give her some exercises to help hair gain control while sitting and rising from the chair. Which of the following exercises is the most appropriate?
A. Lunges
B. Sit-ups
C. Push-ups
D. Squatting

Rationale:
As regular squatting strengthen your lower body targeting glutes and quadriceps also important for hip muscles, calves, hamstrings and obliques. Also strengthen tendons, bones and ligaments around the leg muscles and takes some of the weight off your knees and ankles so helps make your knee more stable. And slowing down the process of being muscles, tendons and ligaments less elastic due to the aging process.

13- Which of the following is the most appropriate exercise to maintain bone mass in theelderly population?
A. Aerobic
B. Aquatic
C. Stretching
D. Progressive resistance

Rationale:
As aerobic training is widely used for the treatment of osteoporosis as it presents a low-risk of fractures due to the low-intensity combined with metabolic stimulus necessary for bone synthesis.

14- Exercise in elderly to keep bone density ?
a-Aerobic
b-Resistive
c-Hydro exe

Rationale:
As aerobic training is widely used for the treatment of osteoporosis as it presents a low-risk of fractures due to the low-intensity combined with metabolic stimulus necessary for bone synthesis.

15- An older adult with osteoarthritis presents for pain management and improved mobilityShe complains of severe pain in both knees while climbing stairs and bending activities.Which of the following types of exercise is the most appropriate management?

A. Aquatic
B. Isokinetic
C. Close kinematic
D. Open kinematic chain

Rationale:
Aquatic exercise is a low-impact activity that takes the pressure off your bones, joints and muscles. Water also offers natural resistance, which can help strengthen your muscle and improve muscular endurance and improve joint use so lessen pain if you have osteoarthritis.

16-Arthrokinematics refers to the movement of joint surfaces and can be described as the convex-concave rule. Which of the following is the most appropriate description of the open-chain extension of the tibiofemoral joint convex-concave rule?

A. Femur is rolling and gliding in the same direction on the tibia
B. Tibia is rolling and gliding in the same direction on the femur
C. Tibia is rolling and gliding in different direction on the femur
D. Femur is rolling and gliding in different direction on the tibia

Rationale:
If the convex femur rolls on the concave tibia, roll and glide occur in opposite direction and when the concave tibia rolls on the convex femur, roll and glide share a common direction. During knee extension, tibia glides anteriorly on femur and more precisely, from 2o degree knee flexion to full extension, tibia rotates externally.

17- A therapist reviews an X-ray of a patient diagnosed as having spondylolisthesis. What definition of spondylolisthesis is helpful for diagnosis?

A. Forward slippage of a vertebra on the vertebra above it
B. Forward slippage of a vertebra on the vertebra below it
C. Posterior slippage of a vertebra on the vertebra above it
D. Posterior slippage of a vertebra on the vertebra below it

Rationale:
Spondylolisthesis is a spinal condition that affects the lower vertebrae. This disease causes one of the lower vertebrae to slip forward onto the bone directly beneath it.

18- Spondylolisthesis :
a- vertebrae above vertebra dislocate posterior
b- vertebrae below vertebra dislocate posterior
c- vertebra above vertebra dislocate anterior
D- vertebrae below vertebra dislocate anterior

Rationale:
Spondylolisthesis occurs when one vertebra slips forward over the vertebra below it.

19- In the geriatric population, ... usually occurs after ..... is present.
A-Spondylolisthesis, spondylolysis
B-Spondylolysis, spondylolisthesis
C-Spondyloschisis, spondylolysis
D-Spondylolisthesis, spondyloschisis

Rationale:
Spondylolysis and spondylolisthesis are different spinal conditions but related to each other, as spondylolysis is a weakness or stress fracture in one of the vertebrae. But in some cases, the stress fracture weakens the bone so much that it is unable to maintain its proper position in the spine and the vertebra starts to shift or slip out of place. This condition is called spondylolisthesis.

20- Sternoclavicular in forced by except:-
A-sternoclaviculariig
B-intra- articular disc
C-intracalviculariig
D-acroimoclaviculariig

Rationale:
As sternoclavicular joint is forced by intra-articular disc, anterior and posterior sternoclavicular lig, interclaviculariig and costoclaviculariig.
But acromioclavicular lig is part of the acromioclavicular joint.

21- What not true about curvature of cervical region
A-cervical curvature lordosis increases preesure on disc
B- cervical curvature lordosis increases pressure on lig
C- cervical curvature lordosis increases pressure on facet jiont
D-Lumber and thoracic disorders not accompiened with cervical curvature

Rationale:
The normal spine is structurally balanced for optimal flexibility and support of the weight of the body, as the lumbar spine has an inward curve called lordosis, the thoracic spine has an outward curve called kyphosis, the cervical spine has

an inward curve called lordosis. These curves work in harmony to keep the center of the gravity aligned over the hips and pelvis so abnormal curvature in the spine can put it out of alignment. Also, degenerative changes of the cervical spine are commonly accompanied by a reduction or loss of the segmental or global lordosis and causing a neck pain.

22- pt complain of groin pain with limited abduction with external rotation you expect the diagnosis is
a- inguinal hernia
b- adductor strain
c- abductor ms sprain
d- external obliques weakness

Rationale:
Adductor strain or injury is a common cause of medial leg and groin pain. The adductor complex includes three muscles, all these muscles primarily adduct the thigh. Adductor longus provides some medial rotation, adductor magnus extends the hip. In open chain activation, the primary function is hip adduction. In closed chain activation, they help stabilize the pelvis and lower extremity during the stance phase of gait. Also they have secondary role in hip flexion and rotation.

23- fixed stiffness of 1st IPJ joint of big toe called
A- Hallux varus
B- Hallux valgus
C- Hammer toe
D- Hallux rigidus

Rationale:
-Hallux rigidus or stiff big toe is degenerative arthritis and stiffness due to bone spurs that affects the metatarsophalangeal joints at the base of the hallux
-Hallux varus is a deformity of the great toe joint where the hallux is deviated medially away from the first metatarsal bone.
-Hallux valgus is also called bunion is a deformity of the joint connecting the big toe to the foot. The big toe bends towards the other toes and the joint becomes red and painful.
- Hammer toe is a deformity that causes your toe to bend or curl downward instead of pointing forward.

24-Which is the following being foot deformity?
a- halluxvalgus
b- genuvarum.
c- Genu valgum

Rationale:

As hallux valgus is a deformity of the joint connecting the big toe to the foot. But genu varum or bow legs is a condition where one or both of legs of your child curve outward at the knees and genu valgum or knocked knee is a condition in which the knees angle in and touch each other when the legs are straightened.

25- If the therapist is evaluating a patient who has been diagnosed as having hammer toes and low metatarsal arch deformity,which of the following physical findings could the therapist detect during the physical evaluation ?
a-metatarsophalangeal joint flexion-proximal interphalangeal joint extension
b- metatarsophalangeal joint hyperextension- proximal interphalangeal joint hyperextension
c- metatarsophalangeal joint abducted-proximal interphalangeal joint flexion
d- metatarsophalangeal joint hyperextension-proximal interphalangeal joint flexion

Rationale:

Hammer toe also referred to as contracted toe or rotated toe, is a common toe deformity that is characterized by your toe bending upward at the middle joint with the end of the toe becoming angled downward causing the toe to no longer lay flat. So to give the shape of hammer toe, metatarsophalangeal joint hyperextend and proximal interphalangeal joint flex.

26-A therapist is scheduled to evaluate a patient with a chronic condition of hammer toes. Where should the therapist expect to find callus formation?
a- The distal tips of the toes
b- The superior surface of the interphalangeal joints
c- The metatarsal heads.
d- All of the above.

Rationale:

- Hallux rigidus: loss of motion of first MTP joint in adults due to degenerative arthritis, osteophyte formation leads to dorsal impingement
- Hallux Valgus: is lateral deviation of the great toe so as to put a valgus deformity on the first metatarsophalangeal (MTP) joint. A deviation of 15-20 degrees is considered abnormal.
- Hallux Varus: deviation of the distal portion of the great toe at the metatarsophalangeal joint to the inner side of the foot away from the second toe
- Hammer Toe: flexion at the PIP joint, extension deformity at DIP joint Deformity can be rigid or flexible

27- Observing a patient in a standing position, the therapist notes that an angulation deformity of the right knee causes it to be located medially in

relation to the left hip and foot. This condition is commonly referred to as:
a-genu varum
b-genu valgum
c-pes cavus
d-none of the above.

Rationale:

- Genu Valgum "knock-knee": it is a condition in which the knees angle in and touch one another when the legs are straightened.
- Genu Varum "Bowed-Leg": it is an exaggerated bending outward of the legs from the knees down that causes the knees to be spread apart when the feet are flat on the floor

28- knock knee, what does it mean:
a. genurecurvatum
b. genuvarum
c. genuvalgum
d. hallux valgus

Rationale:
Genu valgum is called knocked knee.
Genu varum is called bowed leg.
Genu recurvatum is called hyperextended knee
Hallux valgus is a forefoot deformity.

29- A 7 years old young boy, had FRACTURE of lateral condyle of femur. He developed malunion as the FRACTURE was not reduced anatomically. Malunion will produce:
a. genu valgum
b. genu varum
c. genu recurvatum
d. dislocation of knee

Rationale:
Because the malunion in the lateral condyle of femur, the affected knee curve outward which is called genu varum.

30- A 13-year-old boy presented for consultation because his knees tend to knock together and he tends to fall during running. Examination revealed separation of the medial malleoli by 3 inches when standing with the knees in contact with each other and the feet directed forwards. The correct diagnosis is :
a. coxa vara.
b. genu valgum.

c. genu varum.
d. talipescalcaneovalgus.
e. bilateral flat foot.

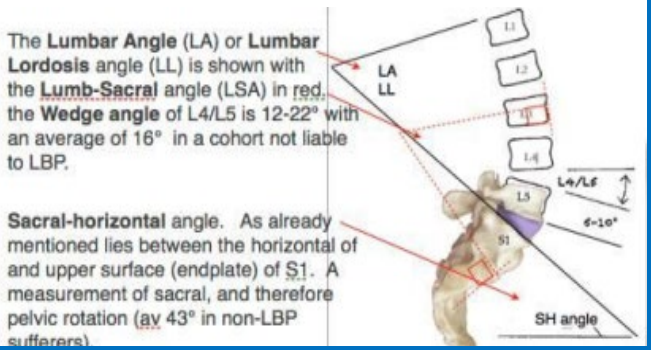
Rationale:

-coxa vara is a deformity of the hip, whereby the angle between the head and the shaft of the femur is reduced to less than 120 degrees.
-genu valgum"knock-knee": it is a condition in which the knees angle in and touch one another when the legs are straightened
- genu varum "Bowed-Leg": it is an exaggerated bending outward of the legs from the knees down that causes the knees to be spread apart when the feet are flat on the floor
- talipescalcaneovalgus in which the front part of the foot is bent upward and turned outward.
-flat foot or fallen arches is a postural deformity in which the arches of the foot collapse, with the entire sole of the foot coming into complete or near-complete contact with the ground.

31-A 10-year-old boy presents to outpatient physical therapy with complaints of diffuse pain in the right hip, thigh, and knee joint. The patient was involved in a motor vehicle accident 3 weeks ago. He is also obese and has significant atrophy in the right quadriceps. The right lower extremity is held by the patient in the position of flexion, abduction and lateral rotation. Which of the following is most likely the source of the patient's signs and symptoms?
a-Greater trochanteric bursitis
b-Avascular necrosis
c-Slipped femoral capital epiphysis
d-Septic arthritis

Rationale:

Slipped femoral capital epiphysis:
It is most common in the adolescent period (boys ages 10-16 y. & girls aged 12-14y).
Males have 2.4 times the risk compared with females.
Obesity is a risk factor because it places more shear forces around the proximal growth plate in the hip at risk.
Prodromal symptoms (e.g. hip or knee pain, limp, decreased range of motion),
Patients often hold their affected hip in passive external rotation.
Gait pattern Antalgic - Shortened stance phase on the affected side, Out-toeing



32-Patient with disc herniation and decrease back motion P.T ASSIST:
A- sterno-costal angle
b-lumbosacral angle
c-cobbs angle
d-cubitus angle
Rationale:
The variations in lumbosacral angles cause changes in the spinal kinematics that may affect the occurrence of disc
herniation. Loss of lumbar lordosis increases the compressive forces on the spine and may associate with the presence of disc
herniation. Lumbosacral alignment must be taken into account when evaluate the pathophysiology of lumbar disc herniation

33- In thoracic outlet syndrome is :
a- superior to operative thoracic
b- posterior to operative thoracic
c- inferior to operative thoracic
d- anterior to operative thoracic

Rationale:
Thoracic outlet syndrome:
It is a syndrome involving compression at the superior thoracic outlet→ involving compression of a neurovascular bundle passing between the anterior scalene and middle scalene.
It can affect the brachial plexus (nerves that pass into the arms from the neck), and/or the subclavian artery or vein (blood vessels that pass between the chest and upper extremity).

34-The thoracic outlet syndrome, is due to :
a- Upper opening chest
b- Lower opening chest
c- Lateral opening chest
d- Anterior opening chet
Rationale:
The thoracic outlet is the ring formed by the top ribs, just below the collarbone. Thoracic outlet syndrome (TOS) occurs when nerves or blood vessels are compressed by the rib, collarbone or neck muscles at the top of the outlet

35- damage as a result of Salter-Harris type IV supracondylar humeral epiphyseal fracture in a young athlete will most likely result in:-
a- refracture at a future time
b- nonunion
c- arrested growth
d- severing of the radial nerve

Rationale:

The Salter-Harris classification system is a method used to grade fractures that occur in children and involve the growth plate, which is also known as the physis or physial plate. The classification system grades fractures according to the involvement of the physis, metaphysis, and epiphysis. The fracture grade has important implications for both prognosis and treatment.Salter-Harris type IV fractures are intra-articular injuries in which the fracture extends through the epiphysis, across the physis and through the metaphysis.

Types IV fractures carry a risk for growth retardation, altered joint mechanics, and functional impairment.

36- damage as a result of Salter-Harris type IV supracondylar humeral epiphyseal fracture in a young athlete will most likely result in:-
a- refracture at a future time
b- nonunion
c- arrested growth
d- severing of the radial nerve

Rationale:

Salter-Harris type IV fractures are intra-articular injuries in which the fracture extends through the epiphysis, across the physis and through the metaphysis.

37- Thumb limited movement which of the following muscle not affected:
a- Flex. Policies bevies
b- Abd. Policies bevies
c- oppons. Pollicis

38-Which tendon is most commonly involved with lateral epicondylitis?
a-Extensor carpi radialis longus
b-Extensor carpi radialis brevis
c-Brachioradialis
d-Extensor digitorum

Rationale:

ECRB is attaches to the lateral epicondyle. The ECRB may also be at increased risk for damage because of its position. As the elbow bends and straightens, the muscle rubs against bony bumps.

39- There are post contusion of the patient thumb, the least movement allow :
a- radial and ulnar deviation
b- thumb adduction and flexion

c- extension and flexion of the wrist
d- supination and pronation

Rationale:
Adduction and Flexion is the movement of thumb that affected with contusion

40- During test of splenius capitis which action not needed :
a) Extension
b) Flexion
c) Rotation

Rationale:
splenius capitis action is extension , rotation And side bending

41- A patient with neck pain is being examined by a physiotherapist. To test the semispinalis capitis muscle, which of the following movements is not required?
a- flexion
b- extention
c- rotation
d- lateral flexion

Rationale:
semispinalis capitis action is extension, rotation and lateral flexion

42- The splenius capitis muscle is a prime mover for :
a- neck flexion
b- neck extension
c- neck rotation
d- neck side bend

Rationale:
The splenius capitis muscle is a prime mover for head extension. The splenius capitis can also allow lateral flexion and rotation of the cervical spine.

43-which of following not needed in testing the trapezius muscle :
a-head flexion
b-head extension
c-head rotation
d-head lateral flexion

Rationale:
trapezius action is rotation, retraction, elevation, and depression of scapula

44- which of following not needed in testing the sternocleidomastoid muscle:
a-head flexion
b-head extension
c-head rotation
d-head lateral flexion

Rationale:
sternocleidomastoid action Unilaterally; cervical rotation to opposite side, cervical lateral flexion to same side
Bilaterally; cervical flexion, raises the sternum and assists in forced inhalation

45- avascular necrosis can be a possible Squeal of FRACTURE of all of the following bones, except:
a-Femur neck
b-Scaphoid
c-Talus.
d-Calcaneum

Rationale:
- AVN of the femoral head is a type of osteonecrosis due to disruption of blood supply to the proximal femur.
- Talus is predisposed to AVN due to ischemia, owing to its unique structure, characteristic extraosseous arterial sources, and variable intraosseous blood supply.
- The most common complication of scaphoid fracture is retrograde blood supply (from distal to proximal).

46- An 11 - year - old child with the diagnosis of Legg - Calvé - Perthes disease referred to the physical therapist, which of the following applies regarding Legg - Calvé - Perthes disease?
A- Tuberculosis of the hip joint
B. Transient synovitis of the hip
C. Slipped proximal femoral epiphysis
D- Avascular necrosis of the femoral head

Rationale:
Legg - Calvé - Perthes disease is a childhood condition that occurs when blood supply to the ball part (femoral head) of the hip joint is temporarily interrupted and the bone begins to die.

47-Avascular necrosis would least likely be a major complicating factor in the rehabilitation of a patient with:
a-Legg-CalvéPerthes' disease
b-a tibial fracture

c- degenerative joint disease of the cervical spine
d- a carpal scaphoid fracture

Rationale:
Legg-Calve Perthes disease (LCPD):
resulting avascular necrosis of the proximal femoral head is from compromise of the tenuous blood supply to this area. LCPD usually occurs in children aged 4-10 years.
Avascular necrosis (AVN) following rotational ankle fractures is most commonly ● described in the talus; however, it can also occur in the tibial plafond.
A scaphoid fracture is a break of the scaphoid bone in the wrist. Complications may include nonunion of the fracture, avascular necrosis of the proximal part of the bone, and arthritis
The disease has an insidious onset and it may occur after an injury to hip; In the vast majority of instances, the disorder is unilateral. Both hips are involved in less than 10% of cases, and the joints are involved successively, not simultaneously.
Other names of LCPD :
Coxa plana
Calve-Perthes disease
Legg-Perthes disease
Avascular necrosis of the femoral head
Osteo-chondritis juveniles

48-A 30-year-old man presented with severe pain in his wrist on the dorsal aspect. He had fallen on an outstretched hand 6 months ago. The pain is aggravated by flexion of the wrist (see report). X-ray of the wrist: Sclerosis at the lateral wrist area Which of the following is the most likely cause of the pain?
A. Wrist sprain
B. Extensor tendonitis
C. Median nerve injury
D. Avascular necrosis of scaphoid
Rationale:

Scaphoid fractures are most commonly caused by a fall on an outstretched hand. Complications may include nonunion of the fracture, avascular necrosis of the proximal part of the bone, and arthritis. Chronic fracture of waist of scaphoid may lead to secondary sclerosis

49-All of the following are associated with supracondylar Fracture of humerus, except:
a- It is uncommon after 15 years of age

b- Extension type FRACTURE is more common than the flexion type
c- Cubits varus deformity commonly results following malunion not sure
d- Ulnar nerve is most commonly involved

Rationale:
supracondylar of humerus is the most common type of elbow fracture in children.
The Median and Radial Nerves lie anterior to the supracondylar humerus region they are at risk for injury, primarily post-trauma either by stretching, piercing or impinging at the fracture ends or being entrapped between two fracture fragments (traumatic or primary nerve injury), whereas the Ulnar Nerve injury is usually secondary to treatment (iatrogenic or secondary nerve injury)

50- The depth of glenoid is increased by what?
a- Labrum
b- scapula
c- cervical
d- ligaments

Rationale:
The cavity surface is covered with cartilage in the fresh state, and its margins, slightly raised, give attachment to a fibrocartilaginous structure, the glenoid labrum, which deepens the cavity

51- Increase glenoid cavity for proximal humerus in shoulder
A. capsule
B. labrum
C. ligament and tendon

Rationale:
as described before

52- Inversion injury at the ankle can cause all of the following except:
a- fracture tip of lateral malleolus
b- fracture base of the 5th metatarsal
c- sprain of extensor digitorum brevis
d- fracture of sustentaculum tails.

Rationale:
Lateral ankle sprains are referred to as inversion ankle sprains or as supination ankle sprains. complex of ligaments on the lateral side of the ankle is torn by varying degrees.
The sustentaculumtali is a triangular projection found on the medial surface of the calcaneus

53- When the ankle is forcibly inverted and plantar flexed, the ligament that is most frequently disrupted is the:
a- deltoid
b- Anterior Talo-fibular ligament
c- posterior Talo-fibular ligament
d- calcaneofibular

Rationale:
Their anatomical location and the mechanism of sprain injury mean that the calcaneo-fibular and posterior talofibular ligaments are less likely to sustain damaging loads

54- The therapist is evaluating a 32-year-old woman for complaints of right hip pain. The patient has injured the strongest ligament of the hip. The therapist places the patient in the prone position on the plinth and passively extends the involved hip. The therapist notes an abnormal amount of increase in passive hip extension. Which of the following ligaments is damaged?
a- Ischio-femoral ligament.
b- Iliofemoral ligament (Y ligament of Bigelow)
c- Pubofemoral ligament
d- Ligamentumteres.

Rationale:
The iliofemoral ligament has a 'Y' shape and prevents hyperextension of the hip.

55-Deformity associated with colles fracture (fracture lower end of radius)
a- Dinner fork deformity
b - Pes planus
c- Talipusequinovarus

Rationale:
A dinner fork deformity, also known as a bayonet deformity, occurs as the result of a malunited distal radial fracture, usually a Colles fracture

56-What the other name of fracture radius:
a-no name
b-pontine fracture
c-bennt fracture
d-colles fracture

Rationale:
- A Colles fracture is a type of distal radius fracture.

- A Pontine ftacture is fracture of the skull base.
- A Bennett fracture is a fracture of the base of the thumb

57-Colles fracture may possibly cause injury to which of the following
a- flexor policies
b- abductor policies
c- adductor policies
d- extensor policies longus

Rationale:
the extensor pollicis longus is in close proximity to the bony architecture of the dorsal distal radius and therefore is at risk for injury.

58- A 55-year-old patient falls on an outstretched hand (see report).
X-ray: Fracture with dinner-fork deformity of the upper limb.
Which of the following bones is most likely fractured?
A. Ulna
B. Radius
C. Humerus
D. Clavicle

Rationale:
The Colles fracture is commonly described as a dinner fork deformity, it is a distal fracture of the radius.

59- Kienbock's disease is due to a vascular necrosis of :
a- femoral neck
b- medial cuneiform bone
c- lunate bone
d- scaphoid bone.

Rationale:
Kienbock's disease is a rare bone disorder that affects the lunate.The disorder prevents the lunate bone from receiving the necessary amount of blood supply. This lack of blood leads to avascular necrosis of the lunate

60-Stress fracture of foot also called soldier's fracture caused from overuse, which is the most affected metatarsals:
a- second metatarsals
b- Fourth metatarsals
c- big toe.
Rationale:

March fractures (soldier's fracture) are metatarsal fractures, most commonly second and third metatarsal fractures caused by an overuse injury. The repetitive impact to the metatarsals with weight-bearing exercises cause microfractures, which consolidate to stress fractures

61- Coxa vara present as abnormality of: -
a-proximal femur
b-distal femur
c-proximal tibia
d-distal tibia

Rationale:
a- Coxa vara is a deformity of the hip, where by the angle between the head and the shaft of the femur is reduced to less than 120 degrees.
b,c- deformity of distal femur or proximal tibia causes genu verum,valgum or genu recarvatum .

62- Cupitus Valgus deformity results from.
a- lateral humeral condyle fracture.
b- medical condylar.
c- supra condylar.

Rationale:
Cubitus valgus is the most common complication of lateral condylar fractures. Medial condylar fracture cause verus deformity.

63- the normal angle between the fumer and the neck of fumer is 126 when the angle increase the deformity is:
a- coxa valga
b- genu Valgus
C- genu recurvatum
D- coxa vera
Rationale:
a- Coxa valga is a deformity of the hip in which the angle between the femoral shaft and the femoral neck is increased compared to age-adjusted values (about 150 degrees in newborns gradually reducing to 120-130 degrees in adults).
b- In genu valgum, the lower extremities turn inward, causing the appearance of the knees to be touching while the ankles remain apart.
c- Genu recurvatum is operationally defined as knee extension greater than 5 degrees.

64-Which is correct about Volkmann's ischemic contracture
a-affect flexor of forearm

b-affect palmar fascia
C-lead to ulnar neuritis

Rationale:
a-Volkmann contracture is a deformity of the hand, fingers, and wrist caused by injury to the muscles of the forearm.
b- Dupuytren's disease is a condition that affects the fascia—the fibrous layer of tissue that lies underneath the skin in the palm and fingers.
c- Ulnar neuritis, or cubital tunnel syndrome, is inflammation of the ulnar nerve in the arm that results in numbness or weakness in the hand due to Direct trauma to the cubital tunnel

65-volkmans ischemic contracture occurs due to:
a-fracture of medial epcondyle of humerus
b-fracture of lat epicondyle of humerus
c-fracture lower third of humerus
d- surgical neck
Rationale:
Volkmann ischemic contracture is classically associated with a supracondylar fracture of the humerus. However, any fracture of the arm or elbow may result in this deformity. The key precipitating event in supracondylar fractures of the humerus is ischemia of the forearm due to acute compartment syndrome

66- One of the possible complications following a fracture is Volkmann's ischemic contracture. This condition:
a- Is caused by an interference with the venous return
b- Is caused by an interference of the nerve supply
c- May occur if the fracture is sustained in the lower extremity
d-None of the above
Rationale:
Volkmann contracture occurs when there is a lack of blood flow (ischemia) to the forearm. This occurs when there is increased pressure due to swelling, a condition called compartment syndrome.

67- Person fall on lat aspect of shoulder last thing occur is
a- dislocation SC joint
b- dislocate acromio clavicular joint
c- # medial clavicle
d- #middle third of clavicle

Rationale:
a- SC joint is faraway of the lat aspect of shoulder (site of fall)
b- acromio clavicular joint located at lat aspect of shoulder (site of injery)

68- Skeletal traction used mostly as a treatment of
a) shaft of femur
b) carpal bones
c) pelvis
d) calcenus

69-Which of the following bones is mostly treated with skola traction after a fracture?
A.Tibia
B. Knee
C. Pelvis
D. Femoral shaft
Rationale:
Skeletal traction is most commonly used to treat fractures of the femur, or thighbone.

70- Congenital dislocation occur in
a- Hip joint
b- knee joint
c- shoulder joint
d- elbow joint

Rationale:
a- Congenital hip dislocation (CHD) occurs when a child is born with an unstable hip. It's caused by abnormal formation of the hip joint during their early stages of fetal development. Another name for this condition is "developmental dysplasia of the hip."
c,d- shoulder and elbow dislocation often occur due to truma or accedint

71-Resisted isometric testing (RIT) is often used in the musculoskeletal examination to indicate the involvement of the contractile tissues. Which of the following RIT findings most likely results from a minor lesion of the contractile tissues?
A. Weak and painful
B. Weak and painless
C. Strong and painful
D. Strong and painless

Rationale:
A. Weak and Painful: Serious impairment, like a secondary deposit or a fracture might be present. However, if a patient is reluctant to replicate the severe pain it may appear as apparent weakness.
B. Weak and Painless: There could be a complete rupture of the muscle or

tendon, but most commonly might be a malfunction of the nerves. This impaired function of the nerve leads to muscle weakness.

C. Strong and Painful: Minor lesion in a part f the muscle or tendon and its attachment.

D. Strong and Painless: No lesion in the contractile structure

72- isometric contraction as a test in minor lesion

a- Strong painful

b- Strong painless

c- Weak painful

d- Weak painless

Rationale:

a. Strong and Painful: Minor lesion in a part f the muscle or tendon and its attachment.
b. Strong and Painless: No lesion in the contractile structure
c. Weak and Painful: Serious impairment, like a secondary deposit or a fracture might be present. However, if a patient is reluctant to replicate the severe pain it may appear as apparent weakness
d. Weak and Painless: There could be a complete rupture of the muscle or tendon, but most commonly might be a malfunction of the nerves. This impaired function of the nerve leads to muscle weakness.

73-Which of the following is the most vulnerable position for dislocation of the hip?

a- 30o hip extension, 30o hip adduction, and minimal internal rotation

b- 30o hip flexion, 30o hip adduction, and minimal external rotation

c- 30o hip flexion, 30o hip abduction, and minimal external rotation.

d- 30o hip extension, 30o hip abduction, and minimal external rotation.

Rationale:

Because it is the loose packed position of the hip joint. Thus, ligaments, tendons and capsule are relaxed.

74- A 28-year-old basketball player underwent knee injury a few days ago and referred tothe Rehabilitation Clinic for assessment. The therapist performs a test in the supine position.The knee joint is held in 90-degree flexion by 1 hand, and the foot is held by the other handand rotates the leg medially and laterally. Which of the following describes this particular test?

A. Ober

B. Faber

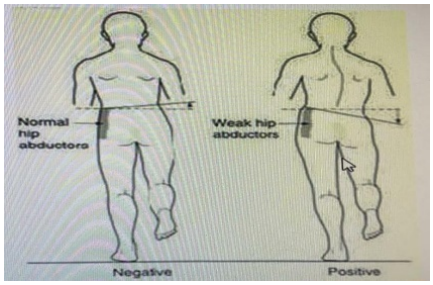
C. Lachman

D. McMurray

Rationale:

A. The patient should be in side-lying with the affected side up.
B. The patient lies supine on the examination table and is asked to place one foot on the opposite knee (placing the hip in flexion abduction external rotation).
C. The patient lies supine with the knee resting on the examiner's knee. One of the examiner's hands stabilizes the femur against the examiner's thigh, and the other hand applies anterior stress.
D. the patient supine and relaxed. The examiner grasps the patient's heel with one hand and the joint line of the knee with the other hand. The knee is flexed maximally, with external tibial rotation (medial meniscus) or internal tibial rotation (lateral meniscus).

75- Which of the following orthopedic tests is suitable for evaluating the weakness of hipabductors (see image)?



A. Ober
B. Thomas
C. Lachman
D. Trendelenburg

Rationale:
A. The Ober's test evaluates a tight, contracted or inflamed Tensor Fasciae Lata (TFL) and Iliotibial band (ITB). Applied from side lying position
B. The Thomas Test is used to measure the flexibility of the hip flexors
C. The Lachman test is a specific clinical exam technique used to evaluate patients with a suspected anterior cruciate ligament (ACL) injury.
D. trendelenburg test investigates stability of the hip and particularly the ability of the hip abductors (gluteus medius and gluteus minimus) to stabilize the pelvis on the femur.

76- The therapist performed Trendelenburg test for patient with right weakness of gluteus medius. The patient stands on right lower limb. The pelvis drop would be observed as
a. Pelvis does not drop as the patient leans by the trunk to the right
b. Pelvis will drop at the left side
c. Pelvis will drop at the right side
d. Pelvis does not drop as the patient leans by the trunk to the left

Rationale:
Patient stands on 1 foot and tries to maintain his position.
-- The unsupported pelvis should rise for negative.
-- If pelvis stays level or descends for positive.

77- Trendelenburg Test for:
a- hip abnormalities
b- knee problems

c- back problems
d- foot abnormalities.

Rationale:
trendelenburg test investigates stability of the hip and particularly the ability of the hip abductors (gluteus medius and gluteus minimus) to stabilize the pelvis on the femur.

78- A football player has twisted in knee and come to your clinic which test is non-important
a- Anterior drawer test
b- Lachman test
c- Varus stress test
d- Trendelenburg test

Rationale:
a,b-used to evaluate patients with a suspected anterior cruciate ligament (ACL) injury.
c- used to evaluate patients with a suspected lateral collateral ligament
d- used to evaluate patients with a suspected hip abduction deformity

79- Pt with lordosis from standing and +ve Thomas test due to:
a- strong lumbar extensors
b- fixed flex deformity
c- hamstring strain

Rationale:
fixed flexion deformity due to it`s positive Thomas test.

80-A therapist performs a test for a patient and was positive with the thigh of the patient raised some inches above the examination table. What is the test name and which muscle is shortened?
a. Ober test – Iliotibial band
b. Straight leg raising test – hamstring
c. Thomas test – Iliopsoas
d. Thomson – gastrocenmies

81-During an evaluation, the therapist taps on the flexor retinaculum of the patient's wrist, which causes tingling in the thumb. What test is this? For what condition does it screen?
A-Phalen's test – carpal tunnel
B-Finkelstein test – de Quervain's disease
C-Tinel's sign – de Quervain's disease
D-Tinel's sign – carpal tunnel

Rationale:
The patient places her flexed elbows on a table, allowing her wrists to fall into maximum flexion. The patient is asked to push the dorsal surface of her hands together and hold this position for 30-60 seconds.

82-Pt has tear with rotation movement Which test not need:
a- drawar test
b- lachmen test
C- Trendulberg test

Rationale:
Trendulberg Test because it`s for Gluteus Medius , It`s not Rotator muscle

83-A 21-year-old football player with a knee injury; the physical therapist during knee assessment is thinking about anterior cruciate ligament injury. Which of the following is the most appropriate test?
A. Lachman
B. McMurray
C. Varus stress
D. Trendelenburg
Rationale:
A. used to evaluate patients with a suspected anterior cruciate ligament (ACL) injury
B. used to evaluate patients with a suspected meniscus injery
C. used to evaluate patients with a suspectedlateral collateral ligament
D. used to evaluate patients with a suspected hip abductor injery.

84- A 26-year-old athlete twists a knee during a practice session.During the assessment, the physiotherapist performs the Lachman test to check the stabilityof which of the following?
A. Anterior cruciate ligament
B. Lateral collateral ligament
C. Medial collateral ligament
D. Posterior cruciate ligament

Rationale:
Lachman test is used to evaluate patients with a suspected ACL ingery

85- PT Lachaman test to asses
A- A.C.L
B- M.C.L

C- P.C.L

86- The pivot test is for:
a- anterior cruciate ligament
b- posterior cruciate ligament
c- medial meniscus
d- lateral meniscus.

Rationale:
a- assessted by pivot test
b- assessted by posterior drawer test
c,d- assessted by apley`s test

87-A patient is placed in supine position with the knee in 90o of flexion. The foot is stabilized by the therapist's body on the examination table. The therapist then wraps his fingers around the proximal tibia so that the thumbs are resting along the anteromedial and the anterolateral margins. The therapist then applies a force to pull the tibia forward. What special test is being performed?
a- Pivot shift
b- Lachman's test
c- Anterior drawer
d- Posterior drawer

Rationale:
a- started from extended knee
b- knee in about 20-30 degrees flexion
d- The therapist then applies a force to pull the tibia backward.

89-A 30-year-old basketball player underwent knee injury a few days ago and referred to the Physical Therapy Clinic for assessment. The therapist performs a test in a prone position with knee flexed to 90 degrees, and longitudinal compression with tibia rotated medially and laterally. Which of the following is this particular test?
A. Clarke
B. Apley's
C. Lachman's
D. Mc Murray's

Rationale:
a- applied from extended knee.
c,d- applied from supine.

90-A therapist is examining a 3-years-old child who is positioned as follows: supine, hips flexed to 90 o, hips fully adducted, and knees flexed. The therapist passively abducts and raises the thigh, applying an anterior shear force to the hip joint. A click at 30 o of abduction is noted by the therapist. What orthopedic test is the therapist performing, and what is its significance?
a- Ortolani's test – hip dislocation
b- Apley's compression/distraction test – cartilage damage
c- McMurray test – cartilage damage
d- Piston test – hip dislocation.

Rationale:
Apley`s compression distraction test &Macmurray Test are knee tests

91- Ortolani's test for:
a- shoulder dislocation
b- hip dislocation
c- elbow dislocation
d- knee ligament.

Rationale:
b/ Ortolani and Barlow tests, which play a big role in the clinical screening for **developmental dysplasia of the hip**

92- A patient is positioned by the therapist with the cervical spine rotated to the right. The patient then extends the neck as the therapist externally rotates and extends the right upper extremity. The patient is ten instructed to hold a deep breath. The radial pulse is palpated in the right upper extremity by the therapist. What type of special test is this, and for what condition is it testing?
a- Adson's maneuver – cervical disc herniation
b- Lhermitte's sign – cervical disc herniation
c- Adson's maneuver – thoracic outlet syndrome
d- Lhermitte's sign – thoracic outlet syndrome

Rationale:
a/ adson`s test can't assess cervical herniation
c/ this is a discreptionoddeson's test to assess TOS.
B,d/ Lhermitte's sign can help in MS cases.

93- posterior shear test for
a- sacroiliac joint
b- hip joint
c- lumbosacral join

Rationale:

Evaluate the Sacroiliac Joint and Help distinguish between pain originating from the pelvic girdle and lower back

94- Thompson test is for
a. Shortening of iliotibial band
b. Integrity of Achilles tendon
c. Shortening of iliopsoas
d. Integrity of quadriceps tendon

Rationale:
a- it's assessed with ober's test
b- It is performed as a clinical test to identify the presence of a complete **Achilles rupture.**
c- it's assessed with Thomas test

95-When you examine shoulder joint by asking patient To abduct shoulder to 90 degrees then lower slowly, this is:
a- Codman's Test (for rotator cuff tear) (also called drop arm test)
b- Infraspinatus test

96- A therapist is screening a patient complaining of pain at the anterior left shoulder region. The pain is increased when the examiner instructs the patient to position the left arm by his side with the elbow flexed at 90o and to actively supinate the forearm against resistance (provided by the examiner). What test is being performed
a- Froment's sign.
b- Yergason's test
c- Waldron test
d- Wilson test.

Rationale:
a- to test ulner palsy palsy.
b- to test biceps tendon pathology with supinate forarm .
c- test for ankle.
d- to test knee osteocondrities.

97- A physical therapist is evaluating a patient who complains of posterior ankle pain. The patient is positioned prone with the feet extended over the edge of the mat. The therapist squeezes the involved gastrocnemius over the middle third of the muscle belly. What test is the therapist performing? What indicates a positive test?
a- Thompson's test – plantarflexion of the ankle
b- Homan's test – plantarflexion of the ankle.
c- Thompson's test – no ankle movement

d- Homan's test – no ankle movement.

Rationale:
b,d- HOMAN'S Test is for dorsi flexor and applied from supine position

98- anterior taleofibular ligament to be assessed resist
a- planter flexion + inversion
b- dorsi + inv
c-dorsi + eversion
d- plant + ev

Rationale:
it`s Anterior drawer testPatient lies supine with their foot relaxed and the examiner stabilizes the tibia/fibula with the foot in 20 degrees of plantar flexion. The examiner grabs the ankle and draws the talus forward in the mortise. Inversion gives an anterolateral stress with increased stress on the ATF.

99- A 45-year-old school teacher with lower back pain. When evaluating the spinal X-ray, the physical therapist observes 75% of anterior displacement of L4 vertebra over L5 vertebra. What is the grade of the displacement?
A. 1
B. 2
C. 3
D. 4

Rationale:
The spondylolisthesis is graded by measuring how much of a vertebral body has slipped forward over the vertebral body beneath it into "5 grades ".
Grade1: 25% of vertebral body has slipped forward.
Grade2: 50%.
Grade3: 75%.
Grade4: 100%.
Grade5: Vertebral body completely fallen off (i.e., spondyloptosis)

100-A 29-year-old woman who is 8 months pregnant presents to an outpatient clinic with complaints of "pain and tingling" over the lateral thigh. She also indicates no traumatic injury. The symptoms increase after she has been sitting for 30 minutes or longer, and the overall intensity of the symptoms has been increasing over the past 2 weeks. The therapist notes that repeated active lumbar flexion does not increase pain, and the patient's lumbar range of motion is normal for a pregnant woman. There is also no motor weakness in the hip or pelvis, and the sacroiliac joint is not abnormally rotated. What is the most probable diagnosis?

a- L3 disc dysfunction
b- Spondylolisthesis
c- L4 disc dysfunction
d- Meralgiaparesthetica.

101-A 30-year-old patient complains of a dull pain in his lower back that extends down the right leg. Pain is aggravated with movement and relieved with rest. Examination reveals that the muscle power and the knee jerk reflex were weaker in the right leg compared to the left leg. Furthermore, sensory disturbances were noticed in the right leg. Which of the following is the most likely source of the symptoms?
A. Sciatic Pain
B. L3-L4 Disc Protrusion
C. Cauda Equina Syndrome
D. Sacral Osteophyte

102-A 48-year-old woman has a pain in her right foot 3 months ago receiving treatment in the clinic. During treatment, the patient says she had severe foot pain in the morning when she started to walk after wake
up. Which of the following is the most likely diagnosis?
A. Acute ankle sprain
B. Chronic ankle sprain
C. Acute plantar fasciitis
D. Chronic plantar fasciitis

Rationale:
Plantar fasciitis is a condition where the plantar fascia, a thick ligament on the bottom of your foot, is irritated. Symptoms include stiffness or pain in the heels or feet. Symptoms might be worse in the morning because of poor blood supply to the heel and foot area when you're at rest.

103- A 52-year-old man has a pain in his right foot 3 days ago and treated in Physical Therapy Clinic. The patient asked about the mechanism of the injury and said he felt down where his foot was in plantar flexion and inversion position. Which of the following is the most likely diagnosis?
A. Medial ankle sprain
B. Lateral ankle sprain
C. Anterior ankle impingement
D. Posterior ankle impingement

Rationale:
The most common mechanism of injury in lateral ankle sprains occurs with forced plantar flexion and inversion of the ankle as the

body's center of gravity rolls over the ankle. The ATFL followed by the CFL are the most commonly injured ligaments.

104- patient with ankle sprain injury have pain and red color after take analgesics and rest the pain subsides which grade of ankle sprain?
a- 1
b- 2
c- 3
d- 4

Rationale:
Grade 1 ankle Sprain: Slight stretching and some damage to the fibers of the ligament mostly, such injuries heal quickly.
105- Patient has lateral ankle minor pain at rest, palpation increase pain?
a- Grade 1
b- Grade 2
c- Grade 3

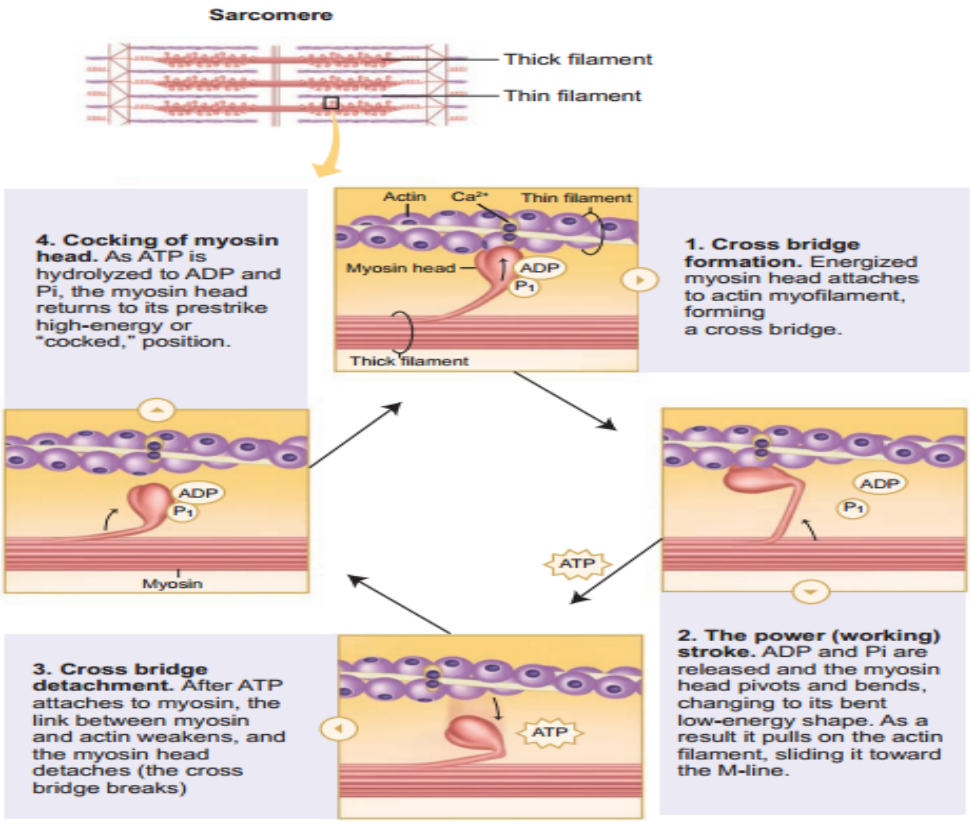
Rationale:
Grade 2: incomplete tear with moderate pain, swelling and bruising. Although it feels somewhat stable, the damaged areas are tender to the touch and walking is painful.

107- A 35-year-old woman with hip/pelvic dysfunction is referred to physical therapy. The physical therapist instructed the patient to do bridging exercise. What is the muscle group working most in bridging exercise?
A. Hip flexors and spine flexors
B. Hip flexors and spine extensors
C. Hip extensors and spine flexors
D. Hip extensors and spine extensors
Notes:

108- A clinical instructor is explaining to his student how a muscle contract. The instructor describes the cycle of cross bridging. He begins stating the first step is that the cross-bridges attach to the thin filament. Which of the following occurs next (in the correct order)?
a- The attachment with the actin filament is lost. The cross-bridge moves into position to attach to the thick filament. Cross-bridge moves, causing the thin filament to move.
b- The cross-bridge moves into position to attach to the thick filament. The attachment with the myosin filament is lost. Cross-bridge moves, causing the thin filament to move.

c- The attachment with the myosin filament is lost. Cross-bridge moves, causing the myosin filament to move. The cross-bridge moves into position to attach to a myosin filament.

d- Cross-bridge moves, causing the actin filament to move. The attachment with the thin filament is lost. The cross-bridge moves into position to attach to an actin filament.



Cross-bridge cycle of muscle contraction

109- A 37-year-old man complained of weakness in his arm. 6 weeks ago, he fell off his motorbike and landed on his shoulder. Examination revealed wasting of the flexor surface of his arm and difficulty with elbow flexion, especially with a supinated elbow are noted. His shoulder, hand movements and elbow extension were normal. Which of the following nerves is most likely injured?

A. Ulnar
B. Radial
C. Median
D. Musculocutaneous

Rationale:

Weakness of Biceps Brachii muscle decreases the ability to flex the forearm against gravity specially when the forearm is in supination, biceps brachii supplied by musculocutaneous nerve .

110- Clinical examination by the physiotherapist reveals the weakness of the fingers and wrist dorsiflexor muscle. Which of the following nerves is involved?
A. Ulnar
B. Radial
C. Median
D. Spinal accessory

Rationale:
The branches of the radial nerve provide motor supply for the posterior muscles of the arm and forearm.
Due to its length, the radial nerve is the most commonly injured nerve of the upper extremity.

111-A 41-year-old patient with posterior hip dislocation after trauma, during assessment, there is nerve damage associated with this injury. Which of the following nerves is affected?
A. Sciatic
B. Femoral
C. Superior gluteal
D. Inferior gluteal

Rationale:
The most common nerve complications associated with posterior hip dislocations are sciatic nerve injury (10% of cases)

112- Which is incorrect about sciatic nerve
a- to test it by SLR
b - maximum stretch occur 90 flexion
c- to stretch tibialpart. peroneal part with inversion
d- to stretch tibialpart. peroneal part with &eversion

115- A PT is performing a positional release technique to the cervical spine. Her goal in treatment is to open the affected foramen to decrease nerve root irritation. Which position is best for that goal?
a- forward flexion, contralateral sidebending, ipsilateral rotation.
b- forward flexion, contralateral sidebending, contralateral rotation.
c- forward flexion, ipsilateral sidebending, ipsilateral rotation.
d- forward flexion, ipsilateral sidebending, contralateral rotation.

Rationale:

maximally open the foramen, the neck should be forward flexed, contralaterallysidebent and contralaterally rotated. Option A would be best opining of the affected facet joint .

116- A PT is evaluating a patient with a know endocrine dysfunction. During muscle testing which joint would be expected to show a painless weakness?
a- hip
b- elbow
c- ankle
d- wrist

Rationale:
endocrine system disease can be manifested with proximal, painless muscle weakness. Of the options listed, the hip is the most reasonable expected weakness.

117- which of the following exercise schematics uses sets of repetitions at 100 percent, 75 percent, and 50 percent of the ten-repetition maximum, in that order for strengthening ?
a- Oxford
b- DeLorme
c- DAPRE
D- DOMS

Rationale:
Oxford technique or the reverse pyramid technique. The Oxford technique is emphasizing training from the high load to the lighter load based on the 10 RM. It starts from the 100 % of 10 RM, 75 % of 10 RM and 50 % of 10 RM for the first, second and third set respectively.

118- which one of the following has the greatest percentage composition of type I slow-twitch muscle fibers?
a-gastrocnemius
b- biceps femoris
c- biceps brachialis
d- Supraspinatus

Rationale:
the supraspinatus is a phasic or postural muscle, which indicate it has:higher composition of slow-twitch oxidative muscle fibers. It responds best to slow-weight, high-repetition exercise.

119- A physical therapist is treating a 24-year-old woman with lumbar muscle spasm. Part of the patient's treatment involves receiving instruction on correct sleeping positions.Which of the following would be most comfortable?
A-Supine with no pillows under the head or extremities
B-Prone with a pillow under the head only.
c- Sidelying with a pillow between flexed knees

Rationale:
The best sleeping position for back pain is on your side with a partial bend in the knees. Keeping the knees bent helps balance the body and reduces pressure on the lumbar spine. Many people find it helpful to put a small pillow between their knees to make this position more comfortable.

121-A patient arrives at an outpatient clinic with an order from the physician for whirlpool and wound care to a lower Extremity wound. The therapist decides to test the temperature in the whirlpool at warm. Which of the following Setting in degrees Celsius is appropriate?
A-27.5o Celsius
B-35.5o Celsius
C-49o Celsius
D-60o Celsius

122- Capsular tightness has limited your patients ability to fully extend her kneeTreatment to restore joint motion
a- anterior glide , external rotation of tibia
b- anterior glide , internal rotation of tibia
c- posterior glide, e1xternal rotation of tibia
d- posterior glide, internal rotation of tibia

Rationale:
Ant. Glide with external rotation of tibia will increase the extension of the Knee
Post Glide with internal rotation increase flexion of the Knee

123- a Boy who was 16 years old with a history of knee sprain 3 months ago, has subjected today to recurrent knee sprain, the best possible intervention is:
a- mini-squat exercises
b- knee immobilization
c- Stretching exercises
d- heat application
Rationale:
The plan of care in the acute setting shoud be directed toward control, inflammation, pain swelling (up to 1 week postinjury)

Immobilization of the part (rest, splint, tape, cast) is preferred to avoid positions of stress to the part.
Stretching and resistance exercises should not be performed at the site of the inflamed tissue.
heat application is not recommended during this stage.

124- Which of the following is not benefit in physiotherapy:
a- increase muscle fiber size
b- increase heart rate
c- increase ligaments and tendons
d- increase the power structure of bones

Rationale:
it can`t be increased by physiotherapy.

125- Variable that may not be considered during application of intermittent traction to cervical spine is
a- Weight of patient
b- Duration of traction
c- diagnosis
d- Angle of application

Rationale:
Variables at Practitioner's Discretion
in Delivery of Cervical Traction
- Static or intermittent
- Angle of application
- Dosage
- Amount of tension
- Duration of traction
- Cycle
- Total cycle duration
- Proportion of time of maximum vs. minimumtension
- Inclusion of ascending or descending steps at initiation or conclusion

126- a patient with left-sided C4 facet impingement. PT plans to use manual cervical traction to attempt to open the left facet. Which position of the cervical spine would maximally open the left facet?

a- flexion, left side-bending, left rotation.
b- flexion, left side-bending, right rotation.
c- flexion, right side-bending, left rotation.
d- flexion, right side-bending, right rotation.

Rationale:

maximally open the facet joint of c4, the neck should be forward flexed, contralaterallyside bent and ipsilaterally rotated.

127- Which of the following is the main rehabilitation goal for medial collateral ligament in the minimum protection phase?
A. Decrease pain
B. Normalize gait
C. Range of motion
D. Increase strength

Rationale:
Guiding principles for the minimum protection phase are that continued collagen maturation requires guarded increases in external forces, muscle strength and coordination will enhance functional stability.

128- A 40-year-old woman is diagnosed with rheumatoid arthritis. The patient complains of severe pain in her right shoulder. She is referred to the Physical Therapy Department for pain management from the Rheumatology Clinic. Which of the following is the most appropriate management regarding grades of mobilization?
A. Large amplitude rhythmic oscillations are performed with in the range
B. Small amplitude rhythmic oscillations are performed at the beginning of the range
C. Small amplitude rhythmic oscillations are performed at the limit of available motion.
D. Large amplitude rhythmic oscillations are performed up to limit of the available motion

Rationale:
Grade 1 joint mobilizations are small-amplitude movements used at the beginning of the joint's range of motion in an attempt to decrease or control patient pain levels.

129- A 60-year-old woman developed severe pain and restriction of the shoulder movements in all directions. She was diagnosed as a case of adhesive capsulitis of the shoulder joint and prescribed an intra-articular steroid injection. Following the injection, she developed wasting at the injection site. How long, in weeks, will it take before recovery from the wasting?
A. 10
B. 15
C. 20
D. 25

Rationale:

A cortisone injection may cause fat cells at the injection site to atrophy. A divot or depression in the skin may appear because the underlying fat cells have deteriorated. The skin's appearance usually goes back to normal in 6 months.

130- A physiotherapist is testing scapulohumeral rhythm in an adult patient with right shoulder pain. If this patient has normal scapulohumeral rhythm, the test will show approximately 1° of scapular motion for which of the following?
A. 2° of glenohumeral motion
B. 4° of glenohumeral motion
C. 6° of glenohumeral motion
D. 8° of glenohumeral motion

Rationale:
The glenohumeral and scapulothoracic joints move simultaneously in 2:1 ratio.

131- a percentage of 1 % of Scapulohumeralmovement is accompanied by :
a- 2% of glenohumeral
b- 4% of glenohumeral
c- 6% of glenohumeral
d- 8% of glenohumeral.
Rationale:
The normal ratio of The glenohumeral and scapulothoracicarticulationduring arm elevation is 2:1 ratio.

132- You wish to mobilize a patient's shoulder using an inferior glide technique. It would be best to use this technique be propositioning the patient's arm in:
a- 95 degrees of abduction with lateral rotation
b- 125 degrees of abduction and internal rotation
c- 55 degrees of abduction and neutral rotation
d- 95 degrees of shoulder flexion and neutral rotation

Rationale:
Glenohumeral joint:
- loose-packed position is 55* of abduction and 30* horizontal adduction.
- close-packed position is abduction and external rotation

133- which of the following is not considered a clinical feature ofknee degenerative joint disease
a- pain increased with downstairs activities
b- sudden onset
c- stiffness at morning
d- osteophytes formation on x ray
Rationale:

symptoms which may include: • pain that increases on weight bearing activities (standing and walking, walking downstairs particularly) • insidious onset of symptoms followed by progressive periods of relapses and remissions • pain and stiffness in the morning • stiffness following periods of inactivity • pain and stiffness that arise after unaccustomed periods of activity • bony deformity (e.g. characteristic Varus deformity may follow from collapse of the medial compartmental joint space) • reduction of the
joint space observed on X-ray, with bony outgrowths or osteophytes

134- Osteoarthritis OA in old age due to:
a- traumatic
b- tear and wear process

Rationale:
Osteoarthritis in old age occurs when cartilage, the tissue that cushions the ends of the bones within the joints, breaks down and wears away. In some cases, all of the cartilage may wear away, leaving bones that rub up against each other which is considered primary osteoarthritis
But traumatic cause is considered a secondary osteoarthritis which occurs at any age. Joint injuries. Injuries, such as those that occur when playing sports or from an accident, can increase the risk of osteoarthritis.

135-Sever old osteoarthritis patient. You recommended for
a- hot backs + exercise + walking
b- walkers
c-cold backs + exercise + walking
d-cane

Rationale:
in severe osteoarthritis , physical therapy and exercises don't have the best effect on cases .So for these cases walkes will be the best thing to reduce the weightbearing on the joint.

136- Pt complains of pain in the hip region from 3 years increasing With time &WB its
A- O.A of hip joint
B- Tight hip flexor
C- Trochanteric bursitis

Rationale:
OA progress with time and the patient complain pain during weight bearing due to the bone friction.
But the symptoms of tight hip flexor are:
- A sharp or sudden pain in the hip, pelvis or groin area.

- Cramping, tender or sore muscles along the upper leg.
- Swelling or bruising on the hips or thigh.
- Pain in an adjacent muscle group, like your glutes or core.
- Decreased strength along the groin area.

The symptoms of Pain on the outside of the hip and thigh or in the buttock:

- Pain when lying on the affected side.
- Pain when you press in or on the outside of the hip.
- Pain that gets worse during activities such as getting up from a deep chair or getting out of a car.

137- In taping an athlete's ankle prophylactically before a football game, in what position should the ankle is slightly positioned before taping to provide the most protection against an ankle sprain?
a-Inversion, dorsiflexion, abduction
b-Eversion, plantarflexion, adduction
c-Eversion, dorsiflexion, abduction

Rationale:
Prophylactic taping is a preventive technique used for the protection, stabilization and care of athletic injuries

138- Pt cannot raise hand in abduction or maintain abduction position due to:
a) Adhesive capsulitis
b) Supra spinatoustendenitis
c) Rotator cuff tear

Rationale:
in case of adhesive capsulitis,the most common limitation in ROM is external rotation and abduction.
Principally, the 2 planes must be affected at the same time.
In case of supraspinatus tendonitis, the most common limitation is internal rotation.

139- The rotator cuff of the shoulder joint is weakest:
a- superiorly
b- anteriorly
c- posteriorly
d- inferiorly.

Rationale:
The supraspinatus is the most common injured muscle of the rotator cuff. If you have pain lifting your arm to the side, overhead, or reaching out to grab something, the supraspinatus may be the affected muscle.

140- patient with shoulder pain during evaluation there is no passive or active ROM in abd. And rot. That may cause by:
a- frozen shoulder
b- supraspinatus tendonitis
c- rotator cuff tear
d- acromioclavicular dysfunction

Rationale:
both passive and active rom are affected mainly in cases of frozen shoulder. There are at least 2 planes affected rotation and abduction.

141- Which of the following pass through the subacromial space?
A. Supraspinatus tenon, glenohumeral ligament, subacromial bursa
B. Supraspinatus tendon, long head of bicep tendon, subacromial bursa
C.Subscapularis tendon, supraspinatus tendon, long head of bicep tendon
D. Long head of bicep tendon, short head of bicep tendon and radial nerve

Rationale:
there are only 4 structures that pass through subacromialbursa ,(Supraspinatus tendon, long head of bicep tendon, subacromial bursa, superior capsule) and if this space is compressed,shoulderimpingment syndrome will occur.

142- A 60-year-old patient is complaining from a painful left shoulder with limited
range of motion that prevents him from changing his clothes independently. Examination reveals that both active and passive range of motion were restricted rotation and abduction particularly. Which of the following is the most likely diagnosis?
A. Bankart Lesion
B. Deltoid Bursitis
C. Adhesive Capsulitis
D. Subacromial Impingement

Rationale:
A Bankart lesion is a lesion of the anterior part of the glenoid labrum of the shoulder. This injury is caused by repeated anterior shoulder subluxations. so limitation will be in all directions. Deltoid busitiscausessharp pain with overhead shoulder motions.subaromialimpingment causes restricted shoulder motion with weakness in movements such as reaching overhead, behind the body, or out to the side.

143- Acute adhesive capsulitis:
a- acute inflammation capsule with fibrous

b- chronic inflammation capsule with fibrous
c- capsule inflammation
d- fibrous capsule

Rationale:
in case of adhesive capsulitis ,there are thickening of the synovial capsule, adhesions within the subacromial or subdeltoid bursa, adhesions to the biceps tendon, and/or obliteration of the axillary fold secondary to adhesions.

144- During examination of shoulder, you found increased motion (laxity) in anterior, posterior and inferior directions, the condition is
a- Global shoulder instability
b- Rotator cuff tear

Rationale:
Shoulder instability usually occurs when the lining of the shoulder joint, ligaments or labrum become stretched, torn or detached, allowing the ball (humeral head) to move either completely or partially out of the socket. But ,rotator cuff tear causes limitation in shoulder motion.

145-A patient with a diagnosis of a rotator cuff tear has just begun active range of motion. The therapist is strengthening the rotator cuff muscles to increase joint stability and oppose the superior shear of the deltoid. Which of the rotator cuff muscles participate least in opposing the superior shear force of the deltoid?
a- Infraspinatus
b- Subscapularis
c-Teres minor
d-Supraspinatus

Rationale:
the line of action of supraspinatus is directed superiorly so will oppose the inferior shear force . And vice versa in case of the infraspinatus, subscapularis and teres minor.

146- A therapist is attempting to gain external rotation range of motion in a patient's right shoulder. The therapist decides to use contract-relax-contract antagonist. In what order should the following rotator cuff muscles contract to perform this movement successfully?
a-Infraspinatus – teres minor
b-Subscapularis – supraspinatus
c-Teres minor – infraspinatus
d-Supraspinatus – subscapularis

Rationale:
because we want to stretch the muscle that prevent external rotation its called agonist and it's subscapularis because it is action internal rotation, so we want to stretch it to gain external rotation.

147- A 32-year-old patient with a chronic shoulder pain. Examination reveals excessiveinternal rotation of the shoulders and winging of the scapula during overhead motion.Which of the following is the best approach for helping this patient?
A. Strengthening of the rhomboids and stretching of the upper trapezius
B. Strengthening of the pectoral muscles and the upper trapezius muscles
C. Stretching of the upper trapezius and strengthening the middle and lower trapezius
D. Stretching of pectoral muscles and strengthening the middle and lower trapezius

Rationale:
winging of scapula mean that there are tightness in pectoral muscles and weakness in serratus anterior, rhomboids, middle and lower trapezius and there is also long thoracic nerve affection.the excessive IR is caused by tightness of pectoral muscles.so we need to stretch the tight muscles and strengthen the weak muscles.

148- A 33-year-old patient with a reconstruction of massive tear of the rotator cuff. ThePhysical Therapist decided to initiate the full active shoulder flexion. Which of the following is the most appropriate prescribed time in weeks to initiate themovement?
A. 1
B. 2
C. 3
D. 6
Rationale
Following a reconstruction of massive tear of the rotator cuff, the full active shoulder flexion is preferred to be initiated after 6 weeks

149-a 43-year-old woman who received rotator cuff repair 5 weeks ago. During passive range of motion, the therapist notes a capsular end feel at 95o of shoulder flexion. What should the therapist do?
a- Continue with passive range of motion
b- Begin joint mobilization
c- Schedule the patient an appointment with the physician immediately.
d- A and B
Rationale:

it is better for clinical decision making to continue cooperation with the referring surgeon for optimization of results.

150- Some tests performs to detect ligament& capsule stability in shoulder joint when positioned in abduction lateral rotation with continued anterior propagation of humeral head this due defect in ligament in
A superior part
B inferior part
C anterior part

Rationale:
anterior apprehension test is positive when there is instability in the glenohumeral ligaments from anterior ,the humeral head will move anteriorly and this mean that there is instability in shoulder joint.

151-A patient presents to an outpatient clinic with complaints of shoulder pain. The therapist observes a painful arc between 70o and 120o of active abduction in the involved shoulder. This finding is most indicative of what shoulder pathology?
a-Rotator cuff tear
b-Acromio - clavicular joint separation
c-Impingement
d-Labrum tear

Rationale:
from 70 to 120 its impingement, last 30 degree of abduction its acromiocalvicular joint separation, if patient can't raise arm in abduction or maintain abduction its rotator cuff tear, if there is no active or passive abduction and lateral rotation it's frozen shoulder)

152- A 48-year-old woman had pain in her right shoulder 3 weeks ago. During treatment, thetherapist abducts the patient's arm away from the body passively, but the patient cannotmaintain this position.Which of the following is the most likely diagnosis?
A. Frozen shoulder
B. Bankart Lesion
C. Rotator cuff rupture
D. Supraspinatus impingement

Rationale:
Rotator cuff injuries often result in a dull ache that can be felt deep in the shoulder. It can occur due to specific movements, such as lifting or lowering the arm. Some people experience discomfort during rest as well. Sleeping is often disturbed by pain, especially when lying on the affected shoulder.

153- partial painful arch of ROM between 90-120 what is the cause ?
a- supraspinatus impingement
b-adhesive capsulitis

Rationale:
painful arch From 90 to 120 means supraspinatus impingment.
Painful arch From 160 to 180 means acromioclavicular injury..

154- In taping an athlete's ankle prophylactically before a football game, in what position should the ankle is slightly positioned before taping to provide the most protection against an ankle sprain?
a-Inversion, dorsiflexion, abduction
b-Eversion, plantarflexion, adduction
c-Eversion, dorsiflexion, abduction

Rationale:
prophylactic ankle taping has been the mainstay for prevention of ankle injuries. Proper ankle taping is generally understood to reduce plantarflexion and inversion of the ankle , and this has been shown to be the most common mechanism associated with ankle sprain injuries. so the ankle will positioned in eversion , dorsiflexion and abduction.

155- A 21-year-old athlete came to the Physical Therapy Clinic with left knee pain. After the examination left patellar tendinitis was found.
Which of the following is the most likely activity predisposing to patellar tendinitis?
A. Jumping
B. Walking
C. Running
D. Stair climbing

Rationale:
Patellar tendinitis, also known as jumper's knee, is most common in athletes whose sports involve frequent jumping — such as basketball and volleyball.

156- in chondromalacia consider in the treatment ?
a- patellar tapping and strenthening of the quadriceps
b- strengthening vastusmedialis only
c- strentheningvastuslateralis only
d- strenthening of hamstring

Rationale:
The best treatment of chondromalacia includes :

Doing exercises to both stretch and strengthen the quadriceps and hamstring muscles ,Taping to realign the kneecap ,Losing weight (if you need to),Using special shoe inserts and support devices ,Wearing the right kind of sport or running shoes

157- Patient has chondromalacia contraindicated to use?
a- mobalization
b- taping , bandage
c- strengh hamstring

Rationale:
the hamstring should be stretched not strengthened.

158-A 25-year-old man had pain in his left knee one week ago. During treatment, there is pain behind kneecap when he walks downstairs. Which of the following is the most likely diagnosis?
A. Knee osteoarthritis
B. Chondromalacia patella
C. Anterior cruciate ligament injury
D. Medial collateral ligament injury

Rationale:
going down puts much more pressure on the knees. When you go down stairs, the force on the kneecap is 3.5x your body weight. This means that if you weigh in the neighbourhood of 80kg, your knees bear a force of 280 kg For such a small surface area

159- power weight lifter is being evaluated by a physical therapist for pain in his knee joint . The therapist notes severe pain during grinding of patella and when climbing stair Which of the following is the most appropriate description for this condition?
A- Jumpers knee
B- Perthes disease
C- Chondromalacia
D- Illiotibial-ban frictional syndrome

Rationale:
Chondromalacia patella (knee pain) is the softening and breakdown of the tissue (cartilage) on the underside of the kneecap (patella). Pain results when the knee and the thigh bone (femur) rub together. Dull, aching pain and/or a feeling of grinding when the knee is flexed may occur.

160- A 30-year-old patient sustained a brachial plexus injury along with a comminuted fracture of the radial head in a car accident. A total elbow

arthroplasty has been advised. Which of the following is a contraindication to the procedure?
A. Irreparable ligaments
B. Elbow flexors paralysis
C. Insufficient bone stock
D. Heterotopic ossification

Rationale:
a successful arthroplasty need sufficient bone stock to catch the implant in place

161- A 60-year-old woman is referred to outpatient physical therapy services for rehabilitation after receiving a left total knee replacement 4 weeks ago. The patient is currently ambulating with a standard walker with a severely antalgic gait pattern. Before the recent surgery the patient was ambulating independently without an assistive device. Left knee flexion was measured in the initial evaluation and found to be 85o actively and 94o passively. The patient also lacked 10o of full passive extension and 17o of full active extension. Which of the following does the therapist need to first address?
a- Lack of passive left knee flexion.
b- Lack of passive left knee extension
c- Lack of active left knee extension
d- Ability to ambulate with a lesser assistive device.

Rationale:
the patient has to achieve passive knee extension before she can gain full active knee extension. Full active knee extension and full flexion are important and should be a major focus of the patient's session, but the question asks for the most serious deficit. Ambulating with a lesser assistive device should be the focus at a later time because the patient's gait is still severely antalgic and obvious instability is still present. Usually a patient is advanced to a lesser assistive device when he or she can ambulate without large gait deviations with the current assistive device

162- A patient presents to physical therapy with complaints of pain in the right hip due to osteoarthritis. Which of the following is not true about this type of arthritis?
a- Causes pain usually symmetrically because it is a systemic condition.
b- Not usually more painful in the morning.
c- This type of arthritis commonly involves the distal interphalangeal joint.
d- Mainly involves weight-bearing joints.

Rationale:

this type of arthritis commonly involves the distal interphalangeal joint are complains of Osteo myelitis not osteoarthritis

163- A 57-year-old patient diagnosed as reflex sympathetic dystrophy and referred to theOutpatient Physical Therapy Clinic.
Which of the following is the most important line of treatment?
A. Skincare to prevent trophic ulcer
B. Vigorous mobilization of stiff joints
C. Interruption of the sympathetic reflex arc
D. Strengthening of the hand muscle to preventing wasting

Rationale:
physical therapy intervention for reflex sympathetic dystrophy is Movement therapy,Graded motor imagery,Range-of-motion exercises,Manualtherapy,Muscle strengthening ,Functional training.

164- A 65 years old patient with well treated coll's fracture, few monthes later came with severe pain in hand with coldness color changes . x ray show big callus around radial head not remember This may be due to :
A- osteoarthritis of wrist
B -prearticular ossification
C- Sudeck's atrophy/reflex sympathetic dystrophy

Rationale:
Sudeck's atrophy /reflex sympathetic dystrophy (RSD) /algo-dystrophy /causalgia
the term complex regional pain syndrome is now being used to describe these pathological states; This is a complication where the patient complains of severe pain on movement or at rest out of proportion to the initial injury. The limb is swollen. The skin appears shiny and discolored and feels cold; in extreme cases this may lead to the limb becoming exquisitely tender and discolored, Osteoporotic and have permanent contractures.

165- Subdeck's atrophy which is not true:-
a- sever burning pain
b- sympathetic hyperactivity
c- trophic changes occur
d- more perspiration

Rationale:
as Sudeck's atrophy is a sympathetic dystrophy not hyperactivity.

166-A 34 - year - old man with increased lumbar lordosis in a standing position, but when the patient sits the lordosis decrease or becomes absent.
Which of the following muscles should be stretched in this case?
Alliopsoas
B Lumbar flexor
C Gluteus maximus
D Rectus abdominus

167-Patient have lumbar lordosis when he stand and disappears when he sits we stretch: -
A-Iliopsoas
B-Gluteus maximus
C-Flex lumbar

169- The most common deformity in geriatric is
a- Kyphosis
b- Scoliosis
c- Kyphoscoliosis
d- Lordosis

170- The most common site of fracture in old patient due to failing is:
a. head of femur
b. trochanteric
c. shift of femur
d. neck of femur.

Rationale:
as it is the most weakest point

171- Severe kypho-scoliosis will lead to
a- Left rather than right ventricular failure
b- Right rather than left ventricular failure
c- Frequent respiratory infections
d- No cardiac abnormalities

172- In contracture of lower lumbar back muscles, the pelvis is affected by
a. Anterior pelvic tilting with upper thoracic kyphosis
b. Anterior pelvic tilting with lordosis
c. Posterior pelvic tilting with lordosis
d. No effect on the pelvis

Rationale:

Anterior pelvic tilt is caused by muscle imbalances in the lower back, hips, and gluteal muscles (glutes)

173- What does the tightness of the low back muscle produce?
A. Lateral pelvic tilt and scoliosis
B. Anterior tilt of the pelvis and lordosis
C. Posterior tilt of the pelvis and lordosis
D. Anterior tilt of the pelvis and kyphosis

174-As which of the following vertebral column or trunk problems the weakness of the abdominal muscles is most likely manifested?
A. Increased lumber lordosis
B. Increased thoracic kyphosis
C. Increased posterior pelvic tilt
D. Decreased posterior pelvic tilt

175- Patient with abd scapula, round shoulder, kyphosis ,ant rotation of pelvic
A. Lordosis
B. Sway back
C. Flat back
D. Kypholordosis

Rationale:
due to abduction scapula with round shoulder " Kyphosis " with Anterior rotation pelvic.

176- Patient has post pelvic tiled, flat lower thoracic & increase upper back kyphosis
A) Flat back
B) Sway back

Rationale:
because it`s not recognized that the patient has excessive hip extension with knee extension and flat lower thoracic

177- A 65 - year - old thin man presented with pain in the upper back. There was a visible hump on his back.
Which of the following is the most appropriate management?
A- Calcium intake
B- Regular painkiller
C- Paraspinal exercises
D- Shortwave diathermy .

Rationale:

Due to excess formation of fat

178-A therapist would diagnose a case with uncompensated and isolated right gluteus medius weakness by just observing the patient while walking. Which of the following is a demonstration of this condition?
A .Drop of the left pelvis during stance on the right.
B.Drop of the right pelvis during the stance on the left .
C.Circumduction of the left pelvis during stance on the right.
D.Lateral flexion of the trunk to the left during the stance on the right.

179-A patient who has suffered a zone 2 rupture of the extensor tendon of the third digit presents to physical therapy. This patient had a surgical fixation of the avulsed tendon. During the period of immobilization, which of the following deformities is most likely to develop?
a- Boutonniére deformity
b- Claw hand.
c- Swan neck deformity
d- Dupuytren's contracture.

180- A 52 - year - old man with right - hand finger pain, an inflexion of the proximal interphalangeal joint, and hyperextension of the distal interphalangeal joint. Which of the following is the most likely finger deformity?
A. Mallet finger
B Trigger finger
C. Swan neck deformity
D. Boutonniere deformity

Rationale:
as it causes injury to the extensor tendon of pip joint.

181- The therapist receives a referral to evaluate a patient with a boutonniere deformity. With this injury, the involved finger usually presents in the position of
a- Flexion of the proximal interphalangeal (PIP) joint and flexion of the distal interphalangeal (DIP) joint.
b- Extension of the PIP joint and flexion of the DIP joint.
c- Flexion of the PIP joint and extension of the DIP joint.
d- Extension of the PIP joint and extension of the DIP

Rationale:
as it causes injury to the extensor tendon of pip.

182- Type of contraction which entire MS acting over two joint muscle
a-active insuffiency
b-passive insuffiency

c-isometric cont

Rationale:
Active insufficiency occurs when a multi-joint muscle shortens over BOTH joints simultaneously, and hence, creates so much slack, that muscle tension is almost completely lost.For example: Knee is flexed, hip is extended, hamstring is shortened and in a weakened position.

183- Muscle acting over two joint can`t contract on two joint at same time
a- active insuffiency
b- passive insuffiency
c- isometric cont

Rationale:
Passive insufficiency is the muscle which stretched over two joints at the same time of contracting the opposite side.
It occurs when the multi-joint muscle is lengthened to its fullest extent at both joints, but also preventing the full ROMof each joint it crosses.

184-A therapist is screening a patient complaining of pain at the anterior left shoulder region. The pain is increased when the examiner instructs the patient to position the left arm by his side with the elbow flexed at 90o and to actively supinate the forearm against resistance (provided by the examiner). What test is being performed
a- Froment's sign.
b- Yergason's test
c- Waldron test
d- Wilson test.

185- A physical therapist is evaluating a patient who complains of posterior ankle pain. The patient is positioned prone with the feet extended over the edge of the mat. The therapist squeezes the involved gastrocnemius over the middle third of the muscle belly. What test is the therapist performing? What indicates a positive test?
a- Thompson's test – plantarflexion of the ankle
b- Homan's test – plantarflexion of the ankle.
c- Thompson's test – no ankle movement
d- Homan's test – no ankle movement.

Rationale:
Tompson test used for detect gastroconemuis injuries and become positive when there is rupture in it so no ankle movement

186- anterior taleofibular ligament to be assessed resist
a- planter flex + inversion
b- dorsi + inv
c-dorsi + eversion
d- plant + ev

Rationale:
it`s Anterior drawer testPatient lies supine with their foot relaxed and the examiner stabilizes the tibia/fibula with the foot in 20 degrees of plantar flexion. The examiner grabs the ankle and draws the talus forward in the mortise. Inversion gives an anterolateral stress with increased stress on the ATF.

187-Which of the following is the most useful orthopedic test to differentiate intra-articular and extra-articular injury?
A. Ober's
B. Gapping
C. Integrity
D. Quadrant

188-The therapist is performing an orthopedic test on a 25-year-old man with the chief complaint of low back pain. The patient has a positive Thomas test. With this information, what might the therapist need to include in the treatment plan?
A- Stretching of the hip abductors
b- Stretching of the hip adductors
c- Stretching of the hip extensors
d-Stretching of the hip flexors

189- shoulder dislocation head humerusfallen: -
A- posterior
B- inferior
C- anterior

Rationale:
because anterior dislocation occur 90%of dislocated shoulder and in it the humeral head moves toward the front of the body

190-A 23-year-old young athlete complains of right shoulder dislocation. Which of the following is the most common direction?
A. Anterior and superior
B. Anterior and inferior
C. Posterior and superior
D. Posterior and inferior

191- Patient has shoulder dislocation and reduction for this dislocation and referral to you for early mobilization to prevent:
a- Stiffness
b- Recurrent dislocation
C- Osteoarthritis

Rationale:
Stiffness so we do early mobilization to reduce acquired weakness and improve functional recovery

192-A 24year old patient with a shoulder dislocation for 3 weeks, has been referred to a physical therapist for early mobilization of shoulder joint. the goal behind early mobilization is: -
a-to prevent stiffness of the shoulder joint
b -to prevent osteoarthritis as a complication of the dislocation
c -to prevent axillary nerve injury
d -to prevent the complication of recurrence of dislocation

193-A 24 - year - old patient with shoulder dislocation for 3 weeks has been referred to a physical therapist for the early mobilization of the shoulder joint. Which of the following is prevented by early mobilization?
A- Tendinitis
B- Nerve injury
C- Vascular insufficiency
D- Stiffness of the shoulder joint

194- Fracture due to repeated minor injury:
a- pathological
b- fatigue
c- repeated stress
d- compression

195- Patient has cruciate ligament injury
a) Strength Quad
b) Hamstring
c) Both

Rationale:
In acl+pcl (cruciate ligament)we should act on muscles that act on knee to stabilize it (quad +hamstring)

196-Talipesequinovarus:
a. Ankle dorsi flexed, everted, adducted

b. Ankle plantar flexed, inverted, abducted
c. Ankle dorsi flexed, everted, abducted
d. Ankle plantar flexed, inverted, adducted

197- Which statement not correct about talipesequinovarus:
a- clubfoot
b- maybe congenital.
c- D.F and eversion.
d- P.F and inversion.

Rationale:
Dorsi flexion and eversion isn`t Talipesequinovarus because taipes (clubfoot)is congenital is where a baby is born with afeet that turn in and under plantarflexion at the ankle combined with adduction and inversion at the subtalar, midtarsal and anterior tarsal joints. Club foot can be described as congenital dislocation of the talo-calcaneal-navicular joint).

198- Valgus position of foot
a- planterflexion
b- dorsiflexion
c- eversion
d- inversion

200- which movement not accompany with hyper extension of cervical region:
a- poking chin
b- decrease flexion of lower cervical region
c- increase activity of SCM AND levator scapulae

Rationale:
because when wakened the deep cervical flexors lengthen as the chin tilts away from the neck often called (chin poking)
and when decrease flexion of lower cervical the lordotic curve of spine loss and increase hyperextention

201- Which of following not risk factor for osteoporosis?
a-enclosed family history
b- Fracture in gym exercises
c-low calcium intake
d-corticosteroid medications

202- Decrease osteoblast cause
a-osteoporosis

b-ricket
c-osteomalacia

Rationale:
because osteomalcia due to vitamins and mineral diffeciency and it is happen in children and adult .rickets caused by vitamins and mineral diffeciency and affect children

203- Patient stay at hospital treated by corticosteroid then go to physiotherapist, this patient will have problem it is
a- myopathy
b- osteoporosis
c- depression
d- decrease appetite

Rationale:
The side effect of corticocosteroid is muscle weakness ,blurred vision ,hair thining ,excessive hair growth ,bruising,slow wound healing,acne,osteoporosis,high blood pressure,stomachirritation,ulcers,glaucomaorcataracts,roundeness of face

204- A 53-year-old patient is referred to the physical therapy for mobility training following a prolonged hospitalization. During the hospital stay, the patient was on corticosteroids for the management of the medical condition. Which of the following is a potential adverse effect of prolonged use of corticosteroids?
A. Myopathy
B. Depression
C. Osteoporosis
D. Decreased appetite

205- A 62-year-old patient is diagnosed with osteoporosis and has no fractures. For the last month, the patient started feeling increased mid and low back pain while breathing and during other functional movements. Which of the following are the most appropriate exercises for this patient?
A. Trunk flexion and rotation
B. Trunk flexion and extension
C. Trunk rotation and abdominal stabilization
D. Trunk extension and abdominal stabilization

Rationale:
stabilization because we should avoid high impact exercise ,activities such as jumping ,running,or jogging ,jerky and rapid movement

206- what is the best position for artificial hip ankylosis
A-Flex 20 , abd 10, ext rotation 5
B-No flex , abd 10, ext rotation 5
C -Abd 5 , ext rotation 10

Rationale:
Surgical Considerations: Arthrodesis, also known as artificial ankylosis
- position of hip fusion:
- neutral abduction, external rotation of 0-30 deg&, 20-25 -deg of flexion;
- avoid abduction and internal rotation;
- this position is design to minimize excessive lumbar spine motion

207-postoperative arthroplasty complication?
a- fracture
b- dislocation
c- infection

208- Position of hip arthroplasty :
a- abd. 30 with flex. Maintained at 70_90
b- Abd. 20 with flex. Maintained at 70_90
c- abd.10 with flex maintained at 70_90
d- abd.30 with flex. Maintained at 20_90

Rationale:
the patient shouldnot lean forward (flexion) so we take least degree in flexion the choice number

209-A 70-year-old patient who had a total hip replacement was referred to the Physical Therapy Clinic for post-operative rehabilitation and education. Which of the following hip movements should **be avoided**?
A. Internal rotation, flexion, and adduction
B. External rotation, flexion, and abduction
C. Internal rotation, extension, and abduction
D. External rotation, extension, and adduction

210-A 76-year-old woman received a cemented right hip arthroplasty (THA) 24 hours ago. The surgeon documented that he used a posterolateral incision. Which of the following suggestions **in inappropriate** for the next 24 hours?
a- Avoid hip flexion above 30o
b- Avoid hip adduction past midline
c- Avoid any internal rotation.
d- Avoid abduction past 15o.

211-Patient with total hip replacement and in acute stage we do:-
A-daily
B-three to four times a weak
C-twice a weak

212- Patient with depression of metatarsal pad and claw toe the clinical picture:
a- Hyperext of metatarsal bone and ext of I.P.JT
B- FLEXION of metatarsal bone and ext of I.P JT
C- ABDUCTION OF metatarsal bone AND FLEXION I.PJT
D- hyper ext of metatarsal bone and flexion of I.P JT

Rationale:
clawing (hyperextension of the metatarsophalangeal joints and flexion of the other phalanges)
• Mallet toe (flexion of the distal interphalangealjoints)
• hammer toe (hyperextension of the metatarso-phalangeal and flexion of the proximal interphalangeal joint)
• Hallux valgus (lateral deviation of the first interphalangea
• Hallux rigidus (stiffness of the first interphalangeal

213- A therapist is evaluating a patient diagnosed with right-knee flexion deformity. Which of the following physical findings is most likely detected during the physical evaluation?
A. Knee flexion, ankle normal, short quadriceps muscle
B. Knee flexion, ankle dorsiflexion, short hamstring muscle
C. Knee flexion, ankle plantar flexion, short quadriceps muscle
D. Knee flexion, ankle plantar flexion, lengthened soleus muscle

214- patient with winged scapula and excessive internal rotation needs to
a- stretch middle and lower trapezius
b- Stretch pectorals major and strength middle trapezius
c- strength pectorals major and Stretch upper trapezius

216-What ligament is most involved in sustaining the longitudinal arch of the foot?
A-Plantar calcaneonavicular ligament
b- Long plantar ligament
c- Plantar calcaneocuboid ligament
d- Anterior talofibular ligament

Rationale:
Planter calcaneo-navicular ligament(spring lig)is strong lig connect bones in foot and ankle

217- triple arthrodesis involves:
a-Calcaneocuboid, talonavicular and talocalcaneal.
b-Tibiotalar, calcaneocuboid and talonavicular.
c-Ankle joint, calcaneocuboid and talonavicular.
d-None of the above

Rationale:
Because triple arthrodesis is fusion procedure of 3 joints (Calcaneocuboid, talonavicular and talocalcaneal.)

218- A patient is referred to physical therapy with a ruptured Achilles tendon. The patient reports that he was moving the lawn and was going down a steep incline when he felt a sharp, sudden pain in the left heel region. Which of the following would be proper location for the Achilles tendon insertion?
a- Talus
b- Calcaneus
c- Cuboid
d- Navicular

Rationale:
Note The Achilles tendon is a thick tendon located in the back of the leg. It connects the gastrocnemius and soleus muscles in the calf to an insertion point at the calcaneus (heel bone). It is the strongest tendon in the body and allows people to push off while walking, running and jumping.

219-Which of the following is the most likely cause of absent plantar flexion on gastrocsoleus squeeze?
A. Ankle sprain
B. Sciatic nerve injury
C. Achilles tendon tear
D. Deep vein thrombosis

Rationale:
Gastroconemuis that responsible for planter flexion and insert in achilles tendon

220- Fracture due to repeated minor injury:
a- pathological
b- fatigue
c- repeated stress
d- compression.

Rationale:

a/A broken bone caused by disease, often by the spread of cancer to the bone.
d/Compression fractures are small breaks or cracks in the vertebrae due to Osteoporosis

221- burst fracture?
a- Humerus
b- Ankle
c- spine

Rationale:
A burst fracture is a type of traumatic spinal injury in which a vertebra breaks from a high-energy axial load

222- Bennett's fracture is
a. Little finger fracture
b. Thumbfracture
c. Middle finger fracture
d. Distal end of radiusfracture

Rationale:
a/ called broken pinky toe
b/A Bennett fracture is a fracture of the base of the thumb resulting from forced abduction of the first metacarpal.
d/ called Broken Wrist

223- Monteggia fracture means which of the following
a- fracture of the radius and subluxation of the lower end of ulna
b- fracture of the ulna with radial head subluxation
c- green stalk fracture with minor angulation fracture of the radio-ulnar joint

Rationale:
a/ called DRUJ or gallize fructure
b/Monteggia injury is a radial head dislocation combined with an ulnar shaft fracture
c/ greenstick fracture occurs when a bone bends and cracks, instead of breaking completely into separate pieces.
occur in children younger than 10 years of age.

224- The angle between longitudinal axis passing throw femural neck and longitudinal axis of fumer is decreased in the case of :
A- genu valgum
B -genu varum
C- genu recurvatum
D- coxa vara

Rationale:
a/ In genu valgum, the lower extremities turn inward, causing the appearance of the knees to be touching while the ankles remain apart.
b/ condition where one or both of your child's legs curve outward at the knees. This creates a wider space than normal between the knees and lower legs.
c/ Genu recurvatum is a term that is used when one hyperextends their knee.

225- the normal angle between the femur and the neck of femur is 126. when the angle increase the deformity is:
a-coxa valga
b-genu Valgus
C- genu recurvatum

Rationale:
a/Coxa valga is defined as the femoral neck shaft angle being greater than 139 °
b/In genu valgum, the lower extremities turn inward, causing the appearance of the knees to be touching while the ankles remain apart.
c/ Genu recurvatum is operationally defined as knee extension greater than 5 degrees.

226- The first thing affected in Rheumatoid arthritis :
a- capsule
b- articular cartilage
Rationale:
Cartilage damage is a key feature of degenerative joint disorders-primarily osteoarthritis (OA)

227- Which of the following is the main reason that vigorous stretching and manipulative techniques are contraindicated in acute stages of rheumatoid arthritis?
A.Flexibility
B.Hypermobility
C.Hypomobility
D.Joint protection

Rationale:
Stretching can help improve flexibility, reduce stiffness, and increase range of motion. Stretching daily is important for relieving RA symptoms.

228- Involvement of PIP joint, DIP joint and the carpometacarpal joint of base of thumb with sparing the wrist is seen in:
a- Rheumatoid arthritis
b- Osteoarthritis

c- Psoriatic arthritis
d- Pseudogout

Rationale:
b/ usually ocuur in large joints
d/Pseudogout most commonly affects the knees. Less often, it affects the wrists and ankles

229- If a patient has longstanding rheumatoid arthritis of the cervical spine, the complication that is most likely to be seen and of concern is:
a- facet joint impingement.
b- disk herniation
c- spinal stenosis
d- subluxation.
Rationale:
Because subluxation happens due to inflammatory process of RA in soft tissue, while a,b, c answers happenes in degenerative problems

230-patient came to clinic with morning pain and stiffness this disease mean
a- systemic degenerative
b- muscle spasm
c- joint infection
Rationale:
b and c answers don't related specially to morning while answer A is correct as systemic degenerative areSome of the most common types of degenerative diseases are cancer, diabetes, Parkinson's, Alzheimer's, rheumatoid arthritis, and osteoporosis. Millions of people around the world are affected by these diseases. In fact, in most countries, certain types of degenerative diseases are consistently among the top causes of mortality

231-A patient is referred to the therapist with a diagnosis of arthritis. What type of arthritis would the therapist expect if the patient present with the following signs and symptoms? (1) Bilateral wrists and knees are involved, (2) pain at rest and with motion, (3) prolonged morning stiffness, and (4) Crepitus.
a- Osteoarthritis
b- Rheumatoid arthritis
c- Degenerative joint disease
d- It is not possible to determine with the given information

Rationale:
a/ pain is associated with movement
b/ prolonged morning stiffness is the most sign of Rheumatoid arthritis

232- The difference between RA and OA
a- OA's degenerative disease
b- RA's is systemic disease & and joint have deformity
c- RA's remission and relapse
d- OA's curable disease

Rationale:
Because hydro static pressure on chest wall and on abdomen wall which acts on diaphragm and there is indirect effect on the shift of blood to the thorax :resulting in decrease vital capacity

233-patient suffering from morning stiffness this is a sign of
a- swelling of the body tissues
b- prolonged stiffness with inflammation.
c- muscle inflammation.
d- bonny projection at the joint

234-The therapist receives a referral to evaluate a patient with a boutonniere deformity. With this injury, the involved finger usually presents in the position of
a- Flexion of the proximal interphalangeal (PIP) joint and flexion of the distal interphalangeal (DIP) joint.
b- Extension of the PIP joint and flexion of the DIP joint.
c- Flexion of the PIP joint and extension of the DIP joint.
d- Extension of the PIP joint and extension of the DIP

Rationale:
a/ usually is due to chronic finger injury
b/ called swan neck

235- A boutonniere deformity of the finger would not consist of
a- hyperextension of the proximal interphalangeal joint
b-overstretch of the extensor digitorumcommunis tendon
c- volar slippage of the lateral bands
d- hyperextension of the distal interphalangeal joint

236- A therapist is evaluating a patient with poor motor coordination. The therapist observes that when the patient is standing erect and still, she does not respond appropriately when correcting a backward sway of the body. With the body in a fully erect position, a slight backward sway should be corrected by the body firing specific muscles in a specific order. Which list is the correct firing order?
A- Bilateral abdominals, bilateral quadriceps, bilateral tibialis anterior.
B- Bilateral abdominals, bilateral tibialis anterior, bilateral quadriceps

C- Bilateral tibialis anterior, bilateral abdominals, bilateral quadriceps
D- Bilateral tibialis anterior, bilateral quadriceps, bilateral abdominals.

237-A 14-year-old girl with right thoracic scoliosis is referred to physical therapy. The therapist should expect which of the following findings?
a-Left shoulder high, left scapula prominent, and right hip high
b-Left shoulder low, right scapula prominent, and left hip high
c-Right shoulder high, right scapula prominent, and right hip high
d-Right shoulder low, right scapula prominent, and left hip high
Rationale:
Scoliosis name due to convex side, so rt side means right convex and left concave

238-A patient with adhesive capsulitis of the gleno-humeral joint should demonstrate the greatest limitation of motion when performing shoulder
a-flexion.
b-abduction.
c-medial rotation.
d-lateral rotation.
Rationale:
External rotation is the main movement to gain close backed position

239-A middle-aged diabetic woman patient diagnosed with adhesive capsulitis of
the shoulder. Which functional limitation most likely occurs in this situation?
A. Bathing
B. Combing
C. Carrying heavy object
D. Reaching hand into back pocket

Rationale:
combing is an action containing flextion and abduction of shoulder

240- patient come with sever elbow pain we can use
a- Heat application
b- Cold application
c- Faradic stimulation

Rationale:
if acute trauma, use cold & if chronic, use heat & if acute inflammation, use cold.

241-Benefits of isotonic exercise include the following except:
a- high mechanical energy

b- Less resistance to fatigue
c- High efficiency
d- Not delayed muscle soreness

Rationale:
Pain of isotonic exercise appears during the activity

242-An athlete developed severe pain in his thighs 48 hours after a long run. Which of the following is the most likely explanation?
A. Sprain
B. Synovitis
C. Muscle cramp
D. Delayed onset muscles soreness

Rationale:
a/ pain appears immediately and swelling may occur
c/ muscle cramp is a sudden and involuntary contraction of one or more of your muscles. It takes some second or minutes
d/ is muscle pain that begins after you've worked out. It normally starts a day or two after a workout.

243- Manipulation is described in the management of
A-osteoporosis
B-spinal dysfunction
C-osteomyelitis

Rationale:
we contra indicated to make any manipulation technique for osteoporosis or osteomyelitis .

244- You wish to mobilize a patient's shoulder using an inferior glide technique. It would be best to use this technique be propositioning the patient's arm in:
a- 95 degrees of abduction with lateral rotation
b- 125 degrees of abduction and internal rotation
c- 55 degrees of abduction and neutral rotation
d- 95 degrees of shoulder flexion and neutral rotation

Rationale:
Glenohumeral joint:
- loose-packed position is 55* of abduction and 30* horizontal adduction.
- close-packed position is abduction and external rotation

245- in shoulder inferior gliding position of the arm
a- slight flexion

b- slight extension
c- 30 degree adduction
d- external rotation

246- An 87-year-old woman presents to an outpatient physical therapy clinic complaining of pain in the left sacroiliac joint. The examination reveals higher left anterior superior iliac spine (ASIS) than the right ASIS, shorter leg length on the left side (measured in supine position), and left side posterior superior iliac spine (PSIS) lower than the right PSIS. In what position should you place the patient to perform the correct sacroiliac mobilization of the left in nominate?
a- Right side lying
b- Supine
c- Prone
d-None of the above

Rationale:
the Sacroiliac joint disappear after 80 years old

247- A physical therapist is beginning an evaluation of a patient with a diagnosis of “knee strain”. Range of motion limitation does not follow the normal capsular pattern of the knee. Which of the following are possible causes of the restriction in range of motion?
a-Ligamentous adhesions
b-Internal derangement
c-Extra-articular lesions
d-All of the above

248- Which of the following durations in weeks are required for high - intensity exercises to regain wasted muscle mass?
A- 02-04
B- 05-07
C- 08-10
D- 11-12

Rationale:
the answer Is variable actually there Is a lot of factors like gender age and muscle type Influence the answer.
Gaining muscle is a slow process. It can take about three to four weeks to see a visible change. You'll see some real results after 12 weeks, but it "all depends on your goals, and what type of strength training you are doing

249- After performing a complete muscle test for a patient with bilateral lower limb weakness,the therapist recorded that the patient has a poor grade of the left hip abductors.Which of the following can the patient do?

A. Not able to abduct the leg
B. Perform a complete range of abduction from a side-lying position
C. Perform a complete range of abduction from the supine position
D. Perform an incomplete range of abduction from the supine position
Rationale:
Supine position for gravity elimination which is convenient with poor grade of muscle test, full ROM is better than partial ROM for better firing and strength of the muscle

250-A 35-year-old woman with a diagnosis of lumbar strain has a physician's prescription with a frequency and duration of 3 sessions/ week for 6 weeks. The physical therapy evaluation reveals radiculopathy into the L5 dermatome of the right lower extremity, increased radiculopathy with lumbar flexion, decreased radiculopathy with lumbar extension, poor posture, and hamstring tightness bilaterally at 60 . What is the best course of treatment?
a- Lumbar traction , hot packs, and ultrasound.
b- McKenzie style lumbar extensions, a posture program, hamstring stretching, and a home exercise program.
c- McKenzie style lumbar extensions, a posture program, hamstring stretching, home program, hot packs, and ultrasound.
d- Lumbar traction, hot packs, ultrasound, and hamstring stretching.

Rationale:
Hot packs are not indicated because there is no mention of abnormal muscle tone. The entire lumbar area is too much surface area for ultrasound. An argument could be made for lumbar traction, but it is paired with heating modalities in all of the answers.

251- A 63-year-old patient with lower back pain that started a year ago was referred to thePhysical Therapy Clinic for assessment and treatment. The therapist found tightness of agroup of muscles that predisposes to the development of the pain.Which of the following is the most likely group of muscles?
A. Quadriceps
B. Hamstrings
C. Gastrocnemius
D. Tibialis anterior
Rationale:
Lower crossed syndrome which is the main cause of mechanical LBP characterized with weak back extensors and hip flexors, while tight hamstring and lumbar flexors

252-A 42-year old receptionist presents to an outpatient physical therapy clinic complaining of low back pain. The therapist decides that postural modification

needs to be part of the treatment plan .What is the best position for the lower extremities while the patient is sitting?
a- 90o of hip flexion, 90o of knee flexion, and 10o of dorsiflexion
b- 60o of hip flexion, 90o of knee flexion, and 0o of dorsiflexion
c- 110o of hip flexion, 80o of knee flexion, and 10o of dorsiflexion
d- 90o of hip flexion, 90o of knee flexion, and 0 of dorsiflexion
Rationale:
10 degree dorsi flexion is better used as postural modification needs to be part of the treatment plan to stretch calf.

253-A 41-year-old patient with low back pain (LBP) stands at work for most of the day. The therapist advised the patient to keep one foot up on a stool to be more comfortable. Which of the following is the most appropriate explanation?
A.Elevating the foot relieves stress from the feet
B.Elevating foot decreases the presser on the knees
C. Elevating the foot allows the pelvis to tilt upward and decreases the lumbar lordosis
D. Elevating the foot relieves stress from the feet and decreases the presser on the knee

254-A physical therapist is asked to evaluate a 37-year-old man with right side sciatica. The therapist performs a passive straight leg raise test of the right lower extremity with the knee and ankle in neutral position. In performing this test on a patient with an L5 disc protrusion, what is the lowest degree at which the therapist would expect to reproduce the patient's symptoms?
a- At 25o of hip flexion.
b- At 65o of hip flexion
c- At 70o of hip flexion
d- At 90o of hip flexion.
Rationale:
Neurologic pain which is reproduced in the leg and low back between 30-70 degrees of hip flexion is suggestive of lumbar disc herniation at the L4-S1 nerve roots.
Pain at less than 30 degrees of hip flexion might indicate acute spondyloithesis, gluteal abscess, disc protrusion or extrusion, tumor of the buttock, acute dural inflammation, a malingering patient, or the sign of the buttock.
Pain at greater than 70 degrees of hip flexion might indicate tightness of the hamstrings, gluteus maximus, or hip capsule, or pathology of the hip or sacroiliac joints.

255-A 30-year-old woman with elbow dysfunction is referred to the physical therapy. The physical therapist assessed the patient and found limited flexion due to casting for a long time. The physical therapist decided to

mobilize the elbow joint to increase the range of motion.
What direction of mobilization should the therapist do following the convexconcave rule?
A. Rotational mobilization
B. Medial-lateral mobilization
C. Anterior-posterior mobilization
D. Inferior-superior mobilization
Rationale:
Ulnar hook end considered as the concave part which follow the same direction of the movement

265-A 43-year-old patient with pain radiating down the posterior aspect of the left hip and thigh is referred to a physiotherapist. According to the referral, the underlying cause is a posterior herniated disk in the lumbar spine. The physical therapist decides to apply mechanical lumbar traction. Which of the following is the best position for lumbar traction?
A. Prone with no pillow
B. Supine with both knees flexed
C. Prone with a pillow under the abdomen
D. Supine with a pillow under the lumbar spine

266-A 65 - year old man with left hip pain for the past year . The physical therapist assesses the patient and notices there is a pain and stiffness in the hip . particularly on internal rotation .
Which of the following is the most likely diagnosis ?
A - Acute hip osteoarthritis
B - Chronic hip osteoarthritis
C-Acute trochanteric bursitis
D- Chronic trochanteric bursitis
Rationale:
a-Acute hip osteoarthritis pain around joint with inflammation signs
B - Chronic hip osteoarthritis.. Crepitation with limited ROM especially external rot
C-Acute trochanteric bursitis
D- Chronic trochanteric bursitis..cant climbing up stairs or weight bearing with affected limb

267- Patient is suffering from morning stiffness; the most likely developing condition is due to:
A-Bones are growing beyond their margins
B- Fasciitis (Refers to an inflammation of the fascia)
C- Accumulation of lactic acid
D- Muscle Spasm

Rationale:
Your sleeping position can cause sustained compression, pinching, oxygen starvation of tissues and decreased blood flow to your muscles. This in turn can result in a build-up of lactic acid which causes stiffness when you wake up the next morning

268-Kumar, a 31 years old motorcyclist sustained injury over his right hip joint. X-ray revealed a posterior dislocation of the right hip joint. The clinical attitude of the affected lower limb will be:
A. External rotation, extension & abduction
B. Internal rotation, flexion & adduction
C. Internal rotation, extension & abduction
D. External rotation, flexion & abduction

Rationale:
Because it is the clinical attitude of the posterior dislocation.

269- patient came to you with pain in the right buttocks and local tenderness in the tibial tuberosity with pain in resistive knee flexion what is your diagnosis?
A) Hamstring strain
B) hip arthritis
C) sports hernia
D) joint inflammation

Rationale:
Hamstring strain Pain in the back of the thigh and lower buttock when walking, straightening the leg, or bending over Tenderness , Bruising

270- A 28-year-old professional weight lifter is undergoing a physical therapy examinationfollowing complaints of pain in the lower abdominal area and groin when lifting weights. Adetailed examination by the physiotherapist reveals pain in the groin on coughing, resistedadduction sit-ups, and weight-bearing. What is the most likely cause of the problem?
A. Hamstring strain
B. Sportsman's hernia
C. Osteoarthritis of hip joint
D. Tightness in the hip flexors(iliopsoas

Rationale:
the symptoms In the question Is related to the answer the other answer Isn't acceptable

271- patient has hand injury without open wound and making tight band only after few 3 week referral to you and have contracture of hand and wrist causes

of these are :
a- Dupuytren's contracture Tendon hand affected
b-Volkaman contracture Tendons hand&wrist
Rationale:
Volkaman is related to restricted blood flow and the case had injury without open wound and making tight band

272-Rheumatoid arthritis
a- cold limb
b- radial deviation
c-pain ,wet ,swelling

273- The first thing affected in Rheumatoid arthritis :
a- capsule
b- articular cartilage

274- Involvement of PIP joint, DIP joint and the carpometacarpal joint of base of thumb with sparing the wrist is seen in:
a- Rheumatoid arthritis
c- Osteoarthritis
d- Psoriatic arthritis
e- Pseudo gout

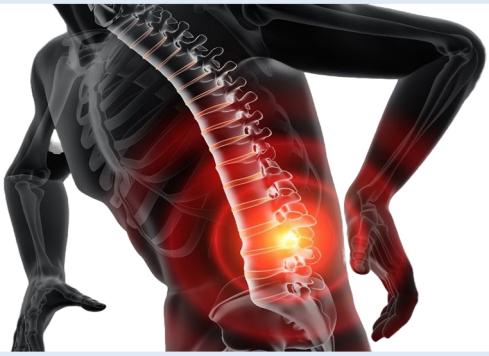
275-If a patient has longstanding rheumatoid arthritis of the cervical spine, the complication that is most likely to be seen and of concern is:
a- facet joint impingement.
b- disk herniation
c- spinal stenosis
d- subluxation.

276-patient came to clinic with morning pain and stiffness this disease mean
a- systemic degenerative
b- muscle spasm
c- joint infection

277-A patient is being evaluated by a physical therapist because of bilateral knee pain. The therapist is attempting to rule out ankle or foot dysfunction as the source of the pain. Which of the following observations is not true in evaluating a patient without foot or ankle problems in the standing position?
a- The talus is situated somewhat medially to the midline of the foot.
b- In quiet standing the muscles surrounding the ankle joint remain silent
c- The first and second metatarsal heads bear more weight than the fourth and fifth metatarsal heads
d- The talus transmits weight to the rest of the bones of the foot.

Rationale:

activity was recorded from the right lateral gastrocnemius, soleus, tibialis anterior, biceps femoris, rectus femoris, rectus abdominus and erector spinae muscles of the lumbar spine





Chapter

1.D Sports Medicine

Question 1

"Baseball " mallet finger ?

1. metacarpal
2. **distal phalanx**

Correct Answer: 2
rationale:
because mallet finger develops after an injury to the extensor mechanism at the DIP joint, resulting in "droop" of the finger into flexion at the DIP joint

Question 2

The blood flow percentages among athletes during exercise will:

1. **increases 15 fold with 60 mHg/ ml**
2. increases 25 fold with 60 mHg/ ml

Question 3

- What is the blood flow for an athlete at rest?

1. 1 to 2 ml/min/110g
2. 6 to 7 ml/min/110g
3. 60 to 80 ml/min/110g

Correct Answer: 1
rationale:
at exercises will be 6 to 7 ml/min/110g and at max activate will be 60to 80ml/min/110g

Question 4

Exercise guidelines for the well-trained, well-controlled insulin dependent athlete with diabetes recommend all of the following except:

1. **decreasing or omitting carbohydrate snacks just following intense exercise or activity.**
2. decreasing or omitting the insulin dose just prior to exercise or athletic activity.
3. decreasing or omitting the insulin dose just following the exercise or athletic activity.
4. ingesting up to 30 grams of additional carbohydrate for every 30 minutes interval of intense exercise or activity.

Correct Answer: 1
rationale:
Because Carbohydrates are the primary nutrient used during prolonged, moderate-to-high intensity exercise



Question 5

An athlete developed recurrent ankle sprain following a twisting injury to the ankle. Loss of which of the following is the most likely cause?
OR long jumping athlete female 23 years , comes to outpatient clinic with complain of instability after touching ground after jump she also says that no previous trauma to knee or hip what is the cause?

1. Sensation
2. **Proprioception**
3. Muscle strength
4. Ligament stretch

Correct Answer: 2
rationale:
Because Proprioception is our body's ability to tell where it is in space. Receptors in our joints send information to our brain about the position of our limbs and provides essential information to enable adjustment of ankle position

Question 6

Patient close his eyes and move his shoulder and then ask him about degree and position of limb so we test

1. light touch
2. fine touch
3. **proprioception**

Correct Answer: 3
rationale:
because Proprioception has been defined as one's ability to integrate the sensory signals from various mechanoreceptors to determine body position and movements in space

Question 7

A 29-year-old football player sustained a twisting injury to the ankle. There was severe pain and swelling at the injury site. He was unable to walk. His MRI ankle confirmed a tear at the deltoid ligament. He had been managed with Plaster of Paris and anti-inflammatory medicines. He had been advised to have post-cast physiotherapy. Which of the following is the most likely factor in applying controlled force in recovery?

1. Pain management
2. Swelling reduction
3. Collagen re-modeling
4. **Proprioception develop**

Correct Answer: 4
rationale:
Because When proprioception training is missed, it is more likely that a patient will re-sprain their ankle even if the patient "feels" like the ankle is back to normal.

Question 8

An athlete with stable fracture of neck of humerus, referred to you to improve mobility, which form of exercise will you choose
Isometric

1. **Resistive exercise in all directions**
2. Pendulum ex

Correct Answer: 1
rationale:
Because there is no need to fear from the resistance movement of the arm
If unstable fracture of neck humerus we choose the isometric exercises to prevent refracture of the humerus.

Question 9

Which of the following is the most appropriate minimum duration, in weeks, for high-intensity resistive exercises to achieve muscle hypertrophy?

1. **8**
2. 12
3. 16
4. 20

Question 10

An athlete with unstable fracture neck of humerus , refereed to you to improve mobility. Which form of exercise will you chose

1. **isometric**
2. resistive exercise in all directions

Correct Answer: 1
rationale:
To prevent refracture of the humreus

Question 11

History of football players has injured by twisted knee and take out of game and after little time it swelling and warm. after few days it locks during climbing stairs and painful and cannot put full weight in walking:

1. patella fracture
2. tibia fracture
3. **MCL rupture**
4. Rupture of something of fibers

Correct Answer: 3
rationale:because mechanism of injury for MCL rupture due to it`s injured by twisted knee, slightly flexed knee is forced into valgus while the tibia externally rotates. and there is pain with swelling, warm
Waddling Gait occurs when there is extreme muscular weakness in the thigh and hip muscles as commonly found in pseudohypertrophic muscular paralysis and muscular atrophy or dystrophy. In this gait, the shoulders are thrown back, the lower section of the spine is lordotic, the pelvis is tilted greatly, and when walking, the subject swings from side and his hip drops with each step.



Question 12

foot player come to the physiotherapist with pain at the left knee joint while the physiotherapist taking history patient said that the injury occurred when he rotates to the right on weight bearing to the left knee. it was flexed 60 degrees. that means that the injury is at

1. medial collateral ligament
2. **semilunar cartilage at the knee**
3. femoral condyles
4. upper shaft of the tibia

Correct Answer: 2
rationale:
Because Semilunar cartilage at the knee (meniscus tears) can happen when a person changes direction suddenly while running, or standing used one leg weight bearing and often occur at the same time.
Waddling Gait occurs when there is extreme muscular weakness in the thigh and hip muscles as commonly found in pseudohypertrophic muscular paralysis and muscular atrophy or dystrophy. In this gait, the shoulders are thrown back, the lower section of the spine is lordotic, the pelvis is tilted greatly, and when walking, the subject swings from side and his hip drops with each step.

Question 13

A 27-year-old football player sustained a knee injury and was referred to the Physical Therapy Clinic. The initial evaluation showed that it appeared as if the "terrible triad" type of injury was present. Which of the following major structures is most likely damaged in this type of injury?

1. Lateral and medial menisci and patellar tendon
2. **Medial meniscus, medial collateral ligament, and ACL**
3. Medial collateral ligament, medial hamstring, and quadriceps muscle
4. Lateral collateral ligament, iliotibial band, and lateral hamstring tendon

Correct Answer: 2
rationale:
terrible triad is an injury to the anterior cruciate ligament, medial collateral ligament, and meniscus..

Question 14

A patient is an outpatient facility because of an injury sustained to the right knee joint. Only the structures within the synovial cavity were compromised during the injury. Knowing this information only, the therapist is not concerned with injury to which of the following structure?

1. Patellofemoral joint
2. **Anterior cruciate ligament**
3. Medial meniscus
4. Femoral condyles

Correct Answer: 2
rationale:
because the two cruciate ligaments, which attach in the intercondylar region of the tibia below and the intercondylar fossa of the femur above, are outside the articular cavity

Question 15

A football player presents to an outpatient clinic with complaints of pain in the right knee after an injury suffered the night before. The physician determines that the anterior cruciate ligament (ACL) is torn. Which of the following is most commonly associated with an injury causing damage to the ACL only?

1. Varus blow to the knee with the foot planted and an audible pop.
2. **Foot planted, medial tibial rotation, and an audible pop.**
3. Valgus blow to the knee with the foot planted and no audible pop.
4. Foot planted, lateral tibial rotation, and no audible pop.

Correct Answer: 2
rationale:
Because most risk biomechanics causes of ACL are result of the abnormal anterior and rotary translation of the tibia and most common symptoms is audible pop sound

Question 16

Which of the following is the most likely cause of anterior cruciate ligament injury ?

1. Forceful knee bending
2. .Lateral twisting of the knee
3. Blow to the lateral side of the knee
4. **Blow to the posterior side of the knee**

Correct Answer: 4
rationale:
Because ACL function is to prevent excessive anterior translation of tibia on femur

Question 17

Which of the following is the most appropriate on the third postoperative day after anterior cruciate ligament reconstruction?

1. Stationary cycle, close chain exercise, elliptical trainer
2. Close chain exercises, balance board, full weight-bearing.
3. **Isometric exercise, partial weight-bearing, ankle pumping**
4. Patellar mobilization, open chain exercises, balance training

Correct Answer: 3
rationale:
Because according to ACL post-operative rehabilitation the open chain exercises, full weight-bearing and balance training not in the early stage of rehabilitation but in late stage.



Question 18

Which of the following is inappropriate exercise for a patient who received an anterior cruciate ligament reconstruction with a patella tendon autograft 2 weeks ago?

1. **Lateral step-ups**
2. Heel slides
3. Stationary bike
4. Pool walking.

Correct Answer: 1
rationale:
Because the graft is still weak and cant adapt with lateral stress in the early stage .

Question 19

A 32-year-old man is referred to physical therapy with the diagnosis of a recent complete anterior cruciate ligament tear. The patient and the physician have decided to avoid surgery as long as possible. The therapist provides the patient with a home exercise program and instructions about activities that will be limited secondary to this diagnosis. Which of the following is the best advice?

1. There are no precautions
2. The patient should avoid all athletic activity for 1 year.
3. The patient should avoid all athletic activity until there is a minimum of 20% difference in the bilateral quadriceps muscle as measure isokinetically.
4. **The patient should wear a brace and compete in only light athletic events.**

Correct Answer: 4
rationale:
We use knee brace to protect knee from stress because ACL resist anteromedial stress

Question 20

A 22-year-old patient was referred to the Physical Therapy Clinic for Anterior Cruciate Ligament (ACL) reconstruction rehabilitation. MMT of knee extension showed that the patient was able to extend his knee against minimal resistance for the full range of motion but not against maximal resistance. What is the appropriate Manual Muscle Testing (MMT) grade for this knee extension?

1. Grade 2
2. Grade 3
3. **Grade 4**
4. Grade 5

Correct Answer: 3
rationale:
Grade 4 consist of Full ROM against gravity, mild to moderate resistance. But grade 5 consist of Full ROM against gravity, maximal resistance

Question 21

Which of the following strains on the knee causes tears of the meniscus?

1. Adduction
2. Abduction
3. Hyperextension
4. **Combined flexion and rotation**

Correct Answer: 4
rationale:
Because most common mechanism of injury is turning or twisting maneuver of the leg in weight bearing which include semi flexion with rotation

Question 22

Which is the most likely structure injured if the patient is complaining of knee locking?

1. **Meniscus**
2. ACL tear
3. OA's
4. MCL

Correct Answer: 1
rationale:
Because If a meniscus tear occurs (The medial meniscus common), a fragment can break away and become stuck in the knee joint (between the articular surfaces of the femur and the tibia), causing the joint to lock and shows an abnormal springy end feel

Question 23

what is the structure that will be tested if you put the patient in a prone lying position with knee flexed 90 degrees and applying perpendicular force to the foot in addition to medially and laterally rotations:

1. **meniscus injury**
2. ligament injury
3. muscular injury
4. capsular injury.

Question 24

A 26-year-old woman slipped and had blunt trauma to the knee joint. She developed severe pain and swelling of the joint. Following regular medication and physiotherapy, her pain gradually subsided, but there were occasional painful episodes of locking of the joint. Which of the following is the most likely injured structure?

1. Bone
2. Capsular
3. Ligament
4. **Meniscus**

Correct Answer: 4
rationale:
Because the trauma causes a meniscal tear and the fragment break away and lead to locking of the joint



Question 25

among the following choices, what is the most appropriate cause of Knee joint medial rotation (internal Rotation)

1. weak vastus medialis
2. weak semitendinosus
3. Weak quadriceps
4. **weak biceps femoris**

Correct Answer: 4
rationale:
the biceps femoris has three main actions. It is participating in the knee flexion movement, lateral rotation of the knee joint (in flexed knee joint position) and extension of the hip joint (by the long head only). So, in case of weakness of the lateral rotators the knee joint will medially rotate due to the muscle imbalance.

Question 26

About Tennis elbow all the following is true except

1. also called lateral epicondylitis
2. **precipitated by ulnar neuritis**
3. usually occur in dominant side
4. occurs due to inflammation in the common Extensor origin.

Correct Answer: 2
rationale:
tennis elbow also called lateral epicondylitis due to (inflammation in the common Extensor origin) caused by overuse of the elbow in the dominant upper limb

Question 27

A physical therapist is speaking to a group of avid tennis players. The group asks how to prevent tennis elbow (lateral epicondylitis). Which of the following is incorrect information

1. **Primarily use the wrist and elbow extensors during a backhand stroke**
2. Begin the backhand stroke in shoulder adduction and internal rotation
3. Use a racket that has a large grip
4. Use a light racket.

Correct Answer: 1
rationale:
Because the cause of tennis elbow is repeated wrist and elbow extensors which cause pain common extensor origin and the pain can spread from the elbow into the forearm and the wrist.

Question 28

paralysis of all intrinsic muscles of the hand except the abductor pollicis Brevis

1. Median nerve
2. **Ulnar nerve**
3. Radial nerve
4. . posterior interosseous nerve

Correct Answer: 2
rationale:
The abdactor policis Brevis muscle is innervated by The Median nerve

Question 29

A patient has traumatically dislocated the tibia directly posteriorly during an automobile accident. Which of the following structures is the least likely to be injured?

1. Tibial nerve
2. Popliteal artery
3. **Common peroneal nerve**
4. Anterior cruciate ligament

Correct Answer: 3
rationale:
The common peroneal nerve is the smaller and terminal branch of the sciatic nerve which is composed of the posterior divisions of L4, 5, S1, 2.
It courses along the upper lateral side of the popliteal fossa, deep to biceps femoris and its tendon until it gets to the posterior part of the head of the fibula. So, it is not affected by the posterior dislocation of the knee joint

Question 30

You are evaluating an athlete who is complaining of pain in the left shoulder region. Your assessment of the shoulder elicits pain in the last 30 degrees of shoulder abduction range of motion. This finding is most congruent with

1. calcific supraspinatus tendinitis.
2. sub acromial bursitis.
3. **Acromioclavicular joint sprain.**
4. thoracic outlet syndrome.

Correct Answer: 3
rationale:
Acromioclavicular joint sprains are common, usually resulting from a fall on the shoulder or, less often, an outstretched arm. Tears (ruptures) may also occur in tendons. hgthe maximum compression of the will occurs in the in the last degree of shoulder abduction movement



Question 31

A 13-year-old girl had fractured her left patella during a volleyball game. The physician determines that the superior pole is the location of the fracture. Which of the following should be avoided in early rehabilitation?

1. Full knee extension
2. 45 o of knee flexion
3. **90 o of knee flexion**
4. 15 o of knee flexion

Correct Answer: 3
rationale:
The goals of treatment are a functional extensor mechanism, articular congruity and full, painless motion of the knee. An nondisplaced fractures of the patella with the normal function of the extensor mechanism can be managed non-surgically, initially with an extension splint. Over the next weeks, as the fracture unites, the range of flexion can be gradually increased, and the quadriceps strengthened in the inner range and the flexion will make tension on the fracture during the contraction of the quadriceps muscle

Question 32

A 24-year-old bicyclist had repetitive injuries in the last 2 years after changing his bike. The physical therapist hypothesis that ,the problem is in the weight distribution of the bicyclist which is nottdistributed evenly after using the new bike.Which of the following is the most appropriate kind of intervention?

1. Fitness
2. Psychological
3. **Biomechanical**
4. Strengthening

Correct Answer: 3
rationale:
The problem in the athlete body weight distribution in the bicycle should be managed by a biomechanical specialist

Question 33

A physical therapist is treating an automobile mechanic. The patient asks for tips on preventing upper extremity repetitive motion injuries. Which of the following is incorrect advice?

1. Use your entire hand rather than just the fingers when holding an object
2. Position tasks so that they are performed below shoulder height.
3. **Use tools with small straight handles when possible**
4. When performing a forceful task, keep the materials slightly lower than the elbow.

Correct Answer: 3
rationale:
in ergonomics the small straight handles will increase the possibility for more injuries

Question 34

A patient with Osgood-Schlatter disease will experience the most difficulty and pain while:

1. **playing basketball or volleyball**
2. sitting for prolonged periods of time
3. walking down stairs.
4. lifting heavy weights

Correct Answer: 1
rationale:

- Osgood-Schlatter disease (also known as Apophysitis of the tibial tubercle, or knobby knees)
- It is an irritation of the patellar ligament at the tibial tuberosity.
- It is characterized by painful lumps Just below the knee and is most often seen in young adolescents.
- Risk factors may include overzealous conditioning (running and jumping) but adolescent bone growth is at the root of it.

Question 35

- A teenager comes to an outpatient facility with complaints of pain at the tibial tubercle when playing basketball. The therapist notices that the tubercles are abnormally pronounced on bilateral knees. What condition does the patient most likely have?

1. Jumper's knees.
2. Anterior cruciate ligament sprain.
3. **Osgood-Schlatter disease**

Correct Answer: 3
rationale:
Osgood-Schlatter disease is a condition that causes pain and swelling below the knee joint, where the patellar tendon attaches to the top of the shinbone (tibia), a spot called the tibial tuberosity. There may also be inflammation of the patellar tendon, which stretches over the kneecap.Osgood-Schlatter disease is most commonly found in young athletes who play sports that require a lot of jumping and/or running

Question 36

A tennis player receives a surgical repair of the annular ligament. Where should the therapist expect to note the most edema?

Radial ulnar joint
Olecranon bursa
Ulno-humeral joint
Lateral triangle



Question 37

The therapist is treating a track athlete who specializes in sprinting and wants to increase his other speed on the track. To accomplish this goal, the plan of care should include activities to develop fast-twitch muscle fibers. Characteristics of this type of fiber include:

1. Fatigues slowly, fiber colors appear red and used more in aerobic
2. **Fatigues quickly, fiber colors appear white and used in anaerobic**
3. Fatigues quickly, fiber colors appear white and used more in aerobic
4. Fatigues slowly, fiber colors appear white, and used more in anaerobic

Correct Answer: 2
rationale:
Fast twitch muscles are good for rapid movements like jumping to catch a ball or sprinting for the bus. They contract quickly, but get tired fast, as they consume lots of energy.

Question 38

Which of the following is the most appropriate criterion to initiate an active range of motion in injury?

1. Analgesia
2. **Stable joint**
3. Non-constant pain
4. Absence of swelling

Correct Answer: 2
rationale:
Because the criteria of motion is the stable joint and ligaments and muscles

Question 39

your interview with an 18 year-old female cross country runner elicits a history of stiffness and diffuse ache in her right knee which is aggravated by prolonged sitting. Based on this information, your physical examination should focus on the likely diagnosis of:-

1. liotibial band friction syndrome
2. Osgood-schlatter disease
3. meniscal tear
4. **patellofemoral syndrome**

Correct Answer: 4
rationale:
The diagnosis of the patellofemoral syndrome prolonged sitting due to the pattela confact with the lateral of patellar groove

Question 40

Which of the following actions places the greatest stress on the patellofemoral joint?

1. When the foot first contacts the ground during the gait cycle.
2. Exercising on a stair stepper machine.
3. Running down a smooth decline of 30o
4. **Squats to 120o of knee flexion.**

Correct Answer: 4
rationale:
The compressive force makes the patellofemoral joint to contact more and make pain and degeneration in the cartilage

Question 41

A physical therapist is evaluating a 17-year-old distance runner with complaints of lateral knee pain. During evaluation, the therapist performs the following test: The patient is placed in supine position with the hip flexed to 45o and the knee to 90o. The therapist then places firm pressure over the lateral femoral epicondyle and extends the patient's knee. Pain is felt by the patient at the point of palpation when her knee is 30o from full knee extension. The positive result of this test suggests which of the following structures as the source of pain?

1. **Iliotibial band**
2. Biceps femoris
3. Quadriceps
4. Lateral collateral ligament.

Correct Answer: 1
rationale:
The ITB is generally viewed as a band of dense fibrous connective tissue that passes over the lateral femoral epicondyle and attaches to Gerdy's tubercle on the anterolateral aspect of the tibia the iliotibial band syndrome one of it is symptoms is knee painYour lateral epicondyle is on the outside of your knee near the bottom of your femur, where the bone widens. Your tense iliotibial band repeatedly rubs against your lateral epicondyle when you flex and extend your knee. The friction causes inflammation in your tendon and pain in your knee.

Question 42

while crossing the finish line of a 100meter race an athlete lunged and fell on her dorsiflexed wrist. She immediately complained of pain and tenderness over the radial aspect of the wrist in the area of the anatomical snuffbox. You suspect a possible fracture of the:-

1. extension flunate
2. capitate
3. **scaphoid**
4. hook of the humate

Correct Answer: 2
rationale:
According the anatomy of scaphoid is forming the primal part of the floor of the anatomical snuff box and the most common bone farcture in the carpal bones and the scaphoid is in the radial side of the hand

Question 43

Athlete (power lifter) complain of pain in groin area after weight lifting and also pain in lower abdomen what is the most likely cause?

1. adductor strain
2. **sports hernia**
3. jumpers knee



Correct Answer: 2
rationale:
Sportsman's hernia can cause debilitating pain in the lower abdominal, inguinal, and groin regions that is either unilateral or bilateral and is exacerbated with exertion or valsalva Once again, resisted double hip adduction will be painful and unchanged with the pelvic ring stabilization belt

Question 44

A 26-year-old football player developed severe pain in his ankle following a twisting injury during a match. He was given ice fomentation and anti-inflammatory medications. He had little pain on rest. Physical examination revealed stress and palpation produce severe pain (see report).
X-ray of the ankle: Unremarkable.
Which of the following is the most likely grade of the sprain?

1. **1**
2. 2
3. 3
4. 4

Correct Answer: 1
rationale:
Because the grade 1 in the ankle sprain the pain is mild pain and minimal and minimal functional loss the grade 1 is mild ligament stretch

Question 45

A 28-year-old professional weight lifter is undergoing a physical therapy examination following complaints of pain in the lower abdominal area and groin when lifting weights. A detailed examination by the physiotherapist reveals pain in the groin on coughing, resisted adduction sit-ups, and weight-bearing.

1. Hamstring strain
2. **Sportsman's hernia**
3. Osteoarthritis of hip joint
4. Tightness in the hip flexors(iliopsoas)

Correct Answer: 2
rationale: same as **Question 43**

Question 46

Following a motor vehicle accident, a patient with chest trauma developed atelectasis. To help in the immediate management of atelectasis, it would not be appropriate to implement:

1. pain reduction techniques
2. segmental breathing
3. incentive spirometry
4. **paced breathing**

Correct Answer: 4
rationale:
Atelectasis is a condition in which there is collapse of lung tissue with loss of volume the treatment include the intensive sprirometry and the sysmptoms of atelectasis is the cest pain and the segmental breathing is the method of treatment of segmental atelectasis

Question 47

A 24-year-old runner complained of sudden onset of pain in the calf. Upon examination by the physical therapist at the same time of injury, calf muscles appeared to be intact, as did the hamstrings. The knee was stable, and no tenderness was present over the knee ligaments. The therapist advised the patient to get some rest, after 2 days of rest, the symptoms disappeared, and the patient has been able to run pain-free for the last year. Which of the following is the most likely injury?

1. **Shin splints**
2. Achilles tendonitis
3. Rupture of plantaris muscle
4. Rupture of the deep portion of soleus muscle

Correct Answer: 1
rationale:
The medial tibial stress syndrome (shin splint) the tenderness during palpation in the origion of the tibialis posterior and soleus muscles especially the distal and middle tibial regions and the best treatment is rest in acute phase

Question 48

Which of the following is the main precaution for exercise after posterior cruciate ligament reconstruction?

1. Avoid walking
2. Avoid knee extension
3. Avoid excessive anterior shear force
4. **Avoid excessive posterior shear force**

Correct Answer: 4
rationale:
An excessive posterior shear force causes posterior displacement of the tibia on the femur, thus disrupting the healing graft.
Throughout the rehabilitation process, limit the number of repetitions of knee flexion to lessen abrasion to the PCL graft



Question 49

A 17-year-old athlete has just received a posterior cruciate ligament reconstruction. The therapist is attempting to explain some of the characteristics of the posterior cruciate ligament. Which of the following is incorrect information

1. The posterior cruciate ligament prevents posterior translation of the tibia on the femur.
2. Posterior bands of the posterior cruciate ligament are their tightest in full knee extension.
3. The posterior cruciate ligament is attached to the lateral meniscus and not to the medial meniscus.
4. **The posterior cruciate ligament helps with medial rotation of the tibia during full knee extension with** open chain activities.

Question 50

Ligament that restricts the excess posterior translation of femur over tibia

1. lateral collateral ligament
2. medial collateral ligament
3. anterior cruciate ligament
4. **posterior cruciate ligament**

Correct Answer: 4
rationale:
LCL is primarily preventing medial translation of the tibia.
MCL also prevents hyperextension of the joint and posterior translation of the tibia.
ACL prevents excessive tibial medial and lateral rotation, as well as varus and valgus stresses.

Question 51

football player fall in full knee flexion and medial rotation injury to

1. ACL
2. **PCL**
3. Medial collateral

Correct Answer: 2
rationale:
if there is meniscus we choose it first

Question 52

player football takes trauma in anterior aspect of tibia with positive posterior drawer sign

1. anterior curicate ligement
2. **posterior curicate ligemnet**
3. medial collteral ligement
4. lateral colatteral ligment

Correct Answer: 2
rationale:
the posterior drawer test is used to assess the integrity of the posterior cruciate ligament.

Question 53

football player fall in full flexion and media rotation?

1. ACL
2. PCL
3. Medial collateral
4. **medial meniscus**

Question 54

A 20-year-old athlete is complaining that his knee locks, and he has to move it around to unlock it. Which of the following is the most likely cause for his knee lock?

1. **Torn meniscus**
2. Chondral Lesion
3. Patellar Subluxation
4. Quadriceps Tendonitis

Correct Answer: 1
rationale:
symptoms of meniscal tear are:
- A popping sensation.
Swelling or stiffness.
Pain, especially when twisting or rotating your knee.
Difficulty straightening your knee fully.
Feeling as though your knee is locked in place when you try to move it.
Feeling of your knee giving way

Question 55

A 21-year-old football player complains of right ankle pain after an ankle sprain. What is the type of sports injury accounts for most sprained ankles?

Eversion
Inversion
Dorsiflexion
Dorsiflexion and Inversion

Correct Answer: 1
rationale:
Lateral (inversion) ankle sprains is the most common type occur due to excessive supination of the rear foot about an externally rotated lower leg soon after initial contact of the rear foot during gait or landing from a jump.

Question 56

Which of the following is the main cause of adductor tendinopathy in sportsmen?

1. Neuropathies
2. **Overuse injury**
3. Muscular weakness
4. Congenital hip deformity

Correct Answer: 2
rationale:
At a histopathological level there are changes in the molecular structure of the tendon, typically collagen separation and collagen degeneration and at a macroscopic level typically see tendon thickening, a loss of mechanical l properties and pain.



Question 57

A 36-year-old football player developed severe knee pain while playing. He was unable to move his leg. Examination revealed his knee had swelling and limited range of motion. There was a spring-like end :

1. Arthritis
2. Inflammation
3. **Meniscal injury**
4. Ligamentous injury

Question 58

An athlete comes to the Physiotherapy Department with a right knee injury is undergoing a test to determine the nature of the injury. The patient is prone with the knee flexed at a right angle. The patient's symptoms are worse when the physiotherapist rotates the tibia medially and laterally while maintaining a distraction force through the knee joint
Which of the following do the symptoms indicate?

1. **-Capsular injury**
2. Meniscal injury
3. Muscular injury
4. Ligamentous injury

Question 59

A 17-year-old football player is referred to the outpatient physical therapy clinic with a diagnosis of a recent third-degree medial collateral ligament sprain of the knee. The patient wishes to return to playing football as soon as possible. Which of the below is the best protocol?

1. **Fit the patient with a brace that prevents him from actively moving the knee into the last available 20o of extension. Prescribe general lower extremity strengthening with the exception of side lying hip adduction.**
2. Do not fit the patient with a brace. All lower extremity strengthening exercises are indicated.
3. Fit the patient with a brace that prevents him from actively moving the knee into the last available 20o of extension. Avoid all open-chain strengthening for the lower extremity.
4. Do not fit the patient with a brace. Prescribe general lower extremity strengthening with the exception of side lying hip adduction.

Question 60

You are treating a 16-year-old female athlete for patellofemoral malalignment syndrome. During competition, the orthosis that would be most helpful in aiding this situation is a:

1. **patellar stabilizing brace.**
2. derotation brace.
3. knee orthosis with a hinge joint.
4. knee orthosis with an offset hinge joint.

Correct Answer: 1
rationale:
patellar stabilizing brace is a great way to stabilize your kneecap, alleviate patellofemoral pain, and protect against movements that might cause more damage.

Question 61

A clinical instructor is explaining to a physical therapy student the function of the screw home mechanism in the knee joint. Part of the therapist's explanation involves teaching the student the movement of the tibia and femur during closed chain activities. When the knee joint is extended in a closed-chain activity, which of the following statements is

1. The femur laterally rotates on the tibia.
2. **The femur medially rotates on the tibia**
3. The tibia laterally rotates on the femur
4. The tibia medially rotates on the femur.

Correct Answer: 2
rationale:
function of the screw home mechanism in the knee joint:-

DURING KNEE EXTENSION		DURING KNEE FLEXION	
OPEN CHAIN	CLOSED CHAIN	OPEN CHAIN	CLOSED CHAIN
TIBIA GLIDES ANTERIORLY ON FEMUR	FEMUR GLIDES POSTERIORLY ON TIBIA	TIBIA GLIDES POSTERIORLY ON FEMUR	FEMUR GLIDES ANTERIORLY ON TIBIA
From 20-30° knee flexion to full extension		From full knee extension to 20-30° flexion	
Tibia rotates externally	Femur rotates internally on stable tibia	Tibia rotates internally	Femur rotates externally on stable tibia

Question 62

A 25-year-old woman badminton player had a severe shooting pain at the common extensor origin of wrist extensors muscles at the dominant hand and also weak handgrip strength when holding the rocket. Which of the following is the most appropriate description of this condition?

1. **Tennis Elbow**
2. Student Elbow
3. Golfer's Elbow
4. Nursemaid Elbow

Correct Answer: 1
rationale:
Tennis elbow is an overuse injury of the common extensor tendon origin. The common extensor tendon originating from the lateral epicondyle of the elbow is directly involved in Tennis elbow.



Question 63

A 19-year-old tennis player had severe training for several hours for the past 2 weeks. He describes continuous pain over the lateral epicondyle and common extensor origin and subsides with rest. Which muscle-strengthening exercise is the most appropriate to be prescribed?

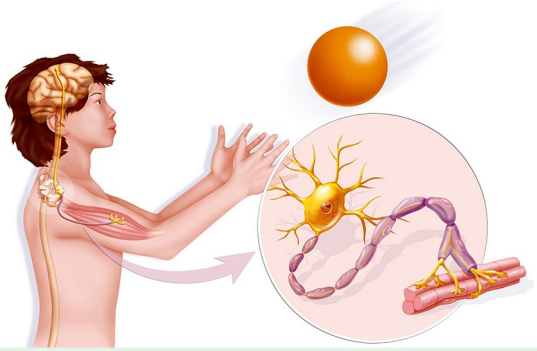
1. flexor
2. **Wrist extensor**
3. Shoulder flexor
4. Shoulder extensor

Correct Answer: 2
rationale:
These muscles share a tendinous origin at the lateral epicondyle: extensor carpi radialis brevis, extensor carpi ulnaris, extensor digiti minimi, and extensor digitorum & become weakness with lateral epicondylitis.

Question 64

Before strength hip flexors we do

1. **Knee extension**
2. Extension hip
3. Internal rotation hip



2 Neuromuscular System

chapter

Question 1

Rate of regeneration of severed nerve is:
1- 0.1 mm /day
2- 1 mm /day
3- 1 cm /day
4- none

Correct Answer: 2
rationale:
1 mm /day is a stated fact.

Question 2

Normal action potential:
1- 30 to -60 mVolt
2- 60 to -90 mVolt
3- 30 to -90 mv
4- 30 to -80 mv

Correct Answer: 3
rationale:
indicating that neurons have a means of generating a constant voltage across their membranes when at rest. This voltage, called the resting membrane potential, depends on the type of neuron being ex-amined, but it is always a fraction of a volt (typically -40 to -90 mV).

Question 3

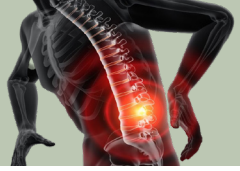
Transfer chemical stimulation to mechanical:
1- Small intestine
2- Stomach
3- Skeletal muscle

Correct Answer: 3
rationale:
transfers nerve impulses which are electrical to mechanical force that produces movement.

Question 4

Light touch loss, in which tract may happened?
1- sensory tract
2- spinothalamic tract
3- motor tract

Correct Answer: 1
rationale:
by excluding the other 2 answers as the spinothalamic tract is responsible for transmitting pain and temperature only.



Question 5

Which of the following is involved in the perception of dull aching pain?

1- spinotectal tract
2- spino-olivary tract
3- spinoreticular tract

Correct Answer: 3
rationale:
The central ascending pathway for sensation consists of two systems: the spinothalamic tract and the spinoreticulothalamic system. The first pathway conducts the sensation of sharp, stabbing pain; the second conveys deep, poorly localized, burning pain.

Question 6

While evaluating a patient who suffered a complete spinal cord lesion, the therapist notes the following strength grades with manual muscle testing: wrist extensors = 3+/5, elbow extensors = 2+/5, and intrinsic muscles of the hand = 0/5. What is the highest possible level of this lesion?

1- C3
2- C4
3- C7

Correct Answer: 3
rationale:
as the intrinsic muscles of the hand are supplied by t1 myotome.

Question 7

In which location is the median nerve frequently or mostly entrapped?

1- Forearm
2- Arm
3- Elbow
4- Wrist

Correct Answer: 4
rationale:
carpel tunned syndrome.

Question 8

Supracondylar ridge of humerus injury to:

1- radial nerve
2- brachial artery
3- radial artery

Correct Answer: 2
rationale:
brachial artery in the same location of the median nerve.

Question 9

Supracondylar ridge of humerus injury to

1- Ulnar n nerve
2- brachial nerve
3- median nerve

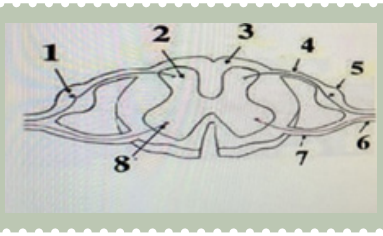
Correct Answer: 3
rationale:
review the course of the nerve.

Question 10

Which of the following is the appropriate label for "8", in this schematic figure of the cross-section of the spinal cord (see image)?

1- Grey matter
2- Spinal nerve
3- Dorsal root ganglion
4- Soma of motor neuron

Correct Answer: 4
rationale:
Soma of motor neuron is the cell body.

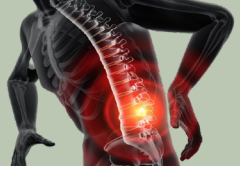


Question 11

A 14-year old girl place excessive valgus stress to the right elbow during a fall from a bicycle. Her forearm was in supination at the moment the valgus stress was applied. Which of the following is most likely involved in this type of injury?

1- Ulnar nerve
2- Extensor carpi radialis
3- Brachioradialis
4- Annular ligament

Correct Answer: 1
rationale:
as It descends the arm medial to the brachial artery. It then passes posterior to the medial epicondyle of the humerus, into a passageway known as the cubital tunnel. It enters the forearm between the humeral and ulnar heads of the flexor carpi ulnaris (FCU) and continues down the forearm between the FCU and FDP.
NOTE:
Cubital valgus fracture means move the forearm away from the humreus which causes fracture of medial epicondyle that causes ulnar nerve injury.
Supracondylar → median nerve injury
Mid humerus → radial nerve injury
Upper humerus → brachial nerve injury
The cause of cubital Varus fracture → supracondylar Fracture



Question 12

A therapist is evaluating a patient in an intensive care unit. The therapist notes no eye opening, no verbal response, and no motor response. On the
67/ A patient has been referred to a therapist complaining of neck pain, nerve root compression, sensation changes over the middle finger and weakness of triceps. What is the nerve root that is compressed?

1- C6
2- C7
3- C5
4- C3

Correct Answer: 2
rationale:
supplies the triceps reflex and the dermatome of the middle finger.

Question 13

pt. has median Nerve compression assist him by:

1- Weak of medial 2 fingers & decrease grasping objects
2- Numbness of 3 ½ fingers & weak opponents MS

Correct Answer: 2
rationale:
The median nerve then enters the hand via the carpal tunnel and provides motor innervation to the flexor pollicis brevis (FPB), abductor pollicis brevis (APB), opponens pollicis, and the lateral two lumbricals. The sensory distribution of the median nerve supplies the palmar aspect of the lateral palm, palmar aspect of the lateral three and a half digits, and the dorsal aspect of the lateral three and a half digits distal to the PIP joint.

Question 14

a college student presents with pain in the shoulder that began the week after finals, but she cannot recall any mechanism of injury. Upon inspection, the examiner notes scapular winging on the dominant side. During evaluation the patient has difficulty with elevation greater than 120 degree. She has normal scapular elevation. What would be the best differential diagnosis ?

1- acute rotator cuff tendonitis
2- shoulder impingement
3- long thoracic nerve lesion
4- accessory nerve lesion

Correct Answer: 3
rationale:
The long thoracic nerve innervates the serratus anterior. A scapular stabilizer. The nerve can be compressed with a backpack strap as it lies along the chest wall. This patient will have scapular winging but good scapular elevation strength. Damage to the spinal accessory nerve will inhibit elevation of the scapula.

Question 15

with affection of phrenic nerve we stretch?

1- sternocleidomastoid
2- trapezius
3- diaphragm
4- masseter

Correct Answer: 1
rationale:
Sternocleidomastoid above phrenic nerve by 1 inch and we can manipulate it by stretching SCM muscle.

Question 16

A patient presents to an outpatient facility with complaints of pain in the groin area (along the medial left thigh). With manual muscle testing of the involved lower extremity a therapist determines the following: hip flexion = 4+/5, hip extension = 4+/5, hip abduction = 4+/5, hip adduction = 2+/5, hip internal rotation = 2+/5, and hip external rotation = 2+/5. Which nerve on the involved side is most likely injured?

1- Lateral cutaneous nerve of the upper thigh.
2- Obturator nerve
3- Femoral nerve.
4- Ilioinguinal nerve.

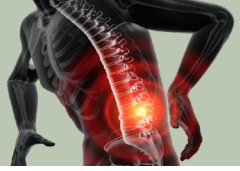
Correct Answer: 2
rationale:
because it supplies all the muscles of the medial thigh which are collectively known as hip abductors, also it sends back sensory information about the media thigh to the cns.

Question 17

A patient presents to an outpatient clinic with an order to evaluate and treat the right forearm and wrist secondary to nerve compression. The patient has the following signs and symptoms: pain with manual muscle testing of pronation, decreased strength of the flexor pollicis longus and pronator quadratus, and pain with palpation of the pronator teres. What nerve is most likely compromised? What is the most likely area of compression?

1- Median nerve – carpal tunnel
2- Ulnar nerve – Guyon's canal
3- Ulnar nerve – pronator quadratus
4- Median nerve – pronator teres

Correct Answer: 4
rationale:
it enters the forearm between the superficial and deep heads of the pronator teres muscle. And It supplies the flexor pollicis longus (FPL), and pronator quadratus.



Question 18

You are performing sensory tests on a patient diagnosed with C6 nerve root impingement. Testing should concentrate on the:

1- 3rd, 4th and 5th fingers
2- ulnar border of the hand
3- thumb and index fingers
4- medial forearm

Correct Answer: 3
rationale:
Thumb and Index fingers is a dermatome for C6 nerve root.

Question 19

Acceleration injury (whiplash injury) may cause:

1- pain in posterior aspect of neck
2- pain at post aspect of back
3- dislocation of vertebral colum
4- tearing of post para spinal ligament

Correct Answer: 1
rationale:
Whiplash injury: is a non-medical term used to describe neck pain following an injury to the soft tissues of the neck especially ligaments, tendons, and muscles. It is caused by an abnormal motion or force applied to your neck that causes movement beyond the neck's normal range of motion,
Symptoms of whiplash injury are:
1. Neck pain
2. Neck swelling
3. Tenderness along the back of the neck, Muscle spasms (in the side or back of your neck)
4. Difficulty moving the neck around
5. Headache
6. Pain shooting from the neck to the shoulder or arm

Question 20

A 19-year-old wrestler fell on his elbow and then complained of pins and needles sensation in his small finger. Which of the following is the most likely cause?

1- Contusion of the ulnar nerve
2- Injury to the brachial plexus
3- Contusion of the median nerve
4- Strain of the brachioradialis muscle

Correct Answer: 1
rationale:
The sensory branch provides sensation to the palmar aspect of the medial hand, fifth digit, and medial aspect of the fourth digit.

Question 21

The muscles that supplied by phrenic nerve C3, C4 and C5 are called:

1- scalene
2- sternocleidomastoid
3- serratus anterior
4- diaphragm

Question 22

when the fracture accurse in neck of humerus what nerve affected?

1- Redial N
2- Median N
3- Ulnar N
4- Subcutaneous nerve

Question 23

Most common compression of radial n in:

1- shaft of humerus
2- neck of humerus
3- distal end

Question 24

Weakness of dorsiflexor of wrist and fingers due to injury which of following:

1- median N
2- ulnar N
3- radial N

Question 25

Which of the following sites of fracture involving the humerus most likely causes radial nerve palsy?

1- Shaft
2- Head
3- Lower end
4- Surgical neck

Question 26

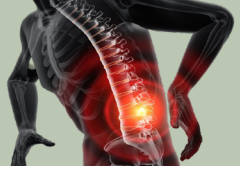
Common site of Radial nerve lesion:

1- wrist
2- elbow
3- shoulder
4- spiral grove of humerus

Question 27

pt has median Nerve compression assess him by:

1- Weak of medial 2 fingers & decrease grasping objects.
2- Numbness of 3 ½ fingers & weak opponents ms.



Question 28

Radial nerve most probable compressed region:

1- elbow
2- finger
3- shoulder
4- wrist

Question 29

which nerve which give back of the arm and forearm?

1- radial
2- ulnar
3- median
4- musculocutaneous

Question 30

Sciatic nerve injury due to:

1- Post dislocate hip
2- Ant dislocate hip

Question 31

Ulnar nerve pass:

1- anterior to medial epicondyle
2- anterior to lateral epicondyle
3- posterior to medial epicondyle
4- posterior to lateral epicondyle

Question 32

Delayed ulnar nerve due to fracture of:

1- olecranon fossa
2- Medial epicondyle
3- Lateral epicondyle

Question 33

Most compression site of ulnar nerve in:

1- shoulder joint
2- spinal groove of humerus
3- elbow
4- wrist

Question 34

ulnar nerve give supply to:

1- whole hand
2- outer 3 +1/2 fingers
3- inner 1+1/2 fingers

Question 35

The ulnar nerve supplies all of the following muscles except:
1- adductor pollicis.
2- flexor pollicis longus.
3- interossei muscles.
4- hypothenar muscles.

Question 36

guyon tunnel Injury of which nerve:
1- ulnar
2- radial
3- median
4- peroneal

Correct Answer: 1
rationale:
Guyon's canal also called ulnar tunnel or ulnar canal, is an anatomical fibro-osseous canal located on the medial side of the hand. It extends between the proximal boarder of the pisiform bone and distally at the hook of the hamate.
The ulnar nerve and ulnar artery pass through the Guyon canal as they pass from distal fore-arm to the hand.

Question 37

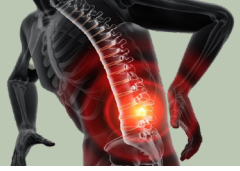
when examine patient for carpal tunnel syndrome, which nerve do you examine?
1- Radial n.
2- Ulnar n.
3- Median n

Correct Answer: 3
rationale:
Median n passes through the carpel tunnel.

Question 38

In carpal tunnel syndrome the entrapped nerve is the:
1- radial
2- median
3- ulnar
4- axillary

Correct Answer: 2
rationale:
5th finger abduction and adduction of the thumb are mediated by ulnar nerve muscles.



Question 39

A 45-year-old patient was complaining of tingling sensation and numbness on his 1st 2nd and the radial half of the 3rd finger of his right hand. Examination showed that his right-hand grip strength was weak compared to his left hand.
No other symptoms were reported. Which of the following is the most likely cause of the symptoms?
1- Wrist arthritis
2- Cervical radiculopathy
3- Carpal tunnel syndrome
4- Ulnar nerve entrapment

Correct Answer: 3
rationale:
Carpal tunnel syndrome is a median nerve compression.

Question 40

A therapist is beginning an evaluation of a 34-year-old woman with a diagnosis of carpal tunnel syndrome. Part of the evaluation consists of grip strength testing. To accurately test strength of the flexor digitorum profundus, where should the grip dynamometer's adjustable handle be placed?
1- 1 inch from the dynamometer's nonadjustable handle.
2- 3 inches from the dynamometer's nonadjustable handle.
3- 1.5 inches from the dynamometer's nonadjustable handle.
4- All of the above are equally effective.

Question 41

Carpal tunnel syndrome causes:
1- claw hand
2- wrist drop.
3- paralysis of all muscles of the hand
4- ape-like hand

Correct Answer: 4
rationale:
the median nerve is the only nerve passing through the carpel tunnel so in carpel tunnel syndrome all the muscles supplied by it which are (the lateral two lumbricals and the thenar muscle group).

Question 42

Site of compression of median nerve in:
1- elbow
2- wrist
3- shoulder

Question 43

The following muscles are innervated by the median nerve except:
1- pronator teres
2- pronator quadratus
3- flexor pollicis longus
4- adductor pollicis

Correct Answer: 4
rationale:
The adductor pollicis is innervated by the deep branch of the ulnar nerve.

Question 44

which muscle doesn't affected in carpal tunnel syndrome
1- flexor pollicis brevis
2- adductor pollicis longus
3- abductor pollicis bervis

Correct Answer: 2
rationale:
Adductor policies longus The muscles that might be less affected by CTS, the "LOAF" muscles...lumbricals, opponens (opposes thumb to other fingers), abductor pollicis brevis (thumb abduction), and flexor pollicis brevis (flexes thumb).

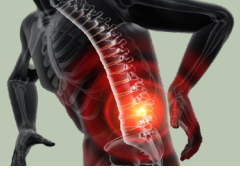
Question 45

A patient is referred to physical therapy with complaints of sensation loss over the area of the radius of the right upper extremity, extending from the elbow joint distally to the wrist. Therapy sessions are focused on assisting the patient in regarding normal sensation. Which of the following nerves is responsible for sensation in this region?
1- Medial antebrachial cutaneous
2- Lateral antebrachial cutaneous
3- Musculocutaneous
4- Both B and C

Question 46

In poliomylitis which is incorrect:
1- Hyper reflex
2- Reflex absent
3- Ms weakness

Correct Answer: 1
rationale:
a disease of the lower motor neurons that affects the gray matter of the spinal cord.



Question 47

Polio virus transfer through:

1- Respiratory tract
2- Skin
3- Hepatic failure
4- GIT

Correct Answer: 4
rationale:
Polio is very contagious, and a person can spread the virus even if they aren't sick or experiencing symptoms. The polio virus enters the body through the mouth, usually from hands contaminated with the stool of an infected person.

Question 48

Sacral dimple at what level?

1- S1
2- S2
3- S3

Question 49

Post polio clinical features:

1- Paralysis
2- weakness
3- atrophy
4- no change

Question 50

Convalescent stage of polio:

1- 2weeks-2years
2- 2weeks-4years
3- 0-2week
4- 2week to through life

Question 51

The spaces between the covered parts of an axon are called:

1- ventricles
2- synaptic clefts
3- nodes of ranvier
4- vescicles

Correct Answer: 1
rationale:
Nodes of Ranvier are gaps in the myelin sheath coating on the neural axon. The myelin allows the electrical impulse to move quickly down the axon. The nodes of Ranvier allow for ions to diffuse in and out of the neuron, propagating the electrical signal down the axon

Question 52

A therapist is evaluating a patient in the intensive care unit. The therapist notices that the patient is moving his hands and fingers in slow, writhing motions. Which of the following terms best describes this type of movement?

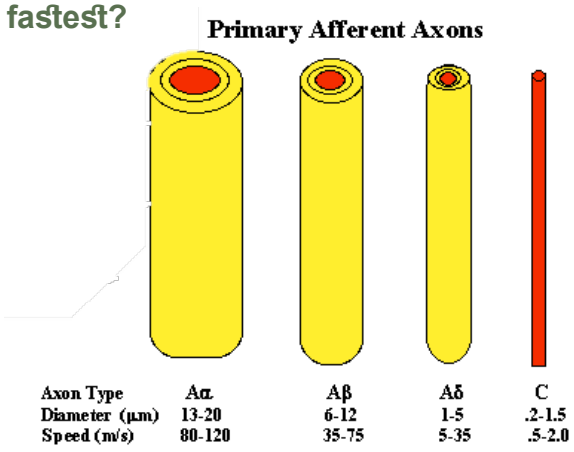
1- Lead-pipe rigidity
2- Ballismus
3- Chorea
4- Athetosis

Correct Answer: 4
rationale:
Athetosis: Athetosis refers to the slow, involuntary, and writhing movements of the limbs, face, neck, tongue, and other muscle groups.
Ballismus: Ballismus is a severe movement disorder that is characterized by spontaneous involuntary movements, muscular weakness and incoordination of movements of the proximal extremities.
Chorea: Chorea is a movement disorder that causes involuntary, irregular, unpredictable muscle movements.

Question 53

Which of the following neural fibers are the largest and fastest?

1- C fibers
2- A fibers
3- A and C are equal
4- None of the above



Question 54

Initiate muscle contraction ?

1- Alpha fibers
2- Gamma fibers

Question 55

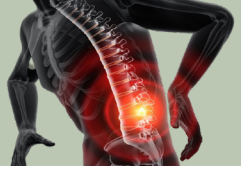
The therapist is evaluating a patient with left-side visual field deficits in both eyes. A lesion at what location may cause this defect?

1- At the optic chiasm.
2- At the right side optic tract.
3- At the left side optic nerve
4- At the right side optic nerve.

Question 56

patient with partial ataxia, physiotherapist make contraction and no movement is placed in limb?

1- rhythmic stabilization
2- rhythmic initiation



Question 57

Electrically insulating material in a nerve fiber is:

1- Nodes of Ranvier
2- Axon
3- Dendrites
4- Myelin sheath

Question 58

Nodes of Ranvier are found in:

1- brain
2- heart
3- axon
4- eye

Question 59

Nodes of Ranvier are found in:

1- Myelinated nerve fibers
2- Non myelinated nerve fibers
3- RBC
4- WBC

Question 60

Ankylosing means :

1- loosed packed position
2- joint stiffness
3- closed packed position

Correct Answer: 2
rationale:
joint stiffness , ankylosing means is a stiffness of a joint due to abnormal adhesion and rigidity of the bones of the joint.
NOTE:
ankylosing spondylitis mean :
known as Bechterew's disease (or syndrome) and Marie- Strümpell disease, is a chronic inflammatory disease of the axial skeleton with variable involvement of peripheral joints and non-articular structures Also Ankylosing is a term meaning stiff or rigid and spondylitis means inflammation of the spine.

Question 61

Ankylosing Spondylitis means:

1- loosed packed position
2- abnormal condition of stiffness and fusion of vertebrae
3- closed packed position
4- excessive flexion

Question 62

WHICH statement incorrect about Brown-Sequard syndrome?
1- ipsilateral babinski sign
2- ipsilateral loss of pain and temp
3- contralateral loss of pain and temp

Correct Answer: 3
rationale:
as pain and temperature are carried out by the lateral spinothalamic tract and it crosses to the other side at the level of each segment.

Question 63

A 79-year-old patient with Brown-sequard syndrome. Which of the following are the most likely causes?
1- Ipsilateral motor loss and complete spinal cord injury
2- Ipsilateral motor loss and incomplete spinal cord injury
3- Contralateral motor loss and complete spinal cord injury
4- Contralateral motor loss and incomplete spinal cord injury

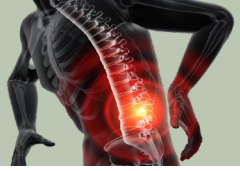
Question 64

The therapist receives an order to treat a 42-year-old man admitted to the hospital 3 days ago with a stable wound to the left lower thoracic spine. The patient is unable to move the left lower extremity and cannot feel pain or temperature differences in the right lower extremity. What is the most likely type of lesion?
1- Anterior cord syndrome
2- Brown-Sequad syndrome
3- Central cord syndrome
4- The patient is equally as likely to have anterior cord syndrome as he is to have

Correct Answer: 2
rationale:
Brown-Sequard Syndrome (BSS), an incomplete spinal cord injury that results from a hemisection to the spinal cord. on either side, typically in the cervical and thoracic spinal cord region. This lesion leads to interruption of normal function of nerve tracts in one half of the spinal cord. It typically presents with paralysis (hemiparaplegia) on the side of the lesion due to compromised lateral corticospinal tracts; loss of proprioception and vibration sense on the same side from damage to the dorsal column and a loss of pain and temperature sensation (hemianaesthesia) contralateral to the lesion owing to injury to the lateral spinothalamic tract.

Question 65

Which of the following is the exercise to reduce pain and improve mobility in patients with ankylosing spondylitis?
1- Aquatic
2- Flexibility
3- Stretching
4- Strengthening



Question 66

A therapist receives an order to evaluate and treat a 76-year-old woman who was involved in a motor vehicle accident 2 days ago. The patient's vehicle was struck in the rear by another vehicle. The patient has normal sensation and strength in bilateral lower extremities but paralysis and loss of sensation in bilateral upper extremities. Bowel and bladder function are normal. The patient most likely has what type of spinal cord injury?

1- Anterior cord syndrome.
2- Brown-Sequard syndrome.
3- Central cord syndrome
4-There is no evidence of an incomplete spinal cord lesion

Correct Answer: 3
rationale:
Central cord syndrome (CCS) is the most common form of cervical spinal cord injury. It is characterized by loss of motion and sensation in arms and hands. It usually results from trauma which causes damage to the neck, leading to major injury to the central corticospinal tract of the spinal cord.

Question 67

A 42-year-old construction worker received a burst fracture in the cervical spine when struck by a falling crossbeam. Proprioception is intact in bilateral lower extremities. The patient has bilateral loss of motor function and sensitivity to pain and temperature below the level of the lesion. This type of lesion is most typical of which of the following syndromes?

1- Central cord syndrome
2- Brown Sequard syndrome
3- Anterior cord syndrome
4- Conus medullaris syndrome

Correct Answer: 3
rationale:
Anterior cord syndrome: It causes loss of movement and loss of sharp/dull and hot/cold feelings below the level of injury with preserved proprioceptive sense.

Question 68

A patient informs his therapist that his problem began 3 months after a bout of the flu. The patient originally experienced tingling of the hands and feet. He also reports progressive weakness to the point that he required a ventilator to breathe. He is now recovering rapidly and is expected to return to a normal functional level in 3 more months. From which of the following conditions is the patient most likely suffering?

1- Parkinson's disease
2- Guillain - Barrè syndrome
3- Multiple sclerosis
4- Amyotrophic lateral sclerosis

Question 69

A 39-year-old patient with lower limb problem, during the assessment, there is a weakness in the gastrocnemius muscle. Where is this muscle located?

1- Back of leg
2- Front of leg
3- Back of thigh
4- Front of thigh

Question 70

therapist is assisting a patient in gaining lateral stability of the knee joint. The therapist is using strengthening exercises to strengthen muscle groups that will increase active restraint on the lateral side of the joint. Which of the following offers the least amount of active lateral restraint?

1- Gastrocnemius
2- Popliteus
3- Biceps femoris
4- Iliotibial band.

Question 71

A posterior lateral herniation of the lumbar disc between vertebrae L4 and L5 most likely results in damage to which nerve root?

1- L4
2- L5
3- L4 and L5
4- L5 and S1

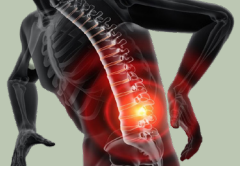
Correct Answer: 1
rationale:

Disc Herniation Level	Traversing Nerve Root	Exiting Nerve root
L1-2	L2	L1
L2-3	L3	L2
L3-4	L4	L3
L4-5	L5	L4
L5-S1	S1	L5

Question 72

On examination of a cross section of the spinal cord of a cadaver, the examiner notes plaques. The finding is most characteristic of what condition?

1- Parkinson's disease
2- Myasthenia gravis
3- Multiple sclerosis.
4- Dementia.



Question 73

Acute Inflammatory Demyelinating Poly radiculo neuropathy is also referred as:

1- Guillain–Barré syndrome
2- Compressive myelopathy
3- Friedreich's ataxia
4- None of the above

Question 74

The failure to release acetylcholine at the neuromuscular junction is the cause of

1- myasthenia Gravis
2- guillain - Barre syndrome
3- lambert - Eaton syndrome
4- locked - in syndrome

Correct Answer: 3
rationale:
lambert - Eaton syndrome: This is very similar to myasthenia gravis, however the target of the attack is different in MG as the acetylcholine receptor on the nerve is affected, whereas in LEMS it's the voltage-gated calcium channel on the nerve which is the trigger to release the acetylecholine at the neuromuscular junction.

Question 75

A 65-year-old man presents to physical therapy with complaints of pain due to compression fractures of the C2 and C3 vertebrae. The patient has an unusually large cranium. He describes his condition by stating, "Much of my bone tissue is continually decreasing, then reforming". The patient also indicates that the condition has caused limb deformity. Which of the following diseases does he have?

1- Paget's disease
2- Achondroplastic dwarfism
3- Osteogenesis imperfecta
4- Osteopetrosis

Question 76

Multiple sclerosis MS which is incorrect:

1- more common in men than women
2- cause of MS unknown
3- more in black countries
4- progressive degenerating

Question 77

Pt with neurological condition with evaluation found is unable to perform movement after all repetition, with adequate rest period comes normal this pt is

1- mythenia graves
2- myopathy
3- hyprertonia

Question 78

A 28-year-old woman complained of progressive worsening of physical functions and was diagnosed by her physician with multiple sclerosis. Which of the following is the most appropriate regarding the immediate referral to physical therapy?

1- Recommended to avoid complications related to inactivity.
2- Delay physical therapy to avoid progression of the symptoms.
3- Delay physical therapy to avoid complications related to inactivity.
4- Recommended to avoid progression of the central nervous system (CNS) impairments.

Question 79

When instructing the family of a 9 year-old boy with Duchene muscular dystrophy, the main emphasis in the lower extremities should be placed on:

1- strengthening the knee extensors and plantar flexors
2- strengthening the plantar flexors and stretching the hip extensors
3- stretching the hip flexors and plantar flexors
4- strengthening the hip flexors and knee extensors

Question 80

A 32-year-old construction worker fell off a ladder. In his effort to prevent the fall, the worker reached for a beam with his right arm. This motion stretched the brachial plexus, resulting in decreased function in the right arm. Full function returned after 2 ½ weeks. What is the most likely type of injury?

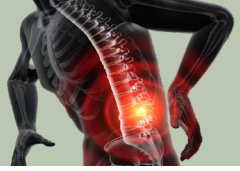
1- Axonotmesis
2- Neurotmesis
3- Neurapraxia
4-None of the above

Correct Answer: 3
rationale:
Neurapraxia: Local myelin damage usually secondary to compression.
Axonotmesis: Axon severed but endoneurium intact (optimal circumstances for regeneration).
Neurotmesis : Complete physiologic disruption of entire nerve trunk.

Question 81

A 31-year-old man has loss of vision in one eye, staggering gait, numbness in bilateral upper extremities, and decreased bowel and bladder control. The episodes of the above symptoms have occurred every few weeks for the past 6 months. Each episode has been slightly worse than the first. What is the most likely condition?

1- Parkinson's disease
2- Guillain Barre syndrome
3- Multiple sclerosis
4- Amyotrophic lateral sclerosis



Question 82

A therapist starts the physical examination of patient who has neurological disorders. The therapist finds that the patient has both upper and lower limb weakness with breathing problems which were exaggerated by the repetition of movements and rest Which of the following is the most likely diagnosis?

1- Myotonia
2- Myopathies
3- Myasthenia gr
4- Motor neuron disorders

Correct Answer:2
rationale:
myopathy exaggerated by the repetition of movements and rest.

Question 83

pt with neurological condition with evaluation found is unable to perform movement without repetitions:

1- myathenia graves
2- myopathy
3- hyprertonia

Question 84

Which of the following nerve injuries is a condition in which degeneration of the axon takes place, but the sheath of the nerve remains intact?

1- Neuropathy
2- Neurapraxia
3- Axonotmesis
4- Neurotmesis

Question 85

Axon is injured but nerve intact:

1- neuropraxia
2- neurotemesis
3- axonotemesis
4- transient ischemia

Question 86

Which of the following nerve injuries is a condition in which both the axon and the sheath of the nerve are destroyed?

1- Neuropathy
2- Neurapraxia
3- Neurotmesis
4- Axonotmesis

Question 87

In order of grade of severity, which of these nerve injuries is the worst?

1- neurotemesis
2- transient ischemia
3- neuropraxia
4- axonotemesis

Correct Answer: 1
rationale:
Neurotemesis: disrupt axon and endoneurium
Axonotemesis: axon and myelin sheet disruption: endoneurium intact
Neuropraxia: focal demyelination axons

Question 88

Motor neuron disease which of the following is not true:

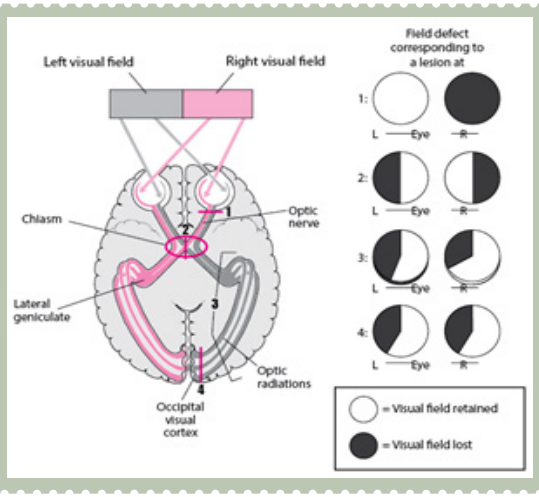
1- Affect anterior horn cell in spinal cord and cause lower motor neuron lesion
2- Affect cranial nerve and Most common facial nerve affected with lower motor neuron lesion
3- Affect brain stem and cause upper motor neuron Lesion and cause lower motor neuron lesion
4- A & B

Question 89

all the following is true about motor neuron disease except

1- Affect anterior horn cell in spinal cord and cause lower motor neuron lesion.
2- Affect corticospinal tract and cause upper motor neuron Lesion.
3- Affect cranial nerve and Most common facial nerve affected with lower motor neuron lesion.
4- Affect brain stem and cause upper motor neuron Lesion and cause lower motor neuron lesion.

Correct Answer: 1
rationale:



Question 90

Hemiplegia is :

1- Loss half field of vision
2- Paralysis halve of the body

Question 91

all the following is true about motor neuron disease except:

1- Affect anterior horn cell in spinal cord and cause lower motor neuron lesion.
2- Affect corticospinal tract and cause upper motor neuron Lesion.
3- Affect cranial nerve and Most common facial nerve affected with lower motor neuron lesion.
4- Affect brain stem and cause upper motor neuron Lesion and cause lower motor neuron lesion.

Question 92

patient have stroke which position increase complication?

1- sitting on chair with head and neck in midline
2- supine with arm abducted and extended
3- side-lying on sound side
4- lateral trunk leans toward affected side

Question 93

A patient who has suffered a recent stroke is being treated by a physical therapist. The patient exhibits increased extensor tone in the supine position along with an exaggerated symmetric tonic labyrinthine reflex (STLR). What is the best position to initiate flexion movements of the lower extremity?

1- Prone position
2- Sidelying position
3- Supine position
4- A and B

Correct Answer: 1
rationale:
Prone Position in STLR increase flexion tone.

Question 94

when you evaluate a CVA patient what is the first thing to notice

1- imbalance and in-coordination
2- visual fields
3- can't express himself

Correct Answer: 3
rationale:



Question 95

Facial nerve nuclei innervations:

1- Bilateral pyramidal innervations
2- Unilateral pyramidal innervations
3- Bilateral no pyramidal innervation

Question 96

Which of the following is the most important to assess first during an evaluation of a patient with a recent stroke?

1- Sensory status
2- Motor control
3- Mental status
4- Ambulation potential

Question 97

While ambulating a stroke patient (right side is the involved side), the therapist notes increased circumduction of the right lower extremity. Which of the following is an unlikely cause of this deviation?

1- Increased spasticity of the right gastrocnemius.
2- Increased spasticity of the right quadriceps.
3- Weak hip flexors.
4- Weak knee extensors

Question 98

A physiotherapist receives a referral for a patient. The referring physician has outlined all clinical findings in the referral, one of which is "Hemiplegia".
What does this term mean?

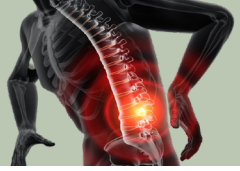
1- Loss of movement in all 4 mbs
2- Loss of movement in 1 side of the body
3- Loss of movement in both upper or lower limb
4- Loss of movement in both lower limb including trunk

Question 99

A 57 - year - old patient presented with a 6 - hour history of progressive left - sided body weakness and difficulty with speech. Over the next 10 hours, the symptoms began to improve and settle. Which of the following is the most likely diagnosis?

1- Stroke
2- Hemiplegia
3- Multiple sclerosis
4-Transient ischemic attack

Correct Answer: 4
rationale:
Transient ischemic attack has same symptoms of stroke but the symptoms don't last more than 24 hours .



Question 100

A therapist is evaluating a patient who had a recent stroke. Which of the following statements is about the Babinski reflex of the plantar surface of the foot?

1- Diagnoses extrapyramidal tract lesion
2- Positive response means flexion of the big toe
3- Done by stroking the lateral aspect of the sole
4- Done by stroking the lateral aspect of the dorsum of the foot

Question 101

When should a physical therapist begin discharge planning for a patient admitted to a rehabilitation unit with a diagnosis of a recent stroke?

1- At the first team meeting
2- At the last team meeting
3- Two weeks before discharge
4- After the initial evaluation by the physical therapist

Question 102

Patient with hypertension with sever stroke when will you expecting that the patient will gain full control of the muscles?

1- week
2- over a month
3- over a year
4- 3 years

Question 103

A physical therapist is treating a 76-year-old woman with left lower extremity hypotonia secondary to a recent stroke. Which of the following is an incorrect method to normalize tone?

1- Rapid irregular movements
2- Approximation.
3- Prolonged stretch
4- Tactile cues

Correct Answer: 1
rationale:
Rapid irregular movements as spasticity is velocity dependent.

Question 104

A 59-year-old patient comes to the Physical Therapy Clinic with left hemiplegia. The physical therapist decides to do positioning for a spastic lower limb in the antispastic pattern. Which of the following positioning does the therapist have to refrain from?

1- Hip, knee and ankle flexion
2- Protraction of hip with inward rotation
3- Retraction of hip without inward rotation
4- Elongation of the trunk on the affected

Question 105

A physical therapist is ordered to evaluate a 74-year-old man who has suffered a recent stroke. The therapist performs a chart review before performing the evaluation. Which of the following is of the least importance to the physical therapist in assessing the patient's chart?

1- Nursing assessment
2- Physician's orders/notes
3- Respiratory assessment
4- Dietary assessment

Question 106

A 69-year-old man with a cerebrovascular accident involving left anterior cerebral artery causes the main motor deficit in which of the following?

1- Ipsilateral leg
2- Ipsilateral arm
3- Contralateral leg
4- Contralateral arm

Correct Answer: 3
rationale:
the ACA supplies the motor fibers of the lower limbs (cortical branch).

Question 107

During evaluation of a hemiplegic patient you found that there is balance deficiency, the cause is:

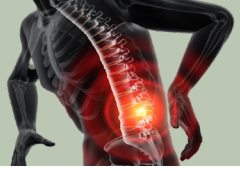
1- weakness of glutei
2- weakness of quadriceps
3- spasticity of planter flexors
4- weakness of dorsi flexors

Correct Answer: 4
rationale:
weakness of dorsi flexors as it affects the ankle strategy.

Question 108

A physical therapist is ordered to evaluate a patient in the intensive care unit who recently suffered a stroke. The evaluation reveals the following signs and symptoms: contralateral hemiplegia (upper extremity involvement greater than lower extremities), homonymous hemianopsia, aphasia (due to the dominant hemisphere being involved), and contralateral loss of sensation in the upper extremities. Which of the following arteries is the most likely location of the lesion?

1- Middle cerebral artery
2- Anterior cerebral artery
3- Internal carotid artery
4- Posterior cerebral artery



Question 109

Which of the following is true about anterior cerebral artery?

1- affect in lower extremity in same side
2- affect in upper extremity in same side
3- affect in lower extremity in opposite side (contra lateral)
4- affect in upper extremity in opposite side

Question 110

Patient with CVA damage LT Anterior cerebral artery affect

1- contralateral leg
2- ipsilateral leg
3- ipsilateralarm

Correct Answer: 1
rationale:
Contralateral leg as It`s the most affected leg in ant. Cerebral Artery.

Question 111

A 68-year-old man who suffered a stroke 4 weeks ago (involving the dominant hemisphere) presents with contralateral hemiparesis and sensory loss (greater in the lower extremity than in the upper extremity), mental confusion, and aphasia. What is the most likely location of the infarction?

1- Middle cerebral artery
2- Internal carotid artery
3- Posterior cerebral artery
4- Anterior cerebral artery

Question 112

A 50-year-old suffering from a stroke was referred to a Physical Therapy Rehabilitation Center. The treating physical therapist found that the patient is having visual disturbance, contralateral homonymous field defect, memory disturbance, and contralateral sensory loss. Which of the following cerebral arteries is most likely affected?

1- Middle
2- Lateral
3- Anterior
4- Posterior

Correct Answer: 4
rationale:
Posterior supplies the thalamus and the visual areas.

Question 113

Meaning of rigidity is:

1- decreased ROM
2- Muscle stiffness

Question 114

Patient has C V A and middle cerebral artery affected which part will have the best complete recovery

1- shoulder
2- elbow
3- hand
4- hip

Correct Answer: 4
rationale:
hip because the MCA affected the UL more than Lower Limb.
Anterior cerebral artery:
* Contralateral monoplegia cortical sensory loss.
* sometimes behavioral abnormalities associated with frontal lobe damage.
Middle cerebral artery:
(Supplies most of the convexity of the cerebral hemispheres)
* Dense contralateral hemiplegia.
* Contralateral homonymous hemianopia.
* Cortical type of sensory loss Speech problems in left hemisphere lesions, with neglect of contralateral side Lesions.
* in the right hemisphere result in parietal damage, visuospatial disturbances and left-sided neglect.
Posterior cerebral artery:
* Visual disturbance Contralateral homonymous field.
* defect Memory disturbance and contralateral sensory loss.

Question 115

Which of the following is a red flag sign of the lower limb?

1- Sprain
2- Osteoarthritis
3- Osteonecrosis
4- Muscle contracture

Question 116

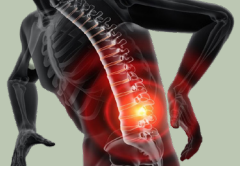
When evaluating a patient with an upper motor neuron lesion, the therapist reads the scientific term "rigidity" written in the patient's file.
What does this term mean?

1- Muscle stiffness
2- Decreased volume of speech
3- Involuntary oscillation of the body
4- Movements that are reduced in speed, amplitude, and range

Question 117

When evaluating Parkinson's pt you observed uniform muscle resistance, this means:

1- cogwheel rigidity
2- lead-pipe rigidity
3- spasticity



Question 118

A therapist is evaluating the range of motion of a patient diagnosed as having Parkinson's disease finds a smooth, uniform resistance to passive movement. Which of the following is the scientific term used to describe the condition?

1- Mutism
2- Spasticity
3- Lead pipe rigidity
4- Cogwheel rigidity

Question 119

Which of the following neurological conditions is associated with orofacial dysfunction?

1- Paraplegia
2- Parkinsonism
3- Diabetic neuropathy
4- Spinal cord lesion at the cervical area

Question 120

A therapist is evaluating a patient with an upper motor neuron lesion. During the assessment, the therapist finds that the patient has a postural tremor associated with the slowness of the movements. Which of the following neurological disorders does this patient most likely have?

1- Stroke
2- Multiple sclerosis
3- Parkinson's disease
4- Motor neuron disease

Correct Answer: 3
rationale:
None of those diseases is characterized by a tremor, except the Parkinson's disease. Parkinsonian tremor classically occurs at rest, but many patients also have postural tremor.

Question 121

A 68-year-old patient with Parkinson's disease with tremor at rest, slowness of movement and rigidity. Which area is most likely affected in the central nervous system?

1- Pons
2- Cerebellum
3- Basal ganglia
4- Medulla oblongata

Correct Answer: 3
rationale:
The pathology of Parkinson's disease (PD) targets the dopamine cells in the substantia nigra pars compacta, that greatly affects the nigrostriatal dopaminergic pathway of basal ganglia.

Question 122

Earliest sign of parkinsonism:
1- Diplopia
2- Tremor
3- Festinating gait

Correct Answer: 2
rationale:
PD is characterized by tremor, rigidity, and bradykinesia, with postural instability appearing in some patients as the disease progresses, all other clinical pictures like festinating gait and diplopia are secondary symptoms to the cardinal symptoms (tremors, rigidity, and bradykinesia).

Question 123

Parkinsonism disease is :
1- cerebral cortex
2- basal ganglion

Correct Answer: 2
rationale:
see Q.121

Question 124

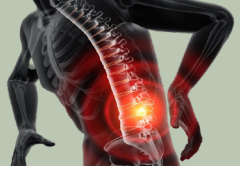
A 66 - year - old man has Parkinson's disease. Which neurotransmitter is responsible for Parkinson's disease?
1- Serotonin
2- Dopamine
3- Glutamic acid
4- Gamma - aminobutyric acid

Correct Answer: 2
rationale:
see Q.121

Question 125

The therapist observes a patient with the latter stages of Parkinson's disease during ambulation. Which of the following characteristics is the therapist most likely observing?
1- Shuffling gait
2- Increased step width
3- Difficulty initiating the first steps
4- A and C

Correct Answer: 4
rationale:
Many people with mid-stage to advanced Parkinson's disease (PD) experience "freezing." Freezing is the temporary, involuntary inability to move. It happens due to bradykinesia.



Question 126

Parkinson's gait, which is not true:

1- Wide base of support with arm swing
2- Propulsion gait
3- Festinating gait
4- Retropulsion

Correct Answer: 1
rationale:
The Parkinson's gait is characterized by shuffling gait (quick-stepping, short-stride walking movements) or festinating gait (like a shuffling gait. Unlike a shuffling gait, a person is always in a stooped position). Their center of gravity is always too far forward, so to avoid falling the person takes short, shuffled steps. When the patient can't catch his COG, he goes in high speed either forward (Propulsion) or backward (Retropulsion).
Additionally, the patient loses the associated movement of the arm swing due to rigidity and those population would also present with a small base of support.

Question 127

parkinsonism patient has complained of falling and decreased balance what will you do

1- lowering COG
2- decrease BOS
3- raction
4- passive movement

Question 128

when evaluating Parkinson's gait?

1- festinating and shuffling gait
2- short BOS

Correct Answer: 2
rationale:
see Q.126

Question 129

Abnormal pattern of gait that not occur in Parkinson is:

1- propulsion gait
2- festination gait
3- retropulsion gait
4- high steppage gait

Correct Answer: 4
rationale:
High stepping gait is a form of gait abnormality characterized by foot drop or ankle equinus due to loss of dorsiflexion.

Question 130

In Treatment of Parkinson pt :

1- strengthening of elongated muscles and stretching of rigid flexors
2- strengthening of elongated muscles and stretching of rigid extensors
3- strengthening of flexors & extensors
4- none of the above

Correct Answer: 1
rationale:
Parkinson's diseases patients suffer from rigidity that affects the flexors of all the body. Which results in flexed posture and postural instability. So, the prober treatment strategy should focus on stretching the rigid flexors and strengthening of elongated extensors.

Question 131

A 45-year-old patient with Parkinson's disease has been referred for physical therapy and mobility exercise. Which of the following exercise programs is the best for this patient?

1- Isometric exercise training.
2- Strengthening of both flexors and extensors.
3- Strengthening of the weak elongated extensors and stretching of the rigid flexors.
4- Strengthening of the weak elongated flexors and stretching of the rigid extensors.

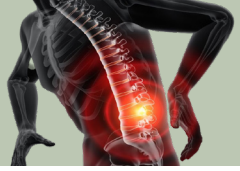
Correct Answer: 3
rationale:
see Q.130

Question 132

An 81 - year - old man with Parkinson's disease is treated by physical therapist. The patient has been ambulating with a cane. He is referred to physical therapy because of a fall at home. The family reports a decrease in gait ability during the past several months. The therapist decides to begin gait training with a rolling walker. Which of the following is the most appropriate treatment?

1- Strengthening of hip flexor and hip extensor muscles.
2- Strengthening of the hip extensor, and stretching of the hip flexors.
3- Prolonged passive stretching of the anterior tibial muscles bilaterally.
4- Strengthening of the hip flexors, and stretching of the gluteal muscles.

Correct Answer: 2
rationale:
see Q.130
Patients with Parkinson's disease usually ambulate with the trunk in flexion. Increased trunk flexion causes a festinating gait to be more pronounced. Therapy should strengthen extensor muscles while stretching the flexors.
A & D isn't the correct answer because in the PD patient the hip flexors are much rigid than the extensors.
C isn't the correct answer because the calf muscle is much rigid than the anterior tibial group.



Question 133

A physical therapist is treating a 72-year-old woman with a diagnosis of Parkinson's disease 3 times/week at an outpatient facility. The patient is taking 500 mg/day of a medication designed to decrease spasticity. The therapist notices that the patient is performing well on certain days and poorly on others. Which of the following ways can the therapist possibly improve the patient's performance on the days when she performs poorly?

1- The therapist can encourage the patient to increase her daily dosage of medication to 600 mg/day on the days she usually performs poorly.
2- The therapist can encourage the patient to decrease her daily dosage of medication to 100 mg/day on the days she usually performs poorly.
3- The therapist can schedule sessions so that there are fewer sessions on the days the patient performs poorly.
4- The therapist should call the physician to suggest another medication.

Correct Answer: 3
rationale:
The therapist should never instruct a patient to change the dosage of the medication. In addition, calling the physician and expressing concerns about the possible effects of the medication observed at therapy would be much more appropriate than calling to suggest another medication.

Question 134

Which of the following is the most likely cause of a decline in performance from over-training?

1- Poor hydration
2- Low motivation
3- Inadequate protein
4- Excessive frequency

Correct Answer: 4
rationale:
Over-training is defined as "excessive frequency, volume, or intensity of training, resulting in a reduction of performance, which is also caused by a lack of proper rest and recovery".

Question 135

As a result of diminished movement associated with Parkinson's disease, a physical therapist might use cueing strategy primarily to help improve:

1- trunk stability and proximal tone
2- trunk rotation
3- upper extremity function
4- Gait

Correct Answer: 4
rationale:
Cueing is using external temporal or spatial stimuli to facilitate movement (gait) initiation and continuation.

Question 136

As a result of diminished movement associated with Parkinson's disease a physical therapist might employ rhythmic initiation primarily to help improve:-

1- trunk stability and proximal tone
2- trunk rotation
3- upper extremity function
4- active and passive ROM

Correct Answer: 2
rationale:
trunk rotation is lacking with PD, helps to improve bed mobility.

Question 137

A 20-year-old woman is referred to a therapist with a diagnosis of recurrent ankle sprains. The patient has a history of several inversion ankle sprains within the past year. No edema or redness is noted at this time. Which of the following is the best treatment plan?

1- Gastrocnemius stretching, ankle strengthening, and ice.
2- Rest, ice, compression, elevation and ankle strengthening.
3- Ankle strengthening and a proprioception program.
4- Rest, ice, compression, elevation and gastrocnemius stretching.

Question 138

Prevention of transmission of pain beyond the dorsal horn:

1- peripheral n injury
2- gate control theory
3- spinal cord injury
4- opiate theory

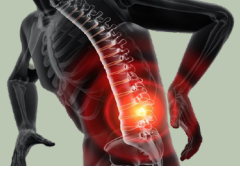
Correct Answer: 2
rationale:
The gate control theory of pain asserts that non-painful input "large-diameter (touch-, pressure-, and vibration- transmitting) afferent nerve fibers" closes the nerve "gates" to painful input "small-diameter", which prevents pain sensation from traveling to the central nervous system (beyond the dorsal horn).

Question 139

If the therapist who has a patient with a chronic pain in the lower back area decides to use TENS, usually the therapist is deciding to reduce pain through:

1- opide theory
2- gate control theory

Correct Answer: 2
rationale:
TENS stimulate Aβ fibers that block the transmission of Aδ and C nociceptive fibers.



Question 140

Prior to performing manual traction in the cervical region, one should perform a test for vertebral artery insufficiency in order to ascertain the adequacy of blood supply to the:

1- vertebrae
2- Upper extremities
3- Spinal canal
4- brain stem

Correct Answer: 4
rationale:
Vertebrobasilar insufficiency (VBI) occurs when there is inadequate blood flow through the posterior circulation of the brain in the vertebral and basilar arteries. The posterior circulation supplies the medulla, cerebellum, pons, midbrain, thalamus, and occipital cortex.

Question 141

All of these are non-equilibrium-coordination tests except

1- finger to finger
2- jumping
3- grasp
4- jogging

Correct Answer: 1
rationale:
Finger to finger is a non-equilibrium coordination test.

Question 142

A physical therapist is assessing a 40-year-old man's balance and coordination. The following instructions are given to the patient: "Stand normally, with your eyes open. After fifteen seconds, close your eyes and maintain a normal standing posture". Several seconds after closing his eyes, the patient nearly falls. What type of test did the patient fail?

1- Non equilibrium test
2- Equilibrium test
3- Romberg test
4- B and C

Correct Answer: 4
rationale:
Equilibrium tests are usually conducted with the patient in a static position such as standing (to test the ability of the patient to maintain balance in this position) or dynamic (to test the ability of the patient to maintain his balance during changing his position). The Romberg test can be considered as a type of equilibrium test.

Question 143

A physical therapist is asked by a co-worker to finish evaluating a patient because an emergency requires the therapist to leave. The co-worker agrees and resumes the examination. The first therapist left notes titled, "sensory assessment". Two wooden blocks identical in appearance but 1 pound different in weight are on the table in front of the patient. What test was the prior therapist most likely performing?

1- Barognosis test
2- Stereognosis test
3- Graphesthesia test
4- Texture recognition

Correct Answer: 1
rationale:
Barognosis is the ability to differentiate between different weights. Stereognosis is the ability to differentiate between different sizes and shapes. Graphesthesia is the ability to identify letters, numbers, or designs traced on the skin. Texture recognition is the ability to differentiate between textures such as cotton, wool, and silk.

Question 144

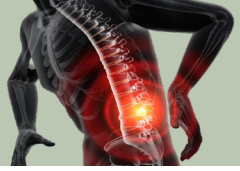
patient can't make two-point discrimination in lower leg what is the site of lesion?
1- brain stem
2- sensory cortex
3- peripheral nerve

Correct Answer: 2
rationale:
Two-point discrimination is a cortical sensation that requires some processing by the cortex to discriminate one stimulus from another.

Question 145

All are not correct about spinal cord except ...
1- Occupies the lumbar cistern
2- Has twelve (12) cervical segments
3- Contains the cell bodies of postganglionic sympathetic efferent neurons
4- Ends at the conus medullaris

Correct Answer: 4
rationale:
A isn't the answer because the spinal cord ends at level of L1:L2 and at the level of lumber cistern there is the cauda equina not the cord itself.
B isn't the answer because the spinal cord has only 8 cervical segments.
C isn't the answer because the postganglionic sympathetic neurons are present in sympathetic chain (sympathetic ganglia) not in the spinal cord.



Question 146

motor unit contain?

1- Motor neuron and their axons & all m.s supplied.
2- Motor neucli in A.H.C & axons supply m.s.
3- Motors neucli in P.H.C &axons supply m.s.

Correct Answer: 1
rationale:
A motor unit is made up of a motor neuron, its axons and all the skeletal muscle fibers innervated by the neuron's axon terminals, including the neuromuscular junctions between the neuron and the fibers.

Question 147

Motor area:

1- Control motor activity of opposite side of the body
2- Control motor activity of same side
3- Receive sensation

Correct Answer: 1
rationale:
In the brain there is a crossed control rule, that means the primary motor cortex normally controls the voluntary movements on the opposite side of the body.
C isn't the correct answer because the primary sensory areas are responsible for receiving different kinds of sensations.

Question 148

While observing a patient with posttraumatic brain injury (TBI), the therapist notes an increase in left ankle plantar flexion during loading response (heel strike to foot flat) of the involved lower extremity. With this particular patient, the left side is the involved side. Which of the following is not likely cause of this deviation?

1- Spasticity of the left gastrocnemius.
2- Hypotonicity of the left tibialis anterior.
3- Leg length discrepancy.
4- Left quadriceps hypertonicity.

Correct Answer: 4
rationale:
Increased spasticity of the left gastrocnemius causes an increase in plantarflexion. Hypotonicity of the tibialis anterior causes a foot drop due to the inability to dorsiflex actively. Leg length discrepancy should be evaluated in the initial assessment. This is a possibility in any patient, regardless of the diagnosis. The least likely cause of this deficit is a hypertonic left quadriceps, which most likely would cause an increase in dorsiflexion on the involved side in an effort to decrease the functional leg length.

Question 149

To prepare a patient with a cauda equina lesion for ambulation with crutches, the upper quadrant muscles that would be most important to strengthen would be the

1- upper trapezius, rhomboids and levator scapulae.
2- deltoid, coracobrachialis, and brachialis.
3- middle trapezius, serratus anterior, and triceps.
4- lower trapezius, latissimus dorsi and pectoralis major.

Correct Answer: 4
rationale:
Muscles needed for crutch use in cauda equina lesion are the shoulder depressors and extensors along with elbow extensors.
The lower fibers of trapezius depress the scapulae.
The latissimus dorsi assists in depression of the arm with the teres major and pectoralis major. It adducts, extends, and internally rotates the shoulder.

Question 150

A therapist is evaluating a patient who suffered brain injury in a motor vehicle accident. The somatosensory cortex is involved, resulting in deficits in sensation in the right upper and lower extremities. The therapist knows from visualizing the homunculus that at least part of the injury is in one of the following locations:

1- Inferior right hemisphere of the somatosensory cortex.
2- Superior right hemisphere of the somatosensory cortex.
3- Inferior left hemisphere of the somatosensory cortex.
4- Superior left hemisphere of the somatosensory cortex.

Correct Answer: 4
rationale:
The right hemisphere is responsible for the left side of the body and vice versa and the presentation of the human body in the primary somatosensory cortex is up-side-down which means the Superior part receives sensations from the lower extremity, the middle part receives sensations from the upper extremity and the inferior part receives sensations from the head.

Question 151

While observing a patient with posttraumatic brain injury (TBI), the therapist notes an increase in left ankle plantar flexion during loading response (heel strike to foot flat) of the involved lower extremity. With this particular patient, the left side is the involved side. Which of the following is not a likely cause of this deviation?

1- Spasticity of the left gastrocnemius.
2- Left quadriceps hypertonicity.
3- Hypotonicity of the left tibialis anterior.
4- Leg length discrepancy.

Correct Answer: 2
rationale:
see Q.148

Question 152

A physical therapist is treating a 17-year-old boy who suffered a traumatic brain injury. The patient has been in stage IV of Rancho Los Amigos Cognitive Functioning Scale for 1 week. Which of the following is an inappropriate treatment approach?

1- Start treatment at the same time each day.
2- Change treatment if the patient shows a decrease in interest.
3- Give the patient many different exercises options.
4- learning new skills.

Correct Answer: 4
rationale:
Level IV of Rancho Los Amigos Cognitive Functioning Scale means that the individual is confused agitated and needs maximal assistance.
Learning new motor skill would be difficult for this patient in this stage of recovery because he is confused and easily agitated. Treatment should include activities that are familiar to the patient.

Question 153

Myotome is :
1- muscle supplied by certain nerve.
2- skin supplied by certain nerve.

Correct Answer: 1
rationale:
Myotome is the muscle supplied by certain nerve, while dermatome is the skin supplied by certain nerve.

TABLE 2: MYOTOMES	
C1, C2	Cervical flexion
C3	Cervical side flexion
C4	Scapula elevation
C5	Shoulder abduction
C6	Elbow flexion and wrist extension
C7	Elbow extension and wrist flexion
C8	Thumb extension
T1	Finger abduction
L1, L2	Hip flexion
L3	Knee extension
L4	Ankle dorsiflexion
L5	Big toe extension
S1	Ankle plantiflexion
S2	Knee flexion

Question 154

Dermatome is :
1- muscle supplied by certain nerve.
2- skin supplied by certain nerve.

Correct Answer: 2
rationale:
see Q.153

Question 155

myotome of this biceps muscle is

1- C3-C4
2- C5- C6
3- T1- T2

Correct Answer: 2
rationale:
C4- Scapular elevation
C5- shoulder abduction.
C6– Elbow flexion Wrist extension
C7 – Elbow extension
C8 – Thumb Extension and wrist ulnar deviation
T1 – Finger abduction

Question 156

pt suffering from weakness of biceps brachials the corresponding myotome is

1- C4
2- C5
3- C6

Correct Answer: 3
rationale:
see Q.155

Question 157

patient with neck pain and he could not extend his elbow because it`s painful this pain referred to

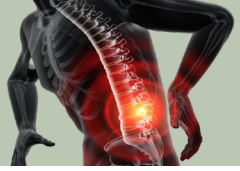
1- C5
2- C6
3- C 7
4- C8

Correct Answer: 3
rationale:
see Q.155

Question 158

pt has lumber pain radiated to

1- L1 , L2
2- L4 , L5
3- S1 , S2



Question 159

A 45-year-old patient with chronic neck pain, during the neurological examination, the physiotherapist finds a dull response of the triceps reflex.
What does this reflex give information about spinal levels?
1- C1-C3
2- C3-C4
3- C4-C5
4- C6-C8

Correct Answer: 4
rationale:
Triceps deep tendon reflex arch is innervated by C7.

Question 160

What is correct about superficial abdominal (upper abdominal) reflex innervations:
1- T7 to T9
2- T11 to T12
3- L1 , L2

Correct Answer: 1
rationale:
Roots involved in superficial abdominal reflex are Thoracic 7th – 12th segments, Thoracic 7th – 9th segments are the roots involved in the upper abdominal reflex, while Thoracic 10th – 12th segments are the roots involved in the lower abdominal reflex.

Question 161

Level of superficial abdominal reflex:
1- T10 – T11
2- L1 – L2
3- L2 – L3

Question 162

Patient complaining from back pain, with examination you find decreased sensation over big toe what level of lesion would you expect:
1- L 3 -L4
2- L4 – L5
3- L5 – S1

Question 163

Function of Golgi tendon organ:
1- detect muscle length
2- detect muscle tension

Question 164

Reflex quadriceps muscle is:

1- L4
2- L5
3- S1
4- S2

Question 165

while you examine a patient you find decreased tendon jerk at Achilles tendon, what is the level of lesion?

1- L4 –L5
2- L5 –S1
3- S1 – S2

Question 166

A 20-year-old patient injured in a car accident has been diagnosed as having spinal cord injury. During the physical evaluation, the therapist finds the ankle jerk is lost. At which of the following levels of the spinal cord is it lost?

1- S3,4
2- L1,2
3- L3, 4
4- S1, 2

Question 167

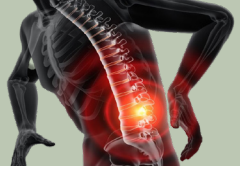
During physical therapy evaluation of a 53-year-old patient with peripheral nerve injury, the therapist asks the patient to walk across the floor. The therapist notices the patient raising one leg to release the toes from the ground. What kind of paralysis does this patient have?

1- Hip flexors
2- Knee flexor
3- Knee extensors
4- Ankle dorsiflexors

Question 168

A physical therapist is attempting to explain the importance of slow stretching to an athlete training to compete in a marathon. The therapist explains that quick stretching often causes the muscle to which is a response initiated by the which are located in the muscle fibers. Fill in the blanks?

1- Relax – Golgi tendon organs
2- Contract – Golgi tendon organs
3- Relax – muscle spindles
4- Contract – muscle spindles



Question 169

What is correct about planter reflex innervations is?

1- L4-L5
2- L2-L3
3- L5-S1
4- L5 -S1.2

Question 170

A 44-year-old man with trauma at the spinal cord, involving L3-L4 segmental level. Which of the following muscles are affected?

1- Abdominal
2- Hip flexors
3- Knee extensors
4- Ankle dorsiflexors

Question 171

Which of the following is the most appropriate technique to prevent injury during stretching exercises?

1- Slow speed
2- Linear force
3- Reduced force
4- Low repetitions

Question 172

for relaxation of parkinson's patient and to facilitate ex .which is correct?

1- Rhithmic initiation
2- Rhithmic stabilization
3- Sit pt on rocking chair

Correct Answer: 1
rationale:
Rhythmic initiation - parkinsonism
Rhythmic stablization - increase stability(Attaxia)

Question 173

which statement is incorrect about Golgi tendon organ?

1- it monitor muscle length
2- its affected by muscle contraction

Correct Answer: 1
rationale:
it monitor muscle length is incorrect it`s correct about muscle spindle.

Question 174

which statement is about Golgi tendon organ?
1- it monitor muscle length.
2- its affected by muscle contraction.
3- its more scattered in tendon.
4- its in the same series of muscle fibers.

Question 175

for relaxation of parkinson's patient and to facillitat ex which incorrect :
1- Rhithmic initiation
2- Rhithmic stabilization
3- Sit pt on rocking chair

Question 176

muscle spindle is stimulated by:
1- Passive stretch
2- Passive movement
3- Active exercises

Question 177

Regarding muscle spindle all following are true except:
1- has capsule.
2- differs according to muscle size.
3- extrafusal muscle fibers are parallel to intrafusal muscle.

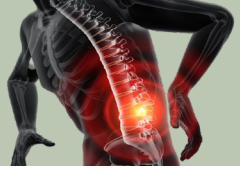
Question 178

Which is incorrect about muscle spindle?
1- Has capsule for each spindle.
2- Detect any tension in ms.
3- Parallel with extrafusal ms fiber.

Correct Answer: 2
rationale:
Detect any tension in muscle is incorrect for muscle spindle it`s correct about Golgi tendon organs .
For remember :
- Golgi Tendon organs has a T so we can remember it`s action about tension of muscle.
- Muscle spindle has L so we can remember it`s action for Length.

Question 179

Chronaxie definition:
1- Minimum time to depolarize the nerve when current intensity is twice the rheobase.
2- Minimum time to depolarize the nerve.



Question 180

A 55-year-old woman presents to the Physical Therapy Department with clonus. Which of the following is the most likely cause?

1- Cranial nerve injury
2- Peripheral nerve injury
3- Lower motor neuron lesion
4- Upper motor neuron lesion

Correct Answer: 4
rationale:
Clonus Mean : Involuntary muscular contractions and relaxations.

Question 181

Where does the main decussation occur in which the pyramidal tract descends from the cerebral cortex to the spinal cord?

1- Pons
2- Dura mater
3- Gray matter
4- Medulla oblongata

Question 182

A 43-year-old man with right biceps brachii rupture presents to physical therapy after a surgical repair. According to the surgeon, the rupture was at the musculotendinous junction. Which of the following has most likely been compromised?

1- Meissner's corpuscles
2- Merkel's disks
3- Ruffini endings
4- Golgi tendon organs

Question 183

Which of the following is Ruffini's end organs?

1- Rapidly adapting
2- Change in muscle length
3- Change in muscle tension
4- Monitor position of the limb in space

Correct Answer: 1
rationale:
what is the function of Ruffini end organ?
One of the encapsulated sensory nerve endings found in the dermis and in subcutaneous tissue, once thought to mediate the sense of warmth, now believed to be a pressure receptor.

Question 184

pyramidal tract make decussation at:
1- medulla
2- midbrain
3- pons
4- cerebral cortex

Question 185

bradykinesia means:
1- slowness of movement
2- fast movement
3- impaired limb sensation

Question 186

A 64-year-old patient with Parkinson's disease treated in the Physical Therapy Clinic. The physical therapist evaluates the patient gait to detect any movement problems. What is bradykinesia?
1- Slow movement
2- Fast movement
3- Rigid movement
4- Normal movement

Question 187

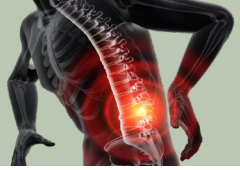
When evaluating a patient with an upper motor neuron lesion, the therapist read the term bradykinesia in the patient's file. What does this term mean?
1- Difficulty swallowing
2- Slowness of movement
3- Decreased volume of speech
4- Sudden episode of immobility

Question 188

Hypotoncity means:
1- decrease tone in muscle
2- Increase tone
3- incoordination

Question 189

All are forms of hypertonia except:
1- spasticity
2- dystonia
3- rigidity
4- hemianopia



Question 190

Hypertonicity means:
1- decrease tone in muscle
2- increase tone
3- incoordination

Question 191

Hypertonia indicates
1- upper motor neuron injury
2- lower motor neuron injury
3- cerebellar lesion
4- autonomic imbalance

Question 192

Difficult speech is called:
1- Dysarthria
2- Dysphagia

Question 193

Difficulty of swallowing means:
1- Dysarthria
2- Dysphagia

Question 194

the term apraxia (clumsy syndrome) refers to?
1- inability to perform purposeful movements
2- swallowing
3- slow motion
4- involuntary movement

Question 195

A 57-year-old paraplegic patient comes to the Physical Therapy Clinic. The therapist gives the patient instructions as a home recommendation.
Which of the following instructions should the patient be warned about?
1- Lift in the chair every 10 minutes
2- Protect the limbs against excessive cold or heat
3- Use a hot water bottle whenever unable to micturate
4- Use a mirror for detection of marks, abrasions, and blisters

Correct Answer: 4
rationale:
in absence of sensation patients should be warned to regularly check for any signs of skin breaking down or injury to avoid complications such as infected wounds and ulcers .

Question 196

The term dyspraxia refers to?
1- inability to perform purposfull movements
2- swallowig
3- slow motion
4- involuntary movement

Question 197

Bradyphrenia mean:
1- slowness of thought
2- slowness of movement

Correct Answer: 1
rationale:
Slowness of thought, seen in Parkinsonis.

Question 198

Which of the following scientific terms describes the condition of the slowness of thought processes and lack of concentration and attention?
1- Akinesia
2- Hypophonia
3- Bradykinesia
4- Bradyphrenia

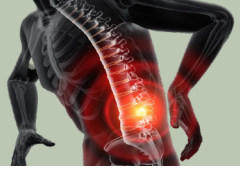
Question 199

The term aphasia refers to:
1- speech problem
2- swallowig
3- slow motion
4- involuntary movement

Question 200

Which of the following is most appropriate in regard to the anatomical position of Broca's speech area?
1- Inferior frontal gyrus
2- Superior frontal gyrus
3- Inferior temporal gyrus
4- Superior temporal gyrus

Correct Answer: 1
rationale:
Wernicke's area Present in Superior temporal gyrus of the brain.



Question 201

Wernicke's area in which gyrus of the brain located?
1- superior part of temporal lobe
2- arcuate fasciculus
3- parietal lobes
4- frontal lobes

Question 202

Which of the following scientific terms is defined as the marked impairment of production and comprehension of language ?
1- fluent aphasia
2- nonfluent aphasia
3- global aphasia

Question 203

You are working with a patient who exhibits a fluent aphasia. This form of aphasia is usually characterized by:
1- normal auditory comprehension
2- very slow speech
3- impaired reading and writing
4- impaired articulation

Question 204

The term used to describe difficulty in reading :
1- dyslexia
2- dysgraphia
3- dyspraxia
4- paraphasia

Question 205

Patient with stroke the neurologist refers him to pt. first for assessment pt. find?
1- Chorea
2- Nystagmus
3- Aphasia

Question 206

A physiotherapist received a referral for a patient. The referring physician outlined all clinical findings in the referral, one of which is "abarognosis". What does this term mean?
1- Double vision
2- Difficulty swallowing
3- Slowness of movement
4- Inability to recognize the weight of an object

Question 207

During the physical evaluation of a hemiplegic patient, the therapist determines that the patient is unable to recognize the weight of any object.
What is the scientific term for this condition?
1- Allesthesia
2- Barognosis
3- Astereognosis
4- Prosopagnosia

Question 208

A physical therapist evaluating a 66 year old female who has a history of severe head trauma following a motor vehicle accident. The patient has difficulty with rapid alternating movements while performing neurologic testing. The BEST term to describe this specific impairment is:
1- ataxia
2- dysmetria
3- dysarthria
4- dysdiadocokinesia

Question 209

patient suffer from double vision means
1- hemianopia
2- anomia
3- diplopia
4- agnosia

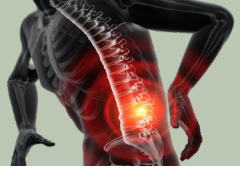
Correct Answer: 3
rationale:
Loss of half of visual field.

Question 210

A physical therapist received a patient with dystonia who was referred by the neurologist. Which of the following statements is the most appropriate definition of dystonia?
1- Difficulty swallowing
2- Slowness of movement
3- Speech related problems
4- Involuntary repetitive movement

Question 211

What is meaning of dysmetria?
1- lack of coordination of movement, the inability to judgment of distance.
2- lack of speak.



Question 212

Ataxia means :

1- in-coordination in movement
2- inability to perform action
3- movement very slow

Question 213

neuron has no axon, no refractor organ but make synapse with second neuron at the ganglion is called

1- preganglionic neurone
2- postganglionic neurone
3- autonomic neuron

Question 214

patient has a complete spinal cord lesion at the C5 level. It is important for this patient to generate a functional cough in order to avoid potential respiratory problems. In this case, effective coughing:

1- should be facilitated by use of phrenic nerve stimulator.
2- should be facilitated by use of glossopharyngeal breathing.
3- can be elicited with manual abdominal pressure provided with the assistance of a caregiver.
4- can be elicited with manual abdominal pressure provided independently by the Patient.

Question 215

C8 SCI independent mobility with which adapted of the following:

1- manual wheel chair
2- machinery wheel chair
3- manual wheel chair with slide board

Question 216

Pt with SCI there is weakness in his lower limb to test him:

1- Give resistance to ms at middle ROM
2- Test hip flexors L1-L2
3- Hip ext L3

Question 217

pt with new SCI (Spinl Cord Injury) reffersd to you for prevention of bed rest complication which of the following not included in your program

1- passive movement.
2- elastic bandaging for prevention of deep venous thrombosis.
3- alternative positioning to avoid bed sources.
4- respiratory ex.

Question 218

In SCI which of the following in not of complications:

1- hyperreflexia
2- systemic hypertension
3- pressure sores
4- orthostatic hypotension

Question 219

To help decrease shear when transferring a patient with a spinal cord injury from bed to chair, it would be best to use:

1- a draw sheet.
2- Skin lubricant on involved skin surfaces.
3- An air mattress.
4- Sheep skin pad.

Question 220

Which of the following spinal levels is the lowest for a spinal injury to result in autonomic dysreflexia?

1- C6
2- T6
3- L1
4- S2-S4

Question 221

In SCI which of the following in not of complications Complete transection of the spinal cord at the C1 level produces all of the following effects except:

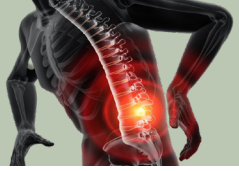
1- Hypotension.
2- Limited respiratory effort.
3- Anaesthesia below the level of the lesion.
4- Areflexia below the level of the lesion.

Correct Answer: 4
rationale:
Complications of SCI are: Skin Breakdown, pneumonia, Osteoporosis and Fractures, Heterotopic Ossification, Spasticity, Urinary Tract Infections, Autonomic Dysreflexia, Deep Vein Thrombosis, Pulmonary Embolism ,Orthostatic Hypotension, Cardiovascular Disease , Syringomyelia , Neuropathic / Spinal Cord Pain , Medication Problems , Hyperthermia ,Hypothermia.

Question 222

Main purpose from semi fowler position?

1- More accesses
2- Gravity assisting position



Question 223

A patient with a spinal cord injury is being treated by physical therapy in an acute rehabilitation setting. The patient has been involved in a motor vehicle accident that resulted in a complete C8 spinal cord lesion. The patient is a 20-year-old man who has expressed concern to the therapist about his future sexual function. Which of the following is the most correct information to convey to the patient?

1- Psychogenic erection is possible, reflexogenic erection is not possible, and ejaculation is possible.
2- Psychogenic erection is not possible, reflexogenic erection is not possible, and ejaculation is not possible.
3- Psychogenic erection is possible, reflexogenic erection is possible, and ejaculation is possible.
4- Psychogenic erection is not possible, reflexogenic erection is possible, and ejaculation is not possible.

Question 224

A therapist is sent to the intensive care unit to evaluate a patient who has suffered a severe recent head injury. While reviewing the patient's chart, he discovers that the patient exhibits decerebrate rigidity. The therapist is likely to find this patient in which of the following positions?

1- The patient will be positioned with all extremities extended and the wrist and fingers flexed.
2- The patient will be positioned with the upper extremities flexed, the lower extremities hyperextended, and the fingers tightly flexed.
3- The patient will be positioned with all extremities flexed and wrist and fingers extended.
4- The patient will be positioned with the upper extremities extended, the lower extremities flexed and the fingers hyperextended.

Question 225

In a patient with head injury, unexplained hypotension warrants evaluation of:

1- Upper cervical spine
2- Lower cervical spine
3- Thoracic spine
4- Lumbar spine

Question 226

A therapist is treating a 35-year-old man who has suffered loss of motor control in the right lower extremity due to peripheral neuropathy. The therapist applies biofeedback electrodes to the right quadriceps in an effort to increase control and strength of this muscle group. The biofeedback can help achieve this goal in all of the following ways except:

1- Providing visual input for the patient to know how hard he is contracting the right quadriceps.
2- Assisting the patient in recruitment of more motor units in the right quadriceps.
3- Providing the measure of torque in the right quadriceps.
4- Providing the therapist input on the patient's ability and effort in contracting the right quadriceps.

Question 227

which nerve give motor &sensory branch supplying facial muscle is?

1- V
2- VI
3- VII
4- VIII

Question 228

A therapist is treating a patient with spinal cord injury. The therapist is discharging the patient after completion of all physical therapy goals. One of the completed long-term goals involved the ability to dress and bathe independently with assistive devices. This would be a most challenging but obtainable goal for which of the following?

1- C5 quadriplegia
2- C7 quadriplegia
3- T1 paraplegia
4- C4 quadriplegia

Correct Answer: 1
rationale:
C5 quadriplegia which means that the lesion isn`t affect the pharanic nerve which supply the diaphragm and it supply musculocutaneous nerve Tricepses muscle.

Question 229

What is the main cause of breathing ineffective in upper thoracic and cervical sci?

a- Recurrent pneumonia.
b- paralysis of abdominals.
c- paralysis of intercostals.

Question 230

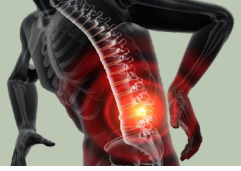
A 48 - year - old woman patient with paraplegia after a road traffic accident with no activity below the level of lesion immediately after the injury. Which of the following is the most likely stage?

1- Stage of spinal shock
2- Stage of gait training
3- Stage of the total recovery
4- Stage of failure of a reflex activity

Question 231

Which of the following is the most appropriate stage of injury to initiate a passive range of motion exercises?

1- Acute
2- Chronic
3- Sub-acute
4- Functional



Question 232

A physical therapist is attempting to increase a patient's functional mobility in a seated position. To treat the patient most effectively and efficiently, the following should be performed in what order? (1) Weight shifting of the pelvis, (2) Isometric contractions of the lower extremity, (3) Trunk range of motion exercises, (4) Isotonic resistance to the quadriceps

1- 1,2,3,4
2- 2,3,1,4
3- 4,3,2,1
4- 3,2,1,4

Question 233

A 20 - year - old patient with incomplete T6 spinal cord injury was referred to the clinic for assessment and treatment. A home program of various strengthening exercises was prescribed to the patient and he was asked to come to the clinic once a week. OB What is the most appropriate assessment tool to track patient improvement over visits?

1- Glasgow Coma Scale (GCS).
2- Functional Independence Measure (FIM).
3- Activities - specific Balance Confidence Scale (ABC).
4- The American Spinal Injury Association (ASIA) Scale.

Question 234

A therapist is treating a patient with an injury at the T8 level and compromised function of the diaphragm. If no abdominal binder is available, what is the most likely position of comfort to allow him to breathe most efficiently?

1- Sitting position
2- Semi-fowler position
3- Standing
4- Supine

Question 235

A physical therapist is reviewing the chart of 24-year-old woman with a diagnosis of L2 incomplete paraplegia. The physician noted that the left quadriceps tendon reflex is 2+. What does this information relay to the therapist?

1- No active quadriceps tendon reflex.
2- Slight quadriceps contraction with reflex testing.
3- Normal quadriceps tendon reflex.
4- Exaggerated quadriceps tendon reflex.

Question 236

Fracture dislocation L4-5 will result in:

1- Quada Equina
2- Flaccid Paraplegia
3- Spastic Quadriplegia
4- Spastic Paraplegia

Question 237

A therapist is testing key muscles on a patient who recently suffered a spinal cord injury. The current test assesses the strength of the long toe extensors. Which nerve segment primarily innervates this key muscle group?

1- L2
2- L3
3- L4
4- L5

Correct Answer: 4
rationale:
L5 dorsal root innervate extensors long toes

Question 238

The following is a long-term goal for a patient with spinal cord injury: independence in performing a manual cough without applying pressure to the abdomen. This goal is the most challenging and obtainable for a patient with a complete Lesion at which of the following spinal cord levels?

1- C5
2- C7
3- T2
4- T10

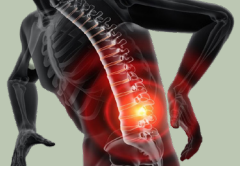
Correct Answer: 2
rationale:
loss of abdominal muscles which will make this task a very challenging one.

Question 239

A 35-year-old woman suffered brain injury in a motor vehicle accident and presents with the following symptoms: an intention tremor, nystagmus, hypotonia, and dysdiadocokinesia. What is the most likely location of the lesion?

1- Basal ganglia
2- Dorsal columns
3- Frontal lobe
4- Cerebellum

Correct Answer: 4
rationale:
all signs point to the cerebellar ataxia as falire to coordinate the the movement of the eye causes nystagmus and tremors in cerebellar ataxia are intention tremors (increases when close to the target).



Question 240

Which of the following abnormal responses is observed after the checking of Babinski sign?

1- Absence of sensation.
2- Loss of coordination in movement.
3- abnormal response of the plantar reflex.
4- Loss of knowledge or inability to perceive objects.

Correct Answer: 3
rationale:
dorsi flexion of the big toe with or without fanning of the other toes.

Question 241

An abnormal Babinski reflex indicates damage to the:

1- Spinal cord
2- Brainstem
3- cerebellum
4- Pyramids

Correct Answer: 4
rationale:
positive Babinski is a sure pyramidal sign.

Question 242

A patient diagnosed with cervical stenosis coupled with a positive Lhermitte`s sign,as elicited by the maneuver in the picture, would most likely indicate:

1- upper motor neuron findings in the lower extremities.
2- lower motor neuron findings in the upper extremities.
3- lower motor neuron findings in the lower extremities.
4- both upper and lower motor neuron finding in the lower extremities.

Question 243

Which of the following is the most effective fall prevention intervention in the elderly population?

1- Medicine review
2- Balancing exercises
3- Individualized program
4- Strengthening exercises

Question 244

Main purpose from semi lying position:

1- Gravity assistance
2- Efficiency

Question 245

A physical therapist is ordered to evaluate a 65-year-old woman who has suffered a recent stroke. The occupational therapist informs the physical therapist that the patient has apraxia. She cannot brush her teeth on command. However, she can point out the toothbrush and verbalize the purpose of the brush. From this information, what sort of apraxia does this patient have? How should the physical therapist approach treatment?

1- Ideomotor apraxia. The physical therapist should speak in short, concise sentences.
2- Ideomotor apraxia. The physical therapist should always give the patient 3 step commands.
3- Ideomotor apraxia. The physical therapist should always give the patient 5 step commands.
4- Ideational apraxia. The physical therapist should speak in short, concise sentences.

Question 246

A 60-year-old woman who has suffered a recent stroke has right-side homonymous hemianopsia. Which of the following statements is true about placement of eating utensils in early rehabilitation?

1- The utensils should be placed on the left side of the plate.
2- The utensils should be placed on the right side of the plate.
3- The utensils should be placed on both sides of the plate.
4- The plate and utensils should be placed slightly to the right.

Question 247

A 52-year-old man with sciatica presents to outpatient physical therapy. The patient indicates that he is experiencing paresthesia extending to the left ankle and severe lumbar pain. Straight-leg-raise test is positive with the left lower extremity. Of the following, which if the most likely source of pain?

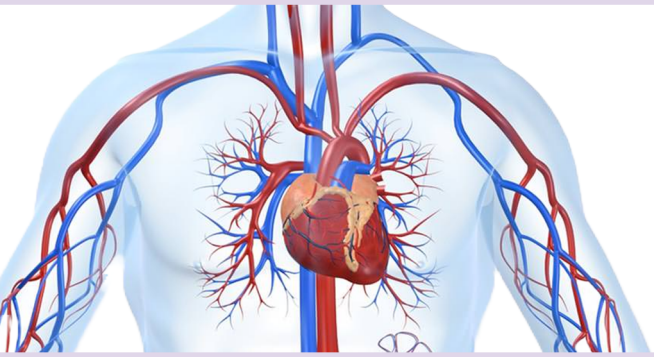
1- A lumbar disc with a left posterior herniation or protrusion.
2- Piriformis syndrome.
3- Sacroiliac joint dysfunction.

Correct Answer: 1
rationale:
A lumbar disc with a left posterior herniation or protrusion If the patient experiences sciatic pain when the straight leg is at an angle of between 30 and 70 degrees, then the test is positive and a herniated disc is likely to be the cause of the pain.

Question 248

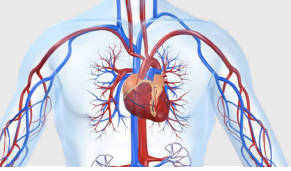
A therapist is obtaining a subjective history from a new patient diagnosed with right-side hemiplegia. The therapist notes that the patient is able to understand spoken language but unable to speak well. Most of the patient's words are incomprehensible. The patient also has difficulty in naming simple objects. What type of aphasia does the patient most likely have?

1- Anomic aphasia
2- Broca's aphasia
3- Crossed aphasia
4- Wernicke's aphasia



Chapter

3 Cardiovascular and Pulmonary Systems



Question 1

Rock-port 1 mile test used to test?

1- muscles
2- cardiovascular fitness
3- CNS

Correct Answer: 2
rationale:
The Rockport walking test is an evaluation you can self-administer to determine your cardiovascular fitness , it evaluates the capacity of your lungs in relation to the volume of exercise you can tolerate. The aim of the test is to measure your VO2 max, the maximum amount of oxygen you can utilize during intense exercise, measured in milliliters of oxygen used per kilogram of body weight per minute (ml/kg/min)

Question 2

What is the goal of the first phase of rehabilitation on Cardiopulmonary

1- help the patient to advance from passive range of motion ROM to active
2- increase endurance- Oxygen consumption

Correct Answer: 2
rationale:
The goal of this first stage is to recuperate and allow your body to begin the healing process. It begins by assessing the patient's physical ability and motivation to tolerate rehabilitation. Therapists and nurses may start by guiding patients through non-strenuous exercises in the bed or at the bedside, focusing on a range of motion and limiting hospital deconditioning and this help increasing endurance and oxygen consumption by capturing atmospheric oxygen and distributing it into the blood stream through diffusion.

Question 3

Which of the following is the most appropriate trunk and head position, in degrees, to start chest physiotherapy?

1- 0
2-15
3- 30
4- 45

Correct Answer: 3
rationale:
This position is best for draining the bottom front parts of your lungs , and enable gravity to assist the mobilization of secretions

Question 4

Muscle that work the most in respiration?

1- Intercostal
2- diaphragm
3- abdominal

Correct Answer: 2
rationale:
The diaphragm, located below the lungs, is the major muscle of respiration. It is a large, dome-shaped muscle that contracts rhythmically and continually, and most of the time, involuntarily. Upon inhalation, diaphragm contracts and flattens and the chest cavity enlarges his movement creates a vacuum in your chest, allowing your chest to expand and pull in air . external intercostal have fibres that are angled obliquely downward and forward from rib to rib. The contraction of these fibres raises each rib toward rib above, assisting in inhalation

Question 5

Muscle that participate in respiration?
1-SCM ms
2-abdominals ms
3-intercostals ms

Correct Answer: 3
rationale:
intercostal muscles are a group of muscles found between the ribs which are responsible for helping form and maintain the cavity produced by the ribs.muscles assist with expansion and contraction during breathing and mainly involved in the mechanical aspect of breathing by helping expand and shrink the size of the chest cavity
external intercostal have fibres that are angled obliquely downward and forward from rib to rib. The contraction of these fibres raises each rib toward the rib above, with the overall effect of raising the rib cage, assisting in inhalation but the role of internal intercostal is in forced expiration , *internal intercostals* pull down on the rib cage and push air out of the lungs

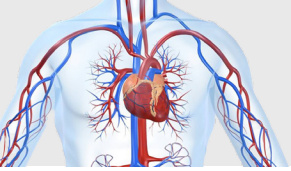
Question 6

Percussion is applied from
1. 1 to 3 min
2. **2 to 4 min**
3. 2 to 5 min
4. 2 to 6 min

Correct Answer: 2 : 2to 5 min. in excessive secretions 3-5 Min
rationale:
percussion is usually done for three to five minutes to allow as much mucus as possible to drain from your lungs and is sometimes followed by vibration over the same area for approximately 15 seconds. The person is then encouraged to cough or huff forcefully to get the mucus out of the lungs

Question 7

At the beginning of the muscle activity which make increase of blood supply to it
1. **sympathetic adrenergic**
2. sympathetic cholerigric
3. parasympathetic



Correct Answer: 2
rationale:
sympathetic nervous system regulates skeletal muscle motor innervation and acetylcholine receptor stability and also plays a crucial role in the regulation of arterial blood pressure and blood flow during exercise, and several important neural mechanisms are responsible for changes in sympathetic vasomotor outflow

Question 8

volkmans ischemic disease most dependable sign for this patient is-

1. **Pain**
2. cyanosed color of hand
3. weak extensors
4. weak flexion

Correct Answer: 1
rationale: volkmann contracture is a permanent shortening (contracture) of forearm muscles, usually resulting from injury, that gives rise to a claw like deformity of the hand, fingers, and wrist. It is more common in children. A similar condition can occur in the foot
Symptoms affect the forearm, wrist, and hand. They include:
- pain
- Decreased sensation
- Paleness of the skin
- Muscle weakness and loss (atrophy
(if said 2 and 4 be good answer

Question 9

Volkmann's ischemic contracture occur due to :

1-fracture of medial condyle of humerus
2- fracture of lateral condyle of humerus
3- fracture lower third of humerus
4-surgical neck fracture

Correct Answer: 3
rationale:
volkmann ischemic contracture is classically associated with a supracondylar fracture of the humerus. However, any fracture of the arm or elbow may result in this deformity.s

Question 10

which is correct about Volkmann's ischemic contracture

1. **affect flexor of forearm**
2. affect palmar fascia lead to ulnar neuritis

Correct Answer: 1
rationale:
In Volkmann contracture, the muscles of the forearm are severely injured usually, the flexors muscles ,This leads to contracture deformities of the fingers, hand, and wrist

Question 11

One of the possible complications following a fracture is Volkmann's ischemic contracture. This condition:

1. Is caused by an interference with the venous return
2. Is caused by an interference of the nerve supply
3. May occur if the fracture is sustained in the upper extremity
4. **None of the above**

Correct Answer: 4
rationale:
Volkmann contracture is a deformity of the hand, fingers, and wrist caused by injury to the muscles of the forearm
Volkmann contracture occurs when there is a lack of blood flow (ischemia) to the forearm. This occurs when there is increased pressure due to swelling

Question 12

Persuading a sedentary patient to become more active, the therapist explains the benefits of exercise. Which of the following is an inappropriate list of benefits?

1. **Increased efficiency of the myocardium to obtain oxygen, decreased high-density lipoprotein (HDL) and decreased cholesterol**
2. Decreased low-density lipoprotein (LDL), decreased triglycerides, and decreased blood pressure.
3. Increased efficiency of the myocardium to obtain oxygen, decreased cholesterol, and decreased LDL.
4. Both 1 and 3 are inappropriate lists

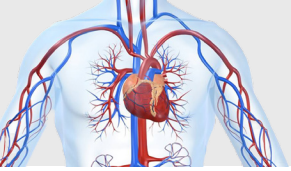
Correct Answer: 1
rationale:
As the cardiac work rises more than the myocardial oxygen consumption, the left ventricular efficiency increases. exercise also stimulates enzymes that help move LDL from the to the liver. From there, the cholesterol is converted into bile BUT exercise increase plasma HDL-cholesterol (HDL-C) levels in a dose–response manner

Question 13

Fracture of Lower third of humerus cause stiffness in elbow physiotherapist must do very gentle stretch to avoid

1. disuse muscle atrophy
2. hypertrophy
3. **muscle and tendon tear**

Correct Answer: 3
rationale:
overstretch on tendon or muscle can cause strain so we must avoid vigorous stretch
NOTE: also to avoid myositis ossificans



Question 14

What is the blood flow for an athlete at resting?

1. 1 to 2 ml/min/110g
2. **6 to7 ml/min/110g**
3. 60 to 80 ml/min/110g

Correct Answer: 2
rationale: 6 to7 ml / min / 110g so with increase size there is increase blood flow

Question 15

Fracture Lower humerus fracture, therapist tend to do very gradual stretch:

1. to prevent disuse atrophy
2. to prevent hypertrophy
3. to stimulate muscle
4. **to prevent muscle and tendon rupture.**

Correct Answer: 4
rationale:
exerting too much energy or going too deep into a stretch can result in a torn muscle. Slowly ease into your stretches.

Question 16

The therapist is stretching the biceps muscle in a patient with a fracture at the lower end of the hummers with severe limitation of the ROM. The therapist needs to perform a very gradual stretch to:

1. avoid disuse atrophy
2. avoid hypertrophy
3. **prevent muscle and tendon tear**
4. stimulate muscle

Correct Answer: 3
rationale:
stretching a muscle slowly prevents it from overextending and being damaged. When stretches occur too quickly, such as with bouncing, the muscle can be seriously damaged

Question 17

The use of compression stockings on the feet and ankles is contraindicated in which patient population?

1. Chronic venous disease
2. Recent total knee replacement
3. Burn patients
4. **Chronic arterial disease**

Correct Answer: 4
rationale:
These garments can exacerbate peripheral arterial disease symptoms by impeding already-limited circulation

Question 18

A patient presents to an outpatient physical therapy clinic with 140 degree kyphoscoliosis curve. What is the therapist's greatest concern?

1. The patient's complaint of low back pain
2. Gait deviation.
3. **Pulmonary status.**
4. Poor upright standing posture

Correct Answer: 3
rationale:
kyphoscoliosis is a thoracic cage deformity that causes extrapulmonary restriction of the lungs and gives rise to impairment of pulmonary functions, as restrictive lung diseases. The condition may be primary (idiopathic) or secondary to neuromuscular disease, spondylitis ...

Question 19

When O2 demand in heart decrease what happened ?

1. decrease HR multiplied by systolic pressure
2. **decrease diastolic pressure**
3. increase coronary arteries
4. increase HR multiplied by systolic pressure

Correct Answer: 2
rationale:
By decreasing oxygen demand , the amount of blood that normally fills the ventricles decreases so diastolic blood pressure decrease

Question 20

Which of the following is the deconditioning effect on heart associated with bed rest?

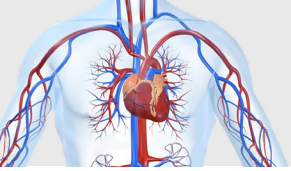
1. **decrease VO2 max .**
2. Decrease heart rate .
3. Increase plasma volume .
4. Increase cardiac output .

Correct Answer: 1
rationale:
(VO2max) is reduced in individuals confined to bed rest,
To counteract this decrease in stroke volume and maintain sufficient cardiac output, a gradual increase in heart rate , Bed rest is also associated with an increase in sodium excretion and, as water tends to follow sodium, this leads to sodium diuresis that cause to the reduction in plasma volume

Question 21

Average skeletal muscle blood flow at resting is:

1. **1-4 ml/min per 100g**
2. 5-8 ml/min per 100g
3. 9-12 ml/min per 100g
4. 13-16 ml/min per 100g



Correct Answer: 1
rationale:
At rest, skeletal muscle blood flows may be 1-4 ml/min per 100g
maximal blood flows may reach 50-100 ml/min per 100g

Question 22

During extreme exercise in a well conditioned athlete blood flow through skeletal muscle can increase to :

1. 15-25 fold rising to 50to 80 ml/min/100g of ms
2. **6-7ml/min/100g of ms**
3. averages 1-2ml/min/100g of ms
4. it's not estimated

Correct Answer: 2
rationale:
Skeletal muscle blood flow increases dramatically, while blood flow to other tissues, especially the abdominal viscera and kidneys, is reduced
as a result of sympathetic vasoconstriction, renal and splanchnic blood flow can both fall to 25% of resting values during heavy exercise in humans

Question 23

Left ventricle more important than Right ventricle at accident why

1. Contains un oxygenated blood
2. transmit blood less o2
3. **Transmit blood to aorta**
4. D-Transmit un oxygenated blood to aorta

Correct Answer: 3
rationale:
Left ventricular contraction forces oxygenated blood through the aortic valve to be distributed to the entire body

Question 24

Right Ventricle of the heart

1. have bulk of muscle
2. **have low oxygen blood**
3. have greater oxygen blood
4. neither of all

Correct Answer: 2
rationale:
right ventricle (RV) pumps oxygen-poor blood through the pulmonary valve (PV) into the main pulmonary artery (MPA). From there, the blood flows through the right and left pulmonary arteries into the lungs

Question 25

voltage of heart

1. -90
2. +90
3. **+60**
4. -30

Correct Answer: 3
rationale:
the average voltage level is 60 mV.

Question 26

The formula for calculating 'maximum heart rate' is:

1. 120 minus age
2. 180 minus age
3. 200 minus age
4. **220 minus age**

Correct Answer: 4
rationale:
The basic way to calculate your maximum heart rate is to subtract your age from 220 (MHR = 220 – age)

Question 27

What is the minimal heart rate to measure target heart rate before exercise?

1. n 20-30
2. **63 or 60**

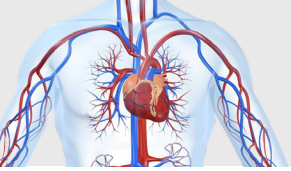
Correct Answer: 2
rationale:
because the normal heart rate is 60-100

Question 28

Which of the following statements is the most appropriate for cardiac output?

1. Volume of blood pumped by each ventricle per beat.
2. Volume of blood in the ventricle at the end of systole.
3. Volume of blood in the ventricle at the end of diastole.
4. **Volume of blood pumped by each ventricle per minute**

Correct Answer: 4
rationale:
Cardiac output is the volume of blood pumped per minute by each ventricle of heart.
cardiac output is the product of heart rate (HR) and stroke volume (SV) and is measured in liters per minute



Question 29

Cardiac output means:output?

1. **blood pumped from ventricle/min**
2. blood pumped from ventricle/beat
3. blood still in ventricle during systolic
4. blood still in ventricle during diastolic

Correct Answer: 1
rationale:
Cardiac output is the volume of blood pumped per minute by each ventricle of heart.

Question 30

Cardiac pacemaker is

1. locating in AV node
2. contain connective tissues
3. contain nerve tissues
4. **in SA node**

Correct Answer: 4
rationale:
the sinus node is sometimes called the heart's "natural pacemaker." Each time the sinus node generates a new electrical impulse; that impulse spreads out through the heart's upper chambers, called the right atrium and the left atrium

Question 31

Cardiac pacemaker:

1. in AV node
2. **create these rhythmical impulse**

Correct Answer: 2
rationale:
SA node is often referred to as a natural pacemaker because it generates a series of electrical pulses at regular intervals

Question 32

Which is true about the pacemaker of the heart?

1. **it is the tissue that has more Rhythmic in the heart**
2. locating in AV node
3. contain connective tissues
4. contain nerve tissues

Correct Answer: 1
rationale:
sinus node called natural pacemaker and continuously generates electrical impulses, thereby setting the normal rhythm and rate in a healthy heart.

Question 33

Which is true about SA node:

1. type of myocardium with higher rhythm
2. **type of myocardium which generate electrical impulse**
3. consist of nerve cell
4. in AV nodee

Correct Answer: 2
rationale:
sinoatrial node (SA node) is a specialized myocardial structure that initiates the electrical impulses to stimulate contraction

Question 34

Which of the following initiate the electricity of the heart ?

1. **SA node**
2. AV node
3. AV bundle
4. Purkinje fibers

Correct Answer: 2
rationale:
sinoatrial node (SA node) is a specialized myocardial structure that initiates the electrical impulses to stimulate contraction

Question 35

Impulses reach atrium early to contract before ventricles through

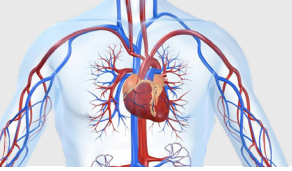
1. Purkinje fibers
2. **AV node**
3. Bundle of His
4. Atrioventicular bundle

Correct Answer: 2
rationale:
the 2 upper chambers of the heart (atria) are stimulated first and contract for a short period of time before the 2 lower chambers of the heart (ventricles). The electrical impulse travels from the sinus node to the atrioventricular node (also called AV node

Question 36

Coronary artery for the heart;

1. one
2. **two**
3. three
4. four



Correct Answer: 2
rationale:
There are two coronary arteries, each containing several branches: Right coronary artery and left coronary arteries , Both of these arteries originate from the root of the aorta, immediately above the aortic valve .the left coronary artery originates from the left aortic sinus, while the right coronary artery originates from the right aortic sinus

Question 37

Coronary arteries arise from aorta at

1. left atrium
2. **base of aorta at level of aortic valve**
3. at a point near to the base of aorta
4. arise at point of left ventricle

Correct Answer: 2
rationale:
There are two primary coronary arteries, the right coronary artery (RCA) and the left main coronary artery (LMCA). Both of these originate from the base of the aorta

Question 38

coronary artery supply heart during

1. systole
2. in between
3. **diastole**
4. not related to systolic and diastolic

Correct Answer: 3
rationale:
Coronary blood flow occurs mostly during diastole because the coronary vasculature has one particular property: it is compressed by the contracting myocardium such that no flow occurs during systole , during systole, intramuscular blood vessels are compressed and twisted by the contracting heart muscle and blood flow to the left ventricle is at its lowest

Question 39

Irreversible risk factor in coronary artery disease

1. Smoking
2. High blood pressure
3. a Lipo –protein
4. **History of occlusive peripheral artery**

Correct Answer: 4
rationale:
Un-modifiable risk factors (irreversible)
age, male sex, race, and family history.
Modifiable risk factors:
smoking, exposure to second-hand smoke, hypertension, hyperlipidemia, high cholesterol (total or LDL-C) levels,(low HDL-C) levels, high triglyceride levels, diabetes, abdominal obesity, sedentary lifestyle, high homocysteine levels, and high levels of C-reactive protein (which indicates inflammation).

Question 40

A patient asks the therapist to explain the function of his medication verapamil (a calcium antagonist). Which of the following points should be conveyed in the therapist's explanation?

1. **Verapamil causes decreased contractility of the heart and vasodilation of the coronary arteries**
2. Verapamil causes decreased contractility of the heart and vasoconstriction of the coronary arteries
3. Verapamil causes increased contractility of the heart and vasodilation of the coronary arteries
4. Verapamil causes increased contractility of the heart and vasoconstriction of the coronary arteries

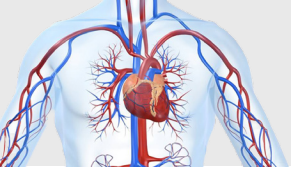
Correct Answer: 1
rationale:
Calcium plays a central role in the electrical stimulation of cardiac cells and in the mechanical contraction of smooth muscle cells in the walls of arteries. Calcium-channel blockers are relatively new synthetic drugs that work by blocking the passage of calcium into the muscle cells that control the size of blood vessels, thereby preventing the muscles of the arteries from constricting so decrease heart rate and contractility, which leads to a reduction in myocardial oxygen demand, which makes them excellent antianginal drugs

Question 41

Base of heart directed

1. forward and anterior
2. forward and posterior
3. backward and posterior
4. **backward and anterior**

Correct Answer: 4
rationale:
the base of heart directed upward, backward, and to the right, is separated from the fifth, sixth, seventh, and eighth thoracic vertebrae by the esophagus, aorta, and thoracic duct



Question 42

In old patient can find

1. stroke volume decrease
2. **stroke volume increase**
3. stroke volume decrease with increase cardiac out put

Correct Answer: 2
rationale:
older people demonstrated significantly higher stroke volume and mean arterial blood pressure but lower heart rate

Question 43

Absolute refractory period in cardiac muscle

1. slower than in skeletal muscle
2. **contain whole systole and beginning of diastole**

Correct Answer: 2
rationale:
the absolute refractory period is much longer in cardiac muscle than in skeletal muscle because of the long duration of the plateau phase in cardiac muscle
Throughout systole and the beginning of diastole, the atria relax and blood flows into them.

Question 44

S.A.Node :

1. Consist of myocardium contractile tissue
2. It's type of myocytes with higher rhythm
3. **type of myocytes which generate electrical impulse and control heart rate**

Correct Answer: 3
rationale:
SA node, also known as the sinus node, represents a crescent-like shaped cluster of myocytes divided by connective tissue that are capable of generating spontaneous action potentials and are responsible for cardiac conduction

Question 45

Parasympathetic action of heart

1. decrease heart rate from 120 ppm to 70 ppm
2. **increase in female more than male**
3. decrease in adults more than child
4. decrease in athelets more than athelets

Correct Answer: 2
rationale:
males have higher sympathetic, and females higher parasympathetic

Question 46

Arterial blood pressure means

1. rdifference between systole and diastole
2. **lateral pressure exerted on arterial wall**
3. max blood comes from ventricles at end of diastole
4. max blood comes from ventricles at beginning of systole

Correct Answer: 2
rationale:
arterial blood pressure is defined as the force that is exerted by the blood on the arterial wall

Question 47

Referral for medical evaluation is advised when resting oxygen saturation falls below:

1. 97%
2. 95%
3. 92%
4. **90%**

Correct Answer: 4
rationale:
If the oxygen level on a pulse ox is less than 90%, that is a concerning sign and a good reason to go to the hospital emergency department , Values under 90 percent are considered low.

Question 48

What is the oxygen saturation in the blood below which signs of oxygen deficiency develop

1. 56%
2. 66%
3. 85%
4. **90%**

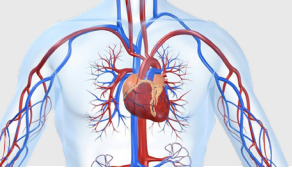
Correct Answer: 4
rationale:
when your blood oxygen falls below a certain level, you might experience shortness of breath, headache, and confusion or restlessness , lower than 92% is considered hypoxic.

Question 49

What the true about apex of heart :

1. **Is downwards of the left side**
2. Is downwards of the right side
3. 3rd degree of intercostals ribs in line of sternum
4. 5th degree of ribs

Correct Answer: 1
rationale:
apex (the most inferior, anterior, and lateral part as the heart lies in situ) is located on the mid-clavicular line, in the left fifth intercostal space.



Question 50

contractility of the heart muscle by age

1. increase
2. **decrease**
3. lost
4. not changed

Correct Answer: 2
rationale:
reduction of ATP production in the aging heart, which is caused by mitochondrial dysfunction. The cardiac contractility requires ATP turnover. The age-related change of metabolism attenuates cardiac contractile function

Question 51

Inner layer of the heart :

1. Endometrium
2. **Endocardium**

Correct Answer: 2
rationale:
endocardium is the innermost layer of the heart. It lines the inner surfaces of the heart chambers, including the heart valves. The endocardium has two layers. The inner layer lines the heart chambers and is made of endothelial cells

Question 52

Outer part of the heart

1. epicardium
2. endocardium
3. **pricardium**
4. myocardium

Correct Answer: 3
rationale:
pericardium , is The outer sac which enclosed the heart and protect it .
Epicardium: is the inner layer of the pericardium.
Myocardium: is the whole major muscle of the heart.
Endocardium: is inner layer which lining valves and heart.

Question 53

A 54-year-old patient with hypertrophic cardiomyopathy was treated in the Cardio Clinic. What is the notable sign of heart muscle?

1. Thin
2. Soft
3. Hard
4. **Thick**

Correct Answer: 2
rationale:
hypertrophic cardiomyopathy is usually caused by changes in genes (gene mutations) that cause the heart muscle to thicken. Hypertrophic cardiomyopathy typically affects the muscular wall (septum) between the two bottom chambers of the heart (ventricles). The thickened wall might block blood flow out of the heart

Question 54

Heart innervation

1. sympathetic
2. parasympathetic
3. pyramidal tract
4. **a+b**

Correct Answer: 4
rationale:
the heart is innervated by sympathetic and parasympathetic fibres from the autonomic branch of the peripheral nervous system. The network of nerves supplying the heart is called the cardiac plexus. It receives contributions from the right and left vagus nerves,as well as contributions from the sympathetic trunk

Question 55

The patient is referred by a physician to begin outpatient cardiac rehabilitation. Which of the following is not a contraindication to enter an outpatient program?

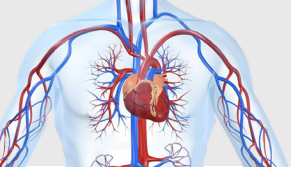
1. Resting systolic blood pressure of 210 mmHg
2. Third-degree atrioventricular block
3. **Resting ST displacement less than 1 mm.**
4. Acute fever

Question 56

56-year-old patient with an ejection fraction of 38%. Blood pressure 110/80 mmHg and 69 / min Heart rate .Which of the following is indicated by the clinical characteristics?

1. **Left heart failure**
2. Right heart failure
3. Myocardial infarction
4. Coronary artery insufficiency

Correct Answer: 1
rationale:
heart failure with reduced ejection fraction also called systolic failure: The left ventricle loses its ability to contract normally. The heart can't pump with enough force to push enough blood into circulation



Question 57

A 64-year-old patient presented with heart valves disorders, there was a blood return from the right ventricle back to the right atrium. Which of the following valves is most likely responsible for prevention?

1. Mitral
2. Aortic
3. **Tricuspid**
4. Pulmonary

Correct Answer: 3
rationale:
tricuspid valve regurgitation is a type of heart valve disease in which the valve between the two right heart chambers (right ventricle and right atrium) doesn't close properly. As a result, blood leaks backward into the upper right chamber (right atrium)

Question 58

The heart valve responsible in prevention blood from return to right atrium from right ventricle is:

1. mitral or bicuspid
2. **tricuspid**
3. pulmonary
4. coronary artery

Correct Answer: 2
rationale:
the tricuspid valve prevents the reverse flow of blood from the right ventricle to the right atrium while bicuspid valve prevents the reverse flow of blood from the left ventricle to the left atrium

Question 59

A 50 - year - old man was presented to the Cardiac Care Unit after undergoing a Coronary Artery Bypass Grafting. According to the patient, he was a chronic smoker and adopted a sedentary lifestyle. On the first post - operative day, he was referred for phase I cardiac rehabilitation. Which of the following is the most appropriate rehabilitation?

1. Running
2. Stair climbing
3. **.Sit to stand activities**
4. Manual

Correct Answer: 3
rationale:
during phase 1 cardiac rehab, patients with coronary artery disease, those who have had a heart attack, those recovering from heart surgery, heart failure and other cardiac diseases, engage in monitored, low-level exercise and receive patient education

Question 60

Effect of pool therapy :

1. increase tidal volume
2. **increase vital capacity**
3. decrease vital capacity

Correct Answer: 2
rationale:
pool exercise affects lung volume measurements as respiratory muscles including diaphragm of swimmers are required to develop greater pressure as a consequence of immersion in water during respiratory cycle, thus may lead to functional improvement in these muscles and also alterations in elasticity of lung

Question 61

51-year-old patient was admitted to the hospital with congestive heart failure and referred to the Cardiac Rehabilitation Unit during hospital admission. What is the suggested rehabilitation program phase?

1. **1st**
2. 2nd
3. 3rd
4. 4th

Correct Answer: 1
rationale:
during phase 1 cardiac rehab, patients with coronary artery disease, those who have had a heart attack, those recovering from heart surgery, heart failure and other cardiac diseases, engage in monitored, low-level exercise and receive patient education

Question 62

Phase I cardiac rehabilitation aim : ?

1. **prevent bed rest complication begin with passive to active movement**
2. increase ex endurance capacity
3. assist in progress functional ADL

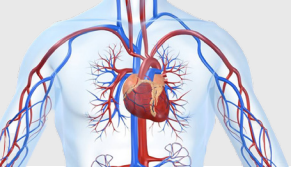
Correct Answer: 1
rationale:
This phase begins in the inpatient setting soon after a cardiovascular event or completion of the intervention. It begins by assessing the patient's physical ability and motivation to tolerate rehabilitation. Therapists and nurses may start by guiding patients through non-strenuous exercises in the bed or at the bedside, focusing on a range of motion and limiting hospital deconditioning.

Question 63

Following an exercise session in a Phase III cardiac rehabilitation program, in which some patient may experience angina pectoris, the therapist employs a gradual and prolonged cool down period primarily to prevent:

1. exertional dyspnea
2. tachycardia
3. **venous pooling**
4. hypertension

Correct Answer: 3
rationale:
some patient experience angina pectoris due to insufficiency blood supply with O2 to the heart , but Generally we make a gradual cooling down period for prevent venous pooling which mean pulling the blood down to the lower extremity



Question 64

main character at cardiac patient :

1. bluish limbs
2. pain at the calf
3. **edema of limbs**
4. abdominal breathing

Correct Answer: 3
rationale:
cardiac edema occurs when the heart's diseased or overworked left ventricle (heart's lower chamber) isn't able to pump out enough of the blood it receives from your lungs.

Question 65

A physician orders stage II cardiac rehabilitation for a patient. The orders are to exercise the patient below 7 metabolic equivalents (METS). Which of the following is a contraindicated activity?

1. Riding a stationary bike at approximately 5.5 mph
2. **Descending a flight of stairs independently**
3. Ironing
4. Ambulate independently at 5-6 mph

Correct Answer: 2
rationale:
metabolic equivalent for task (MET) is a unit that estimates the amount of energy used by the body during physical activity, as compared to resting metabolism. The unit is standardized so it can apply to people of varying body weight and compare different activities
one MET is defined as 3.5 mL O2 uptake/kg per min, which is the resting oxygen uptake in a sitting position). Less than 5 METS is poor, 5–8 METS is fair, 9–11 METS is good, and 12 METS or more is excellent

Question66

Which make rapid decrease in heart rate ?

1. preganglionic fiber
2. postganglionic fiber
3. **fiber near to heart**

Correct Answer: 3
rationale:
Heart rate is an important physiologic regulator of total cardiac output. Through strong influences of the autonomic nervous system, the heart rates in a newborn can range from 70 to 220 beats/min and profoundly impact cardiac output.Fibers from the autonomic divisions alter the intrinsic heart rate by changing the course of spontaneous diastolic depolarization on pacemaker cells in the SA node. Activation of the cardiac sympathetic fibers will increase the heart rate, whereas stimulation of cardiac parasympathetic fibers will decrease the heart rate.

Question 67

Which of the following is the most likely cause of cardiac arrest following an electric shock?

1. Low heart rate
2. Low blood pressure
3. **Ventricular fibrillation**

Correct Answer: 3
rationale:
if a current from outside the body passes through the heart, it can mask these impulses and disturb the heart's rhythm. This irregular heartbeat is called arrhythmia and can even manifest as a total disorganization of the rhythm, known as ventricular fibrillation

Question 68

Which of the following is associated with an absent P-wave and a pulse of 150 beats per minute?

1. Atrial flutter
2. **Atrial fibrillation**
3. Ventricular flutter
4. Ventricular fibrillation

Correct Answer: 2
rationale:
lack of visible P waves preceding QRS complexes suggests a lack of sinus beats; this may occur with sinus dysfunction or in the presence of fibrillation or flutter waves

Question 69

Patient after exercise experienced low heart rate, low blood pressure, headedness, his skin is pale and cold. What is the most likely?

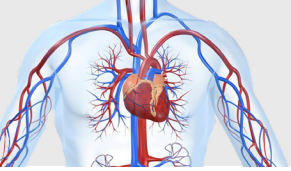
1. Angina
2. cardiogenic shock
3. myocardial infarction.

Correct Answer: 2 cardiogenic shock because skin is pale
rationale: cardiogenic shock, also known as cardiac shock, happens when your heart cannot pump enough blood and oxygen to the brain and other vital organs. This is a life-threatening emergency. It is treatable if diagnosed right away, so it's important to know the warning signs , pressure, fullness or a squeezing pain in the center of your chest that lasts for more than a few minutes. Pain spreading to your shoulder, one or both of your arms, your back, or even your teeth and jaw , pale skin . Shortness of breath

Question 70

Which type of atrioventricular (AV) block is present, given the following information about the patient's rhythm strip: P waves are normal and have a QRS complex following P-R intervals that are longer than 0.2 seconds, and the heart rate is 82 beats per minute?

1. **First-degree AV block**
2. Second-degree AV block – type 1
3. Second-degree AV block – type 2
4. Third-degree AV block



Correct Answer: 1
rationale:
first-degree atrioventricular (AV) block is a condition of abnormally slow conduction through the AV node. It is defined by ECG changes that include a PR interval of greater than 0.20 without disruption of atrial to ventricular conduction. This condition is generally asymptomatic and discovered only on routine ECG

Question 71

A 50-year-old man has a persistent cough, purulent sputum, abnormal dilation of bronchi, more frequent involvement of the left lower lobe than the right, hemoptysis, and reduced forced vital capacity. What is the most likely pulmonary dysfunction?

1. Chronic bronchitis
2. Emphysema
3. Asthma
4. **Bronchiectasis**

Question 72

Correct Answer: 4
rationale:
bronchiectasis is a long-term condition where the airways of the lungs become widened, leading to a build-up of excess mucus that can make the lungs more vulnerable to infection. The most common symptoms of bronchiectasis include: a persistent cough that usually brings up phlegm (sputum) shortness of breath

Question 73

A patient presents to a clinic with decreased tidal volume (TV). What is the most likely cause of this change in normal pulmonary function?

1. Chronic pulmonary disease
2. **Restrictive lung dysfunction**
3. Both of the above
4. None of the above

Correct Answer: 2
rationale:
restrictive lung disease, a decrease in the total volume of air that the lungs are able to hold, is often due to a decrease in the elasticity of the lungs themselves or caused by a problem related to the expansion of the chest wall during inhalation

Question 74

A patient presents with tachypnea, cor pulmonale, hypoxemia, rales on inspiration, and decreased diffusing capacity. What is the probable cause?

1. Restrictive lung dysfunction
2. Chronic obstructive pulmonary disease
3. Neither of the above
4. **A and B**

Correct Answer: 4
rationale:
rales or crackles are respiratory sounds often heard in chronic obstructive pulmonary disease (COPD) as well as in restrictive conditions, such as heart failure, lung fibrosis and pneumonia. 1 Forgacs proposed that crackles heard during inspiration were related to sudden opening airways

Question 78

A patient with restrictive lung disease secondary to circumferential thoracic burns demonstrate decreased ability to expand the lower rib cage and push the abdominal wall anteriorly. In this case, the therapist should consider the use of facilitation techniques to enhance the function of the:

1. rectus abdominis
2. anterior scalenus
3. internal intercostals
4. **diaphragm**

Correct Answer: 4
rationale:
Diaphragm breathing used for restrictive pulmonary disease because its action is in conjunction with the intercostal muscles, it lowers the pressure in the thoracic cavity. This enables air to enter the lungs. When breathing out, the diaphragm relaxes along with the intercostal muscles, allowing air to leave. Air may be forced out faster by increasing abdominal pressure using the transverse abdominis muscle.

Question 79

Which of the following is the most appropriate description of the expiratory flow rate in chronic obstructive pulmonary disease?

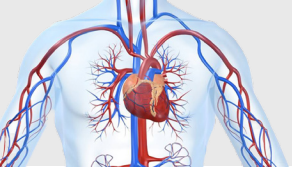
1. Variable
2. Increase
3. **Decrease**
4. Unchanged

Correct Answer: 3
rationale:
patients with chronic obstructive pulmonary disease (COPD) exhibit increases in lung volume due to expiratory airflow limitation. Increases in lung volumes may affect upper airway patency and compensatory responses to inspiratory flow limitation (IFL) during sleep

Question 80

You are evaluating a patient with chronic obstructive pulmonary disease. You would not expect to find:

1. the presence of hypercapnia
2. increased total lung capacity
3. **increased vital capacity**
4. abnormal ventilation/perfusion ratio



Correct Answer: 3
rationale:
Increased Vital capacity couldn`t be found in COPD , but there is almost hypercapnia , increased of Respiration rate lead to Increased in total lung capacity , abnormal ventilation /perfusion ratio .

Question 81

COPD we focus on

1. inspiration
2. expiration
3. **alternative expiration then inspiration**
4. alternative inspiration then expiration

Correct Answer: 3
rationale:
early in the disease, people with COPD may feel short of breath when they exercise. As the disease progresses, it can be hard to breathe out (exhale) or even breathe in (inhale)

Question 82

best exercise for COPD

1. **walking**
2. cycling
3. Swimming
4. running

Correct Answer: 1
rationale:
this low impact activity (meaning it's easy on joints) can improve the body's ability to use oxygen, build endurance, strengthen muscles, and enhance an overall sense of well-being. A regular walking routine also can make it easier for someone with COPD to be more self-sufficient and better able to tolerate exercise)

Question 83

A young woman with Chronic Obstructive Pulmonary Disease is referred for pulmonary rehabilitation. She complains of multiple episodes of shortness of breathing and coughing. Which of the following is the most appropriate for improving health status?

1. Cycling
2. **Walking**
3. Running
4. Swimming

Correct Answer: 2
rationale:
this helps your body use oxygen more efficiently and, with time, can improve your breathing

Question 84

Benefits from ex?

1. Larger lung volume
2. Normal tidal volume
3. **Larger vital capacity**
4. Maximum minute ventilation

Correct Answer: 3
rationale vital capacity is the maximum amount of air that can be breathed out after breathing in as much air as possible. Taking part in regular aerobic exercise has been shown to increase a person's vital capacity

Question 85

A physical therapist is treating a 65-year-old man with pneumonia. The patient questions the benefits of the flow incentive spirometer left in the room by the respiratory therapist a few minutes ago. Which of the following is an appropriate response to the patient's question?

1. **"It gives visual feedback on the lung performance"**
2. "It helps you maintain current lung volumes"
3. "You need to ask the respiratory therapist this question"
4. A and B are correct responses.

Correct Answer: 1
rationale:
an incentive spirometer is a device that will expand your lungs by helping you to breathe more deeply and fully

Question 86

Which of the following are indications for pulmonary suctioning?

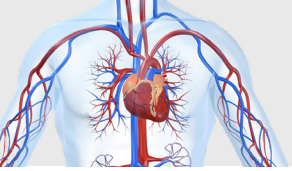
1. Unproductive coughs
2. Breath sounds or wet rales
3. **Respiratory distress**
4. All of the above

Correct Answer: 3
rationale:
clinical indications for suctioning include respiratory distress due to increased copious, retained secretions. Signs of respiratory distress may include increased respiratory rate, tachycardia, gasping and difficulty talking

Question 87

Which of the following exercises does not increase strength of the muscles of forceful inspiration?Unproductive coughs

1. Active cervical flexion exercises
2. Active glenohumeral extension exercises
3. Shoulder shrugs.
4. **Crunches.**



Correct Answer: 4
rationale:
crunches, also known as abdominal crunches, are a core exercise designed to target your abdominal muscles—specifically your rectus abdominis (the six-pack muscle) and your obliques the abdominal muscles act on the abdomen and the abdominal rib cage and are expiratory

Question 88

Which of the following is the most appropriate criterion for discontinuation of chest physiotherapy in an Intensive Care Unit patient?

1. **Clear chest X-ray**
2. Absent chest pain
3. Absent fever for 12 hours
4. Breathing rate of less than 20/min

Correct Answer: 1
rationale:
chest X-rays can detect cancer, infection or air collecting in the space around a lung, which can cause the lung to collapse. They can also show chronic lung conditions, such as emphysema or cystic fibrosis, as well as complications related to these conditions.

Question 89

During a case conference, a respiratory therapist indicates that the patient has a low expiratory reserve volume. What does this mean?

1. The volume of air remaining in the lungs after a full expiration is low.
2. The volume of air in a breath during normal breathing is low.
3. The volume of air forcefully expired after a forceful inspiration is low.
4. **The amount of air expired after a resting expiration is low.**

Correct Answer: 4
rationale:
Expiratory reserve volume extra volume of air that can be expired with maximum effort beyond the level reached at the end of a normal expiration s.

Question 90

foul smelling sputum?

1. Lung abscess
2. **Pneumonia (bacterial pneumonia)**

Correct Answer: 2
rationale:
foul-smelling sputum is associated with anaerobic infections due to aspiration, lung abscess, and necrotizing pneumonia.

Question 91

Which of the following is associated with foul - smelling sputum and finger clubbing ?

1. Cancer .
2. **Pneumonia** .
3. Lung abscess
4. Respiratory failure

Correct Answer: 2
rationale:
although some cases can be inherited or idiopathic, acquired nail clubbing can be a sign of systemic disease, especially pulmonary disease (e.g., malignancy, pneumonia, bronchiectasis, emphysema).

Question 92

A 19 - year old otherwise healthy patient presents to the Emergency Department with chest pain and shortness of breathing. Chest X - ray shows air rim of 3 cm. Which of the following is the most appropriate management?

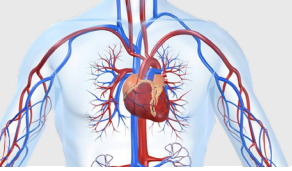
1. Discharge
2. **Aspiration**
3. Chest tube
4. Antibiotics

Correct Answer: 2
rationale:
The British Thoracic Society guidelines divide pneumothoraxes into small and large based on the distance from visceral pleural surface (lung edge) to chest wall, with less than 2 cm being small and more than 2 cm large.
The following is based on the BTS guidelines for the treatment of pneumothorax; local protocols may differ:
-asymptomatic small rim pneumothorax: no treatment with follow-up radiology to confirm resolution
-pneumothorax with mild symptoms (no underlying lung condition): needle aspiration in the first instance
-pneumothorax in a patient with background chronic lung disease or significant symptoms)intercostal drain insertion(small drain using the Seldinger technique

Question 93

A therapist is sent to provide passive range of motion to a patient in the intensive care unit. The chart reveals that the patient is suffering from pulmonary edema. The charge nurse informs the therapist that the patient is coughing up a thin white sputum with a pink tint. Which of the following terms best describes this sputum?

1. Purulent.
2. **Frothy**
3. Mucopurulent
4. Rusty



Correct Answer: 2
rationale:
mucus that contains bubbles and is foamy is commonly referred to as frothy sputum. Frothy sputum can sometimes be a sign of: chronic obstructive pulmonary disease (COPD

Question 94

A 66 year-old man presented with dyspnea and chronic fatigue. He was being managed with oral steroids and an inhaler for his asthma. Physical examination revealed reduced left lower chest movement. Chest physiotherapy has been recommended. Which of the following is the most appropriate method to improve chest wall mobility?

1. Purse lipped breathing
2. Lateral basal expansion
3. **Deep breathing exercises**
4. Posterior basal expansion

Correct Answer: 3
rationale:
if you take a deep breath in, allow the diaphragm to descend and the chest wall to expand, the thoracic spine will move into extension. As you exhale deeply and relax, the thoracic spine will flex

Question 95

Which of the following is the most appropriate description of expiratory reserve volume?

1. Air that remains at the end of normal respiration
2. Total amount of air in a lung after normal inspiration
3. Total amount of air in a lung after maximum inspiration
4. **Total amount of air exhaled after normal resting expiration**

Correct Answer: 4
rationale:
expiratory reserve volume is the additional volume of air that can be expired after a normal or tidal expiration

Question 96

At a team meeting. the respiratory therapist informs the rest of the team that the patient, just admitted to the subacute floor, experienced breathing difficulty in the acute care department. The respiratory therapist describes the breathing problem as a pause before exhaling after a full inspiration. Which of the following is the therapist describing?

1. Apnea
2. Orthopnea
3. Eupnea
4. **Apneusis**

Correct Answer: 4
rationale:
apneustic breathing It is characterized by regular deep inspirations with an inspiratory pause followed by inadequate expiration

Question 97

A therapist is asked to evaluate a patient in the intensive care unit. The patient is comatose but breathing independently. During the assessment of range of motion in the right upper extremity the therapist notices that the patient is breathing unusually. The pattern is an increase in breathing rate and depth followed by brief pauses in breathing. The therapist should notify the appropriate personnel that the patient is exhibiting which of the following patterns?

1. Biot's.
2. **Cheyne-Stokes**
3. Kussmaul's
4. Paroxysmal nocturnal dyspnea

Correct Answer: 2
rationale:
cheyne-Stokes respiration is characterized by alternating apnea and hyperventilation during sleep
This can be clinically observed and documented with a cyclic variation of breathing pattern with a change in saturation from 90% to 100%

Question 98

Which of the following statements is not a common physiologic change of aging?

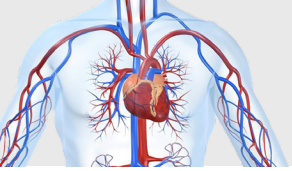
1. Blood pressure taken at rest and during exercise increases
2. Maximal oxygen uptake decreases
3. **Residual volume decreases**
4. Bone mass decreases

Correct Answer: 3
rationale:
functional residual capacity and residual volume increase with age, resulting in a lower vital capacityn

Question 99

A 55-year-old patient with a cardiac problem and used artificial pacemaker to regulate the heartbeat instead of the sinus node, which is the heart's natural pacemaker. Where is the sinus node located in the heart?

1. Left atrium
2. **Right atrium**
3. Left ventricle
4. Right ventricle



Correct Answer: 2
rationale:
The sinoatrial node (SA node or sinus node) is a group of cells known as pacemaker cells, located in the wall of the right atrium of the heart.
Conduction pathway :starts when your sinoatrial (SA) node creates an excitation signal.
The excitation signal travels to: Your atria (top heart chambers), telling them to contract.
-The atrioventricular (AV) node, delaying the signal until your atria are empty of blood.
-The bundle of His (center bundle of nerve fibers), carrying the signal to the Purkinje fibers.
-The Purkinje fibers to your ventricles (bottom heart chambers), causing them to contract

Question 100

In angina pectoris chest pain present in

1. pain at pectoral region
2. pain at base of the heart
3. pain referred to shoulder

Correct Answer: 3
rationale:
In angina pectoris:pain is best described as clenching a fist over the breastbone (sternum). It is dull crushing pain in the center of the chest. with potential radiation to the left shoulder and arm.

Question 101

patient during use of treadmill complains of chest pain the physiotherapist call for cardiologist because he suspect angina pectoris as the pain was

1. sever strapping pain in chest
2. **pain at chest radiating to left shoulder &arm**
3. pain at apex of heart
4. pain in chest radiating to both medial &lateral pectoral region

Correct Answer: 2
rationale:
Angina pain is often described as squeezing, pressure, heaviness, tightness or pain in the chest. It may feel like a heavy weight lying on the chest.
People experiencing angina feel discomfort or pain as a result of the heart getting less oxygen than it needs. Along with left arm pain, symptoms might occur in the shoulders, neck, back, or jaw.

Question 102

In angina pectoris chest pain present in

1. pain at pectoral region
2. pain at base of the heart
3. **pain referred to shoulder**

Correct Answer: 3
rationale: In angina pectoris:pain is best described as clenching a fist over the breastbone (sternum). It is dull crushing pain in the center of the chest. with potential radiation to the left shoulder and arm

Question 103

patient with angina pectoris, pain distribution

1. apex of heart
2. chest only
3. **chest with left shoulder and arm**

Correct Answer: 3
rationale:
Angina occurs when the heart muscle doesn't get as much blood and oxygen as it needs.it usually causes uncomfortable squeezing , pressure,heaviness ,tightness or fullness, dull crushing pain in the center of the chest.Some may also complain of suffocation, choking or breathlessness. The discomfort is diffused and deep-seated, with potential radiation to the left shoulder and arm.And it may spread to neck, jaw ,abdomen , back or ear lobes.
Note : Stabbing, pricking or other sharp and well-localised and transient chest pains are rarely due to angina. These could be due to indigestion or a muscle spasm

Question 104

Its benefit before provide ex program must do WARM UP

1. pooling of venous blood
2. increase O2 demand of heart
3. **decrease O2 deficiency**

Correct Answer: 3
rationale:
Benefits of a warm up :Performance may be improved, as an appropriate warm-up could result in :

1. A gradual increase in blood flow as local vascular beds dilate .
2. Increased body / muscle temperature which decreases blood viscosity and causing a more rapid increase in transport of the enzymes required for energy systems and muscle contraction .
3. Increased speed(faster) of contraction and relaxation of warmed muscles.
4. Allows the heart rate to get to a workable rate for beginning exercise.

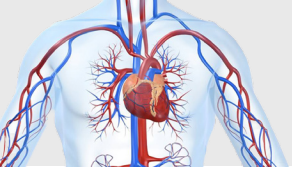
Benefits of a cool down :

1. bring your heart rate and blood pressure to its normal level
2. Prevent blood pooling in veins and reduce the chances of dizziness or fainting.
3. Keeps capillaries dilated to flush muscles with oxygenated blood

Question 105

The therapist works in a cardiac rehabilitation setting. Which of the following types of exercises are most likely to be harmful to a 64-year-old man with a history of myocardial infarction?

1. Concentric
2. Eccentric
3. Aerobic
4. **Isometric**



Correct Answer: 4
rationale:
Performing isometric exercises places too much load on the left ventricle of the heart for many cardiac patients

Question 106

when detect target heart rate in unstable angina we don`t make

1. valsalva maneuver
2. **isometric ex**
3. heavy arm ex
4. isotonic ex

Correct Answer: 2
rationale:
Isometric exercise is defined as a muscular contraction without movement (eg, handgrip) and imposes greater pressure than volume load on the left ventricle in relation to the body's ability to supply oxygen. Cardiac output is not increased as much as in isotonic exercise because increased resistance in active muscle groups limits blood flow. Resistance exercise combines both isometric and isotonic exercise (such as free weight lifting).
Performing the Valsalva maneuver causes an increase in intrathoracic pressure, leading to a reduction in preload to the heart

Question 107

Therapist measure target heart rate to determine

1. frequency of exercise
2. **intensity of exercise**
3. duration of exercise
4. type of exercise

Correct Answer: 2
rationale:
Training or target heart rate: is one of the methods to detect exercise intensity not type of exercise.

Question 108

Physical Therapist detect alarm signs for cardiac patient during training program through

1. **monitor H.R & B.P**
2. Blood oxygen level
3. O2 level

Correct Answer:
rationale:
The therapist should monitor Blood pressure and heart rate during the training session.

Question 109

when detect target heart rate in unstable angina we do

1. valsalva maneuver
2. isometric ex
3. **isotonic ex**
4. heavy arm ex

Correct Answer: 3
rationale: Training or target heart rate: is one of the methods to detect exercise intensity not type of exercise.
Mode of exercise: interval which is better for cardiac patients and continuous.
Types of exercise: aerobic & circuit weight training & respiratory training.

- isotonic exercise A form of exercise that consists of continuous and sustained movement of the arms and legs; isotonic exercises are beneficial to the cardio respiratory systems and include running and bicycling.

Question 110

Which of the following minutes of running are most commonly required in a cardiovascular fitness field test?

1. 10
2. **12**
3. 14
4. 16

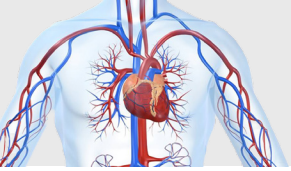
Correct Answer: 2
rationale:
Cardio fitness measures how well your body can perform a rhythmic, dynamic activity at a moderate to high intensity for extended periods of time.
There are :Treadmill tests ,The three-minute step test and The Rockport walk test .
The 12-minute run/walk test measures how far a person can cover (run, walk, jog) in 12 minutes. Ideally you want to take this test on a standard running track, or a location where you can measure distance run without having to stop.

Question 111

A therapist is ordered by a physician to treat a patient with congestive heart failure in an outpatient cardiac rehabilitation facility. Which of the following signs and symptoms should the therapist not expect?

1. Stenosis of the mitral valve
2. Orthopnea
3. **Decreased preload of the right heart**
4. Pulmonary edema

Correct Answer: 3
rationale:
preload is defined as the stretch of myocardium or end-diastolic volume of the ventricles and most frequently refers to the volume in a ventricle just before the start of systole.
-In heart failure, there is a compensatory increase in blood volume that serves to increase ventricular preload and thereby enhance stroke volume by the Frank-Starling mechanism.
-Preload increases with exercise (slightly).



Question 112

knee A 34 - year - old woman with rheumatoid arthritis has been on disease - modifying anti - rheumatic drugs ogs but has had little improvement ultrasound confirmed hypertrophic synovitis.She had undergone synovectomy. Which of the following is the most appropriate duration, in days, for post - operative immobilization?

1. **1-2**
2. 3-4.
3. 5-6
4. 7-8

Correct Answer: 1
rationale:
A synovectomy is a surgical procedure used to treat synovitis and some other conditions that affect the synovium, a thin membrane that lines the inside of synovial joints . In a synovectomy procedure, much of the synovium is removed.In a number of orthopedic injuries and conditions , it produces too much synovial fluid, which contains an enzyme that, in large quantities, 'eats away' at the articular cartilage on the joint surface."
Non surgical treatment of rheumatic conditions like inflammatory arthritis include:Oral drugs known as DMARDs (disease-modifying antirheumatic drugs) and steroid injection (in some cases).
Physical therapy will usually begin after one or two days.

Question 113

The therapist is working in an outpatient cardiac rehabilitation facility. A 50-year-old healthy man inquiries about the correct exercise parameters for increasing aerobic efficiency. Which of the following is the most correct information to convey to this individual?

1. Exercise at 50-85% of maximal volume of oxygen utilization (VO2)
2. Exercise with heart rate between 111 and 153 beats/minute.
3. Exercise at approximately 170 beats/minute.
4. **1 and 2 are correct**

Correct Answer: 4
rationale:
Choices 1 and 2 are correct exercise parameters for a healthy person.
Choice 2 has the patient exercising at 65-90% of his age-adjusted maximal heart rate. Choice 3 is the patient's age-adjusted maximal heart rate

Question 114

A physical therapist is treating a patient who is participating in cardiac rehabilitation. Because the patient complains of chest pain, the therapist attempts to assess heart sounds with a stethoscope. Which of the following is true about the first sound during auscultation of the heart?

1. the first sound during auscultation of the heart?
2. The first sound is of the closure of the aortic and pulmonic valves.
3. The first sound is of the closure of the mitral and tricuspid valves.
4. The first sound is of the beginning of ventricular systole.
5. **2 and 3**

Correct Answer: 5
rationale:
The first sound of heart (lub) corresponds with the closing of the AV valves (mitral and tricuspid valves). The second sound (dup) corresponds to closing of the semilunar valves (aortic and pulmonic valves). Therefore, the first sound is indicative of the onset of ventricular systole, and the second sound is indicative of the onset of ventricular diastole. The first sound is usually lower in pitch and longer than the second.

Question 115

Bursed lip tech used in

1. Asthma
2. **emphysema**
3. Segmental breathing
4. normal person

Correct Answer: 2
rationale:

- Emphysema due to its restrictive disease, in segmental breathing used for after block.
- In people with emphysema, the air sacs in the lungs (alveoli) are damaged. Over time, the inner walls of the air sacs weaken and rupture — creating larger air spaces instead of many small ones.
- Pursed lip breathing works by moving oxygen into your lungs and carbon dioxide out of your lungs. This technique helps to keep airways open longer so that you can remove the air that is trapped in your lungs by slowing down your breathing rate and relieving shortness of breath

Question 116

therapist patient with emphysema and has barrel chest which statement document it

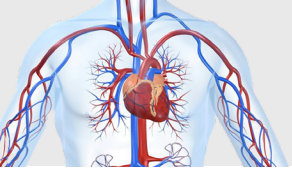
1. depressed ribs angel
2. increase transverse diameter
3. **increase ant-post diameter of chest**
4. decrease intercostal space

Correct Answer: 3
rationale:
Some people who have chronic obstructive pulmonary disease (COPD) such as emphysema develop a slight barrel chest in the later stages of the disease.
Barrel chest refers to an increase in the anterior posterior diameter of the chest wall resembling the shape of a barrel.

Question 117

Lung doesn't collapse because?

1. **dead space and compliance of lung**
2. inspiratory reserve volume
3. expiratory reserve volume



Correct Answer: 1
rationale:
There are factors prevent the lungs from collapsing: surfactant and the intrapleural pressure. Dead space represents the volume of ventilated air that does not participate in gas exchange.

Question 118

A therapist is evaluating a patient with chronic emphysema and mention in the file that the patient is showing a barrel-shape chest deformity. On which of the following, the therapist based this comment?

1. Depression of the ribs
2. **Widening of the intercostal spaces**
3. Narrowing of the intercostal spaces
4. Decreasing the anterior diameter of the chest

Correct Answer: 2
rationale:
Barrel chest occurs because the lungs are chronically overinflated with air. This makes breathing less efficient and aggravates shortness of breath.
The ribs may angle outward at the joints where they attach to the spine and become fixed in their most expanded position.avv

Question 119

therapist patient with emphysema and has barrel chest which statement document it

1. depressed ribs angel
2. increase transverse diameter
3. **increase ant-post diameter of chest**
4. decrease intercostal space

Correct Answer: 3
rationale:
Barrel chest : refers to an increase in the anterior posterior diameter of the chest wall resembling the shape of a barrel.

Question 120

Barrel chest deformity:

1. Flattened epigastric angle
2. decrease intercostal space
3. **Take position of inspiration**
4. Take position of expiration

Correct Answer: 3
rationale:
Barrel chest due to emphysema because the lungs are chronically overinflated (= take position of inspiration with air). There are two main causes of the barrel chest phenomena in emphysema:
1- Increased compliance of the lungs leads to the accumulation of air pockets inside the thoracic cavity.
2- Increased compliance of the lungs increases the intrathoracic pressure.This increase in pressure allows the chest wall to naturally expand outward

Question 121

patient with emphysema and has barrel chest which statement document it

1. depressed ribs angel
2. increase transverse diameter
3. decrease ant post diameter
4. **increase intercostal space**

Correct Answer: 3
rationale:
In a barrel-shaped chest, the anteroposterior diameter is equal to or greater than its lateral(transverse) diameter . The ribs become more horizontal and dorsal kyphosis is present in the majority of cases. Other findings are prominent sternum, elevated clavicles, shortened neck, and widened intercostal spaces. Barrel-shaped chest is usually seen in advanced emphysema. Aging can also produce barrel-shaped deformity of the chest without any lung diseased.

Question 122

All of the following conditions are obstructive lung disorder except:

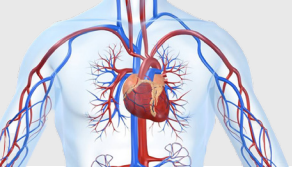
1. chronic bronchitis
2. **pneumonia**
3. emphysema
4. cystic fibrosis

Correct Answer: 2
rationale:
Obstructive lung diseases are are characterized by an obstruction,inflammation and narrrowing in the air passages, with slow and shallow exhalation:Chronic obstructive pulmonary disease (COPD),chronic bronchitis,brochiolitis,Asthma,Bronchhiectasis,cystic fibrosis.
Restrictive conditions are defined by difficulty filling the lungs with air during inhalation.
Intrinsic restrictive disorders are those that occur due to restriction in the lungs (often a “stiffening”) and include: pneumonia,Pneumoconioses,Acute respiratory distress syndrome (ARDS) and Sarcoidosis
Extrinsic restrictive disorders refer to those that originate outside of the lungs. These include impairment caused by:pleural effusion,pneumothorax and pleurisy.

Question 123

A 45-year-old woman with bronchiectasis is referred for chest physiotherapy. She has reduced breath sounds on the right lower chest with crepitus. Which of the following is the main reason to accurately identify the area involved, before the start of the session?

1. Better results
2. **Secretions removal**
3. Teaching technique
4. Prevention of disease spreading to other Areas



Correct Answer: 2
rationale:
Bronchiectasis is a long-term condition where the airways of the lungs become widened, leading to a build-up of excess mucus that can make the lungs more vulnerable to infection. The most common symptoms of bronchiectasis include: a persistent cough that usually brings up phlegm (sputum) shortness of breath

Question 124

Old pt comes with fever; productive cough ;and pulmonary congestion diagnosis is

1. **cystic fibrosis**
2. pulmonary odema
3. heart disease
4. bacterial & viral pneumoni

Correct Answer: 1
rationale:
Symptoms of cystic fibrosis include:lung infections that can cause fever,coughing with thick mucus.The thick and sticky mucus associated with cystic fibrosis clogs the tubes that carry air in and out of your lungs.
Note:people with bacterial pneumonia usually experience wet cough whereas viral pneumonia experice dry cough.

Question 125

In pulmonary edema, there is

1. Lower limb ischemia
2. **Lower limb edema and frothy secretions**
3. Thin, clear sputum

Correct Answer: 2
rationale:
In pulmonary edema, there is :
Chest pain and cough that produces frothy sputum that may have blood in it.Coughing up phlegm that looks pink or has blood in it.

Question 126

In acute bronchitis there is

1. Dry cough
2. Cyanosis
3. **Extended clear mucus secretions**
4. Edema in lower limbs

Correct Answer: 3
rationale:
Bronchitis symptoms are: Cough and Production of mucus

Question 127

complication of fracture rib is

1. **Surgical emphysema**
2. Bronchitis
3. Bronchiactasis

Correct Answer: 1
rationale:
Surgical emphysema is another term for subcutaneous emphysema. It occurs when air or gas enters the subcutaneous tissue, which is the deepest layer of the skin. People can develop surgical emphysema from injury to the thoracic cavity and infection

Question 128

bronchitis is:

1. dry cough
2. wheezing sound
3. peripheral cyanosis
4. **cough with expectoration**

Correct Answer: 4
rationale:
Symptoms of bronchitis (an infection of the main airways of the lungs (bronchi), causing them to become irritated and inflamed) :Cough,Production of mucus (Expectoration or sputum), which can be clear, white, yellowish-gray or green in color — rarely, it may be streaked with blood. Also,there is fatigue ,Shortness of breath ,slight fever and chills and Chest discomfort.

Question 129

in bronchitis there is?

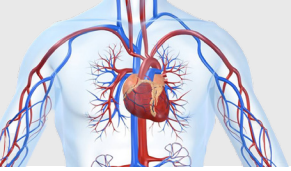
1. dry cough
2. cyanosis
3. **extended clear mucus secretions**
4. edema in lower limbs

Correct Answer: 3
rationale:
In bronchitis: Cough,Production of mucus (Expectoration or sputum), which can be clear, white, yellowish-gray or green in color — rarely, it may be streaked with blood.

Question 130

in early stage of bronchitis there is

1. dry cough
2. **bronchi spasm + expectoration**
3. hymopsis
4. peripheral cyanosis



Correct Answer: 2
rationale:
Symptoms of bronchitis: Cough,Production of mucus (Expectoration). Bronchospasm is a chief characteristic of asthma and bronchitis.

Question 131

A 24-year-old woman presented with frequent history of cough, dyspnea and wheezing, which was worse after exposure to dust.
Which of the following is indicated by this clinical feature?

1. Pneumonia
2. Bronchiectasis
3. **Bronchial asthma**
4. Chronic obstructive pulmonary disease

Correct Answer: 3
rationale:
Asthma is antigen antibody reaction aggravated by Asthma triggers as Airborne allergens(such as pollen, dust mites, or particles of cockroach waste) or smoke in air
Asthma signs and symptoms include :Coughing or wheezing attacks (Wheezing when exhaling, is a common sign of asthma in children),Shortness of breath ,Chest tightness or pain and Trouble sleeping (parasympathetic at night makes brochoconstriction). Bronchospasm is a chief characteristic of asthma and bronchitis.

Question 132

A 48-year-old patient is suffering from chest wheezing, dyspnoea, secretions, and coughing for many years. What does this clinical feature indicate?

1. Asthma
2. Pneumonia
3. Emphysema
4. **Chronic bronchitis**

Correct Answer: 4
rationale:
Chronic bronchitis is an obstructive lung disease characterized by Cough, Production of mucus (Expectoration or sputum, which can be clear, white, yellowish-gray or green in color). Note : The most important cause of chronic bronchitis is cigarette smoking. Air pollution and your work environment may also play a role. This condition causes a cough that's often called smoker's cough. It also causes you to cough up mucus, wheeze, and have chest discomfort.

Question 133

chronic bronchitis symptoms

1. nocturnal dysnea
2. frothy sputum
3. bronchospasm
4. **productive cough**

Correct Answer: 4
rationale:
For either cute or chronic bronchitis :symptoms include :
-Cough.
-Production of mucus (sputum), which can be clear, white, yellowish-gray or green in color — rarely, it may be streaked with blood.
-Fatigue.
-Shortness of breath.
-Slight fever and chills.
-Chest discomfort.

Question 134

Chronic bronchitis symptoms is:

1. nocturnal dyspnea
2. frothy sputum
3. **broncospasm and productive cough**

Correct Answer:
rationale:
Symptoms of bronchitis: Cough,Production of mucus (Expectoration). Bronchospasm is a chief characteristic of asthma and bronchitis.

Question 135

Which of the following is defined as Pulmonary ventilation?

1. Gas exchange between arterial blood and tissues
2. Gas exchange between alveolar air and venous blood .
3. **Gas exchange between atmospheric air and air present inside the lungs**
4. process by which 02 is used by the living cells for the oxidation of the organic substances .

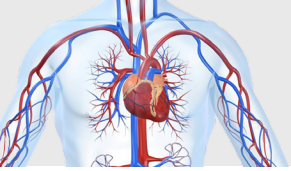
Correct Answer: 3
rationale:
Pulmonary ventilation is commonly referred to as breathing. It is the process of air flowing into the lungs during inspiration (inhalation) and out of the lungs during expiration (exhalation). Air flows because of pressure differences between the atmosphere and the gases inside the lungs.

Question 136

pulmonary ventilation means :

1. gas exchange between alveoli and veinous blood
2. **exchange between atmospheric air and lung air**
3. utilizing O2 by cells which essential for all vital process

Correct Answer: 2
rationale:
Pulmonary ventilation is commonly referred to as breathing. It is the process of air flowing into the lungs during inspiration (inhalation) and out of the lungs during expiration (exhalation). Air flows because of pressure differences between the atmosphere and the gases inside the lungs



Question 137

A 60-year-old man underwent cardiac surgery yesterday. His SpO2, is currently 84, and crackles are heard on breathing. Which of the following is the most appropriate mouth position to perform forced expiration?:

1. Any
2. **Open**
3. Close
4. Alternate

Correct Answer: 2
rationale:
A huff (also called the forced expiration technique [FET] when combined with breathing control) is a maneuver used to move secretions, mobilised by thoracic expansion exercises, downstream towards the mouth.
The technique is generally taught in sitting with open mouth .

Question 138

when evaluation inspiration capacity using spirometer :

1. **max inspiration after normal expiration**
2. max inspiration after max expiration
3. normal inspiration after normal expiration

Correct Answer: 1
rationale:
Inspiratory capacity is the volume of air that can be inspired following a normal, quiet expiration and is equal to
.tidal volume + inspiratory reserve volume

Question 139

pt 32 years old with sever spinal deformity and difficulty in respiration is due to :

1. ms weakness
2. usage of spinal orthosis
3. **decrease lung capacity**

Correct Answer: 3
rationale:
Difficulty in respiration/Dyspnea due to spinal deformity means decrease of lung capacity. spinal deformity as :In severe scoliosis, the rib cage may press against the lungs, making it more difficult to breathe.

Question 140

Pt has 32 years with spinal deformity & dyspnea due to :

1. Muscle weakness
2. Spinal orthosis
3. **Decrease lung volume**

Correct Answer: 3
rationale:
decrease lung volume is means dyspnea .

Question 141

Which of the following is a common reason for breathlessness occurring in scoliosis patients?

1. Anxiety
2. Spinal orthotic device
3. Weak respiratory muscles
4. **Reduced lung capacity due to spinal deformity**

Correct Answer: 4
rationale:
While most people with scoliosis have a mild form of the disorder, scoliosis may sometimes cause complications, including: Breathing problems. In severe scoliosis, the rib cage may press against the lungs, making it more difficult to breathe.

Question 142

Which of the following is most likely to develop during exercise?

1. Larger lung volume
2. **Larger diffusion capacities**
3. Tidal volume within normal
4. Maximum minute ventilation

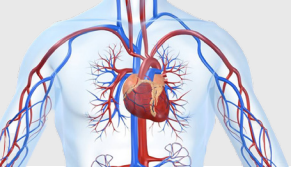
Correct Answer: 2
rationale:
DLCO ("Diffusing Capacity of the Lungs Test"):
Pulmonary diffusing capacity for oxygen increases progressively with increasing severity of exercise; it exceeds the at high levels of exercise by amounts greater than can be accounted for by the difference in diffusivity of the test gases.

Question 143

Which of the following is the most suitable for airway management?

1. Jaw lift
2. Chin lift
3. Jaw thrust
4. **Modified jaw thrust**

Correct Answer: 4
rationale:
Airway management is the required to maintain or restore an individual's ventilation to lungs from mouth and nose. The head-tilt/chin-lift and jaw-thrust are also basic airway management maneuvers. By preventing the tongue from obstructing the upper airway, these maneuvers are used to avoid airway obstruction in unconscious individuals. The head-tilt/chin-lift is not recommended in individuals for which cervical spine injury, or the injury of the upper spine in the neck, is a concern. On the other hand, the jaw-thrust maneuver is suitable when cervical spine injury is suspected. Obese patients need more head elevation than typical when we open the airway.



Question 144

Patient has lung cancer and make right lung pneumoectomy this leading to except:

1. decrease vital capacity
2. decrease residual volume
3. **increase tidal volume**
4. rt shift of trachea

Correct Answer: 3
rationale:
Pulmonary function and gas exchange deteriorate after pulmonary resection. The vital capacity, tidal volume, and functional capacity decrease after pulmonary resection because of loss of effective lung volume and, therefore, affect the setting of the ventilator.

Question 145

The amount of air remaining in the lung after quiet expiration:

1. **functional residual capacity**
2. total lung capacity
3. vital capacity
4. inspiratory capacity.

Correct Answer: 1
rationale:
Functional residual capacity is the volume of air remaining in the lungs after this normal, quiet expiration and is equal to (expiratory reserve volume + residual volume).

Question 146

Patient complain from leg pain after walking 200 feet's pain is removed by rest diagnosis is:

1. **peripheral vascular disease**
2. nerve lesion
3. calf muscle spasm.

Correct Answer: 1
rationale:
Peripheral artery disease (PAD) patient may experience intermittent claudication (The most characteristic symptom of PAD is claudication which is a pain in the lower extremity muscles brought on by walking and relieved with rest), or pain when you walk or climb stairs that goes away with rest.

Question 147

patient coming to you suffering from pain at calves due to walking 200 feets this patient has?

1. calf strain
2. **arterial insufficiency**
3. thrombosis
4. varicose vein

Correct Answer: 2
rationale:
Peripheral artery disease (PAD) is a condition where plaque builds up in the arteries that carry blood to areas like your legs, arms, and internal organs.Damage of the artries is due to High BP,High cholestrol,DM and smoking.If you have PAD, you may experience intermittent claudication (The most characteristic symptom of PAD is claudication which is a pain in the lower extremity muscles brought on by walking and relieved with rest)., or pain when you walk or climb stairs that goes away with rest. This is because your muscles aren't getting enough blood. This is due to blood vessels that have become narrowed or blocked. Other symptoms of PAD include: blue or pale skin ,weak pulse in legs or feet and slow wound healing.

Question 148

Which of the following is the most appropriate intervention to reduce discomfort associated with pregnancy-induced varicose veins?

1. Weight reduction
2. Stretching exercises
3. Muscle strengthening
4. **Dependent position reduction**

Correct Answer: 4
rationale:
Non-pharmacological interventions including include avoidance of prolonged standing and straining, external compression devices (e.g. bandages, support stockings, intermittent pneumatic compression devices) and leg elevation can provide relief from discomfort and may improve symptoms in some women

Question 149

Which of the following statements is NOT true about vertebrobasilar insufficiency (VBI)?

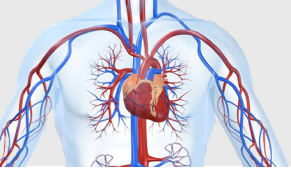
1. it refers to temporary symptoms due to decreased blood flow to brain.
2. symptoms include headache, dizziness, nausea and vertigo.
3. cervical manipulation is contraindication.
4. **cervical mobilization especially rotation, extension or longitudinal distraction, and traction considered as the effective treatment of VBI.**

Correct Answer: 4
rationale:
Vertebrobasilar insufficiency (VBI) is a contraindication to cervical manipulation and high–grade mobilisations, particularly rotations, extensions or longitudinal distractions and traction.

Question 150

A patient with Buerger's disease has been referred to a physical therapist to assist in reducing leg pain. Which of the following goals of treatment should be planned?

1. **Improving venous return**
2. Establishing collateral circulation
3. Increasing strength of the calf muscles
4. Reducing muscle spasm to reduce pain



Correct Answer: 1
rationale:
Buerger's disease (thromboangiitis obliterans) affects blood vessels in the body. In most cases, the first symptom is extreme pain of the lower arms and legs while at rest. Blood vessels swell, which can prevent blood flow, causing clots to form.This can lead to pain, color and temperature skin chenges,tissue damage,and even gangrene (the death or decay of body tissues). In some cases, amputation may be required.
There are treatment approaches for Buerger's disease, but are less effective than quitting smoking. Options include:
-Medications to dilate blood vessels, improve blood flow or dissolve blood clots
-Intermittent compression of the arms and legs to increase blood flow to your extremities

Question 151

Burger disease lead to leg pain-so your goal to

1. open collaterals
2. **increase venous return**
3. decrease leg spasm to decrease pain

Correct Answer: 2
rationale:
same as the previous

Question 152

Burger's disease associated with

1. **smoking**
2. elderly people
3. pregnancy women
4. athlete performing max effort

Correct Answer: 1
rationale:
The exact cause of Buerger's disease is unknown, however tobacco use is strongly linked to its development. Researchers believe that chemicals in tobacco may irritate the lining of the blood vessels, causing them to swell

Question 153

burger's disease is a disease of unknown cause, which is precipitated by:

1. emotional stress
2. hereditary factor
3. obesity
4. **cigarette smoking**

Correct Answer: 4
rationale:
The exact cause of Buerger's disease is unknown, however tobacco use is strongly linked to its development. Researchers believe that chemicals in tobacco may irritate the lining of the blood vessels, causing them to swell

Question 154

burger's disease patient complains of pain after walking in calf muscle in treatment your goal is to:

1. decrease ms spasm
2. strength hamstring
3. increase venous return
4. **open collaterals**

Correct Answer: 4
rationale:
Upon walking the patient with Buerger's Disease experiences Intermittent Claudication (Pain in calf upon walking)

Question 155

Burger's disease affect

1. veins
2. artery
3. nerve
4. vessels

Correct Answer: 4
rationale:
Buerger's disease (also known as thromboangiitis obliterans) affects blood vessels in the body, most commonly in the arms and legs. The legs are affected more often than the arms. It affects the vessels of the Arteries or some veins.

Question 156

In Raynaud's disease it is?

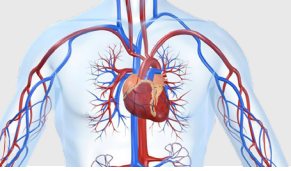
1. **vascular insufficiency disease**
2. intermittent claudication
3. thrombosis in lower limb
4. varicose veins

Correct Answer: 1
rationale:
Raynaud's disease, or Raynaud's phenomenon, is a form of vasculitis in which the arteries of the fingers or toes suddenly narrow in response to cold,A person with Raynaud's phenomenon can experience three phases of skin color changes: white (pallor), blue (cyanosis) and red (rubor).
Raynaud's disease refers to vasospasm — severe, temporary narrowing of the arteries that occurs in response to cold or stress in the absence of any other medical condition. Most cases are women

Question 157

Ice massaging is contraindicated with

1. inflammation
2. **Raynaud's disease**
3. Muscle spasm
4. Acute burn



Correct Answer: 2
rationale:
If you have Raynaud's Disease (i.e. your hands change colour from white to blue to red) do not use Ice cube massage.

Question 158

Arteriosclerosis is:

1. **A condition in which an artery wall thickens**
2. A condition in which a vein wall thickens
3. Hardening (and loss of elasticity) of medium or large arteries
4. Hardening (and loss of elasticity) of medium or large veins

Correct Answer: 1
rationale:

- Arteriosclerosis is a broader term for the condition in which the arteries narrow and harden, leading to poor circulation of blood throughout the body . Atherosclerosis is a specific type of arteriosclerosis. Atherosclerosis is thickening of the walls of the arteries.
- Arteriolosclerosis means hardening of the arterioles, which are small arteries. It affects primarily the inner and middle layers of the walls of arterioles.
- Atherosclerosis affects medium-sized and large arteries, and is the primary cause of heart disease and stroke. Atherosclerosis begins when damage occurs to the inner layer (intima) of your artery wall.

Question 159

Which of the following scientific terms is defined as chronic inflammation of the innermost layer of the arteries?

1. Aneurysm
2. Angina pectoris
3. **Atherosclerosis**
4. Deep arterial thrombosis

Correct Answer: 3
rationale:
Atherosclerosis begins when damage occurs to the inner layer of your artery wall. This layer is called the intima. The surface of your intima is lined with endothelial cells. This thin lining, called the endothelium, is the barrier between your blood and your artery wall.

Question 160

after pt do ex at gym. to check the tolerance of patient...we take heart rate from

1. femoral artery 10 sec *6
2. brachial artery 15 sec*4
3. **radial artery 30 sec * 2**
4. carotid artery 60 se

Correct Answer: 3
rationale:
radial artery 30 sec*2 , it`s the more accurate and easier

Question 161

The most common type of stroke is and its primary precipitating factor is

1. **Atherothrombotic, atherosclerosis**
2. Atherothrombotic, hypertension
3. Hemorrhage, atherosclerosis
4. Hemorrhage, hypertension

Correct Answer: 1
rationale:

- Atherothrombotic stroke is the most common. It occurs when a blood clot forms on an atherosclerotic plaque within a blood vessel in the brain and blocks blood flow
- Most ischemic strokes are caused by atherosclerosis, which is also called hardening of the arteries. Over time, this can completely block blood flow.
- Blood vessels damaged by high blood pressure can narrow, rupture or leak. High blood pressure can also cause blood clots to form in the arteries leading to the brain, blocking blood flow and potentially causing a stroke.

Question 162

The anatomical snuffbox contains a portion of this structure:

1. sdeep radial nerve
2. **radial artery**
3. basilic vein
4. lunate bone.

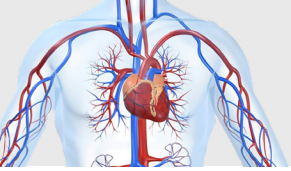
Correct Answer: 2
rationale:
The main contents of the anatomical snuffbox are the radial artery, a branch of the radial nerve, and the cephalic vein. .

Question 163

patient complains of cystic fibrosis in LT lingula

1. **put patient on RT sidelying**
2. put patient on LT sidelying
3. half lying
4. prone lying

Correct Answer: 1
rationale:
put the patient on the Rt side-laying to ease the secretions by this anti-gravity postion and because lingula in left lung.



Question 164

A patient with cystic fibrosis has a great amount of secretion at the lingula.
What is the ideal position that the therapist should put the patient in to drain this secretion?

1. Prone
2. Supine
3. Semi-lying
4. **Side-lying on the right side**

Correct Answer: 4
rationale:
put the patient on the Rt side-laying to ease the secretions by this anti-gravity postion and because lingula in left lung.

Question 165

Acute respiratory distress most important

1. pain
2. **decrease accumulation of fluid (postural drainage)**

Correct Answer: 1
rationale:

- Acute respiratory distress syndrome (ARDS) occurs when fluid builds up in the tiny, elastic air sacs (alveoli) in your lungs. The fluid keeps your lungs from filling with enough air, which means less oxygen reaches your bloodstream and the organs.
- Treatment for ARDS typically aims to increase blood oxygen levels, provide breathing support (Ventilation), and treat the underlying cause of the disease.
- Postural drainage is a technique used to mobilise large amounts of secretions in people with respiratory conditions. Postural drainage relies on the effects of gravity.

Question 166

the patient is placed in a supine position with the head flat, postural drainage is optimal for the

1. apical segments of the upper lobes
2. posterior segments of the upper lobes
3. **anterior segments of the upper lobes**
4. lingular segments of the left upper lobes

Correct Answer: 3
rationale:
Upper Lobe
ANTERIOR SEGMENTS - The patient should lie flat on his back with his arms relaxed to his side; the knees should be slightly flexed over a pillow.

Question 167

in thrombosis we do

1. apical segments of the upper lobes
2. posterior segments of the upper lobes
3. **anterior segments of the upper lobes**
4. lingular segments of the left upper lobes

Correct Answer: 3
rationale:
Upper Lobe
ANTERIOR SEGMENTS - The patient should lie flat on his back with his arms relaxed to his side; the knees should be slightly flexed over a pillow.

Question 168

) A 5-year-old child with cystic fibrosis presents for postural drainage and percussion of the middle lobe of the right lung. Which of the following is the most suitable position?

1. **Supine on a wedge with right shoulder elevated on pillow with head lower than pelvis**
2. Prone with right shoulder elevated on pillow And the head on the same
3. plane as pelvis
4. Supine on a wedge with left shoulder elevated on pillow with the head lower than pelvis
5. Prone with left shoulder elevated on pillow and the head on the same plane

Correct Answer: 1
rationale:
- Middle Lobe of the right lung
- LATERAL SEGMENT and MEDIAL SEGMENT The patient should lie on his back with his body quarter turned to the left maintained by a pillow under the right side from shoulder to hip and the arms relaxed by his side; the foot of the bed should be raised 35cm (14in) from the ground. The chest is tilted to an angle of 15°..

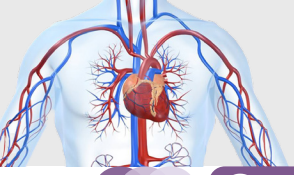
Question 169

A therapist is treating a patient with cystic fibrosis who has just walked 75 feet before experiencing significant breathing difficulties. In an effort to assist the patient in regaining her normal breathing rate, the therapist gives a set of instructions. Which of the following set of instructions is appropriate?

1. “Take a slow deep breath through pursed lips and exhale slowly through your nose only”
2. “Take small breaths through your nose only and exhale quickly through pursed lips”
3. **“Breath in through your nose and exhale slowly through pursed lips”**

Correct Answer: 3
rationale:
technique; “Breath in through your nose, and exhale slowly through pursed lips.” indication for
- =cystic fibrosis w/breathing difficulty
- =emphysema w/SOB

Pursed lip breathing helps control shortness of breath, and provides a quick and easy way to slow your pace of breathing, making each breath more effective.



Question 170

Cerebral embolism begin first from

1. **heart**
2. lung
3. vessels of lower limb

Correct Answer: 1
rationale:
Blood clots that lead to embolic stroke can form anywhere. They usually come from the heart or arteries of the upper chest and neck. After breaking free, the clot travels through the bloodstream to the brain. When it enters a blood vessel that's too small to allow it to pass, the clot becomes stuck in place.

Question 171

Pulmonary embolism begin first from

1. heart
2. lung
3. **vessels of lower limb**

Correct Answer: 3
rationale:
DVT is the most common cause of a pulmonary embolism. Other less frequent sources of pulmonary embolism are a fat embolus (often linked to the breaking of a large bone), amniotic fluid embolus, air bubbles, and a deep vein thrombosis in the upper body.

Question 172

in thrombosis we do

1. fat embolism
2. **pulmonary embolism**
3. arterial embolism
4. fracture embolism

Correct Answer: 2
rationale:
DVT is the most common cause of a pulmonary embolism.A pulmonary embolism occurs when a part of the clot breaks off and travels to the lungs, a potential life threat

Question 173

Basic Airway Management (Manual / Non equipment) all the following EXEPT:

1. head tilt
2. chin lift
3. jaw thrust
4. **laryngeal mask airway (LMA)**

Correct Answer: 4
rationale:
The head-tilt/chin-lift and jaw-thrust are basic non-invasive airway management maneuvers. Advanced airway management involves the use of specialized medical equipment The most commonly used device is the laryngeal mask airway (LMA) that is used as a temporary method to maintain an open airway during administration of anesthesia or as an immediate life-saving measure in a difficult airway situation

Question 174

Which statement incorrect about prevention of D.V.T:

1. use elastic stocking
2. abdominal binder
3. **put patient with head elevated**

Correct Answer: 3
rationale:
-Compression stockings are thought to reduce the risk of DVT by exerting graduated pressure on the leg, with the pressure being greatest at the ankle. This, when combined with muscular activity in the limb, is thought to displace blood from the superficial venous system to the deep venous system.
-Prophylaxis of deep vein thrombosis after major abdominal surgery. Abdominal binder is used helps to reduce postoperative pain, distress, and hemorrhage after surgeries .So, in theory, abdominal binders may actually decrease DVT risk because they may help increase mobility after surgery.

Question 175

Which of the following measures/procedures are not of clinical importance for D.V.T patients

1. Frequent turning
2. **Raising head above the level of the body**
3. Wearing compressive socks
4. Passive range of motion exercises

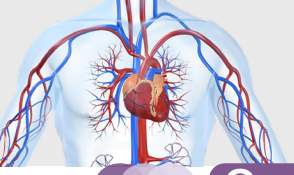
Correct Answer: 2
rationale:
Compression stockings are thought to reduce the risk of DVT by exerting graduated pressure on the leg, with the pressure being greatest at the ankle. This, when combined with muscular activity in the limb, is thought to displace blood from the superficial venous system to the deep venous system..

Question 176

PWhich of the following is NOT considered a strong risk factor for DVT?

1. **Bed rest lasting longer than 3 days.**
2. Hip/knee replacement.
3. MI within the last 3 months.
4. Major Trauma/lower limb fracture.

Correct Answer: 1
rationale:
Notes: Bed rest lasting longer than 3 days is NOT a strong risk factor, it is a WEAK risk factor.
Immboiltiy and prolonged bed rest is a risk factor of DVT.
One of the risks of hip replacement or knee replacement surgery is deep vein thrombosis (DVT).
MI was significantly associated with VTE (venous thromboembolism) but not after adjusting for VTE risk factors.
DVT of lower limb veins is one of the most common complications following surgeries for lower limb.



Question 177

patient needs to increase ROM you can use hold relax or contract relax technique .....what different between two techniques??

1. **isotonic contraction**
2. verbal comments
3. direction of movement
4. none of the above

Correct Answer: 1
rationale:
Isotonic contraction used in contract – relax followed by isometric contraction and we use Isometric contraction in hold relax technique

Question 178

A 40 years old man, was admitted with FRACTURE shaft femur following a road traffic accident. On 2nd day he became disoriented. He was found to be tachypnoeic, and had conjunctival petechiae. Most likely diagnosis is:

1. Pulmonary embolism
2. Sepsis syndrome
3. **Fat embolism**
4. Haemothorax

Correct Answer: 3
rationale:
It is a common phenomenon. It is most commonly seen in patients with multiple fractures and in fractures involving lower limb especially femur.
Fat originates from the site of trauma, particularly from the injured marrow of the fractures bone and the suggestion that the fat arises from the plasma.
Few of these patients develop clinical features similar to ARDS and are known as fat embolic syndrome.

Question 179

patient with fracture right ribs 7-10 perform secretion clearance techniques feels suddenly pain at chest , dyspnea, unable to breathe and trachea shifted to left side , the cause is

1. Pulmonary emboli
2. **Pneumothorax**
3. Copious mucus

Correct Answer: 2
rationale: Ipneumothorax. Because rib fracture cut sheath around lung
Pneumothorax - is an accumulation of air or gas in the pleural space (the space between visceral and parietal pleura of the chest cavity). It can happen secondary to trauma (traumatic pneumothorax). When mediastinal shifts accompany it, it is called a tension pneumothorax. Pneumothorax is the most frequently reported cause of tracheal deviation from pressure buildup.
Symptoms:
-Sharp, stabbing chest pain that worsens when trying to breath in.
-Shortness of breath.
-Bluish skin caused by a lack of oxygen.
-Fatigue.
-Rapid breathing and heartbeat.
-A dry, hacking cough.

Question 180

patient with a recent history of rib fractures suddenly becomes short of breath during secretion removal techniques. The patient looks panicked and complains of sharp pain in the left chest. A quick screen shows a deviated trachea to the right among other signs and symptoms. The MOST LIKELY explanation for the above is:

1. angina.
2. **Pneumothorax.**
3. pulmonary emboli.
4. mucous plugging of an airway

Correct Answer: 2
rationale:
pneumothorax. Because fracture cut sheath around lung

Question 181

If a chest tube gets dislodged during physical therapy treatment and the therapist fails to cover the defect, the patient could develop:

1. pulmonary embolism
2. pulmonary edema
3. **pneumothorax**
4. aspiration pneumonia

Correct Answer: 3
rationale:
Whatever its source, an air leak must be addressed and resolved before the chest tube is removed. A large, persistent leak with no evacuation outlet can lead to tension pneumothorax, in turn causing cardiac tamponade—a life-threatening emergency.

Question 182

Before starting the intervention of a bedridden patient, the physical therapist noticed some signs of pain, swelling and tenderness in the right Calf He suspected a deep vein thrombosis Which of the following immediate actions should be taken?

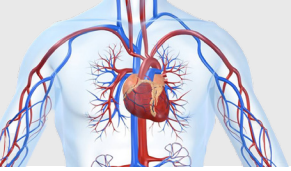
1. Just apply a gentle passive range of motion in the bed
2. **Inform the physician before intervening with the patient**
3. Try to walk the patient exercises with partial to non - weight bearing
4. Ask the patient to perform some active range of motion in the bed

Correct Answer: 2
rationale:
It's important to call your doctor right away or go to the emergency room if you have symptoms of a DVT. Don't wait to see if your symptoms go away. Get treatment right away to prevent serious complications

Question 183

patient with sever back pain, after examination you find disc herniation; which one of the following will be the least to prove the condition :

1. MRI
2. CT
3. **plain X-ray**
4. mylogra



Correct Answer: 3
rationale:
plain X-Ray is the less approval for disc herniation and the best is MRI

Question 184

A therapist working at a hospital calls another therapist in an outpatient facility to provide a brief history of a patient who is scheduled to leave the hospital and receive outpatient therapy. The acute care therapist states that the patient received an injury to the somatic sensory association cortex in one hemisphere. From this information only, the outpatient therapist knows that the patient will most likely

1. ignore someone talking to him or her on the involved side
2. **be unable to find one of his or her extremities.**
3. have no trouble putting on clothes.
4. be unable to understand speech

Correct Answer: 2
rationale:
The somatosensory cortex is a region of the brain which is responsible for receiving and processing sensory information from across the body, such as touch, temperature, and pain. Symptoms that may be experienced as a result of damage:Numbness,Inability to Localize Sensations ,Inability to Judge Weight and Pressure and Phantom Limb Pain is experinced with amputess and there is changes in somatosensory cortex.

Question 185

pseudo claudication is due to compression of:

1. Femoral artery
2. Femoral nerve.
3. **Cauda Equina.**
4. Popliteal artery

Correct Answer: 3
rationale:
Pseudoclaudication" Syndrome Produced by Compression of the Cauda Equina.
Pain in the lower extremities associated with occlusive arterial disease is a common occurrence. On relatively rare occasions, pain in the legs, precipitated by walking, is caused by compression of the cauda equina by hypertrophic ridging or by a herniated lumbar disk.

Question 186

A patient's peripheral skin color progresses from blue to white to red. This would be most characteristic of:

1. chronic venous insufficiency
2. acute venous insufficiency
3. acute arterial insufficiency
4. **vasomotor disorders**

Correct Answer: 4
rationale:
,Vasomotor disorders in the limbs include Raynaud's phenomenon, acrocyanosis, livedo reticularis, vasomotor paralysis, and erythromelalgia.
Raynaud's phenomenon is classically described with a triphasic color change of the digits with initial white or pallor (ischemic phase), then blue or cyanosis (deoxygenation phase), followed by red or erythema (reperfusion phase).
NOTE: Chronic venous insufficiency skin color is normal or cyanotic in dependency
Acute venous insufficiency skin colour is like chronic
Acute arterial insufficiency skin colour is Pale , shiny , dry

Question 187

Congestive Heart Failure is referred to the Physical Therapy Department for gait training. Which assessment test is the most reliable to calculate exercise capacity

1. Chair-stand
2. **6-minute walk**
3. Time up and go
4. Physical performance

Correct Answer: 2
rationale:
Exercise capacity can be assessed by the following tests :Maximal test (exercise stress test; EST),Cardiopulmonary exercise test (CPET),Submaximal test (six-minute walk test; 6MWT),-Submaximal treadmill test and Incremental shuttle walk test (ISWT).
The six-minute walk test (6MWT is a self-paced walking test well suited to a hospital or community setting.
For patients with more severe classes of heart disease, 6MWT results correlate well with VO2 peak, and only one 6MWT is recommended.
In addition to measuring exercise capacity, more specific functional testing can be used. Some examples such as Timed up and go test. More recently, combined measures such as the Short Physical Performance Battery (standing balance, walking speed, and chair stand) and the Functional Difficulty Questionnaire have been validated in cardiac surgery patients.

Question 188

A physical therapist is reviewing the chart of a 49-year-old woman who recently suffered a myocardial infarction. The lab reports reveal that this particular patient has a hematocrit of 41%. How should the therapist proceed?

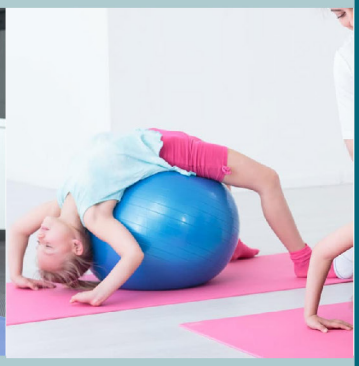
1. **Continue with the evaluation and treatment.**
2. Do not perform this evaluation due to the hematocrit level
3. Inform nursing of this lab report
4. Check nursing notes to determine the last time the patient received a beta-blocker

Question 189

How should you open the airway of an unconscious casualty?:

1. **head tilt and chin lift.**
2. jaw thrust.
3. head tilt and jaw thrust.
4. lift the chin





Chapter

5.A Paediatric

Question 1

A 9-month old boy presents to the Physical Therapy Department with Erb's palsy. Which of the following is the most likely cause?

1- Sacral plexus injury
2- Cranial nerve injury
3- Lumbar plexus injury
4- Brachial plexus injury

Question 2

A 6-month-old baby boy is refered by a neurosurgeon for management. He has an adducted shoulder with a flaccid and intermally rotated arm and a pronated forearm. The boy's mother has gestational diabetes and difficult labor.
Which of the following is the most likely brachial plexus injury?

1- Erb's palsy
2- Klumpke's palsy
3- Horner's syndrome
4- Brachial plexus syndrome

Question 3

A 6-month-old patient with Erb's palsy, leading to loss of elbow flexion power and supination of the forearm, is being treated in the Physical Therapy Pediatric Clinic. Which nerve root is affected?

1- C3-C4
2- C5-C6
3- C7-C8
4- T1-T2

Question 4

An injury of one of the following roots of brachial plexus cause erb's palsy:

1- C5 toT1
2- T1&T2
3- C8&T1
4- C5 & C6

Question 5

An infant with Erb's palsy presents with the involved upper extremity in which of the following positions?

1- Hand supinated and wrist extended
2- Hand supinated and wrist flexed
3- Hand pronated and wrist extended
4- Hand pronated and wrist flexed

Question 6

Rolling begin at:

1- 4 moths
2- 6 monthes

Question 7

In erb's palsy the most affected muscle:
1- elbow flexors and supinators.
2- shoulder flexor and abductors
3- medial rotators
4- a and b

Question 8

In Erb's paralysis, the deformity is called:
1- claw hand
2- ape hand
3- wrist drop
4- policeman's tip position

Question 9

Policeman's tip or Porter's tip position is seen in:
1- upper Plexus Paralysis
2- erb-Duchenne paralysis
3- klumpke's paralysis
4- both a and b

Question 10

In Klumpke's paralysis, the deformity is called:
1- claw hand
2- ape hand
3- wrist drop
4- policeman's tip position

Question 11

A 5-year-old patient with Erb's palsy has been referred to the Rehabilitation Centre. During the evaluation, the therapist recognizes it as Erb- Duchenne or upper arm paralysis. Which of the following is the most likely reason?
1- Cord
2- Lower trunk
3- Upper trunk
4- Brachial plexus division

Question 12

infant with Erb's palsy he can full recover with good biceps & deltoid at:
1- 3 months
2- 6 months
3- 9 months
4- 12 months

Question 13

A 5-year-old child is diagnosed as having Erb's palsy complicated with Erb engram problems. Which of the following kinds of physical problems does this child have, during the physical evaluation?

1- Spasms of the external shoulder rotators
2- Bad pattern of motion, scapular limitation
3- Weakness of the internal shoulder rotators
4- Weakness of the external shoulder rotators

Correct Answer: 4
rationale:
erbs Ingram due to recovery of middle fiber of deltiod more than anterior make this then weakness in internal rotators.

Question 14

A 3 - year - old child with poliomyelitis treated in the Physical Therap Pediatric Clinic Which of the following viruses is mostly responsible for poliomyelitis?

1- Herpes
2- Lansing
3- Varicella
4- Cytomegalovirus

Question 15

To test righting - equilibrium in child

1- Sudden push him from side
2- Turn his head
3- Tilt the child to one side

Question 16

A 35-year-old patient with left side hemiplegia with moderate spasticity was referred to the Outpatient Clinic of physiotherapy for rehabilitation; the therapist treats the upper limb. Which of the following is the suggested approach?

1- Bobath
2- Effleurage
3- Brunnstrom
4- Proprioceptive neuromuscular facilitation

Question 17

Which of the following management techniques is associated with inhibition of abnormal movement patterns?

1- Bobath
2- Margaret
3- Temple-Fay
4- Proprioception neuromuscular facilitation

Question 18

To assist head control in infant what position first?

1- head control
2- righting reaction
3- crawling
4- rolling

Question 19

First motor action could be seen in pediatric is:

1- head control
2- righting reaction
3- crawling
4- rolling

Question 20

Baby with poor head control your first choose of treatment will be OR to assist head control in infant what position first?

1- supine
2- side lying
3- prone
4- sitting

Question 21

What is the average head circumference of a normal term newborn?

1- 33 - 38cm
2- 33 - 35 cm
3- 32 - 35 cm
4- 32 - 38 cm

Question 22

How many weeks gestation (fetal development) is considered normal for a full term baby?

1- 30 weeks
2- 32 weeks
3- 36 weeks
4- 37 weeks

Question 23

Baby from prone can support chest at any month?

1- 1 month
2- 3 month
3- 5 month
4- 7 month

Question 24

At what weeks of development Pincer grasp is seen?

1- 4 weeks
2- 6 weeks
3- 16 weeks
4- 36 weeks

Correct Answer: 1
rationale:
Pincer grasp is developing at 8-9 months then be fine at 10-11 months.

Question 25

Head control is possible in an infant by which month?

1- 2 months
2- 3 months
3- 6 months

Question 26

What is the mean age for a baby in months to start developing the ability to walk upstairs?

1- 12
2- 18
3- 24
4- 30

Question 27

Child can go up and down stairs what age?

1- 12-18 months
2- 18-24 months
3- 24-30 months

Question 28

In normal milestone baby can jump by both legs in swing at

1- 12 months
2- 24 months
3- 18 months
4- 30 months

Question 29

Child can walk tip toe what age?

1- 3 years
2- 3.5 years
3- 4 years

Question 30

At any age the child can jumb in one leg holding on

1- 18 month
2- 24 month
3- 30 month

Correct Answer: 3
rationale:
24 months can jump on 2 legs , from 2- 3 years can hopping using one leg.

Question 31

What is the mean age for a baby to develop the ability to hop on 1 foot while holding on, according to normal developmental milestones?

1- 36 months
2- 18 months
3- 24 months
4- 30 months

Question 32

Baby recognize his mother at what age in month?

1- One months
2- Two months
3- Three months
4- Four months

Question 33

Which of the following is the most likely age, in months, for an infant to differentiate between family and strangers?

1- 5 months
2- 6 months
3- 7 months
4- 8 months

Question 34

All the following is true about spasticity except:

1- Usually occurs cerebral palsy infant
2- Occur in sci
3- associated with lower motor neuron legion
4- can cause hip dislocation

Correct Answer: 3
rationale:
spasticity is seen to be a positive feature of upper motor neuron syndrome . Hypotonia usually occurs with LMN disease, whereas UMN disease may be characterized by hypertonia or spasticity.

Question 35

An infant who can recognize tone of voice, sit independently and pulls to stand through kneeling is most likely to be:

1- 2-3 Months of age
2- 4-5 Months of age
3- 6-7 Months of age
4- 8-9 Months of age

Question 36

Child can kick the ball in:

1- 12 month
2- 24 month
3- 30 month

Question 37

Which of the following ages, in months, do most children display a hand preference?

1- 06
2- 12
3- 18
4- 24

Question 38

At any age the child can walk independably:

1- 12 month
2- 18 month
3- 24 month
4- 30 month

Question 39

In which age by month child can walk and balance?

1- 12 month
2- 18 month
3- 24 month
4- 14 month

Question 40

At any age baby hand preference:

1- 6 months
2- 12 months
3- 18 months
4- 24 months

Question 41

Deficiency of this make deformity in newborn?

1- Vitamin A
2- Vitamin D
3- Vitamin B
4- Vitamin C

Question 42

Fear from prone position in pediatric known as:

1- vomiting
2- irritation
3- cot deaths or crib deaths

Correct Answer: 3
rationale:
3 to 4 months You'll notice a definite improvement in head control by this time. Your baby will be able to raise his head to 45 degrees while on his tummy and keep it up steadily. By 6 months, your baby will be able to hold his head steady and erect, and he'll flex it forward when he's pulled into a sitting position. He may be ready for a jogging stroller at this point.

Question 43

You have evaluated a nine month old who can't assume or maintain quadruped position without assistance his parent insist that child has already begin to walk with assistive you suspect that what parents say is:

1- protective ext. downward
2- spontaneous stepping
3- positive supporting reaction
4- negative supporting reaction

Question 44

What the most primitive reflex of human body?

1- Symmetrical tonic neck reflex
2- sucking reflex
3- asymmetrical tonic reflex
4- tonic lybrinthine

Question 45

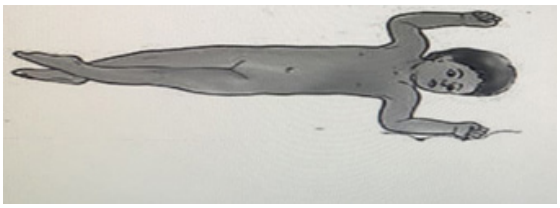
baby supine lying abducting both shoulders with 90 elbow flexion lower limbs extended and adducted at hips and extended at knees and planter flexed ankles , what is this reflex

1- Tonic labrynthinen reflex
2- Moro reflex
3- STNR
4- ATNR

Question 46

When placed supine, the baby adopts a position with abducted shoulders, flexed elbows, adducted extended hips, and extended knees and ankles (see image). What is the term for this reflex?

1- Moro
2- Tonic labyrinth
3- Symmetric tonic neck
4- Asymmetric tonic neck



Question 47

An infant is being examined by a physical therapist. The therapist is resisting movement of the right upper extremity and notices involuntary movement of the left upper extremity which of the following is displayed by the infant?

1- Landau reaction
2- Startle reflex
3- Moro reflex
4- Associated reaction

Question 48

Hand during the evaluation of an infant, the therapist observes that with passive flexion of the head the infant actively flexes the arms and actively extends the legs. Which of the following reflexes is being observed?

1- Protective extension
2- Optical righting
3- Symmetrical tonic neck
4- Labyrinthine head righting

Correct Answer: 3
rationale:
This is due to its correlation with the TLR (Tonic Labyrinthine Reflex) - as an infant begins to integrate the TLR, the STNR emerges to help progress to the next phase of development, crawling. The symmetric tonic neck reflex is commonly referred to as the “crawling reflex” because it allows your baby to make the transition from laying down to getting up on their hands and knees.
- When your baby has reached the STNR stage, you may notice that when their neck and arms straighten, the legs will bend. When the neck and arms bend, the legs will straighten. These movements are important for crawling and taking their first steps.

Question 49

What is the cause of hip dysplasia?

1- mother malnutrition
2- idiopathic

Correct Answer: 2
rationale:
The exact cause or causes of hip dysplasia are not known.

Question 50

A therapist is evaluating an infant with the mother present. The therapist suddenly seems to temporarily lose grip of the infant, causing him to be startled and begin to cry. The infant's mother is noticeable upset but is reassured that startling the infant was part of the assessment. Which of the following may the therapist have been assessing?

1- Landau response
2- Symmetric tonic neck reflex
3- Labyrinthine head righting.
4- Moro reflex

Correct Answer: 4
rationale:
moro reflex is arms abducted and extended follows by flexion. The Moro reflex may be a survival instinct to help the infant cling to its mother. If the infant lost its balance, it helps the infant to embrace its mother and regain its hold on the mother's body.

Question 51

In treating a 5 years with spastic diaplegia as part of the program relaxation might be accomplish:

1- Use of rhythmic stabilization
2- slow rocking
3- inhibition of parasympathetic
4- facilitation of sympathetic

Question 52

Hypotoncity means:

1- decrease tone in muscle
2- Increase tone
3- incoordination

Correct Answer: 1
rationale:
Having less than normal tone or tension.

Question 53

What is not true about spina bifida?

1- inadequate of folic acid intake
2- diagnostic of fatal by ultrasound
3- the injury related of degree of effect of sternum
4- major affected of thoracic region

Correct Answer: 4
rationale:
it's more affected in lumbar spine in lower back bone.

Question 54

Following the reduction of developmental dysplasia of the hip in an infant, the physical therapist should instruct the parents to maintain that hip in a stable position of:

1- flexion and adduction
2- flexion and abduction
3- extension and adduction
4- extension and abduction

Correct Answer: 2
rationale:
Hip dysplasia has been associated with abduction contracture. Positioning the thighs together stretches of the hip capsule on the side opposite the abduction contracture. resulted in resolution of dysplasia .
- flexion with abduction together put the head of the femur in the right position so it helps in resolution of dysplasia.

Question 55

A 4-year-old child with spasticity treated in the Physical Therapy Pediatric Clinic, during spasticity assessment the therapist found there is a considerable increase in muscle tone, passive movement difficulty. What is the grade of patient spasticity according to the Modified Ashworth scale?

1- 1
2- 2
3- 3
4- 4

Correct Answer: 3
rationale:
According to modified ashworth scale grade 3 is Considerable increase in muscle tone, passive movement difficult.
- While grade 2 is a marked increase in muscle tone but limb easily flexed.

Question 56

Which of the following scales is used in the assessment of pediatric spasticity?

1- Borg
2- Berg balance
3- Motor dependency
4- Modified Ashworth

Correct Answer: 4
rationale:
The modified Ashworth scale is the most universally accepted clinical tool used to measure the increase of muscle tone.

Question 57

The physical therapist is reading the physician's interpretation of an x-ray that was taken of the left humerus of a 7-year-old patient. The physician notes in the report the presence of an incomplete fracture on the convex side of the humerus. Which type of fracture is the physician describing?

1- Comminuted
2- Avulsion
3- Greenstick
4- Segmental

Correct Answer: 3
rationale:
Greenstick fractures result from the bending of a bone. Any force that bends a long bone, such as an arm or leg bone, without fully breaking it can cause a greenstick fracture. Instead of snapping into two pieces, the bone cracks on one side ,is most common in young children.

Question 58

A therapist is scheduled to treat a patient with cerebral palsy who has been classified as a spastic quadriplegic. What type of orthopedic deformity should the therapist expect to see in a patient's feet?

1- Talipes equinovalgus
2- Talipes equinovarus
3- Clubfeet
4- B & C are correct

Correct Answer: 4
rationale:
the most common injuries of foot in quadriplegic patient is club foot as quadriplegia in lower limb is planter flexion and inversion in the foot.

Question 59

A 9-year-old patient with cerebral palsy was referred to physical therapy for assessment and treatment. The referral note describes the patient as having severe torticollis to the left side. Which of the following is the most likely cause of this torticollis?

1- Left sternocleidomastoid muscle weakness
2- Left sternocleidomastoid muscle shortening
3- Right sternocleidomastoid muscle weakness
4- Right sternocleidomastoid muscle shortening

Correct Answer: 2
rationale:
torticollis is referred to side bending to the same side of the muscle And rotation to the opposite side of the muscle as the action of sternocleidomastoid.

Question 60

Which of the following is inappropriate for a physical therapist to include in the treatment plan of an infant with a gestational age of 27 weeks and Down's syndrome?

1- Bottle feeding
2- Encourage sidelying position
3- Tactile stimulation with the entire hand rather than the fingertips of the examiner
4- Prone positioning

Correct Answer: 1
rationale:
babies usually use a bottle feeding in the age of 6 weeks.

Question 61

Which of the following sources of stimulation is least effective in obtaining functional goals when treating an infant with decreased muscular tone?

1- Vestibular
2- Weight-bearing
3- Cutaneous (scratch , quick stretch ,painful stimulation)
4- Vibratory

Correct Answer: 3
rationale:
vestibular input affect muscle tone via vestibulospinal tract, controlling golgi tendon organ and muscle spindle activation.
Vibration which is repetitive touch, transmit with proprioception to the cerebral cortex (Area 3,1,2) to improve body awareness and postural control.

Question 62

Which of the following is the most important goal in treating pediatric patients with postural reaction deficits?

1- Age-appropriate responses
2- Automatic responses
3- Conscious responses
4- Lower extremity control before upper extremity control

Correct Answer: 2
rationale:
- Automatic responses is related to the position which is responsible for: Protective , equilibrium , Righting ; all of them is called a postural reaction.
- Age appropriate responses : response related to the age and disappear after growing than that Age like : ASTR . STR , LTR.
- Conscious Responses: is the response related to conscious and active movement.

Question 63

Which of the following statements is true in comparing infants with Down's syndrome to infants with no known abnormalities?

1- Motor milestones are reached at the same time with both groups.
2- Postural reactions are developed in the same time frame with both groups.
3- Postural reactions and motor milestones are developed slower in patients who have Down's syndrome, but with the same association as with normal infants.
4- Postural reactions and motor milestones are not developed with the same association with patients who have Down's syndrome as with normal infants.

Correct Answer: 3
rationale:
patient with down syndrome develop the same milestone as normal but in later time because of their hypotonia.

Question 64

A physical therapist is evaluating a 5-day old infant with cerebral palsy. The infant has an abnormal amount of extensor tone. Which of the following is correct positioning advice for the family and nursing staff?

1- Keep the infant in supine position.
2- Keep the infant in prone position
3- Keep the infant in sideling position.
4- B and C are correct

Correct Answer: 4
rationale:
we can't put the child in supine position while he has an extensor tone cause of tonic labyrinthine supine will induce more tone. instead of that put it in prone to enhance tonic labyrinthine prone to make flexion and side lying used either .

Question 65

Which of the following statements best describes lower extremity positioning in standing during the first 2 years of life of a child with no dysfunction?

1- Femoral anteversion, femoral external rotation, foot pronation.
2- Femoral anteversion, femoral internal rotation, foot supination.
3- Femoral retroversion, femoral external rotation, foot pronation.
4- Femoral retroversion, femoral internal rotation, foot supination.

Correct Answer: 3
rationale:
At Early childhood, the child walks with hip abduction, external rotation and foot pronation to give him wide base of support at the beginning to learn standing and walking.

Question 66

A child presents to physical therapy with a diagnosis of right Sever's disease. What joint should be the focus of the therapist's evaluation?

1- Right knee joint
2- Right hip joint
3- Right wrist joint
4- Right ankle joint

Correct Answer: 4
rationale:
Right ankle joint Sever's disease is a common cause of heel pain in growing kids, especially those who are physically active.

Question 67

A patient comes to the therapist because she has noted a pronounced tuft of hair on the center of her spinal column in the lumbar area. The therapist notes no loss in motor or sensory function. This patient most likely has what form of spina bifida?

1- Meningocele
2- Meningomyelocele
3- Spina bifida occulta
4- Syringomyelocele

Correct Answer: 3
rationale:
spina bifida occulta, pina bifida occulta, or hidden spina bifida, is a mild form of spina bifida caused by a gap forming between the vertebrae in your spinal cord. They may have a dimple, patch of hair, or a red mark at the base of the spine.

Question 68

A physical therapist in the rehabilitation unit is ordered to evaluate and treat a 3-year-old girl with cerebral palsy. The patient's supportive family is present during the evaluation. When should the physical therapist explain the treatment plan and possible functional outcomes to the family?

1- During the evaluation.
2- After the evaluation.
3- After the first full treatment session.
4- After the first rehabilitation team conference meeting.

Correct Answer: 2
rationale:
A comprehensive evaluation of the child with CP is the cornerstone of treatment planning, Once the decision to treat has been made, multiple considerations come into play in determining the choice and timing of treatment.

Question 69

When reviewing a patient's chart, the therapist determines that the patient has a condition in which the cauda equina is in a fluid-filled sac protruding from the back. What form of spina bifida does the patient most likely have?

1- Meningocele
2- Meningomyelocele
3- Spina bifida occulta
4- None of the above

Correct Answer: 2
rationale:
myelomeningocele is a severe type of spina bifida in which the membranes and the spinal nerves protrude at birth, forming a sac on the baby's back.

Question 70

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2- Meningomyelocele
3- Spina bifida occulta
4- None of the above

Correct Answer: 2
rationale:
myelomeningocele is a severe type of spina bifida in which the membranes and the spinal nerves protrude at birth, forming a sac on the baby's back.

Question 71

A physical therapist is beginning an evaluation of a 5-year-old boy. The mother indicates that she pulled the child from a seated position by grasping the wrists. The child then experienced immediate pain at the right elbow. The physician's orders are for right elbow range of motion and strengthening. Which of the following is the most likely diagnosis?

1- Radial head fracture
2- Nursemaid's elbow
3- Erb's palsy
4- Ulnar coronoid process fracture

Correct Answer: 2
rationale:
Nursemaid's elbow, babysitter's elbow, or pulled elbow is a dislocation of the elbow joint Caused by a sudden pull on the extended pronated forearm such as by an adult tugging on an uncooperative child or by swinging the child by the arms during play. The technical term for the injury is radial head subluxation.

Question 72

Spina bifida is characterized by:

1- incomplete closing pia mater
2- hyperextension of knee joint
3- lateral curvature of the spine

Correct Answer: 1
rationale:
spina bifida is incomplete closing of the spine and the membranes around the spinal cord during early development of pregnancy .

Question 73

which is not correct of spina bifida?

1- motor paralysis according to level of lesion
2- commonly occur thoracic spine
3- due to decrease folic acid
4- detected by sonography

Correct Answer: 2
rationale:
pina bifida can occur at any level of the spine, but it's most common in the lower back.

Question 74

Congenital hip dysplasia is caused by:

1- trauma at birth
2- malnutrition
3- lack of vitamins

Correct Answer: 1
rationale:
is widely accepted that hip dysplasia develops around the time of birth because the hip socket is shallower at birth than at any time before or after birth.

Question 75

Spastic diplegia means:

1- primarily affecting the legs but arms are less involved.
2- only legs are involved.
3- only arms are involved.

Correct Answer: 1
rationale:
Spastic diplegia is a form of cerebral palsy marked by stiff limbs, legs are usually more affected than the arms.

Question 76

A therapist is ordered to fabricate a splint for a 2-month-old infant with a congenital hip dislocation. In what position should the hip be placed while in the splint?

1- Flexion and adduction
2- Extension and adduction
3- Extension and abduction
4- Flexion and abduction

Correct Answer: 4
rationale:
Hip dysplasia has been associated with abduction contracture. Positioning the thighs together stretches of the hip capsule on the side opposite the abduction contracture. resulted in resolution of dysplasia flexion with abduction together put the head of the femur in the right position so it helps in resolution of dysplasia.

Question 77

Following the reduction of developmental dysplasia of the hip in an infant, the physical therapist should instruct the parents to maintain that hip in a stable position of:

1- flexion and adduction
2- Flexion and abduction
3- Extension and adduction
4- Extension and abduction

Correct Answer: 2
rationale:
Hip dysplasia has been associated with abduction contracture. Positioning the thighs together stretches of the hip capsule on the side opposite the abduction contracture. resulted in resolution of dysplasia.
flexion with abduction together put the head of the femur in the right position so it helps in resolution of dysplasia.

Question 78

A type of cerebral palsy where there is a damage to the extrapyramidal system:

1- spastic diplegia
2- hemiplegic cerebral palsy
3- ataxic cerebral palsy
4- athetoid cerebral palsy

Correct Answer: 4
rationale:
Extrapyramidal Cerebral Palsy indicates the injury is outside the tract in areas such as the basal ganglia, thalamus, and cerebellum athetoid cerebral palsy the affection is in the cerebellum and basal ganglia .

Question 79

Which of the following is used to describe the term "spastic diplegia"?

1- Paralysis of all limbs
2- Paralysis of 1 limb
3- Paralysis of 2 limbs
4- Paralysis of 3 limbs

Correct Answer: 3
rationale:
Spastic diplegia is a form of cerebral palsy marked by stiff limbs, legs are usually more affected than the arms.

Question 80

A child with spastic diplegia is now independent in using a walker. Additional goals desired by the physical therapist include increasing the child's energy-efficiency and velocity while ambulating. In this case, the selection of walker that is most likely to help improve these factors is:

1- a standard anterior walker with no wheels.
2- an anterior rollator walker with two wheels.
3- a posterior rollator walker with two wheels.
4- a posterior rollator walker with four wheels.

Correct Answer: 4
rationale:
The benefits of the posterior rollator walker with four wheels:

1. Facilitate more upright posture.
2. Decrease hip and trunk flexion during the stance phase.
3. Decrease double stance time.
4. Increase walking velocity.
5. Increase stability and interaction with the other children.
6. Decrease energy expenditure than anterior walker.

Question 81

A 6 year-old child with myelomeningocele at the L5 level and minimal complications should be expected to ambulate with:

1- a reciprocating gait orthosis with a Rollator walker.
2- a shoe insert to maintain proper foot alignment and no ambulatory aid.
3- a knee-ankle-foot orthosis with crutches.
4- an ankle foot orthosis with crutches or a Rollator walker.

Correct Answer: 4
rationale:
Start with rollator walker and progress to crutches. An independent gait is possible for some, usually with gluteus medius lurch and lumber lordosis.

Question 82

A 3-month-old infant has a heart condition known as tetralogy of Fallot. This condition presents with which of the following signs?

1- Atrial septal defect, pulmonary valve stenosis, aorta abnormally located to the right, and right ventricular hypertrophy.
2- Atrial septal defect, pulmonary valve stenosis, aorta abnormally located to the left, and right ventricular hypertrophy.
3- Ventricular septal defect, pulmonary valve stenosis, aorta abnormally located to the right, and right ventricular hypertrophy.
4- Ventricular septal defect, pulmonary valve stenosis, aorta abnormally located to the right, and left ventricular hypertrophy.

Correct Answer: 3
rationale:
ventricular septal defect , pulmonary valve stenosis , Aorta abnormally located to the right and right ventricular hypertrophy.

Question 83

Just after birth, you observe a full-term infant in the neonatal intensive care unit. In the supine position, the shoulders are abducted and externally rotated, elbows and fingers are flexed, hips are abducted and externally rotated and knees are flexed. This posturing would seem to indicate that the:

1- upper extremity tone is abnormal.
2-l ower extremity tone is abnormal.
3- tone is abnormal in both upper and lower extremities.
4- tone is normal in both upper and lower extremities.

Question 84

The therapist is evaluating a patient with a diagnosis of cerebral palsy. The therapist notes that all of the extremities and the trunk are involved. Further assessment also reveals that the lower extremities are more involved than the upper extremities and that the right side is more involved than the left. This patient most likely has which classification of cerebral palsy?

1- Spastic hemiplegia.
2- Spastic triplegia.
3- Spastic quadriplegia.
4- Spastic diplegia.

Correct Answer: 4
rationale:
spastic diaplegia is usually affection of lower extremities more than upper extremities and trunk.

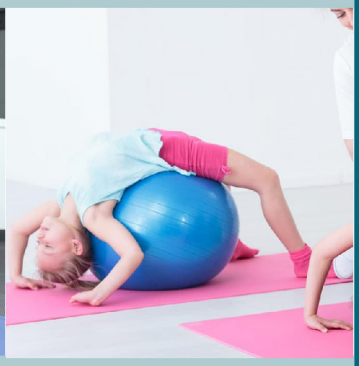
Question 85

Child came to u with Erb's palsy c5, c6 diagnosed by erb engram where is the affection:

1- weakness of external rotators
2- Spasticity of external rotators
3- abnormal pattern of movement + weakness of ant deltoid + biceps spasm
4- Weakness elbow extensors

Correct Answer: 3
rationale:
see Q.81





Chapter

5.B Endocrine

Question 1

Which of the following is widely considered the most accurate body composition assessment?

1- Hydrostatic weighting
2- Electrical impedance
3- Anthropometric measurement
4- None of the above

Correct Answer: 1
rationale:
Hydrostatic weighting is one of the most accurate ways in water while you -body fat Orting the test • you will be submerged on a scale. An administrator will calculate your body fat percentage by comparing your land weight with your underwater weight.

Question 2

A patient reports to therapy stating that his "sugar is too high" for exercise. What is the minimal blood glucose level that is considered too high for a diabetic patient to begin exercise?

1- 300 mg/dl
2- 400 mg/dl
3- 300 g/dl
4- 400 g/dl

Correct Answer: 1
rationale:
the minimum level that considered high to perform an Exercise.

Question 3

To decrease the risk of hypoglycemia in a patient with type I insulin-dependent diabetes, which of the following is inappropriate?

1- Eat or drink a snack high in carbohydrates 30 minutes before exercises.
2- Exercise muscles that have not had an insulin injection recently.
3- A carbohydrate snack for each 30-45 minutes of exercise.
4- Exercise at the peak time of insulin effect.

Correct Answer: 4
rationale:
the peak is the time during which insulin is at its "peals" or maximum effectiveness at lowering blood glucose and it increases your insulin sensation.

Question 4

In controlled diabetes, physiotherapist orders patient to do exercise to?

1- increase blood glucose uptake by ms
2- NO effect
3- increase insulin metabolism

Correct Answer: 1
rationale:
physical exercise induces rapid increase in the rate of glucose uptake in the Contracting skeletal is The enhanced membrane glucose transport Capacity is caused by recruitment of glucose transporters (Glut 4) to the sarcolemma) and T - tubules Hunger.

Question 5

Why is it recommended for a patient with controlled diabetes to practice exercise training and walking?

1- Decrease insulin secretion
2- Increase patient tolerance
3- Increase metabolism of insulin
4- Increase glucose uptake by the muscles

Correct Answer: 4
rationale:
the exercise improves blood glucose control in type 2 diabetes reduces Cardiovascular risks factors Contributes to weight loss and improves well being.

Question 6

A diabetic patient is exercising vigorously in an outpatient clinic. The patient informs the therapist that he or she received insulin immediately before the exercise session. If the patient goes into a hypoglycemic coma, which of the following is not a likely sign?

1- Pallor
2- Shallow respiration
3- Bounding pulse
4- Dry skin

Correct Answer: 4
rationale:
hypogly Comic Coma Sings are pallor, shakiness, shallow respiration , Bounding pulse , sweeting hunger and fatigue.

Question 7

Disease in children and early childhood due to malfunction of calcium phosphorus metabolism:

1- rickets
2- osteomyelitis

Correct Answer: 1
rationale:
Rickets is a condition that affects bone development in children it causes bone pain, poor growth and soft, weak bones that can lead to bone deformities. Adults Con Expectance a similar condition, which is known as osteomalacia or soft bone.

Question 8

A patient has been diagnosed with systemic lupus erythematosus. Which of the following is not a sign of this autoimmune disease?

1- Increased photosensitivity
2- Oral ulcers
3- Butterfly rash.
4- Increased number of white blood cells

Correct Answer: 4
rationale:
white blood cells can be low due to lupus, immunosuppressive therapy. Or the presence of a virus, High WBCS Counts may signal infection but also occur when individuals are corticosteroids such as prednisone.

Question 9

After arriving at the home of a home health patient, the primary nurse informs the therapist that she has activated emergency medical services. The nurse found the patient in what appears to be a diabetic coma. Which of the following is most likely not one of the patient's signs?

1- Skin flush
2- Rapid pulse
3- High blood pressure

Correct Answer: 3
rationale:
Flushed skin occurs when the blood vessels just below the skin widen and fill with more blood and this what happen in diabetic coma and deep, rapid breathing but high blood pressure no if it is low blood pressure it will be true.

Question 10

Which of the following durations, in months, is the time frame for the presence of an increased level of hormones post-partum?

1- 03-05
2- 06-09
3- 10-12
4- 13-15

Correct Answer: 2
rationale:
There is Consensus that the past partum period begging upon birth of the infant. The end is less well defined, but is considered the six to eight weeks after birth because the effects & pregnancy on may systems have largely returned to the pregnancy state bay this time.

Question 11

A 50-year-old patient presents with hot flushes irritability, joint pain and lack of sleep. Which of the following is the most appropriate management?

1- Vitamins
2- Hysterectomy
3- Hormone replacement therapy
4- Oestrogen-progesterone preparation

Correct Answer: 3
rationale:
As hormone replacement therapy is medication that contains female hormones, you take the medication to replace the estrogen that your body stops making during manopause . Hormone replacement therapy is most often used to treat Common manoposud symptoms including het plashes and vaginal discomfort and joint pain.

Question 12

Rickets is due to deficiency of vitamin:

1- C
2- B
3- K
4- D

Correct Answer: 4
rationale:
Vitamin D helps your child's body absorb calcium and phosphorus from food. Not enough vitamin D makes it difficult. To maintain proper calcium and phosphorus levels in bone which can cause rickets.

Question 13

Which of the following is the minimum separation, in cm, of the rectus abdominis muscles in pregnancy that is considered clinically significant?

1- 1 Cm
2- 2 Cm
3- 3 Cm
4- 4 Cm

Correct Answer: 2
rationale:
rectus muscles are normally fused at the midline with no more than 1 to 2 cm separating them (1). Rectus diastasis (RD) (diastasis recti, divarication of the rectus muscles) is an acquired condition in which the rectus muscles are separated by an abnormal distance along their length, but with no fascial defect.

Question 14

Which of the following is the maximum inter rectus abdominis distance in cases of diastasis recti?

1- 1 Cm
2- 2 Cm
3- 3 Cm
4- 4 Cm

Correct Answer: 2
rationale:
Diastasas recti occurs due to increasing intra-abdominal pressure in which the forces applied to the linea alba cause it to stretch resulting in a widening of the interrectus distance. Most studies have agreed that the minimum interrectus distance to designate a diastasis is 2 cm. 6 Diastasis recti is most common following pregnancy; however, obesity and prior abdominal operations can also be the cause.

Question 15

Visceral diseases of the abdomen and pelvis are likely to refer pain where?

1- shoulders
2- back
3- arms
4- legs

Correct Answer: 2
rationale:
It may be a problem with chest and belly organs such as infection in the kidneys may cause pain in your back or sides like kidney stones abdominal aortic aneurysms, gallstone and prostate problems.

Question 16

Bone pain at night is a red flag symptom of what disease:

1- cáncer
2- diabetes
3- osteoporosis
4- Paget's disease

Correct Answer: 1
rationale:
16 osteosarcoma, the most common type of bone cancer, often starts in the long bones (the legs or the arms) but it can occur in any bone. Bone cancers are broken down into separate types based on the type of cell where the cancer began.

Question 17

During exercise pregnancy percentage of blood flow affect fetus growth:

1- 20
2- 40
3- 50
4- 60

Correct Answer: 3
rationale:
Changes in blood flow during pregnancy are likely to play a role in development of venous insufficiency and thromboembolic events. Venous insufficiency and varicose disease were observed in 43% and 72.7% of pregnant women, respectively. Additionally, 50% of pregnant women complained of lower limb edema.

Question 18

What is not correct about bone?

1- contain 90% of inorganic tissues
2- compose of 2 types sponge, compact
3- bone is hard
4- has blood supply , nerve endings

Correct Answer: 1
rationale:
Bone is consists of 40 % inorganic Component (hydrory a pollate), 25 % water and 35 % organic component (Proteins), 90 % organic component collagen and 10% non-collagens proteins.

Question 19

Obesity is a risk factor for:

1- Osteoporosis
2- Osteoarthritis
3- Both
4- Neither

Correct Answer: 2
rationale:
As being overweight increase the load placed on Joint such as the line which increases stress and could the possibly hoston the breakdown of cartilage .

Question 20

Postnatal diastasis recto? The most effective to diagnosis?

1- MRI
2- CT
3- US

Correct Answer: 3
rationale:
There has been a long-standing controversy about diastasis recti diagnostic criteria and its relation to pelvic floor dysfunction (PFD). This study aimed to establish ultrasound diagnostic criteria for diastasis recti and investigate the correlation between it and PFD in early postpartum females.
NOTE:
if there is ‘’fingers’’, we choose it.

Question 21

Postnatal diastasis recto? The most effective to diagnosis?
1- MRI
2- CT
3- US
4- Fingers

Correct Answer: 4
rationale:
Place your fingers on your belly button, pointing towards your pelvis, and press down. Lift your head up about an inch while keeping your shoulders on the ground. If you have diastasis recti, you will feel a gap between the muscles that is an inch wide (~ 2 fingers) or greater.

Question 22

Which of the following is the most common adverse effect of a prolapsed uterus?
1- Hernia
2- Dyspnea
3- Vaginal cancer
4- Abnormal pregnancy

Correct Answer: 4
rationale:
Your uterus (or womb) is normally held in place inside your pelvis with various muscles, tissue, and ligaments. Because of pregnancy, childbirth or difficult labor and delivery, in some women these muscles weaken. Also, as a woman ages and with a natural loss of the hormone estrogen, their uterus can drop into the vaginal canal, causing the condition known as a prolapsed uterus.

Question 23

Ehag for Bee Which of the following musculoskeletal problems is common during the third trimester?
1- Heel spur
2- Back pain
3- Hip arthritis
4- knee ligament injury

Correct Answer: 2
rationale:
The third trimester is a time of great anticipation. In a few short weeks, your little one will finally be here.
Pain can occur in seemingly every part of your body during this time. From your back to your hips to your stomach, there are many places that may be sore and uncomfortable.

Question 24

Which of the following is the most likely cause of pregnancy gait including increased external hip rotation?

1- Pain
2- Cosmesis
3- Better cadence
4- Improved support

Correct Answer: 1
rationale:
This external rotation of the femur (wide knees) helps to increase the pelvic diameter more side to side, which can help the baby enter, or engage, into the pelvis.and reduce the pressure on the uterus which may lead to reduce pain on the back and give most cog.

Question 25

Which of the following is the most likely cause of posterior pelvic pain associated with prolonged sitting in pregnant women?

1- Muscular
2- Radicular
3- Sacroiliac
4- Disc prolapse

Correct Answer: 1
rationale:
(PPP) pain felt at or near the sacroiliac joints of your pelvis as a result of sacroiliac joint dysfunction. These are joints located at the 2 dimples of the lower back. The pain often feels deep within your lower back and can occur on one or both sides of your back. In some cases, pain radiates down to the buttock and the back of the thigh.

Question 26

Which of the following is the most likely cause of back pain in pregnant women?

1- Stress
2- Weight gain
3- Muscle weakness
4- Ligamentous laxity

Correct Answer: 4
rationale:
It is estimated that relaxin increases by 10 fold during pregnancy, causing ligamentous laxity. This increased laxity in the ligaments causes instability in the joint of the pelvis, and misalignment of the spine, resulting in pain and discomfort. Sacroiliac joint pain (SIJP) is the most common pain during pregnancy.

Question 27

Which of the following fibers are responsible for lifting the pelvic floor?
1- Circular
2- Anterior
3- Posterior
4- Anteroposterior

Correct Answer: 4
rationale:
The 2 major muscular supporting structures: the upper, with the pelvic diaphragm, and the lower, with the perineal membrane (urogenital diaphragm) anteriorly and anal sphincter posteriorly.

Question 28

First line of treatment for fistula:
1- Surgery
2- Hygine
3- Physiotherapisy
4- Medications

Correct Answer: 1
rationale:
Surgery is almost always necessary to cure an anal fistula. The surgery is performed by a colon and rectal surgeon. The goal of the surgery is a balance between getting rid of the fistula while protecting the anal sphincter muscles.

Question 29

Contraindication of exercise in pregnancy?
1- gestational diabetes
2- pain
3- musculoskeletal discomfort

Correct Answer: 1
rationale:
Exercise has been proven to be beneficial in improving pregnancy outcomes in women with gestational diabetes mellitus (GDM).

Question 30

A 23-year-old woman presented with incontinence. She developed the condition 3 months ago after childbirth. She experienced leakage when laughing or minor straining. Her urine routine examination and culture were unremarkable. Which of the following is the most appropriate exercise management?

1- Perineal flexibility
2- Adductor flexibility
3- Pelvic floor training
4- Abdominal exercises

Correct Answer: 3
rationale:
People with pelvic floor disorders may experience: Urinary problems, such as urinary incontinence, an urgent need to urinate, painful urination or incomplete emptying of their bladder. Constipation, straining or pain during bowel movemen the main pelvic floor muscles.

Question 31

In developing the plan of care for a 28-year-old pregnant woman which of the following muscles should be the focus of the strengthening exercises to maintain a strong pelvic floor?

1- Piriformis, obturator internus, and pubococcygeus
2- Obturator internus, pubococcygeus and coccygeus
3- Rectus abdominis, iliococcygeus, and piriformis
4- Iliococcygeus, pubococcygeus, and coccygeus

Correct Answer: 4
rationale:
PeLevator ani: The levator ani composes the bulk of your pelvic floor muscles and consists of three separate muscle components: pubococcygeus, puborectalis and iliococcygeus.
Coccygeus: The coccygeus is the smaller muscle component in your pelvic.

Question 32

Which of the following is the most common cause of orthostatic hypotension during pregnancy?

1- Anemia
2- Reduced blood flow
3- Reduced heart rate
4- Inferior vena cava compression

Correct Answer: 4
rationale:
Pregnant patients in the supine position have compression of the inferior vena cava and aorta by the gravid uterus, which leads to decreased venous return and thus hypoperfusion.

Question 33

Contraindication to exercise in pregnancy:

1- Diabetes in pregnancy
2- Pain
3- Musculoskeletal discomfort

Correct Answer: 1
rationale:
There are some potential risks to the fetus resulting from maternal exercise including hypoxia, hyperthermia, and abnormal heart rate changes.

Question 34

Postnatal women complain of some urine leakage during cough we do:

1- Biofeedback
2- Strengthening
3- Interferential

Correct Answer: 1
rationale:
It is a painless process that uses special sensors and a computer monitor to display information about muscle activity. This information or "feedback" is used to gain sensitivity, and with practice, control over pelvic floor muscle function.

Question 35

What is important regarding women health?

1- Stroke
2- hair loss
3- heart disease

Correct Answer: 3
rationale:
Heart disease, the No. 1 killer of women in the United States, causes 1 in 3 deaths, taking a woman's life nearly every minute, the American Heart Association (AHA) reports. Though heart disease also strikes men, women are more likely to die from the disease and experience unique causes, symptoms, and outcome.

Question 36

Pregnant woman has a gait pattern in hip external rotation as:

1- comfortable for her
2- Increase cadence

Correct Answer: 1
rationale:
Compensated for change your body mechanics.

Question 37

Which of the following is the used period, in hours, to record diurnal signs and symptoms pattern?

1- 12
2- 24
3- 36
4- 48

Correct Answer: 2
rationale:
diurnal cycle (or diel cycle) is any pattern that recurs every 24 hours as a result of one full rotation of the planet Earth around its axis.

Question 38

Physical therapist is speaking to a group of pregnant women about maintaining fitness level during pregnancy.
Which of the following statements contain incorrect information?

1- Perform regular exercise routines at least three times per week.
2- Perform at least fifteen minutes per day of abdominal exercises in supine position, during the second and third trimesters.
3- Increase caloric intake by 300 per day.
4- Exercise decreases constipation during pregnancy.

Correct Answer: 2
rationale:
During pregnancy, your blood volume increases by 30 to 50 percent to nourish your growing baby, your heart pumps more blood each minute and your heart rate increases.

Question 39

A physical therapist who is pregnant has been studying the use of transcutaneous electrical nerve stimulation during labor and birth to decrease pain perception. Which of the following is the most effective technique in this situation?

1- Place the electrodes over the upper abdominals during the first stages of labor and over the lower abdominals during the later stages.
2- Place the electrodes over the paraspinals at the L5 level and S1 level throughout labor and delivery.
3- Place the electrodes in a V pattern above the pubic region during labor and delivery.
4- Place the electrodes over the paraspinals at the L1 and S1 level initially during labor, and over the pubic region during the later stages.

Correct Answer: 4
rationale:
by stimulating the body to produce more of its own natural painkillers, called (endorphins). And It also reduces the number of pain signals sent to the brain by the spinal cord.

Question 40

A 30-year-old woman who had a full-term infant 4 weeks ago presents to physical therapy with diastasis recti. The separation was measured by the physician and found to be 3 cm. Which of the following exercises is most appropriate to minimize the separation?

1- Sit-ups while using the upper extremities to bring the rectus abdominis to midline
2- Bridges while using the upper extremities to bring the rectus abdominis to midline
3- Dynamic lumbar stabilization exercises in quadruped position.
4- Gentle head lifts in supine position while using the upper extremities to bring the rectus abdominis to midline.

Correct Answer: 4
rationale:
If you have diastasis recti, you should avoid any exercise that significantly increases your intraabdominal pressure and/or places a great deal of tension on the abdominal muscles and linea alba and bring rectus abdominis to middle.

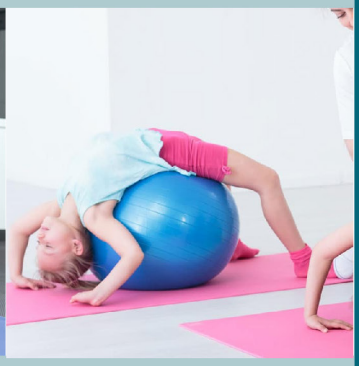
Question 41

Which of the following is least likely in a woman in the eighth month of pregnancy?

1- Center of gravity anteriorly displaced.
2- Heart rate decreased with rest and increased with activity compared to heart rate prior to the pregnancy.
3- Edema in bilateral lower extremities.
4- Blood pressure increased by 5% compared with blood pressure before pregnancy.

Correct Answer: 2
rationale:
Because Exercises where you are lying on your back (especially late in pregnancy this exercises should be avoided in the second and third trimester.





Chapter

5.C Burn

Question 1

Twenty-four hours following a major thermal burn, a patient experiences syncope while the physical therapist works on positioning as part of edema management. The syncope is most likely a result of

1- orthostatic hypotension
2- extreme pain
3- intravascular hypovolemia
4- massive infection

Question 2

Burn affecting epidermis and superficial part of dermis:

1- superficial partial thickness
2- Deep partial thickness
3- Full thickness

Question 3

A patient is referred to physical therapy services for care of a burn wound on the left foot. The majority of the wound is anesthetic. There is significant eschar formation over the dorsum of the involved foot, and moderate subcutaneous tissue damage is present. What is the most likely classification of this burn?

1- Electrical
2- Superficial partial thickness
3- Deep partial thickness
4- Full thickness

Question 4

which is incorrect about burn in shoulder and pectoral region:

1- put airplane splint
2- put in ext and abduction
3- put in flex add int rot

Question 5

Burn at shoulder & arm avoid

1- Airplane splint
2- Abduction
3- Adduction & internal rotation

Question 6

Patient has third degree burn in the anterior aspect of the neck the contraindication in treatment of this case is:

1- put the neck in hyper extension
2- put the neck in flexion to avoid deformity
3- put the neck in side bending.

Question 7

A 35 years old patient with third degree burns affecting the shoulder and axilla, is referred to rehabilitation. Which one of the following treatment approaches is contraindicated for this patient?

1- always mobilize the shoulder into flexion, abduction and external rotation.
2- positioning in adduction and internal rotation assist in healing.
3- airplane splint
4- position with shoulder flexion and abduction

Question 8

To prevent contracture in a newly admitted 8 year-old with anterior neck burns, it would be best to position the neck in:

1- hyperflexion
2- slight flexion
3- neutral
4- extension

Question 9

A 45 year old male presents to the burn unit with partial thickness burns over the entire right arm, left arm, front of head, and front of chest. Approximately. What percentage of his body is burned?

1-.31.5%
2-.36%
3-.40.5%
4-.45%

Question 10

A therapist is asked to estimate the percentage of a patient's body that has been burned. The patient is a 32- year-old man of normal size. Burns are located along the entire anterior surface of the face. The patient also burned the entire anterior portion of the right upper extremity in an attempt to guard himself from flames. Using the rule of nines, what percentage of the patient's body is burned?

1- 9%
2- 18%
c - 4.5%
4- 27%

Question 11

A physical therapist is treating a patient with significant burns over the limbs and upper trunk. Which of the following statements is false about some of the changes initially experienced after the burn?

1- This patient initially experienced an increase in the number of white blood cells.
2- This patient initially experienced an increase in the number of red blood cells
3- This patient initially experienced an increase in the number of free fatty acids
4- This patient initially experienced a decrease in fibrinogen.

Question 12

What is the major concern of the physical therapist treating a patient with an acute deep partial thickness burn covering 27% of the total body? The patient was admitted to the intensive care burn unit 2 days ago.

1- Range of motion
2- Fluid retention
3- Helping the family copes with the injured patient.
4- Home modification on discharge.

Question 13

A therapist is treating an acute full-thickness burn on the entire right lower extremity of a 27-year-old man. What movements need to be stressed with splinting, positioning, and exercise to avoid contractures?

1- Hip flexion, knee extension, and ankle dorsiflexion
2- Hip extension, knee flexion, and ankle plantarflexion.
3- Hip extension, knee extension, and ankle dorsiflexion
4- Hip flexion, knee extension, and ankle plantarflexion

Question 14

Burn in the body's response to thermal insult from external agent such as:

1- heat and cold
2- chemicals
3- electricity and radiation
4- all of the above

Question 15

Which of the following is the best and first treatment for a wound with black eschar over 90% of the wound bed?

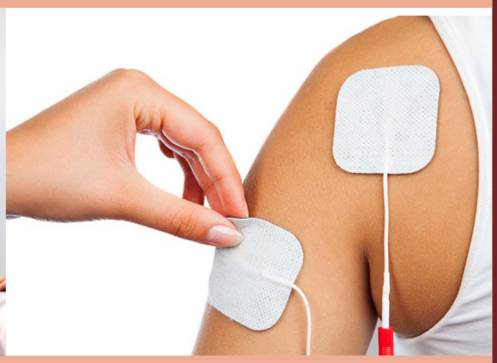
1- Lidocaine
2- Dexamethasone
3- Silvadene
4- Elase

Question 16

Which of the following is the next response of the leukocytes after emigration into the blood stream in the acute inflammatory stage?

1- Margination
2- Pavementing
3- Adhesion
4- Engulfment





Chapter

5.A Therapeutic Exercises

Question 1

Patient walking on slippery surface we teach him to do:

1- COG keeping close to the body

2- Wide base of support

3- Walking fast

Correct Answer: 1

rationale:

The closer the body's center of gravity is to its base of support, the more stable it is.

Question 2

the convex-concave role of the knee joint in open chain movement

1-Tibia roll and glide on the femur in the same direction.

2-Tibia roll and glide on the femur in a different direction.

3-Femur roll and glide on the tibia in the same direction.

4-Femur rolls and glides on the tibia in a different direction.

Correct Answer: 1

rationale:

choose it in closed chain

According to the convex /concave rule, the tibia is concave, and the femur is convex during the movement when the convex surface moves the gliding occurs opposite when the concave surface moves the gliding movement occurs in the same direction. Also, the open chain means that the distal part isn't touching the floor and that is why it is named opened.

Question 3

Which of the following is associated with the property to assume a greater length than the pre-stretch state?

1- Elasticity

2- Plasticity

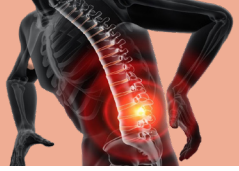
3- Flexibility

4- Extensibility

Correct Answer: 3

rationale:

Flexibility is the ability of a joint or series of joints to move through an unrestricted, pain-free range of motion. Stretching exercises are typically performed to increase flexibility and to allow an improved joint range of motion. Decreased flexibility may also lead to abnormal stress on structures and tissues distant from the initial site of inflexibility, elasticity is the ability of a deformed material body to return to its original shape and size when the forces causing the deformation are removed, and plasticity is s the ability of the tissue to undergo permanent deformation to substitute another defect. Tissue extensibility relates to tissue flexibility, which is related to the strength and power that your muscles can generate.



Question 4

All true of the following is true about passive movement except:

1- moving parts should be grasped gently.
2- unmoving parts should be supported.
3- sequence proximal to distal as it is promoting venous return.

Correct Answer: 3
rationale:
The passive movement is done from distal to proximal to maintain Venus return and lymph flow.

Question 5

which statement is incorrect about the passive movement to

1- passive movement segment must hold gently.
2- proximal part must be supported.
3- passive movement from proximal to distal to increase lymph and venous return.

Correct Answer: 3
rationale:
The passive movement the therapist does it from distal to proximal to improve venous and lymphatic flow and it is the same direction of fascia fibers.

Question 6

Which of the following is most commonly associated with a systematic exercise plan planned performance of bodily movements, postures, and physical activities?

1- Flexibility
2- Endurance
3- Therapeutic
4- Coordination

Correct Answer: 3
rationale:
Therapeutic exercises are movements and physical activities designed to restore function and flexibility, improve strength and decrease pain. If you are receiving physical therapy for an injury or a chronic condition, your physical therapist may incorporate therapeutic exercises in your treatment plan. Specific exercises like flexibility, endurance, and coordination were used.

Question 7

Accessory movement is not:

1- necessary to completion of joints ROM.
2- Passively occur.
3- under voluntary control.

Correct Answer: 3
rationale:
Accessory Movements Definition - Movements within the joint and surrounding tissues that are necessary for normal range of motion that cannot be performed actively by the patient - ARTHROKINEMAT- so the movement is under voluntary control on the other hand the physiological movements were under voluntary muscle control.

Question 8

Type of contraction in which muscle lengthening to provide tension
1- eccentric cont
2- concentric cont
3- isotonic cont

Correct Answer: 1
rationale:
- which MS lengthening to provide tension.
- Concentric contraction which MS shortening to provide tension.
- Isotonic contraction include both concentric and eccentric contraction.

Question 9

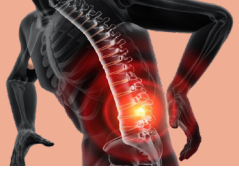
Joint play ROM which occurs not named with:
1- necessary to completion of joints ROM.
2- Passively occur.
3- under voluntary control.

Correct Answer: 3
rationale:
joint play: small movements within a synovial joint that are independent of voluntary muscle contraction. These movements measure not more than 1/8 inch in any plane and follow the contour of the opposing joint surfaces. By nature, joint play is an involuntary movement that is inherent to the musculoskeletal system and cannot be introduced by voluntary muscles. A joint play is another name for the accessory movement.

Question 10

During underwater exercises which will be difficult
1- moving the limb horizontally.
2- moving the limb toward the bottom (against the bouncy force of water).
3- moving the limb to the surface.

Correct Answer: 2
rationale:
Buoyancy is an upward force exerted by a fluid in a direction opposite to a downward force such as gravity and thus it counters (partially or fully) the weight of an object immersed in the fluid. Moving the limb to the surface was used in active assisted training while moving horizontally was used to work in gravity eliminated position.



Question 11

A 20-year-old athlete man sustained an injury to his quadriceps during a jump, Which of the following is the most appropriate preventative exercise program?

1- Aerobic
2- Static stretching
3- Ballistic stretching
4- Dynamic stretching

Correct Answer: 1
rationale:
Aerobic exercises reduce the risk of many conditions. It can reduce pain and improve function in people with arthritis.
Aerobic exercise keeps your muscles strong, which can help you maintain mobility as you get older. Exercise can also lower the risk of falls and injuries from falls in older adults. And it can improve your quality of life. Ballistic exercise is a high-impact exercise that increases the relevance of injury. Stretching is not preventive exercise even static stretching is possible for such a scenario, but it is not optimal.

Question 12

Which of the following types of stretching is associated with repeated movements and momentum to gain desired range of motion?

1- Static
2- Ballistic
3- Dynamic

Correct Answer: 2
rationale:
Ballistic stretching is a form of stretching that uses body momentum from bouncing to force the body's joints to move with a larger ROM than normal In ballistic stretching, the movement is done using a bouncing motion to force the joint to close more. Static stretching is the safest type of stretching and include applying stretch of constant intensity and velocity.

Question 13

If muscles co contract in equal force. Co-contraction of the biceps brachii and the triceps will result in what type of movement:

1- flexion
2- extension
3- isometric
4- supination

Correct Answer: 3
rationale:
- co-contraction is a form of isometric exercise that specifically focuses on contracting muscles on the opposite side of joints against each other to stabilize the joint/hold position.
- In flexion bicepse will contract more than tricepse and the opposite will happen in extension.

Question 14

What is the best way to first exercise the postural (or extensor) musculature when it is extremely weak to facilitate muscle control?

1- Isometrically
2- Concentrically
3- Eccentrically
4- Isokinetically

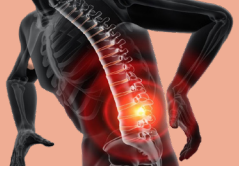
Correct Answer: 1
rationale:
isometric exercises are static strength training workouts, these exercises provide muscular strength only in particular muscles as performed without moving the surrounding joints. These workouts are done in only one position, and thus, they improve strength in only one position. They aim to improve your body's stabilization to a great extent and help contract a particular muscle.
- Concentric exercise include exercise that shortens the muscle.
- Eccentric exercise include exercise that lengthen the muscle .
- Isokinetic exercise include exercise that have constant velocity.

Question 15

A 65 - year - old man with Congestive Heart Failure is referred to the Physical Therapy Department for gait training.
Which assessment test is the most reliable to calculate exercise capacity?

1- Chair - stand
2- 6 - minute walk
3- Time up and go
4- Physical performance OD
5- Heart rate

Correct Answer: 2
rationale:
- Exercise capacity can be assessed by the following tests: Maximal test (exercise stress test; EST), Cardiopulmonary exercise test (CPET),Submaximal test (six-minute walk test; 6MWT),-Submaximal treadmill test and Incremental shuttle walk test (ISWT).
- The six-minute walk test (6MWT is a self-paced walking test well suited to a hospital or community setting.
- For patients with more severe classes of heart disease, 6MWT results correlate well with VO2 peak, and only one 6MWT is recommended.
In addition to measuring exercise capacity, more specific functional testing can be used. Some examples such as Timed up and go test. More recently, combined measures such as the Short Physical Performance Battery (standing balance, walking speed, and chair stand) and the Functional Difficulty Questionnaire have been validated in cardiac surgery patients.
- Heart rate is not an option when dealing with patients that take medications to control their heart rate and of course patient with congestive heart failure should take heart rate regulators.



Question 16

A 28-year-old man comes to the clinic with weakness in the left elbow muscles. The physical therapist decides to start the treatment plan by strengthening exercises of the elbow muscles and asks the patient to carry weight in hand and hold it without any movement in the elbow joint.
What is the type of contraction at the elbow flexion muscles?

1- Isometric
2- Eccentric
3- Isokinetic
4- Concentric

Correct Answer: 1
rationale:
isometric contraction is a muscle contraction without motion. Isometric contractions are used to stabilize a joint, such as when a weight is held at waist level neither raising nor lowering it.
- Concentric exercise include exercise that shortens the muscle.
- Eccentric exercise include exercise that lengthen the muscle .
- Isokinetic exercise include exercise that have constant velocity.

Question 17

Which of the following is the general recommendation regarding exercise for normal older adults?

1- Low intensity
2- High intensity
3- Extreme intensity
4- Moderate intensity

Correct Answer: 4
rationale:
Moderate exercise doesn't need to be athletic or difficult. Studies have shown that simply walking at a brisk pace for 30 minutes or more on most days can lead to significant health improvements. Add simple strengthening exercises two or three times a week and the benefits are even greater.
- Vigorous exercises are not recommended for older adults.

Question 18

A 37-year-old presented woman with vaginal prolapse, which developed 6 months ago following childbirth. She underwent an episiotomy during the difficult labor. Which of the following is the most appropriate initial exercise plan for successful muscle re-education?

1- Functional exercises.
2- Lumbar stabilization.
3- Manual stretch facilitation.
4- Isolated muscle contraction training.

Correct Answer: 4
rationale:
Because Isolation exercises are those that focus on one specific muscle group and thus be more specific for improve pelvic floor muscle, Pelvic floor exercises is the most suitable for this case .

Question 19

A67 y female has acute sever RH.ARTHRITIS Physiotherapy in future do strengthening ex which one is better

1- isometric ex
2- isotonic ex
3- iso kinetic ex
4- immobilization

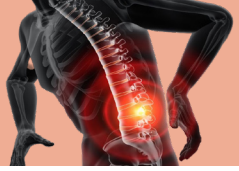
Correct Answer: 1
rationale:
People with rheumatoid arthritis (RA) often have more than one joint affected by the condition. The same two joints on opposite sides of the body are frequently involved with RA. RA impacts small joints, which are found in the wrists, hands, and feet.
Isometric exercises are strength-training workouts in which the muscles contract, but the body doesn't move. Isometrics involve tensing a muscle, then relaxing it. This type of training is especially useful if the usual weight lifting causes your joints to hurt.
- Isometric exercises is the safest type of exercise that patient can do.

Question 20

A 65-year-old patient who suffered a stroke has been referred to rehabilitation with a request to improve stability. Which therapeutic technique should be selected to improve the patient's stability?

1- Slow reversal hold
2- Rhythmic initiation
3- Placing and holding
4- Alternating isometrics

Correct Answer: 1
rationale:
This technique helps to increase strength of weak agonist muscles, active ROM develops coordination by smooth reversible movements helps to reduce fatigue of exercised muscle and increase endurance.
Slow reversal (SR) involves dynamic contraction of the antagonist slowly followed by dynamic contraction of the agonist. Slow reversal-hold (SRH) employs dynamic contraction followed by isometric contraction of the antagonist ,finally followed by the same contraction sequence of the agoinst.
- Rhythmic stabilization is used to improve mobility.
- Placing and holding is used to improve controlled mobility.
- Alternating isometrics is used to improve skill.



Question 21

A 56-year-old man presents to the Physical Therapy clinic after removing a plaster cast, which is covering the right knee joint. During the assessment of the right knee joint by a physical therapist, a hypo mobility has identified.
Which of the following statements describes the hypo mobility?
1- Joint mobility
2- Joint stiffness
3- Decrease mobility
4- Increase mobility

Correct Answer: 3
rationale:
hyopmobility: a decrease in the range of movement of which a body part and especially a joint is capable and therefore decrease mobility of joint. Joint stiffness may be identified as amobility of joint.

Question 22

Which of the following is most effective in mobility in patients with asthma?
1- Hypnotics
2- Beta agonist
3- H2 receptor antagonist
4- Beta adrenergic blockers

Correct Answer: 2
rationale:
medications that relax muscles of the airways, causing widening of the airways and resulting in easier breathing. H2 receptor antagonist was used for GERD while Beta adrenergic blockers were used for heart diseases in the time hypnotics were used for sleep disorders.

Question 23

Which of the following training exercises is defined as overall lengthening of the muscle, as it develops tension and contracts to control motion against the resistance of an outside force (negative work)?
1- Eccentric
2- Isometric
3- Concentric
4- Open chain

Correct Answer: 1
rationale:
eccentric contraction is the motion of an active muscle while it is lengthening under load. Eccentric training is repetitively doing eccentric muscle contractions.
Concentric involves muscle shortening while isometric contraction involves no change in muscle length
- Isokinetic involves muscle contraction under fixed velocity.

Question 24

Type of contraction that make shortening of muscle length:

1- eccentric
2- isometric
3- concentric
4- isokinetic exercise

Correct Answer: 3
rationale:
Concentric contraction occurs when the total length of the muscle shortens as tension is produced. Eccentric involve muscle lengthening. Isometric contraction involves no length change while isokinetic involves muscle contraction with controlled speed.

Question 25

If the proximal bone of the joint is fixed and the distal part of the limb moves in sequence :

1- ACL training
2- Closed-chain exercise
3- Open-chain exercise

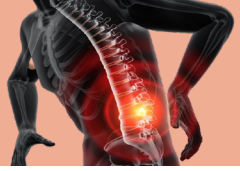
Correct Answer: 3
rationale:
In open kinetic chain exercises, the segment furthest away from the body — known as the distal aspect, usually the hand or foot — is free object and not fixed. The opposite happened with the closed kinematic chain.

Question 26

Which of the following is most commonly associated with a systematic exercise plan, planned performance of bodily movements, postures, and physical activities?

1- Flexibility
2- Endurance
3- Therapeutic
4- Coordination

Correct Answer: 3
rationale:
Therapeutic exercise involves movement prescribed to correct impairments, restore muscular and skeletal function and/or maintain a state of well-being.
It Exercise is a physical activity that is planned and is performed with the goal of attaining or maintaining physical fitness. Physical fitness is a set of traits that allows an individual to perform physical activity.



Question 27

Which of the following is the most appropriate precaution during close kinematic chain training?

1- Avoid resistance to the distal tibia.
2- Knees do not move posteriorly to the toes.
3- Avoid extension of knee between 45° to 15°.
4- Avoid strengthening of quadriceps between 60° to 90°.

Correct Answer: 2
rationale:
To avoid genu recurvatum and possible Osgood shlutter injury or avulsion fracture.

Question 28

Which of the following is the most appropriate description of the therapeutic exercises for timing, sequence and strength of muscles?

1- Mobility
2- Flexibility
3- Endurance
4- Coordination

Correct Answer: 4
rationale:
Mobility exercise is to maintain ROM.
Flexibility exercise is to improve ROM.
Endurance exercises is to improve ability of muscle to contract over long period of time.
Coordination exercises is to improve timing, sequence and strength of muscles.

Question 29

21-year-old female patient presents with neck pain and stiffness that has gradually worsened over the last two weeks. Upon examination, the patient is noted to have pain with left side bending with left rotation and reports pain on the left at the C5-6 junction. Hypo mobility is also noted with right side-gliding of C6. Which of the following techniques will be most appropriate to decrease pain?

1- Closing technique for the mid- thoracic spine.
2- Closing manipulation in extension for C5-C6.
3- Gapping manipulation in flexion for C5-C6.
4- Flexion/opening manipulation for mid-thoracic spine.

Correct Answer: 2
rationale:
Because of hypomobility gapping technique is the most suitable to open the vertebrae
But the question is about technique that decrease pain so ex should work on Closing manipulation in extension for C5-C6.
Gapping manipulation in flexion for C5-C6 is more suitable to improve ROM which is not the question.

Question 30

Regarding closed chain exercise, what of the following?

1- No resistance to distal part.
2- avoid strengthening from 45 to 15 knee extension.
3- avoid strengthening 90 - 120 flexion .
4- Knees doesn't go posterior to toes.

Correct Answer: 4
rationale:
To avoid genu recurvatum and possible Osgood shlutter injury or avulsion fracture.

Question 31

Generally the ROM not depends on:

1- bony structure.
2- elasticity of muscle , ligament and tendon.
3- tightness of muscle, ligament and tendon.
4- bulk of the soft tissue.

Correct Answer: 4
rationale:
the joint to move need the muscle to move it and any abnormality in the muscles affect on the ROM. Bulk of soft tissue is the least factor that ROM depend on it.

Question 32

During weight bearing, a soft tissue contracture resulting in supination of the forefoot will be compensated by:

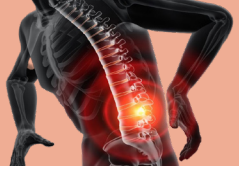
1- pronation of the forefoot.
2- pronation of the rear foot.
3- supination of the rear foot.
4- pronation of the forefoot and rear foot.

Correct Answer: 2
rationale:
- Movement of the forefoot is compensated by movement of the rear foot so that the planter surface is in contact with the ground.
- pronation of the forefoot and rear foot occurs in flat foot and as result to genu valgum.
- supination of the rear foot occurs as compensation to pronation of the forefoot and vice versa

Question 33

After strengthening exercises for elderly, what should be done to prevent OA in elderly?

1- Stretching
2- non weight bearing
3- flexibility
4- resistive training



Correct Answer: 2
rationale:
- Stretching and flexibility were prescribed before the exercise starting as part of warm up Resistive exercises are prohibited for elderly as indicated by ACSM guidelines.
- Non-weight bearing is the optimal exercises to prevent OA in elderly as there is no weight on the knees while exercising.

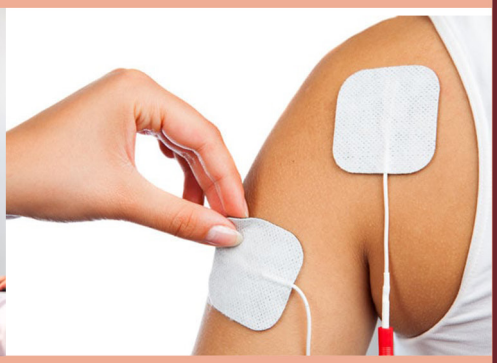
Question 34

What is the most appropriate preventive intervention in elderly? OR What is the most appropriate intervention to prevent falling in elderly

1- Medical review.
2- balance ex.
3- individualized program.
4- strengthening ex.

Correct Answer: 3
rationale:
Balance depends on integration of 3 body systems
Balance: and is exercised by balance exercises
Somatosensory: is exercised by coordination and proprioceptive training
Vestibular: is exercised by vestibular exercises
So the most correct choice is individualized exercise program





Chapter

5.B Electro & Hydro therapy

Question 1

Sudden after using E.S in back pain there's irritable sensation you must:

1- decrease pulse duration
2- continue treatment
3- switch off the device

Correct Answer: 3
rationale:
The correct answer is switch off the device because patients with sensitive skin is one of contraindications of E.S.

Question 2

You apply TENS for a patient and you find sever pain, what would you do?

1- decrease frequency
2- increase wave length
3- decrease pulse width
4- switch off devic

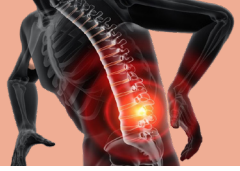
Correct Answer: 1
rationale:
The correct answer is decreace frequency because TENS is a modality that has an analgesic effect which control pain so helps to relief pain so it can't be the cause of pain but if there is pain you should decreace the intensity (frequency) as it may be not suitable for the patient.

Question 3

Electrotherapy has physiological effect of heat that not dependent on which of the folloing:

1- type of the wave
2- size of the area to be treated
3- duration of the application
4- way application

Correct Answer: 1
rationale:
The correct Answer is type of the wave,The extent of the physiological effect of electrotherapy depends on various factors:
1-The size of the area heated
2- The depth of absorption of specific radiations
3- The duration of heating
4-The intensity of irradiation
5-The method of applicatio



Question 4

Applied electrical stimulation to area for long lasting period cause:
1-skin irritation
2-weakness of ms
3-edema occure

Correct Answer: 1
rationale:
The correct Answer is skin Irritation, electrical stimulation is a modality that has an analgesic effect which used to relief pain and shouldn't be used for long over the skin as it causes skin Irritation and burns, and it also should be used at low frequency (1-10)Hz, low amplitude, and low pulse width (50-200us).

Question 5

All of the following are considered as low frequency current except:
1- faradic current
2- diadynamic
3- interferential
4- galvanic

Correct Answer: 3
rationale:
The correct answer is interferential, because IFC uses two frequencies to deliver stimulation to the affected area, one around 4000Hz and the other varies up to 400Hz, Therefore, it is higher in frequency compared to faradic, galvanic and diadynamic which have a frequency of 50:100 HZ.

Question 6

Burn is a response to thermal insult from external agent such as:
1- heat and cold
2- chemicals
3- electricity and radiation
4- all of the above

Correct Answer: 1
rationale:
The correct answer is heat and cold, Thermal Burns are considered 2 typee, one due to external heat sources which raise the temperature of the skin and tissues and cause tissue cell death or charring(hot burns), and latter occurs when ice or other extremely cold objects contact and damage the skin tissue(cold burns).

Question 7

Immediately after switch on TENS, there's severe numbness and exaggeration of symptoms, your response:

1- switch off the machine
2- increase wave length
3- decrease frequency
4- decrease pulse time

Correct Answer: 1
rationale:
The correct answer is switch off the machine, that's because Transcutaneous electrical nerve stimulation (TENS) is a method of pain relief involving the use of a mild electrical current, Some people may be sensitive to or feel uncomfortable with the prickling/tingling sensation generated by the TENS unit so in this condition TENS isn't recommended.

Question 8

Burns caused due to constant direct currents are called:

1- Galvanic burns
2- Chemical burns
3- Electrolytic burns
4- All of the above

Correct Answer: 4
rationale:
The correct Answer is all of the above, this which called electrical burns which occur when electricity comes in contact with the skin and cause damage to the different tissues which may be mild, moderate or severe damage.

Question 9

In electrical stimulation intensity for muscle contraction (shortening) depend on :

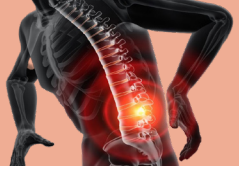
1- shape of current and electrical device used
2- Duration of ttt
3- Size of electrodes
4- Intensity and frequency of current

Correct Answer: 4
rationale:
the intensiy of contraction of muscle depend on the intensity & frequency of the current.

Question 10

Type of laser used in PT ttt :

1- High
2- Soft
3- Mid



Correct Answer: 3
rationale:
The correct Answer is Mid laser, Therapy lasers tend to fall into a particular category of laser light known as 3A or 3B & are often referred to as 'mid laser' sources.
More recently, the terms Low Level Laser Therapy (LLLT) and Low Intensity Laser Therapy (LILT) have been adopted.
- Ohshiro & Calderhead suggest that LLLT involves treatment with a dose that causes no detectable temperature rise in the treated tissues and no macroscopically visible change in tissue structure – essentially, the energy can cause in increase in temperature and a change in tissue structure, but that is not the intention with therapy laser which is applied at levels below that needed to achieve these more overt effects (c/f surgical laser).

Question 11

Which of the following is the smallest current that produces a muscle contraction?
1- Rheobase
2- Chronaxie
3- Direct current
4- Alternating current

Correct Answer: 1
rationale:
The correct answer is Rheobase, Because Rheobase is operationally defined as the minimal electrical current that is necessary to elicit an action potential when current is injected into a cell, In considering the effectiveness of an intracellularly applied current pulse in inducing an action potential.

Question 12

Electrical stimulation duration to produce smooth muscle contraction?
1- 1
2- 2
3- 3
4- 4

Correct Answer: 3
rationale:
It take about 3s of electrical stimulation to produce smooth muscle contraction.

Question 13

Pt treat by phototherapy what this modality?
1- low laser
2- fluid therapy
3- ultrasound

Correct Answer: 1
rationale:
the correct answer is Low Laser , because it has form of light amplifier,it provides enhancement of particular properties of light energy.
- Laser light will behave according to the basic laws of light, in that it travels in straight lines at a constant velocity in space. It can be transmitted, reflected, refracted and absorbed. It can be placed within the electromagnetic spectrum according to its wavelength/frequency which will vary according to the particular generator under consideration.

Question 14

For strengthening the muscle we do electrical stimulation for:
1- 0 pulse
2- 5 pulse
3-16 pulse
4- 10 pulse

Correct Answer: 3
rationale:
The correct answer is 16 pulse, The more pluse give more strength to the muscles.

Question 15

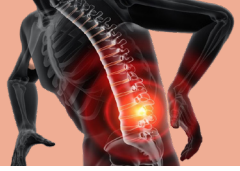
Which type of laser is used by a physical therapist as its depth of penetration is sufficient to produce biological effect in the tissues?
1- Soft
2- Mid
3- Power
4- Deep

Correct Answer: 2
rationale:
The correct answer is Mild, because mild laser has highly wavelength and lowest frequency so it has the greatest depth of penetration.

Question 16

If there is tendonitis in this muscle the best method for treatment:
1- US with deep friction massage
2- T N E S
3- SWD

Correct Answer: 1
rationale:
The correct Answer is US with deep fiction massage as we described above.



Question 17

An elderly cancer patient presents with chronic osteoarthritis in both knees, She has received four sessions of radiotherapy, Which of the following is strongly contraindicated?

1- Laser
2- Hot packs
3- Cold packs
4- Paraffin wax

Correct Answer: 1
rationale:
The correct answer is laser, that's because cancer is one of contraindications of Laser as laser produce heat that helps the spread of cancer cells and tumors in the body.

Question 18

All the following are from the characteristic of laser except:

1- coherence
2- collimation
3- attenuation
4- monochromaticity

Correct Answer: 3
rationale:
The three main characteristics of laser is that it is coherent, collimation and monochromatic.
1-Laser is a light of single wavelength or colour.
2- is a narrow beam that is emitted in a specifi direction.
3-Laser lights are in phase in space and time.

Question 19

In radiating energy when we use it 30 inch from the patient then change it to 15 inch so there will?

1- radiating energy will become 4 times
2- no changes in it
3- will be increased 2 times

Correct Answer: 1
rationale:
the correct answer is radiating energy will become 4 times, that's
because The law of I1/d2 (Where: I = Intensity, d = Distance between the source and the point of calculation)
- So to double the distance from an Infrared source decreases the intensity to (1/2)2 = (1/4) of its original value.
- This is true that if there is halving the distance will quadruple the intensity.

Question 20

Which modality has greater effect in decrease pain in knee joint in fatty patient :

1- ultra sonic wave
2- infrared
3- hot pack
4- paraffin wax

Correct Answer: 1
rationale:
The correct Answer is Ultra-sonic waves because it`s deep than others and fatty tissue is a good absorber for u.s waves .

Question 21

Best treatment of supraspinatus tendinitis is:

1- infrared
2- shortwave
3- ultrasound with deep friction massage

Correct Answer: 3
rationale:
In patients which symptomatic calcific tendinitis of the shoulder, ultrasound treatment helps resolve calcification and is associated with short term clinical improvement.

Question 22

A patient with supraspinatus tendonitis is undergoing ultrasound therapy, When the shoulder joint is in which position, the supraspinatus tendon is best accessed?

1- Adduction and internal rotation
2- Adduction and external rotation
3- Abduction and external rotation
4- Slight abduction and internal rotation

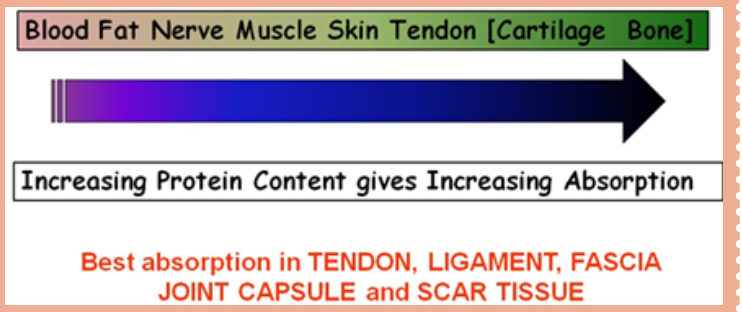
Correct Answer: 4
rationale:
Slight abduction and internal rotation makes supraspinatus tendon more superficial

Question 23

When you treat knee with elevated layer of fat you can use:

1- IR
2- U.S
3- S.W
4- TENS

Correct Answer: 2
rationale:
The correct answer is U.S due to present of fatty tissue.



Question 24

Which of the following tissues absorbs the least amount of an ultrasound beam at 1 MHz?
1- Bone
2- Skin
3- Muscle
4- Blood

Correct Answer: 4
rationale:
The Correct Answer is the blood Vessels is the least absorbs tissues for U.S as it described before.

Question 25

U.S can be mostly absorbed by skeletal muscle as it mostly contains:
1- adipose tissue
2- Connective tissue
3- Nerve endings
4- Proteion

Correct Answer: 4
rationale:
The correct answer is Proteion, as we described above.

Question 26

U.S can be mostly absorbed by skeletal muscle as it mostly contain:
1- adipose tissue
2- connective tissue
3- nerve endings

Correct Answer: 2
rationale:
The correct answer is connective tissue as it higher in protein and low water.

Question 27

One of the common features of ultrasound is that it produces the greatest rise in temperature in the skeletal muscles because it is mostly formed by which of the following?

1- Protein
2- Adipose
3- Endothelial cells
4- Connective tissues

Correct Answer: 1
rationale:
Skeletal muscles contain great amount of protein which known as the best absorber of Ultrasound which produces heat so it rise the temperature in the body.

Question 28

Which of the following is the most appropriate characteristic of a beam of therapeutic ultrasonic waves from transducer?

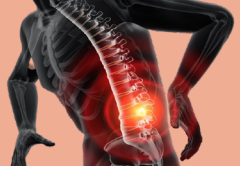
1- Roughly cylindrical
2- Uniform intensities
3- More scattered in the far-field
4- Extreme irregular near the zone

Correct Answer: 3
rationale:
There is 10 Important Properties of Ultrasonic Waves:
Property 1: Ultrasonic waves vibrate at a frequency greater than the audible range for humans (20 kilohertz).
Property 2: They have smaller wavelengths. As a result, their penetrating power is high.
Property 3: They cannot travel through vacuum.
Property 4: Ultrasonic waves travel at the speed of sound in the medium. They have maximum velocity in a denser medium.
Property 5: In a homogeneous medium, they travel at a constant velocity.
Property 6: In low viscosity liquids, ultrasonic waves produce vibrations.
Property 7: They undergo reflection, refraction and absorption.
Property 8: They have high energy content. They can be transmitted over a large distance without much loss of energy.
Property 9: They produce intense heat when they are passed through objects.
Property 10:Like sound waves, ultrasonic waves are longitudinal waves that produce alternate compressions and rarefactions.

Question 29

Which of the following results from changes to quartz in an ultraviolet apparatus?

1- Arsenium
2- Theraktin
3- Tridymite
4- Zirconate



Correct Answer: 3
rationale:
Tridymite is a high-temperature polymorph of silica and usually occurs as minute tabular white or colorless pseudo-hexagonal crystals, or scales, in cavities in felsic volcanic rocks. Its chemical formula is SiO2.

Question 30

Microwaves are strongly absorbed by:
1- Water
2- Fat
3- Protein
4- Muscle tissue

Correct Answer: 1
rationale:
Water is one of the most promising dielectric materials for microwave absorption due to its superior broadband dielectric loss.

Question 31

Modality can increase the temperature of fat?
1- ultraviolet
2- ultrasonic
3- S.W
4- FES

Correct Answer: 2
rationale:
The correct Answer is U.S as we described above.

Question 32

A 63-year-old woman presents to physical therapy with a diagnosis of herpes zoster. The physician informs the physical therapist that the L5 dorsal root is involved and that a transcutaneous electrical neuromuscular stimulation(TENS) unit should be used to help control the pain. Where should the TENS unit electrodes be placed?
1- Posterior thigh
2- Lateral hip/greater trochanter area
3- Anterior thigh
4- Anterior lateral tibia

Correct Answer: 4
rationale:
The correct Answer is Anterior Tibia due to it's involved in L5 nerve supply, The L5 nerve supplies the nerves to the muscles that raise the foot and big toe.

Question 33

Which part of ultrasound produce ultrasonic waves?

1- Head of ultrasound

2- Transducer

Correct Answer: 2

rationale:

An ultrasonic transducer is a device used to convert some other type of energy into an ultrasonic vibration. There are several basic types, classified by the energy source and by the medium into which the waves are being generated.

Question 34

T.E.N.S frequency :

1- 10 -70 HZ

2- 12 -20HZ

3- 5 - 50Hz

4- 1 - 250 Hz

Correct Answer: 4

rationale:

the correct Answer is 1 – 250 Hz , the rate of delivery of these pulses (the pulse frequency) will normally be variable from about 1 or 2 pulses per second (pps) up to 200 or 250 pps.

Question 35

Which of the following is the therapeutic frequency of shortwave diathermy?

1- 27.12 kHz

2- 27.12 MHz

3- 27.12 GHz

4- 27.12 THz

Correct Answer: 2

rationale:

Shortwave diathermy involves the therapeutic application of high-radiofrequency (usually 27.12 MHz [λ = 11.06 m]) electrical currents. Hyperemia, sedation, and analgesia are the basic physiologic effects.

Question 36

Frequency that can make titanic muscle contraction:

1- 50

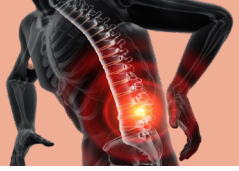
2- 30

3- 20

Correct Answer: 1

rationale:

The correct Answer is 50 as we described above to stimulate innervated muscle at least should be 50 Hz.



Question 37

A 29-year-old woman comes to the clinic after 1 month of right foot tendon transfer, The physical therapist decides to start with the faradic-type current for training a new muscle-action. What is the frequency of faradic-type current?

1- 50-100 Hz
2- 150-200 Hz
3- 250-300 Hz
4- 350-400 Hz

Correct Answer: 1
rationale:
Faradic current is a short duration interrupted direct current with a pulse duration ranging from 0.1 to 1 ms with a frequency of 50 to 100 Hz. Faradic current is surged to produce tetanic contraction and relaxation of the muscle.

Question 38

When using ultrasound in treating chronic bursitis of hip, the most benefit might occur if the ultrasound frequency and dosage were:

1- 1 MHz and 1.5 Watts/cm2
2- 1 MHz and 0.5 Watts/cm2
3- 3 MHz and 1.5 Watts/cm2
4- 3 MHz and 0.5 Watts/cm2

Correct Answer: 1
rationale:
The correct answer is 1 MHz to reach in deep tissue & 1.5 Watts /cm2 which is an adequate intensity for bursitis in hip region , so the choice is 1 MHz and 1.5 Watts/cm2.

Question 39

Which of the following is the most appropriate maximum dosage, in Wcm-2, of Ultrasound therapy in chronic inflammatory conditions:

1- 1.0
2- 1.5
3- 2.0
4- 2.5

Correct Answer: 2
rationale:
An apparently effective treatment window of pulsed ultrasound, 0.16-0.50 W/cm(2) (SATA), proved illusory on closer scrutiny. Conclusions: without a known dose-response relationship users of therapeutic ultrasound can only guess at what dosage might be effective for a patient and how to modify it. Considerably more clinical research is necessary to justify the dosages used in treatment at present.

Question 40

33 - year - old patient comes to the Physical Therapy Clinic with chronic lateral elbow pain. The physical therapist decides to use ultrasonic therapy for treatment. How is burn prevented during ultrasonic therapy treatment?

1- Using big head
2- Using small head
3- moving the head of the ultrasound
4- Constant the head of the ultrasound

Correct Answer: 3
rationale:
Ultrasound physical therapy has a low risk of causing complications. But, exposure to low-intensity ultrasound for a long time may cause superficial burns on the skin. So, medical practitioners usually ensure that the ultrasound probe is in motion when in contact with your skin.

Question 41

A patient comes with severe low back pain. After examination, the therapist plans to apply Transcutaneous Electrical Nerve Stimulation (TENS). Which of the following is important and should be asked about before starting the treatment?

1- Fracture
2- Allergies
3- Hypertension
4- Cardiac pacemaker

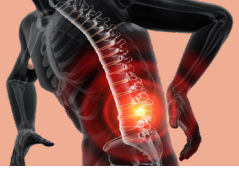
Correct Answer: 4
rationale:
Cardiac pacemaker is the correct answer because it is one of contraindications of TENS because electrical stimulation could cause electromagnetic interference with pacemakers and implanted cardioverter defibrillators.

Question 42

Which of the following theories support the use of a transcutaneous electrical nerve stimulation (TENS) unit for sensory level pain control?

1- Gate control theory
2- Sensory interaction theory
3- Central summation theory
4- None of the above

Correct Answer: 1
rationale:
The gate control theory is also thought to be involved when using a TENS machine. The electrical current stimulates nerve fibres that carry signals relating to touch. The signals travel to the spinal cord where they temporarily block the transmission of pain sensations to the brain.



Question 43

TENS stimulation is :

1- Presynaptic inhibition
2- Presynaptic excitation
3- Post synaptic inhibition
4- Post synaptic excitation

Correct Answer: 1
rationale:
The correct answer is presynaptic inhibition , TENS unit reduces pain through nociceptive inhibition at the presynaptic level in the dorsal horn, thus limiting its central transmission. The electrical stimuli on the skin preferentially activate low-threshold, myelinated nerve fibers. The afferent input from these fibers inhibits propagation of nociception carried in the small un-myelinated C fibers by blocking transmission along these fibers to the target or T cells located in the substantia gelatinosa (laminae 2 and 3) of the dorsal horn.

Question 44

When treating chronic pain which type of TENS used:

1- brief
2- conventional
3- burst
4- acupuncture

Correct Answer: 4
rationale:
The correct answer is acupuncture like a tens:
- In acute stage we use: conventional or high frequency.
- In sub-acute stage we use: burst.
- In chronic stage we use: low frequency (acupuncture).

Question 45

Patient has chronic brachialgia which is the best TENS use:

1- conventional
2- low frequency
3- burst
4- intense

Correct Answer: 2
rationale:
The correct answer is Low frequency , Because:
- Conventional TENS (low-intensity, high-frequency)
- Acupuncture-like TENS (high-intensity, low-frequency)
- Intense TENS (high-intensity, high-frequency)

Question 46

Burst tens stimulate:
1- A delta fiber
2- A beta fiber
3- both of them

Correct Answer: 3
rationale:
Antidromic activation of peripheral nerves by TENS generates nerve impulses that have been shown to collide and extinguish afferent impulses arising from peripheral structures. Peripheral blockade of nociceptive impulses is more likely when TENS activates A-delta fibres (i.e. intense TENS).

Question 47

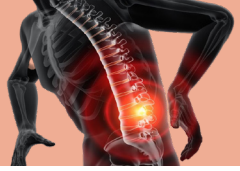
Patient with elbow pain for 5 months and just stopped playing tennis, which type of TENS would you use:
1- Conventional TENS
2- Acupuncture like TENS
3- Brief intense TENS
4- Bust TENS

Correct Answer: 2
rationale:
The correct answer is Acupuncture like Tens , described as above, Traditional TENS (Hi TENS) Usually use stimulation at a relatively high frequency (90 - 130 Hz The stimulation is delivered at 'normal' intensity - definitely there but not uncomfortable.
- Acupuncture (Lo) TENS Use a lower frequency stimulation (2-5 Hz) with wider (longer) pulses (200-250ms - still not at the patients' threshold, but quite a definite, strong sensation.
burst TENS interrupting the stimulation outflow at rate of 2 - 3 bursts/second. The stimulation intensity will need to be relatively high .
- Extract: Acupuncture TENS is used for any chronic pain

Question 48

What is the best modality to use prior to stretching to prepare the muscle?
1- hot pack
2- cold aapplication

Correct Answer: 1
rationale:
Deep heating is thought to lessen nerve sensitivity, increase blood flow, increase tissue metabolism, decrease muscle spindle sensitivity to stretch, cause muscle relaxation, and increase flexibility.



Question 49

A 19-year-old man presented to the Physical Therapy Department with severe pain (8 out of 10 on the pain scale) and swelling in the front of the knee and around the patella. Which of the following is the most appropriate initial treatment?

1- TENS, rest, ice, compression and exercise
2- Heat, TENS, rest, compression and elevation
3- Heat, exercise, rest, compression and elevation
4- Protection, rest, ice, compression and Elevation

Correct Answer: 4
rationale:
RICE treatment is a first-aid treatment for soft tissue injuries like sprains, strains, and bruises for example, sprained ankles, sprained knees, or muscle strains. It's best used immediately after sustaining an injury and should be stopped within two days (48 hours).

Question 50

A 75-year-old patient with severe osteoarthritis has been admitted to a nursing home. Which of the following is the best modality that the therapist should recommend to assist in reducing knee pain?

1- Cane
2- Walker
3- Hot modalities, exercises and walking program
4- Cold application, exercises and walking Program

Correct Answer: 3
rationale:
Hot modalities Provide moist heat to the body which helps in relieving the excess heat. This may also assist in relaxing the muscles and loosening up the joints for adequate comfort and pain relief, Simple home exercises and stretches can help ease some common types of knee pain.

Question 51

What is the best modality to use after muscle soreness?

1- Ice aplication
2- hot pack

Correct Answer: 1
rationale:
Ice application or Cold therapy is also known as cryotherapy. It works by reducing blood flow to a particular area, which can significantly reduce inflammation and swelling that causes pain, especially around a joint or a tendon. It can temporarily reduce nerve activity, which can also relieve pain.

Question 52

A 19 - year - old man presents to the Physical Therapy clinic after removing a plaster cast, which is covering the right elbow joint. The physical therapist decides to start with stretching exercises to increase the range of motion. Which of the following modalities is used for the preparation of stretching?

1- Ice massage
2- Application of local heat
3- High - intensity exercises
4- Application of cold therapy

Correct Answer: 2
rationale:
Heat can soothe and heal muscles by increasing blood flow and is more beneficial for chronic pain or injuries, Heat is also cited as increasing flexibility and thus reducing the chance of injury, and reducing energy cost of muscle contraction by reducing internal friction.

Question 53

Which of the following is the most appropriate combination of superficial heating modalities?

1- Hot pack, wax bath, hydrotherapy, and infrared
2- Microwave, hot pack, hydrotherapy, and hot water bath
3- Hydrocollator, paraffin wax bar, hydrotherapy, and ultrasound
4- Electric heat pad, fluid therapy diathermy and infrared

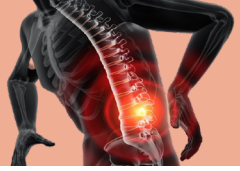
Correct Answer: 1
rationale:
Moist hot packs are perhaps the most widely used form of superficial heating in the clinical setting.

Question 54

When patient coming for you has limited ROM due to edema in addition to leg elevation, you can use which type of electrical stimulation:

1- faradic
2- TENS
3- Galvanic
4- didynamic

Correct Answer: 1
rationale:
The faradic current increase metabolism and remove waste products and bring more blood supply and nutrients to muscle thus increase demand of oxygen and nutrients to the muscle so it reduces joints stiffness and increase range of motion.



Question 55

Baby with flaccid klumps' palsy need stimulation by:

1- interrupted galvanic stimulation
2- reciprocal stimulation
3- Diadynamic

Correct Answer: 1
rationale:
The correct Answer is interrupted galvanic stimulation , due to we use galvanic stimulation with denervated muscles.

Question 56

Decrease edema in ankle joint by :

1- Galvanic Stimulation
2- Interferential
3- Russian current
4- Monophonic rectangular

Correct Answer: 1
rationale:
The correct answer is Galvanic stimulation because Monophasic Rectangular is a type of direct galvanic current or interrupted direct current or High voltage interrupted galvanic stimulation use for decrease edema next to it Faradic current.

Question 57

Decrease edema in ankle joint by:

1- H.V.G.S (high voltage galvanic stimulation)
2- Interferential
3- Mono phasic rectangular
4- Russian current

Correct Answer: 1
rationale:
the correct answer is H.V.G.S as we described before, the available evidence suggests that high volt stimulation administered using negative polarity, pulse frequency of 120 pulses/s, and intensity of 90% visual motor contraction may be effective at curbing edema formation.

Question 58

Therapist use electrical method for pumping action with limb elevated and compressed to:

1- decrease edema
2- increase circulation
3- increase venous return

Correct Answer: 1
rationale:
The correct Answer is Decrease edema, because we do pumping action with compressed limb elevated.

Question 59

A therapist instructed to provide electrical stimulation to a patient with a venous stasis ulcer on the right lower extremity, What is the correct type of electrical stimulation to promote wound healing?

1- Biphasic pulsed current
2- Direct current
3- Interferential current
4- Transcutaneous electrical stimulation

Correct Answer: 2
rationale:
Direct current can help speed wound healing by increasing capillary density and perfusion, improving wound oxygenation, and encouraging granulation and fibroblast activity. Several manufacturers make the high-voltage pulsed current simulator used for this therapy.

Question 60

Which current used for wound healing?

1- TENS
2- faradic
3- diadynamic
4- high voltage galvanic

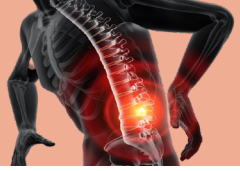
Correct Answer: 4
rationale:
High voltage Galvanic can help speed wound healing by increasing capillary density and perfusion, improving wound oxygenation, and encouraging granulation and fibroblast activity. Several manufacturers make the high-voltage Galvanic current simulator used for this therapy.

Question 61

Contraction of muscle by electrical stimulation which not occur:

1- increase demand of oxygen
2- increase metabolic rate
3- Increase waste out
4- vasoconstriction of blood vessels

Correct Answer: 4
rationale:
the correct answer vasoconstriction of blood vessels , as described above.



Question 62

The tissue of body that consider the best conductor to the electric:

1- nerve
2- fat
3- muscles
4- bones

Correct Answer: 1
rationale:
Nerve cells generate electrical signals that transmit information. Although neurons are not intrinsically good conductors of electricity, they have evolved elaborate mechanisms for generating electrical signals based on the flow of ions across their plasma membranes.

Question 63

All of the following are physiological effect of electrical stimulation except:

1- Vasoconstriction of blood vessels
2- Increase blood flow
3- Increase metabolic rate

Correct Answer: 1
rationale:
The correct answer Vasoconstriction of blood vessels,Vasoconstriction of blood vessels isn't physiological effect of electrical stimulation BUT vasodilatation, increase blood flow, increase metabolic rate, increase temperature of the skin, increase stimulation of the neural receptors.

Question 64

The rate of electrical nerve stimulation through the skin:

1- 10-70 cycle/second or Hz
2- 35-120 cycle/second or Hz
3- 50-150 cycle/second or Hz
4- 15-200 cycle/second or Hz

Correct Answer: 3
rationale:
The correct answer 50-150 cycle/second or HZ, described as above, Not below can`t stimulate & not above to avoid fatigue of muscle.
NOTE:
- 50-150 Hz ----- N. stimulation.
- Below 50 Hz-----no stimulation.
- Above 50 Hz ----- causes fatigue.

Question 65

Electric stimulation which is not true:

1- Vasodilatation
2- Increase metabolic rate
3- Decrease muscle spasm
d- Vasoconstriction

Correct Answer: 1
rationale:
The correct answer is Vasoconstriction of blood vessels, electrical stimulation causes Vasoconstriction only and does't cause vasodilation.

Question 66

Heating with (E.S) which is not correct?

1- V.D
2- Decrease metabolic-rate
3- Decrease muscle spasm

Correct Answer: 2
rationale:
The correct answer is decrease metabolic rate , described as above, The physiological effects of heat therapy include pain relief and increases in blood flow, metabolism, and elasticity of connective tissues.

Question 67

Electrical nerve stimulation frequency is:

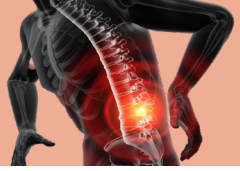
1- 23-120
2- 50-150
3- 10-70

Correct Answer: 2
rationale:
The correct answer is 50 to 150 HZ not below can`t stimulate & not above to avoid fatigue of muscle.

Question 68

A 58-year-old patient comes to the Physical Therapy Clinic with lower back pain and hypertension. The physical therapist decides to use heat modalities to reduce pain. What factor reduces blood pressure after treatment?

1- Increased blood viscosity
2- Generalized vasodilatation
3- Generalized vasoconstriction
4- Increased peripheral resistance



Correct Answer: 2
rationale:
The correct answer is Generalized vasodilatation, The vasodilation causes a decrease in systemic vascular resistance (SVR) and an increase in blood flow, resulting in a reduction of blood pressure.

Question 69

Which of the following is the most beneficial effect of Russian current?
1- Better pain - relieving
2- Better facilitator of healing
3- Reduce inflammation signs
4- Stimulation of deep muscles

Correct Answer: 4
rationale:
Stabilizing spinal muscles with exercises: the Russian current (RC) is reported as an alternative, because it promotes muscle contraction, providing muscle strengthening and hypertrophy.

Question 70

Which of the following is an example of monophasic current?
1- Faradic
2- Russian
3- Diadynamic
4- Interferential

Correct Answer: 3
rationale:
Diadynamic current is an another example for low frequency current rarely used in UK but in mainland Europe has stronger following. It is monophasic sinusoidal current was developed in the 1950's,by Bernard (a French Dentist), So it is called also Bernard current.

Question 71

SWD transfers energy by:
1- convection
2- conversion
3- conduction
4- radiofrequency

Correct Answer: 2
rationale:
Shortwave diathermy uses high-frequency electromagnetic energy to generate heat by conversion.

Question 72

A 35-year-old man fell asleep with his arm on the side of a chair. In the morning, he observed absent sensation on the dorsum of the hand and an inability to extend the wrist. His nerve conduction studies and electromyography confirmed neurapraxia of the radial nerve in the arm. Which of the following is the most appropriate type of electrotherapy indicated for this condition?

1- Faradic current
2- Interrupted direct current
3- Electrical muscle stimulation
4- Transcutaneous nerve stimulation

Correct Answer: 4
rationale:
The electrical impulses can reduce the pain signals going to the spinal cord and brain, which may help relieve pain and relax muscles. They may also stimulate the production of endorphins, which are the body's natural painkillers.

Question 73

Your plan of care includes use of iontophoresis in the management of calcific bursitis of the shoulder. To administer this treatment using the acetate ion, the current characteristics and polarity should be:

1- monophasic twin peaked pulses using the positive pole
2- monophasic twin peaked pulses using the negative pole
3- continuous monophasic using the positive pole
4- continuous monophasic using the negative pole

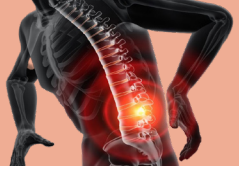
Correct Answer: 4
rationale:
The correct Answer is Continuous monophasic is a direct galvanic current and we using the negative pole When we use ionophoresis it must be direct current or continuous monophasic, and must put the chemical substance in the same polarity of the electrode, so we choose negative pole due to the substance is acetate ion.

Question 74

What is your best first treatment/ intervention of chronic tennis elbow patient?

1- immobilization
2- range of motion exercises ROM
3- pulsed current / US
4- splint

Correct Answer: 3
rationale:
The correct answer is pulsed current / US, The present study demonstrated the ultrasound therapy and pulsed current are helpful to relieve pain for tennis elbow patient.



Question 75

Tennis elbow player has pain in elbow joint 5 month ago and there is production of pro-inflammatory cytokines .what can we use?

1- Pulsed us
2- Continuous us
3- Pulsed with lidocain
4- Continuous with lidocain

Correct Answer: 3
rationale:
The correct Answer is Pulsed with Lidocain.
NOTE:
Phonophoresis:
1- The use of US to drive medication through the skin into the deeper tissues, Local analgesics (lidocaine) and anti-inflammatory drugs {dexamethasone, salicylates) are often used.
2- Method of application is similar to direct contact technique, except tha a medicinal agent is used as part of coupling medium:
- Mode: pulsed 20%
- Treatment intensity: 1-3 w/cm2.
- Treatment time: 5-10 minutes.
- 0.5-0.75 w/cm2, using a medication that is prepared in a medium that will allow transmis sion of the US Gel mediums or transdermal patches have good transmissivity; avoid pastes and creams.

3- Goals and indications:
4- pain modulation
- decreasing inflammation in the subacute and chronic musculoskeletal conditions.

Question 76

A 58-year-old patient with lower back pain treated by TENS to reduce pain,after the third session the therapist evaluated the treatment procedure and found that the patient is an adaptation to the intensity and frequency of TENS therapy. Which of the following should be done to complete the treatment?

1- Change the treatment agent
2- Decrease intensity and frequency of TENS
3- Increase the intensity and frequency of TENS
4- Continuous with the same intensity and frequency of TENS

Correct Answer: 3
rationale:
The correct answer is Increase the intensity and frequency of TENS, As we should put in concederd that intensity and frequency should be increased every session to prevent adaptation.

Question 77

U.S has these frequencies:

1- 1 or 3 MHZ
2- 1 or 5 MHZ
3- 1 or 5 HZ

Correct Answer: 1
rationale:
The Correct Answer is 1 or 3MHZ, More research with various ultrasound machines needs to be performed to see if 3-MHz ultrasound is more appropriate than 1-MHz ultrasound for producing thermal effects in tissues 2.5 cm deep. Researchers can also explore the maximum depth to which 3-MHz ultrasound can adequately heat various 37.

Question 78

Maximum ultrasound frequancy:

1- 3 MHz
2- 5 MHz
3- 2 MHz

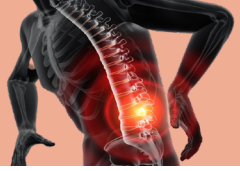
Correct Answer: 1
rationale:
The correct answer is 3MHZ More research with various ultrasound machines needs to be performed to see if 3-MHz ultrasound is more appropriate than 1-MHz ultrasound for producing thermal effects in tissues 2.5 cm deep. Researchers can also explore the maximum depth to which 3-MHz ultrasound can adequately heat various .

Question 79

A patient is being treated with iontophoresis, driving dexamethasone, for inflammation around the lateral epicondyle of the left elbow. The therapist is careful when setting the parameters and with cleaning the site of electrode application to prevent a possible blister. This possibility is not as strong with some other forms of electrical stimulation, but with iontophoresis using a form of , precautions must be taken to ensure that the patient does not receive a mild burn or blister during the treatment session. Fill in the blank:

1- Alternating current
2- Direct current
3- Pulsed current
4- Transcutaneous electrical nerve stimulation

Correct Answer: 3
rationale:
The correct answer is Direct Current, as we described above.



Question 80

Patient has fracture upper tibia and fixated by plate and screw. Complain from complication of knee limitation and pain. Which modalities are contra-indicated?

1- ultra sonic
2- Faradic current
3- Short wave
4- Ice application

Correct Answer: 3
rationale:
The correct Answer is Short wave
Notes: Short wave because it causes deep heating for a large area not specific like US, and it may cause pain due to increase the heat absorbed by metals in plate and screw.

Question 81

An elderly obese woman is diagnosed with a fracture in the neck of the right femur. She is referred to Physiotherapy Department for pain management after 10 weeks of hemiarthroplasty. Which of the modalities is contraindicated?

1- Hot packs
2- Whirlpool
3- Cryotherapy
4- Shortwave diathermy

Correct Answer: 4
rationale:
The correct answer is shot wave diathermy, SWD is contraindicated in areas with metal implants and in patients with pacemakers or implanted deep brain stimulators.

Question 82

Patient with a contaminated surgical wound on lumbar region, what would you use to clean the wound:

1- infrared
2- ultraviolet
3- shortwave
4- low laser

Correct Answer: 2
rationale:
UVC (200–280 nm) is highly antimicrobial and can be directly applied to acute wound infections to kill pathogens without unacceptable damage to host tissue. UVC is already widely applied for sterilization of inanimate objects.

Question 83

Which of the following is a consequence of deep heat applied to an impaired arterial supply area?

1- Shock
2- Burns
3- Scalds
4- Gangrene

Correct Answer: 2
rationale:
The correct answer is Burns, as application of deep heat may cause skin irritation that leads to burns.

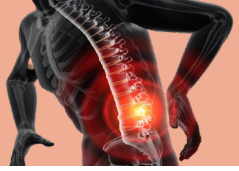
Question 84

Patient coming with industrial burn involved the dorsum muscle of the left hand after study there is dis-continues chronoxia227/m. so we could use:

1- continues direct current
2- faradic current
3- interrupted direct current

Correct Answer: 1
rationale:
The correct Answer is continues direct current , due to it`s appear in discontinues chronoxia which mean that it`s denervated muscle.
NOTE:
- Rheobase:
1- the smallest current that will produce a muscle contraction if the stimulus is of infinite duration; in practice an impulse of 1ms (0.1 sec) is used.
2- In denervation, the Rheobase may be less than that of innervated muscle, and it often rises as re-innervation commence. These changes are not however sufficiently predictable to be reliable guides.
3- The Rheobase varies considerably in different muscles and according to the skin resistance and temperature of the part, while a rise may be due to fibrosis of the muscle.

- Chronaxie:
1- The duration of the shortest impulse that will produce a response with a current of double the rheobase.
2- The chronaxie of innervated muscle is appreciably than that of denervated muscle, the former being less and the latter more than 1ms if the constant voltage stimulator is used.
3- With the constant-current stimulator, the values are higher, but bear a similar relationship to each other.
4- Chronaxie is not a satisfactory method of testing electrical reactions as partial denervation is not clearly shown, the chronaxie being that at the predominant state of the fibers; for example, the chronaxie of a muscle with 25% of its fibers innervated would be the same as that of a completely denervated muscle.



Question 85

Which of these modalities give lowest deep heating?
1- US
2- SWD
3- UV
4- IR

Correct Answer: 4
rationale:
The correct answer is IR, as IR is a superficial heating modality.

Question 86

To treat patient with deep heat, which is not form of deep heat:
1- SWD
2- MWD
3- IR (infra-red)
4- US (ultra sound)

Correct Answer: 3
rationale:
The corrected Answer is IR is a superficial heating.

Question 87

Which of the following physiological effects is considered as an indirect effect of heating?
1- Muscle relaxation
2- Increased blood flow
3- Increased metabolism
4- Stimulation of the neural receptors

Correct Answer: 1
rationale:
Direct effects of heating is increasing blood flow so decrease blood pressure and increasing metallic rate and stimulation of the neural receptors.

Question 88

When use iontophoresis; it is a type of:
1- low frequency TENS
2- HVGS
3- Direct current
4- Russian current

Correct Answer: 3
rationale:
The correct answer is, Direct Current as described above.

Question 89

Use of continuous ultrasound at 1.5 Watts/cm2 will result in:

1- no change in nerve conduction velocity.
2- increase in motor nerve conduction velocity and decrease in sensory nerve conduction velocity.
3- decrease in motor nerve conduction velocity and increase in sensory nerve conduction velocity.
4- increase in both motor and sensory nerve conduction velocity.

Correct Answer: 4
rationale:
The correct Answer is Increase in both motor & sensory nerve conduction velocity.

Question 90

What is the type of electrical current that is used during Iontophoresis?

1- Ultrasonic
2- Direct current
3- Galvanic current
4- Low-frequency TENS

Correct Answer: 2
rationale:
The correct answer is Direct Current, as we described above.

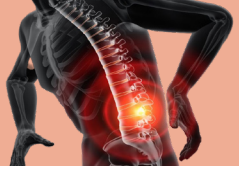
Question 91

The therapist has given a patient an ultraviolet treatment. The patient calls the therapist the next day with complaints of peeling and itching. These signs and symptoms resolve three days later (a total of 4 days after treatment). What dosage did the patient receive?

1- Sub-erythemal dose
2- Minimal erythemal dose
3- First-degree erythemal dose
4- Third-degree erythemal dose

Correct Answer: 3
rationale:

DOSE	LATENT PERIOD (HOURS)	APPEARENCES	PIGMENTATION	DESQUAMATION
E1	Up to 12	Slightly pink	Nil	Nil
E2	4-6	Red	Slight	Powdery
E3	1-4	Fiery, red, & painful	Marked	In sheets
E4	As E3 but with the formation of blisters			



Question 92

After exposure to U.V.R patient of come with Erthyrema due to release?

1- histamine
2- melanin
3- bradykinin
4- endorphin

Correct Answer: 1
rationale:
The correct Answer is histamine is a responsible for Erthyema.

Question 93

After exposure to U.V.R patient of come with pigmentation due to release:

1- histamine
2- melanin
3- bradykinin
4- endorphin

Correct Answer: 2
rationale:
The correct Answer is melanin is a responsible for a pigmentation.

Question 94

Pt has pain due to deep muscle strain we use:

1- Pulsed U/S 1MHZ
2- Pulsed 3MHZ
3- Continous U/S 1 MHZ
4- continous 3 MHZ

Correct Answer: 1
rationale:
The correct Answer is Pulsed U/S 1MHZ as described above.

Question 95

What is best for scar tissue:

1- ultra sound
2- infrared
3- shortwave

Correct Answer: 1
rationale:
The correct Answer is ultra sound waves penetrate well more than Infrared and shortwave for Scar Tissue.

Question 96

pt has infected lumber wound use:

1- Ultra Voilet
2- U/S
3- TENS
4- faradic

Correct Answer: 1
rationale:
The Correct Answer is ultraviolet used for infected wound and with chronic wound we use HVPC.

Question 97

A therapist is treating a patient with an infected wound at the lower lumbar area, Which of the following heating modalities assists in destroying bacteria and fungi that are commonly found in wounds?

1- Short wave
2- Ultrasound
3- Ultraviolet ray
4- Low-level laser therapy

Correct Answer: 3
rationale:
The correct answer is ultraviolet rays, as it helps in killing germs and bacteria.

Question 98

Which of the following can make hypersensitivity?

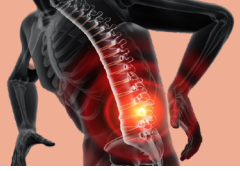
1- TENS
2- Infrared
3- UV
4- US

Correct Answer: 3
rationale:
A small percentage of the population have a skin condition that makes them particularly sensitive to the sun's UV rays. A minimal dose of UV radiation is sufficient to trigger an allergic reaction leading to rash or severe sunburn.

Question 99

What is the best electrical modalities chosen to treat infective wound?

1- low level laser
2- UV
3- Galvanic Stimulation



Correct Answer: 2
rationale:
UVC (200–280 nm) has a significant antimicrobial effect and can be used as efficient bactericide agent for treatment of acute wound infections and killing pathogens without undesirable injury to host tissue.

Question 100

What happens if you do not wear goggles when looking into ultraviolet radiation?
1- Loss of vision
2- Infection
3- Conjunctivitis
4- Pain

Correct Answer: 3
rationale:
Corneal damage, cataracts and macular degeneration are all possible chronic effects from UV exposure and can ultimately lead to decreased vision. 3. The eyelid skin is the thinnest of the whole body. UV damage induces dryness, wrinkles, accentuated skin furrows, sagging, loss of elasticity and mottled pigmentation.

Question 101

A 48-year-old patient with pelvic floor muscle weakness treated by physical therapy agent to increase muscle power. Which of the following physical therapy agents is used?
1- TENS
2- Infrared
3- Ultrasonic
4- Interferential

Correct Answer: 4
rationale:
The correct Answer is interferential It effectively provides pain relief and promotes healing for injured muscles.

Question 102

contraindication in TENS stimulation ?
1- Heart pacemaker
2- Infected wounds

Correct Answer: 1
rationale:
A number of patients are excluded from electrical stimulation treatment because there is concern that electrical stimulation could cause electromagnetic interference with pacemakers and implanted cardioverter defibrillators.

Question 103

Patient with groin pain 2 weeks ago used:

1- cold + interferential
2- hot + interferential
3- pulsed US + interferential
4- continue US + interferential

Correct Answer: 3
rationale:
The correct Answer is continue US+ Interferential
Notes: continue US + interferential due to after 2 weeks ago it means that It's chronic so we use Us continues waves to make a thermal effect used for treatment chronic deep pain.

Question 104

A therapist is treating a patient with a strain at one of the deep muscles and wants to reduce pain. Which of the following is the most appropriate form of ultrasound that the therapist should use?

1- Pulsed 1 MHz
2- Pulsed 3 MHz
3- Pulsed 5 MHz
4- Non-pulsed 1 MHz

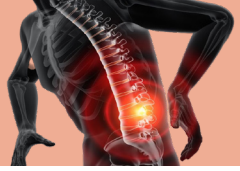
Correct Answer: 1
rationale:
More research with various ultrasound machines needs to be performed to see If 3-MHz ultrasound is more appropriate than 1-MHz ultrasound for producing thermal effects in tissues 2.5 cm deep. Researchers can also explore the maximum depth to which 3-MHz ultrasound can adequately heat various.

Question 105

A 25-year-old patient comes to the Physical Therapy Clinic with scar tissue at the right leg. The physical therapist decides to use ultrasonic therapy for treatment. Which of the following occurs for the scar tissue after treatment?

1- Painful
2-Pliable
3- Inflamed
4- Hardness

Correct Answer: 2
rationale:
Some collagen breaks down at the site of the wound and the blood supply reduces. The scar gradually becomes smoother and softer. Although scars are permanent, they can fade over a period of up to 2 years. It's unlikely they'll fade any more after this time.



Question 106

After applying a hot pack to a patient you observe hyperemia in the treatment area after a few minutes. As the therapist you should:

1- remove the hot pack immediately and assess for a burn.

2- keep the hot pack in place as this is indicative of heat dissipation secondary to vasodilation.

3- elevate the part to minimize edema formation.

4- remove the hot pack immediately as the patient is having a potentially dangerous histamine reaction.

Correct Answer: 2

rationale:

The correct Answer is keep the hot pack to a patient because it`s the action of the heat to make hyperemia due to vasdilation.

Question 107

A 40-year-old woman presented to the physical therapy clinic with hand pain. The physical therapist decides to use Paraffin wax for treatment. What happens to the hand after treatment?

1- Increases in pain

2- Stiffness of joints

3- Local hyperemia

4- Reflex vasoconstriction

Correct Answer: 3

rationale:

Paraffin wax is a form of deep heat therapy that causes vasodilatation of superficial capillaries and arterioles that leads to local hyperemia and reflex vasodilatation.

Question 108

Muscle reeducation In comparing the use of cold pack and hot pack treatments, which of the following statements is false?

1- Cold packs penetrate more deeply than hot packs.

2- Cold increases the viscosity of fluid and heat decreases the viscosity of blood.

3- Cold decreases spasm by decreasing sensitivity to muscle spindles and heat decreases spasm by decreasing nerve conduction velocity.

4- Cold decreases the rate of oxygen uptake, and heat increases the rate of oxygen uptake.

Correct Answer: 3

rationale:

Cold decreases spasm by decreasing sensitivity to muscle spindles and heat decrease spasm by decreasing nerve conduction velocity, cold: decrease spasm by decrease muscle tension by increasing in the threshold of activation of the muscle spindle Decreased muscle spasm, the muscle relaxation probably results from a decrease in firing rates of the efferent fibers in the muscle spindle.

Question 109

A patient recently diagnosed with Multiple Sclerosis presents to a physical therapy clinic. The patient asks the therapist what she needs to avoid with this condition. Which of the following should the patient avoid?

1- Hot tubs
2- Slightly increased intake of fluids
3- Application of ice packs
4- Strength training

Correct Answer: 1
rationale:
The correct Answer is Hot Tubs , any hot application is avoided for multiple sclerosis.

Question 110

A 58-year-old woman presented with severe hip pain. She has been taking steroids for her dermatomyositis. A CT scan hip confirmed avascular necrosis of femoral neck with an impacted fracture. There is a 2 cm leg-length discrepancy. Which of the following is the most appropriate intervention to reduce hip pain?

1- Shoe raise
2- Walking stick
3- Shortwave diathermy
4- Modified seating system

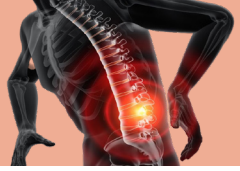
Correct Answer: 3
rationale:
Diathermy are treatments that deliver energy to treat specific areas of the body. These treatments are typically used for the following purposes: Relieve pain, stiffness and muscle spasms. Reduce joint contractures.

Question 111

Use of shortwave diathermy and microwave diathermy is not contraindicated in which of the following conditions?

1- On a patient who has a pacemaker.
2- Over the site of a metal implant.
3- On a patient who has hemophilia.
4- Using pulse shortwave over an acute injury.

Correct Answer: 4
rationale:
Shortwave diathermy has the following precautions or contraindicatios:
1- Malignancy
2- Sensory loss
3- Tuberculosis
4- pregnancy
5- Metallic implants or foreign bodies.
6- Application over moist dressings.
7- Ischemic areas or arteriosclerosis.
8- Thromboangiitis obliterans.



Question 112

SWD is contraindicated in:

1- Infected wounds
2- Peripheral vascular disease
3- Hematoma

Correct Answer: 2
rationale:
The correct Answer is peripheral Vascular disease as we described above.

Question 113

Short wave (SWD) contraindications:

1- Phlebitis
2- Sprain
3- Muscle strain
4- Hematoma

Correct Answer: 1
rationale:
The correct Answer is phlebitis as we described above.
Shortwave diathermy has the following precautions or contraindications:

1- Phelebitis
2- Malignancy
3- Sensory loss
4- Tuberculosis
5- pregnancy.
6- Metallic implants or foreign bodies.
7- Application over moist dressings.
8- Ischemic areas or arteriosclerosis.
9- Thromboangiitis obliterans

Question 114

SWD contraindication is:

1- hematoma
2- arterial insuffiency
3- sinusitis

Correct Answer: 2
rationale:
Contra-indications to short-wave diathermy:

1- Arterial disease
2- Metal in the tissues
3- Disturbed skin sensation
4- Pregnancy
5- Tumors
6- X-ray therapy
7- venous thrombosis or Phlebitis
8- It is unsafe to apply short-wave diathermy to patients who are unable to understand the degree of heat.

Question 115

Cryotherapy contraindication:
1- peripheral artery disease
2- acute injury
3- acute inflammation

Correct Answer: 1
rationale:
The Correct Answer is peripheral artery disease, The following conditions are contraindications to whole body cryotherapy: Pregnancy, severe Hypertension (BP> 180/100), acute or recent myocardial infarction, unstable angina pectoris, arrhythmia, symptomatic cardiovascular disease, cardiac pacemaker, peripheral arterial occlusive disease, venous thrombosis.

Question 116

Which of the following is the main contraindication to deep heat?
1- Pain
2- Metal
3- Spasticity
4- Contracture

Correct Answer: 2
rationale:
The correct answer is metal, as it causes burns and radiated pain.

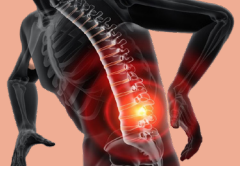
Question 117

Which of the following is contraindicated to ultrasound at 1.5 watts/cm2 with a 1 MHz sound head?
1- Over a recent fracture site.
2- Over non-cemented metal implant.
3- Over a recently surgically repaired tendon.
4- Over the quadriceps muscle belly.

Correct Answer: 3
rationale:
Over recently surgically repaired tendon is contra-indicated, over cemented plate contraindicated but the other can we use US with it.

Question 118

When using electrical stimulation with a unit that plugs into the wall, the therapist must take many different safety precautions. Which of the following precautions probably would not increase safety to the patient and therapist?
1- Never placing the unit in close proximity to water pipes while treating the patient.
2- Never using an extension cord when using a plug-in unit.
3- Always adjusting the intensity of stimulation during the off portion of the cycle.
4- Both A and C are measures that are not likely to increase safety.



Question 119

Which of the following precautions during electromagnetic application?

1- sinusitis
2- phlebitis
3- pregnant and pacemaker
4- capsular injury

Correct Answer: 3
rationale:
The Correct Answer is pregnant and pacemaker as they are contraindicated of electromagnetic.

Question 120

A patient is receiving electrical stimulation for muscle strengthening of the left quadriceps. One electrode from one lead wire, 4 x 4 inches in size, is place on the anterior proximal portion of the left quadriceps. Each of two other electrodes from one lead wire is 2 x 2 inches in size. One of the electrodes is placed on the inferior medial side of the left quadriceps and one on the inferior lateral side of the left quadriceps. This is an example of what type of electrode configuration?

1- Monopolar
2- Tripolar
3- Quadripolar
4- Bipolar

Question 121

A physician has ordered a specific type of electrical stimulation that utilizes a frequency of 2500 Hz with a base frequency at 50 Hz to achieve fused tetany. What type of electrical stimulation has the physician ordered?

1- Iontophoresis
2- Transcutaneous electrical nerve stimulation
3- Intermittent flow configuration
4- Russian stimulation

Correct Answer: 4
rationale:
Russian Stimulation Therapy is a form of Electrotherapy. Its unique name raises a lot questions; it was originally developed by the Russian Olympic team for the enhancement of muscle mass. Russian Stimulation Therapy is mostly used for muscle mass and edema reduction in North America..

Question 122

How to check shortwave before application?

1- Neon light
2- Warmth of electrodes

Question 123

A 33-year-old patient comes to the Physical Therapy Clinic with left shoulder pain. The physical therapist decides to use short-wave diathermy. What should the therapist use to ensure the short-wave diathermy is working?

1- Neon tube
2- Water bath
3- Sound meter
4- Energy meter

Correct Answer: 1
rationale:
Neon tube is a sealed glass tube with a metal electrode at each end, filled with one of a number of gases at low pressure. A high potential of several thousand volts applied to the electrodes ionizes the gas in the tube, causing it to emit colored light. The color of the light depends on the gas in the

Question 124

Infrared can cause burn, what's distance should be between patient and the device:

1- 75 cm
2- 30 cm
3- 10 cm
4- 40 cm

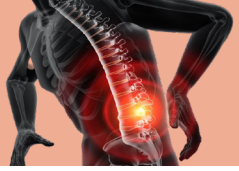
Correct Answer: 1
rationale:
Infrared usually use tungsten material to the lamp with luminous light. The distance between patients and device are 45 to 75cm. The wavelength is 800 nm and perpendicular to the area of pain. The process of therapy takes time between 10 and 15 minutes and can be done repeatedly depending on the cases.

Question 125

A 35-year-old patient presented to the Physical Therapy Clinic with elbow pain. The physical therapist decides to use ultrasonic therapy for treatment. Which of the following the therapist has to use to check the functionality of ultrasonic therapy?

1- Lamp
2- Water bath
3- Sound meter
4- Energy meter

Correct Answer: 2
rationale:
Ultrasonic baths are a cleaning method that uses ultrasound and a liquid to clean objects. Because it is reliable, it is often used for the final cleaning of components and tools. Ultrasonic baths use Cavitation bubbles induced by high-frequency pressure (sound) waves to agitate a liquid.



Question 126

The most affective distance between IR and patient is:

1- 30 cm
2- 90 cm
3- 70 cm

Correct Answer: 3
rationale:
The correct Answer is 70 cm because IR is a superficial heating.

Question 127

Which of the following statements is false about treatment with infrared lamps?

1- Near infrared heats deeper than far infrared.
2- Infrared lamps heat both sides of an extremity at one time.
3- The therapist can change the intensity of the heat by changing the angle between.
4- The beam and the body part being treated.

Correct Answer: 2
rationale:
The correct Answer is Infrared lamps heat both sides of an extremity at one time because it affect only on the side which radiated on.

Question 128

Ultrasonic under water technique the head will be:

1- one inch from the part
2- 3 inch from the part
3- keep in the constant motion

Question 129

A therapist is treating a patient with a venous insufficiency ulcer over the medial malleolus. The wound is moist and not infected. The involved lower extremity is swollen, and the patient reports no pain around the wound. The physician has ordered wound care 3 times a week. Which of the following is the best treatment?

1- Warm whirlpool
2- Unna boot dressing between therapy sessions
3- Intermittent compression pump
4- B and C

Question 130

Physical therapist is ordered to evaluate a patient in the late stages of amyotrophic lateral sclerosis. In the patient's chart is an electromyography report and nerve conduction velocity test. What should the physical therapist not expect to find in these test results?

1- Fibrillation potentials
2- Polyphasic motor unit potentials.
3- Decreased sensory evoked potentials
4- A and B only

Question 131

Coupling media of ultrasonic which of the following of fastest effect:

1- aquatic gel
2- glycerol
3- liquid paraffin
4- crrogel

Correct Answer: 1
rationale:
The correct Answer is Aquatic gel is the fastest effect for coupling media.

Question 132

Coupling media of ultrasonic which of the following of least effect:

1- aquatic gel
2- glycerol
3- liquid paraffin
4- crrogel

Correct Answer: 3
rationale:
The correct Answer is liquid paraffin is the least effect for coupling media of Ultrasonic.

Question 133

For denervated ms we use at least:

1- 10 m.sec
2- 50 m.sec
3- 100 m.sec
4- 300 m.sec

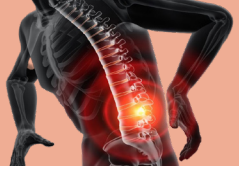
Correct Answer: 3
rationale:
100 msec at least with denervated muscle, if he doesn't say at least we should choose 300 ms or more.

Question 134

In Bell`s palsy we use:

1- continues direct frequency current
2- T.E.N.S
3- Interrupted direct current

Correct Answer: 3
rationale:
The correct answer is interrupted direct current , because it`s innervated muscle.



Question 135

Which current used for stimulated Denervated Muscle?

1- TENS
2- Faradic
3- Diadynamic
4- Direct galvanic

Correct Answer: 4
rationale:
The correct Answer is Direct galvanic is the current which can stimulate denervated Muscle directly.

Question 136

One of these electrical stimulation methods does not stimulate denervated muscle:

1- high voltage interrupted long duration
2- TENS
3- galvanic
4- faradic

Correct Answer: 2
rationale:
TENS current plays its role in a gate control theory which needed nerves and receptors to make its role for modulating pain.

Question 137

Pt of bells palsy long time treated with (E.S) you:

1- M.S weakness
2- no change
3- fibrosis in the affected side

Correct Answer: 3
rationale:
The correct Answer is fibrosis in the affected side due to long use of electricity to stimulate the muscles.

Question 138

When applying electrical muscle stimulation how many contraction occur per session?

1- 10 cont
2- 16 cont
3- 1 cont
4- 5 cont

Question 139

Aphysical therapist is performing electro -myographic testing. During a maximal output test of the patient's quadriceps muscle, 35% of the motor unit potential is poly-phasic. What is the significance of this finding?

1- It is normal in the quadriceps
2- It is normal in the triceps brachii, not in the quadriceps
3- It is normal in the biceps brachii, not in the quadriceps
4- It is abnormal in any muscle

Question 140

During immersion technique in paraffin wax, the temperature will be:

1- 37-41
2- 41-43
3- 43-47
4- 47-51

Correct Answer: 2
rationale:
The correct Answer is 41 to 43 , the paraffin wax Temp. from 40 to 44 C.

Question 141

Wax bathes are available in many variations of size and shape. The melted was needs to be maintained at temperature of: (kept in a temperature that is):

1- 30°- 35°C
2- 40°- 44°C
3- 50° - 53°C

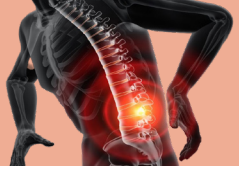
Correct Answer: 2
rationale:
The correct Answer is 40 -44 C , as we described above.

Question 142

Which of the following is the ideal temperature, in degrees, centigrade for a hydrocollator to avoid burns?

1- 40-49
2- 50-59
3- 70-79
4- 90-99

Correct Answer: 3
rationale:
Your Hydrocollators are designed to maintain a constant temperature between 160° F – 165° F (70° C – 79° C). At 160° F, your Hydrocollator's immersion is hot enough to kill any bacteria or living organisms making that an ideal temperature to maintain your hot packs.



Question 143

A Patient arrives at an outpatient clinic with an order from the physician for whirlpool and wound care to a lower Extremity wound. The therapist decides to test the temperature in the whirlpool at warm. Which of the following Setting in degrees Celsius is appropriate?

1- 27.5o Celsius
2- 35.5o Celsius
3- 49o Celsius
4- 60o Celsius

Question 144

Ice massaging is contraindicated with:

1- Inflammation
2- Raynaud's disease
3- Muscle spasm
4- Acute burn

Correct Answer: 2
rationale:
with Raynaud's, arteries to your fingers and toes become narrow and briefly limit blood supply when exposed to cold or stress. Over time, these small arteries can thicken slightly, further limiting blood flow. Cold temperatures are most likely to trigger an attack.

Question 145

Which of the following pathological conditions is contraindicated for the application of ice?

1- Spasticity
2- Muscle spasm
3- Raynaud's disease
4- Late-stage of recovery following nerve injury

Correct Answer: 3
rationale:
The correct answer is Raynaud's disease, with Raynaud's, arteries to your fingers and toes become narrow and briefly limit blood supply when exposed to cold or stress. Over time, these small arteries can thicken slightly, further limiting blood flow. Cold temperatures are most likely to trigger an attack.

Question 146

Therapist is treating a new patient with a diagnosis of lateral epicondylitis. The therapist decides to use iontophoresis driving dexamethasone. Dexamethasone is an---------- and---------- is administered with the.

1- Analgesic, anode.
2- Analgesic, cathode
3- Anti-inflammatory, anode
4- Anti-inflammatory, cathode

Question 147

A therapist should consider using a form of treatment other than moist heat application on the posterior lumbar region of all of the following patients except:

1- Patient with a history of hemophilia
2- Patient with a history of malignant cancer under the site of heat application.
3- Patient with a history of Raynaud's phenomenon
4- Patient with a history including many years of

Correct Answer: 3
rationale:
The correct answer is patient with a history of Raynaud's phenomenon,
Warm water soaks- Research has found that for individuals with Raynaud's, soaking your hands in hot water for five minutes every four hours increased blood flow for at least 2 hours and decreases the number and occurrence attacks.

Question 148

Apist routinely places ice on the ankle of a patient with an acute ankle sprain. Ice application has many therapeutic benefits. Which of the following is the body's first response to application of ice?

1- Vasoconstriction of local vessels
2- Decreased nerve conduction velocity
3- Decreased local sensitivity
4- All occur simultaneously

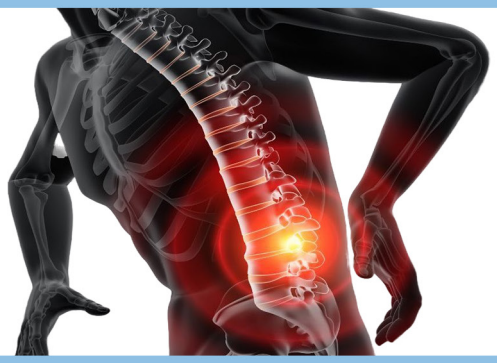
Correct Answer: 1
rationale:
Normally, the vessels that supply blood to the skin constrict or narrow in response to cold temperatures. This reaction, called "vasoconstriction," decreases blood flow to the skin, which helps to minimize heat loss from the warm blood and therefore preserve a normal internal or "core" .

Question 149

Tool provide physiotherapist with visual and auditory signals to monitor muscle recovery?

1- Biofeedback
2- EMG

Correct Answer: 2
rationale:
The correct answer is EMG ELectromyography (EMG) is a diagnostic procedure to assess the health of muscles and the nerve cells that control them (motor neurons). EMG results can reveal nerve dysfunction, muscle dysfunction or problems with nerve-to-muscle signal transmission.



Chapter

6.A Quiz Bank



ANATOMY

Question 1

Which of the following exercises is an anaerobic exercise?

1- Pull-ups
2- Push-ups
3- Weight lifting
4- Treadmill running

Question 2

A physical therapist attempts to palpate the lunate by moving his finger immediately distal to Lister's tubercle. Which wrist motion will allow the therapist to facilitate palpation of the lunate?

1- extension
2- flexion
3- radial deviation
4- ulnar deviation

Question 3

When evaluating forearm muscles, the therapist asks the patient to flex the elbow with the forearm in a neutral position. Which of the following muscles is the therapist evaluating?

1- Anconeus
2- Biceps brachi
3- Brachioradialis
4- Flexor digitorum profundus

Question 4

A 56-year-old man presents to the clinic after 1 month of right forearm pain. Physical therapist tests a muscle that does flexion, pronation and supination the forearm, and it inserts into the distal radius. Which of the following muscles is being assessed in this case?

1- Brachialis
2- Biceps brachii
3- Triceps brachii
4- Brachioradialis

Question 5

A physical therapist reviews a physician's examination of a patient scheduled for physical therapy. The examination identifies excessive medial displacement of the elbow during ligamentous testing. Which ligament is typically involved with medial instability of the elbow?

1- annular
2- radial collateral
3- ulnar collateral
4- volar radioulnar

Question 6

Which of the following muscles is responsible developing "winging of the scapula"?

1- Teres major
2- Subscapularis
3- Levator scapulae
4- Serratus anterior

Question 7

Which of the following joints of the upper limb do protraction and retraction take place?

1- Glenohumeral
2- Sternoclavicular
3- Acromioclavicular
4- Scapulothoracic interface

Question 8

Which of the following characteristics is unique to the cervical spine?

1- Flat erticular surfoces
2- Facet joints angled at 45°
3- Joints of Luschka provide lateral stability
4- Opposite movement at atlantoaxial joint

Question 9

A 28-year-old patient with quadriceps muscles weakness, during evaluation, the therapist assessed the muscle. which arises from the anterior inferior ilac spine. Which of the following is the most appropriate muscle?

1- Vastus lateralis
2- Rectus femoris
3- Vastus medialis
4- Vastus intermedius

Question 10

You are a home therapist treating a patient who underwent a total hip replacement four weeks ago. You notice that the patient arches his lumbar spine in supine. He states that it is uncomfortable and doesn't remember having the problem before. The patient is unable to maintain a comfortable supine position due to:

1- Poor abdominal strength
2- Tight hamstrings muscles
3- Tight iliopsoas muscle
4- Tight piriformis muscle

Question 11

Which of the following is the most likely cause of loss of consciousness following an electric shock?

1- Pain
2- Fright
3- Low heart rate
4- Low blood pressure



CARDIOPULMONARY

Question 1

Which of the following statements is the most appropriate for "cardiac output"?

1- Volume of blood pumped by each ventricle per beat.
2- Volume of blood in the ventricle at the end of systole.
3- Volume of blood in the ventricle at the end of diastole.
4- Volume of blood pumped by each ventricle per minute.

Question 2

The electrical signal responsible for heart contraction begins in the sinoatrial (SA) node, it is located at which of the following?

1- Top of the left atrium
2- Top of the right atrium
3- Posterior surface of the left ventricle
4- Posterior surface of the right ventricle

Question 3

In patient with coronary heart disease which of these is irreversible:

1- past history of peripheral vascular disease
2- smoking
3- hypertention
4- hyperlipidemia

Question 4

A 72 year-old is recovering at home from a myocardial infarction and percutaneous transluminal coronary angioplasty. The physical therapist, decide to use pulse oximetry to monitor his responses to exercise and activity. An acceptable oxygen saturation rate (SaO2) to maintain throughout the exercise period is:

1- 82%
2- 75%
3- 92%
4- 85%

Question 5

Which of the following is described as covering the entire muscle?

1- Actin
2- Epimysium
3- Perimysium
4- Endomysium

Question 6

Which of the following is another name of "tuberculosis" of the spine?

1- Still's disease
2- Pott's disease
3- Little's disease
4- Legg-Calvé Perthes disease

Question 7

A 62 year-old patient has chronic obstructive pulmonary disease. Pulmonary test results include all of the following except increased:

1- Functional residual capacity
2- Residual volume
3- Total lung capacity
4- FEV1/FVC ratio

Question 8

A physical therapist prepares to perform manual vibration as a means of airway clearance with a patient diagnosed with chronic obstructive pulmonary disease. When performing vibration the most appropriate form of manual contact over the affected lung segment is:

1- contact with a cupped hand.
2- contact with the entire palmar surface of the hand.
3- contact with the ulnar border of the hand.
4- contact with the distal phalanx of the middle finger.

Question 9

Which of the following is the main advantage of semi- fowler position for chest physiotherapy?

1- Efficacy
2- Better access
3- Gravity assists
4- Patient comfort

Question 10

As a general rule of prescribing exercise for any patient, the therapist must calculate each patient's target heart rate. Why is this important?

1- Determine the type of exercise.
2- Determine the intensity of exercise.
3- Determine the duration of exercise.
4- Determine the freguency of exercise.

Question 11

A physical therapist designs an exercise program for a patient rehabilitating from a lower extremity injury. The single most important factor in an exercise program designed to increase muscular strength is:

1- the recovery time between exercise sets.
2- the number of repetitions per set.
3- the duration of the exercise session.
4- the intensity of the exercise.



Question 12

A physical therapist monitors a patient's vital signs while exercising in a phase I cardiac rehabilitation program. The patient status post myocardial infarction and has progressed without difficulty while involved in the program. Which of the following vital sign recordings would exceed the typical limits of a phase I program?

1- heart rate elevated 18 beats per minute above resting level.
2- respiration rate of 25 breaths per minute.
3- systolic blood pressure decreases by 20 mmHg from resting level.
4- diastolic blood pressure less than 100 mmHg.

Question 13

An older person with diagnosis of congestive heart failure (CHF) should recognize the symptoms of exertional intolerance. You will know this client is properly informed if she can tell you these symptoms are:

1- dizziness, visual blurring especially with turns and quick movements.
2- Overwhelming weakness with difficulty in standing up and walking.
3- Severe, uncomfortable chest pain with shortness of breath.
4- Shortness of breath at rest and with limited activity, and sudden weight gain.

Question 14

The lung is inflated at all times, under normal conditions, without collapsing, because of which of the following?

1- Forced vital capacity
2- Expiratory reserve volume
3- Inspiratory reserve volume
4- Dead space volume and lung compliance

Question 15

A patient with"Buerger's disease" has been referred to aphysical therapist to assist in reducing leg pain. Which of the following goals of treatment should be planned?

1- Improving venous return
2- Establishing collateral circulation
3- Increasing strength of the calf muscles
4- Reducing muscle spasm to reduce pain

Question 16

A middle-aged man is admitted to the Stroke Unit following basal ganglia bleed. After 2 days, he is referred for chest physiotherapy. Radiological examination confirms left basal lobe collapse. Which of the following is the most appropriate auscultation location for the posterior segment of the basal lobe"?

1- Between C7-T3
2- Between L5-S1
3- Between T3-T10
4- Between 6th intercostal space

Question 17

Patients with " cardiovascular disease " usually show an improvement in exercise tolerance with physical training. This improvement is a result of which of the following ?

1- Reduction of the maximum oxygen consumption .
2- ecreased oxygen uptake by the skeletal muscles or both.
3- Reduction of the amount of oxygen consumed at maximal effort .
4- Increased ability to use oxygen to deliver energy for physical work.

Question 18

Which of the following is the main aim of acute respiratory distress syndrome treatment?

1- Pain relief
2- Patient comfort
3- Reduction in respiratory rate
4- Removal of retained secretions

Question 19

A "cerebral embolism" can cause serious cerebrovascular accidents. Which of the following is the main site of origin for cerebral emboli?

1- Lungs
2- Heart
3- Superficial leg veins
4- Cervical lymphatic system

Question 20

Which of the following is the most likely cause of difficult expiration in chronic obstructive airway disease?

1- Reduced lung elasticity
2- Rapid shallow breathing
3- Reduced airway diameter
4- Expiratory muscles fatigue

Question 21

A patient presents with a large plantar ulcer that will be debrided in the whirlpool. The foot is cold, pale, and painless. The condition that would most likely result in this clinical presentation is:

1- Acute arterial insufficiency
2- chronic arterial insufficiency
3- Chronic venous insufficiency
4- Deep venous thrombosis

Question 22

A 65-year-old man with "Congestive Heart Failure" is referred to the Physical Therapy Department for gait training. Which assessment test is the most reliable to calculate exercise capacity?

1- Chair-stand
2- 6-minute walk Time up and go
3- Physical performance



ORTHOSI & PROSTHESIS

Question 1

During knee extension, there are fundamental arthokinematic movements that occur inside the joint. Which of the following describes the "glide/slide fundamental arthokinematic movements"?

1- Multiple points on 1 articular surface contact single points on another surface.
2- Multiple points on 1 articular surface contact multiple points on another surface.
3- Single point on 1 articular surface contacts multiple points on another surface.
4- Single point on 1 articular surface rotates on a single point on another surface.

Question 2

The main goal of an ergonomics program is to reduce which of the follwing?

1- Risk of fall and trauma
2- Cardiovascular diseases
3- Musculoskeletal disorders
4- Neurological and psychological disorders

Question 3

Hip joint stability depends on which of the following?

1- On the bony articulation.
2- On bony articulations, then ligaments and muscles.
3- On ligaments, then bony articulations and muscles.
4- On muscles, then ligaments and bony articulations.

Question 4

Patient wearing a below knee prosthesis, complain of pain during walking response of therapist will be:

1- keep wearing
2- refer to physician
3- refer to orthotics or prosthesis specialist
4- say him it's normal initially

Question 5

Which of the following statements defines orthosis?

1- Device that replaces an absent body part.
2- Device that supports an existing body part.
3- Practitioner who designs and fabricates prosthesis.
4- Field of design and fabrication of the devices to replace a body part.

Question 6

One joint hip flexor contracture result:

1- no effect on lumber
2- anterior pelvic tilting with kyphosis
3- posterior pelvic tilting
4- anterior pelvic tilt with lordosis

Question 7

In cases of fractures of C4 and C5, the maximum stabilization of the cervical spine is most lkely achieved with which of the following?

1- Soft collar
2- Hard collar
3- Halo orthosis
4- Milwaukee orthosis

Question 8

A patient was recently referred to a therapist with a new above-knee prosthesis for gait training. The patient complains of severe pain at the distal end of the stump after the first day of training. Which of the following is the most appropriate therapist response?

1- Direct the patient to try to adjust the socket.
2- Direct the patient to make an appointment with the physician.
3- Direct the patient not to wear the prosthesis until modifications are made.
4- Direct the patient to wear the prosthesis at all times because the body will eventually get used to it.

Question 9

A physical therapist reviews the medical chart of a patient diagnosed with a fracture of the lower thoracic spine. The chart indicates the patient has worn an anterior control throaco-lumbar-sacral-orthosis (TLSO) for eight weeks. What is the primary purpose of the anterior control TLSO?

1- prevent thoracic flexion
2- prevent thoracic extension
3- prevent lumbar flexion
4- prevent lumbar extension

Question 10

Following major surgery of the right hip, a patient ambulates with a Trendelenburg gait. Examination of the right hip reveals abductor weakness and ROM limitations in flexion and external rotation. As a part of the intervention, the therapist opts to include functional electrical stimulation to help improve the gait pattern. Stimulation should be initiated for the:

1- Right abductors during swing on the right
2- Right abductors during stance on the right
3- Left abductors during stance on the right
4- Left abductors during swing on the right

Question 11

Which of the following will be resulted when a physical therapist advises a patient. who uses a cane to hold it with the contra-lateral hand?

1- Increasing patient falling risk.
2- Reduce energy expenditure in the normal person.
3- Reduce the floor reaction force on the opposite foot.
4- Increase the floor reaction force on the opposite foot.



Question 12

A 57-year-old patient had right side hemiplegia 5 months ago. Examination reveals that the physical therapist finds the patient has difficulty in weight-bearing on the right leg and, as a result, shortens the end of the stance phase of the affected leg and directly goes into swing phase. Which of the following is the most likely cause of the problem?

1- Weakness or contracture of hip flexors.
2- Weakness or contracture of hip extensors.
3- Weakness or contracture of the dorsiflexors.
4- Spasticity or contracture of the plantar flexors.

POSTURE & AXES

Question 1

Which of the following shoulder motions is easily visualized in the frontal plane?

1- Circumduction
2- Flexion and extension
3- Abduction and adduction
4- External and internal rotation

KNOWLEDGE

Question 1

Which of the following body mass index values is considered overweight?

1- 18-20
2- 21-24
3- 25-29
4- 30-34

Question 2

A physical therapist attempts to quantify the amount of assistance a patient needs to complete a selected activity. Categories of assistance include maximal, moderate, minimal, stand-by or independent. This type of measurement is best classified as:

1- interval
2- nominal
3- ordinal
4- ratio

Question 3

A patient management model is a comprehensive approach that is designed to guide the therapist in the management of the patient. Which of the following refers to"the projection of the optimal level and time frame for improvement"?

1- Prognosis
2- Evaluation
3- Examination
4- Intervention

Question 4

A frozen shoulder complains of a limited ability to reach his hand into his back pocket. "Which of the following is the most appropriate description"?

1- Disability
2- Activity limitation
3- Participation restrictions
4- Impairment in body functions

Question 5

Which of the following factors is classified as a yellow flag?

1- Psychosocial factors.
2- History of carcinoma.
3- Bladder and bowel malfunction.
4- Neurological signs and unexplained weight loss.

Question 6

A 24-year-old man comes to the clinic with right knee pain that started 3 weeks ago. The physical therapist applies a measurement of the ground reaction force of the knee joint. Where does the ground reaction force pass at heel strike during normal gait?

1- Lateral
2- Central
3- Anterior
4- Posterior

Question 7

An athlete comes to the Physiotherapy Department with a right knee injury is undergoing a test to determine the nature of the injury. The patient is prone with the knee flexed at a right angle. The patient's symptoms are worse when the physiotherapist rotates the tibia medially and laterally while maintaining a distraction force through the knee joint.
Which of the following do the symptoms indicate?

1- Capsular injury
2- Meniscal injury
3- Muscular injury
4- Ligamentous injury

Question 8

A 72 year-old woman is being treated for depression following the death of her husband. She is currently taking antidepressant medication (tricyclics) and has a recent history of a fall. You suspect the precipitating cause of the fall can be attributed to side effects of her medication resulting in:

1- Cardiac arrhythmias
2- Dyspnea
3- hyperalertness
4- Postural hypotension



Question 9

An 82 year-old patient and his caregivers should understand the common side effects of the medication that he is taking. He is continually in and out of congestive heart failure and has been taking digitalis (Digoxin) to improve his heart function. You will know he and his caregivers understand the adverse side effects of this medication if they tell you they will contact the patient's physician if he demonstrates:

1- confusion and memory loss.
2- Involuntary movements and shaking.
3- Slowed heart rate.
4- Weakness and palpitations.

Question 10

A 92 year-old woman presents with hot, red, and edematous skin over the shins of both lower extremities. She also has a mild fever. The MOST likely cause of her symptoms is:

1- Cellulitis
2- Dermatitis
3- Herpes simplex infection
4- Scleroderma

Question 11

A physical therapist prepares to administer the Berg Balance Scale to a patient rehabilitating from a cerebrovascular accident. Which of the following tools is considered a necessary piece of equipment when administering this outcome measure?

1- reflex hammer
2- goniometer
3- stopwatch
4- stethoscopie

Question 12

A physical therapist attempts to obtain information on the ability of noncontractile tissue to allow motion at a specific joint. Which selective tissue tension assessment would provide the therapist with the most valuable information?

1- active range of motion
2- active-assistive range of motion
3- passive range of motion
4- resisted isometrics

POSTURE & AXES

Question 1

A patient suffers from burns, which include the epidermis layer of the skin and extend to the upper layer of the dermis. Which of the following should be the classification of this burn?

1- Sub-dermal
2- Full-thickness
3- Deep partial-thickness
4- Superficial partial-thickness

ENDOCRINE

Question 1

Which of the following is the most likely cause pregnancy gait including increased external hip rotation?

1- Pain
2- Cosmesis
3- Better cadence

Question 2

A 27-year-old woman who is a 3-month pregnant, presented at the clinic. She has experienced multiple episodes of high blood pressure and was recently diagnosed with gestational diabetes. Her hemoglobin is well within normal limits. Her gynecologist is interested in controlling her diabetes with regular exercise. Which of the following is the most likely contraindication to exercise in pregnancy?

1- Diastasis recti
2- Placenta previa
3- Severe anemia
4- Gestational diabetes

Question 3

A 33-year-old man is undergoing conditioning rehabilitation exercises to increase the fitness level. What is the expected autonomic response during the exercise?

1- Active sympathetic nervous system to increase the availability of oxygen to the muscles.
2- Active parasympathetic nervous system to increase the availability of oxygen to the muscles.
3- Active sympathetic nervous system to increase the availability of oxygen to the digestive system.
4- Active parasympathetic nervous system to increase the availability of oxygen to the digestive system.

Question 4

A 60-year-old obese, hypertensive, and diabetic patient was referred for a physical fitness program and reduction of body weight. Which of the following is the most appropriate fitness program?

1- 15-minute fast running
2- 50-minute slow running
3- 20-minute brisk walking
4- 30-minute brisk walking



ORTHOPEDIC

Question 1

A physical therapist working on an oncology unit reviews the medical chart of a patient prior to initiating an exercise program. The patient's cell counts are as follows: hematocrit 24 ml/dL, white blood cells 2,500 x 103 mm3, platelet 80,000 mm3, hemoglobin 7 gm/dL. Based on the patient's blood counts, which of the following would be the most accurate statement regarding the patient's allowable exercise level?

1- no exercise is allowed
2- light exercise is allowed
3- active exercise is allowed
4- resistive exercise is allowed

Question 2

A six month-old child was referred to physical therapy for right torticollis. The MOST effective method to stretch the muscle is by positioning the head and neck into:

1- extension, left side-bending, and right rotation.
2- extension, right side-bending, and left rotation.
3- flexion, left side-bending, and left rotation.
4- flexion, right side-bending, and left rotation.

Question 3

A right hemiplegic patient, who had a left CVA 3 months ago, presented for evaluation. The right upper limb power was +3/5 and lower limb power was -3/5. Which of the following is the most appropriate resisted training to include?

1- Isometric
2- Isokinetic
3- Open kinematic
4- Manual resisted

Question 4

Which of the following is the main aim of rehabilitation after "patellofemoral dislocation"?

1- Aerobics
2- Stretching
3- Muscle strengthening
4- Reduce the chances of recurrence

Question 5

A 45-year-old patient presented with a 3-week history of severe knee pain. The pain is aggravated by walking. Examination revealed of the patient's knee, there was an empty end feel of the range of motion (see lab results). Normal Value Result 30 x 10 Test Total leukocyte count 4 - 11X 10 S/L Which of the following is the most likely diagnosis?

1- Meniscal tear
2- Hamstring strain
3- Infective arthritis
4- Degenerative arthritis

Question 6

A physical therapist performs the talar tilt test on a 22-year old female rehabilitating from an inversion ankle sprain. Which ligament does the talar tilt test examine?

1- anterior tibiofibular
2- calcaneofibular
3- deltoid
4- posterior tibiofibular

Question 7

Which of the following is another name of "fracture of the distal end of the radius"?

1- Colles
2- Galeazzi
3- Bennett's
4- Monteggia

Question 8

On which of the following nerves cubitus valgus deformity causes a stretch?

1- Ulnar
2- Radial
3- Median
4- Peroneal

Question 9

A 2-year-old child, spastic diplegic, is brought to the Paediatric Rehabilitation Unit with club foot deformity. Which supporting device should be recommended with exercises?

1- Pavlov harness
2- Milwaukee brace
3- Ankle foot orthosis
4- Denis-Browne splint

Question 10

A 42-year-old woman presented to the clinic with right foot pain 2 weeks ago. During the assessment, she says she had severe foot pain in the morning when she started walking after waking up. Which of the following is the most likely diagnosis?

1- Ankle sprain
2- Stress fracture
3- Plantar fasciitis
4- Ankle impingement

Question 11

In degenerative joint disease which is not occure?

1- increased wih wight bearing on the joint
2- geadual onset
3- stiffness at evening
4- increaser after prolonged period of inactivity



Question 12

A 5-year-old child presents for pain management of congenital anterior dislocation of her hip. Which of the following is the most common cause of anterior dislocation?

1- Increased angle of neck and shaft of femur
2- Decreased angle of neck and shaft of femur
3- Increased angle of femoral neck and condyles of femur
4- Decreased angle of femoral neck and condyles of femur

Question 13

A 40-year-old woman with hip dysfunction is referred to the physical therapy. The physical therapist assessed the patient and found tightness in the gluteus medius muscle. Which of the following is recommended to stretch this muscle?

1- Adduction
2- Extension
3- Abduction
4- External rotation

Question 14

A middle-aged diabetic patient enters into the Physiotherapy Clinic with right lower limb pain. The patient complains of deep pain in the right trunk, hip and knee joint while walking and stairs climbing. However, he feels comfortable in sitting and lying position. Which of the following is the most appropriate clinical prediction?

1- Hip joint pathology
2- Knee joint pathology
3- Sacroiliac joint dysfunction
4- Lumber spine dysfunction

Question 15

A 26-year-old weightlifter presented with severe low back pain, radiating to his loft lower limb. He developed the pain during competition 3 days ago. He had a positive straight leg raise test with pain aggravated by spinal flexion (see report) Lumbosacral X-ray: Unremarkable. Which of the following is the most likely cause of pain?

1- Muscular
2- Sacroiliitis
3- Discogenic
4- Zygapophyseal

Question 16

A 40-year-old man presents following a fall. He reports severe radiating pain on the "lateral side of his thigh "for 3 days "Which of the following is the most likely diagnosis"?

1- Ischial bursitis
2- Piriformis syndrome
3- Trochanteric bursitis
4- Quadriceps contusion

Question 17

A 66-year-old patient referred to the Physiotherapy Clinic for treatment and management of a long standing problem in the left knee. According to history, the patient's knee remained stiff and painful for about half an hour, along with aches at the end of the day.
Which of the following is the most likely cause of the problem?

1- Osteoarthritis
2- Torn meniscus
3- Ruptured anterior cruciate
4- Torn medial collateral ligament

Question 18

A 38-year-old patient complains of right shoulder pain. At the Physical Therapy Clinic, a forceful flexion of both shoulder and elbow aggravates the symptoms. Which of the following is the most likely diagnosis?

1- Bicipital tendonitis
2- Subdeltoid bursitis
3- Adhesive Capsulitis
4- Anterior Capsule Strain

Question 19

A diagnosis of bicipital tendinitis has been made following an evaluation of a patient with shoulder pain. The BEST shoulder position to expose the tendon of the long head of the biceps for application of phonophoresis would be:

1- Abduction
2- external/lateral rotation and extension
3- Horizontal adduction
4- internal/medial rotation and abduction

Question 20

A 47-year-old woman comes to the clinic for assessment of pain in her right shoulder since 2 weeks ago. During the assessment, the pain is aggravated when the patient initiates the abduction of the arm. Which of the following muscles is most likely causing this pain?

1- Teres minor
2- Teres major
3- Infraspinatus
4- Supraspinatus

Question 21

A patient is recovering from stroke and demonstrates good recovery in his lower extremity (out-of-synergy movement control). Timing deficits are apparent during gait. Isokinetic training can be used to improve:

1- initiation of movement
2- rate control at varying movement speeds
3- rate control at slow movement speeds
4- both reaction and movement times



Question 22

A patient who is 3 months post CVA is being treated in physical therapy for adhesive capsulitis of the right shoulder, Today, the patient complains of new symptoms including burning pain in the right upper extremity that is increased by the dependent position along with lowered pain threshold and heightened sensitivity to light touch. The right hand is mildly edematous and the skin is dry and warm to touch. The intervention that should be AVOIDED in this case:

1- Stress loading activities with weight bearing in the affected limb
2- Passive manipulation of the shoulder
3- Positional elevation, compression, and gentle massage
4- Active assistive ROM exercises of the shoulder

Question 23

Which of the following types of stretching is associated with repeated movements and momentum to gain a desired range of motion?

1- Static
2- Ballistic
3- Dynamic
4- Therapeutic

Question 24

Which of the following stretching techniques is associated with the application of short-duration recurrent stretching?

1- Cyclic
2- Static Ballistic
3- Progressive

Question 25

A 40-vear-old patient with right above-knee amputation due to peripheral vascular disease comes to the Rehabilitation Clinic with neglected hip flexor contracture.
which of the following is the ideal way to stretch this group of muscles?

1- Hold-relax techniques
2- Active stretching techniques
3- Passive stretching techniques
4- Resisted movement for the agonist

Question 26

A patient rehabilitating from a radial head fracture is examined in physical therapy. During the examination, the physical therapist notes that the patient appears to have an elbow flexion contracture. Which of the following would not serve as an appropriate active exercise technique to increase range of motion?

1- contract-relax
2- hold-relax
3- maintained pressure
4- rhythmic stabilization

Question 27

Which of the following is a term for when the muscle cannot simultaneously contract maximally across 2 joints?

1- Active inversion
2- Passive inversion
3- Active insufficiency
4- Passive insufficiency

Question 28

A physical therapist works with a patient status post stroke on a mat program. The therapist assists the patient in lateral weight shifting activities while positioned in prone on elbows. Which therapeutic exercise technique would allow the patient to improve dynamic stability with this activity?

1- alternating isometrics
2- approximation
3- rhythmic initiation
4- timing for emphasis

Question 29

If the proximl bone of joint fixed the distal part of joint move in sequence so this is:

1- ACL Training
2- Closed Chain Ex
3- Open Chain Ex
4- None Of above

Question 30

A 51-year-old patient with “right hemiplegia” was referred to the Outpatient Physical Therapy Clinic with spasticity of plantar flexors affecting walking. Which of the following is the most appropriate exercise training?

1- Balance
2- Anaerobic
3- Endurance
4- Strengthening



PEDIATRIC

Question 1

What muscles should the therapist expect to be affected when the therapist is evaluating children with "Erb's palsy"?

1- Internal rotators
2- Popliteus muscle
3- Trapezius muscle
4- Elbow flexors and supinator

Question 2

physical therapist identifies that an infant is unable to roll from prone to supine. Which reflex could interfere with the infant's ability to roll?

1- asymmetrical tonic neck reflex
2- Moro reflex
3- Galant reflex
4- symmetrical tonic neck reflex

Question 3

A physical therapist assesses a one-month-old infant. During the treatment session the therapist strokes the cheek of the infant causing the infant to turn its mouth towards the stimulus. This action is utilized to assess the:

1- Moro reflex
2- rooting reflex
3- startle reflex
4- righting reflex

Question 4

A physical therapist completes a developmental assessment on a 7-month old infant. Assuming normal development, which of the following reflexes would not be integrated?

1- asymmetrical tonic neck reflex
2- Moro reflex
3- Landau reflex
4- symmetrical tonic neck reflex

Question 5

One of the most common congenital anomalies that is discovered immediately after birth is "spina bifida". which of the following is characteristic of spina bifida?

1- Enclosure of pia mater
2- Enclosure of vertebral body
3- Hyperextension of knee joint
4- Lateral curvature of the spine

Question 6

Which of the following is the mean age for a baby to develop the ability to support the chest in a prone position?

1- 1 month
2- 3 months
3- 4 months
4- 6 months

Question 7

An appropriate fine motor behavior that should be established by 9 months of age would be the ability to:

1- build a tower of 4 blocks.
2- hold a cup by the handle while drinking.
3- pick up a raisin with a fine pincer grasp.
4- Transfer objects from one hand to another.

Question 8

Which of the following is the most common age, in months for a child to sit with a straight back?

1- 3
2- 4
3- 5
4- 6

Question 9

Which of the following ages, in months, do most children display a hand preference?

1- 06
2- 12
3- 18
4- 24

Question 10

Which of the following is associated with a reduced range of motion secondary to spasticity?

1- Periarticular
2- Fibrotic contracture
3- Myostatin contracture
4- Pseudomyostatic contracture

Question 11

A physical therapy plan of care for a newborn with Erb-Klumpke Palsy would NOT include:

1- Age appropriate movements of the upper extremity.
2- Gentle ROM after immobilization.
3- Partial immobilization of limb across abdomen.
4- Splinting the shoulder in abduction and internal rotation.



PNF

Question 1

Which of the following is an objective of Proprio Neuromuscular Facilitation techniques?

1- Peripheral nerve stimulation
2- Improving subject righting reflex
3- Training the patient's equilibrium reaction
4- Placing specific demands to secure desired responses

Question 2

Which of the following grades occurs egains resistance duringa muscle testing procedure?

1- Poor
2- Good
3- Trace
4- Normal

ELECTRO & HYDRO

Question 1

A 25-year-old man with left lateral epicondylitis is treated in the clinic. Low-level laser therapy is used by the physical therapist to treat this patient. "What are the typical wave lengths of low-level laser therapy"?

1- 100-500 nm
2- 600-1000 nm
3- 1500-2000 nm
4- 2500-3000 nm

Question 2

A 35-year-old man presents with a 6-month history of trigeminal neuralgia. Which of the following is the most appropriate pain management?

1- Laser
2- Ultrasound
3- Ultraviolet rays
4- Shortwave diathermy

Question 3

Which of the following is most likely to benefit from laser?

1- Cancer tissues
2- Infected tissues
3- Epileptic patients
4- Post-traumatic joint disorder

Question 4

A 10 year-old presents with pain (4/10) and limited knee ROM (5/95degree) following surgical repair of the medial collateral ligament and anterior cruciate ligaments. In this case, The modality which can be used with precaution is:

1- premodulated interferential current.
2- continuous shortwave diathermy.
3- high rate transcutaneous electrical stimulation.
4- low dose Ultrasound.

Question 5

A 63-year-old woman referred to the Physical Therapy Clinic for post-operative total knee arthroplasty rehabilitation. The physical therapist decided to use hydrotherapy as part of the rehabilitation . What is the feature that makes the water suitable to decrease the weight on the knee?

1- Buoyancy
2- Viscosity
3- Thermodynamic
4- Hydrostatic pressure

Question 6

Which of the following is most commonly used to reduce pain?

1- Biofeedback
2- Electromyography
3- Electrical muscle stimulation
4- Transcutaneous nerve stimulation

Question 7

A 56-year-old man comes to the Physical Therapy Clinic with low back pain for more than 3 months. During the assessment, the physical therapist decides to reduce the patient's pain by using an electric heating pad. Which type of heat transfer is done between pad and skin?

1- Radiation
2- Convection
3- Conversion
4- Conduction

Question 8

The use of a physical agent, heat transfer, or flow of the thermal energy depends on the application technique used by the physical therapist. What is the form of heat transfer when a cold pack is applied?

1- Radiation
2- Conduction
3- Convection
4- Conversion



Question 9

A 26-year-old patient with chronic inflammatory pelvic pain has been given a course of antibiotics with non-steroidal anti- inflammatory medications, but there has been no significant relief. Which of the following is the most appropriate method of shortwave diathermy application?

1- Co-planar
2- Cross-fire
3- Mono-polar
4- Contra-planar

Question 10

Which medical condition is most commonly prevented by using goggles while receiving ultraviolet therapy?

1- Infection
2- Painful eyes
3- Loss of vision
4- Conjunctivitis

Question 11

Your examination reveals muscle spasms of the deep hip rotators, which are compressing the sciatic nerve and producing pain in the posterior hip region. The MOST effective setting of ultrasound in this case is:

1- 1 MHz continuous at 1.0 W/cm2
2- 1 MHz pulsed at 1.0 W/cm2
3- 3 MHz continuous at 1.0 W/cm2
4- 3 MHz pulsed at 1.0 W/cm2

Question 12

A 42-year-old woman recently diagnosed with multiple sclerosis presents to a Physical Therapy Clinic. Which of the following is the most likely cause of "pseudo exacerbation" of the symptoms?

1- Hot bath
2- Strength training
3- Application of ice packs
4- Slightly increased intake of fluids

Question 13

A 58-year-old woman presented with severe hip pain. She has been taking steroids for her dermatomyositis. A CT scan hip confirmed avascular necrosis of femoral neck with an impacted fracture. There is a 2 cm leg-length discrepancy. Which of the following is the most appropriate intervention to reduce hip pain?

1- Shoe raise
2- Walking stick
3- Shortwave diathermy
4- Modified seating system

Question 14

A 45-year-old man sustained a gun shot wound to the knee. He developed foot drop and loss of sensation on the dorsum of the foot. Nerve conduction studies and electromyography confirmed neurotmesis, the common peroneal nerve at the knee with denervation potentials observed in tibialis anterior muscle. Which of the following is the most appropriate treatment for restoration of muscle power?

1- Faradic current
2- Interferential current
3- Interrupted direct current
4- Transcutaneous nerve stimulation

Question 15

A patient with Bell's palsy was referred to the rehabilitation department with a request to start electrical stimulation of the facial muscle. The patient is referred back to the physician after one month of treatment with complications of prolonged electrical stimulation. Which of the following is a common complication of prolonged electrical stimulation?

1- Burn
2- Irritation of the skin
3- Fibrosis of the affected muscles
4- No side effect from prolonged electrical stimulation

Question 16

A patient with a purulent venous insufficiency ulcer near the left medial malleolus is seen at home by a physical therapist. The MOST important goal/intervention for the therapist to try to achieve with this patient is:

1- elastic wraps and daily exercise
2- daily warm water baths and exercise
3- daily walking for 30-60 minutes
4- compression therapy with exercises

Question 17

A physical therapist prepares a patient education program for an individual with chronic venous insufficiency. Which of the following would not be appropriate to include in the patient education program?

1- wear shoes that accommodate to the size and shape of your feet.
2- observe your skin daily for breakdown.
3- wear your compression stockings only at night.
4- keep you feet elevated as much as possible throughout the day.



SPORTS

Question 1

A 53-year-old patient is noted to have a "baseball finger" injury. Which of the following is the actual injury?

1- Fracture of proximal phalanx of finger.
2- Fracture of the metacarpal of the finger.
3- Avulsion of the extensor tendon from the base of distal phalanx.
4- Tenosynovitis of the flexor digitorum profundus tendon to the involved finger.

Question 2

A weight lifter exhibits marked hypertrophy after embarking on a strength training regime. Hypertrophy can be expected to occur following at least:

1- 1-2 weeks of training
2- 2-3 weeks of training
3- 3-4 weeks of training
4- 6-8 weeks of training

Question 3

A 25-year-old man comes to the clinic with left knee pain before one week ago. During the assessment, there is a positive sign of the valgus stress test. Which of the following is the most likely diagnosis?

1- Medial meniscus injury
2- Lateral meniscus injury
3- Medial collateral ligament injury
4- Lateral collateral ligament injury

Question 4

A 16-year-old adolescent presented with foot slap on the right side after a fracture of the proximal fibula. Which of the following nerves is most likely injured?

1- Tibial
2- Femoral
3- Obturator
4- Common peroneal

NEUROREHABILITATION

Question 1

Which of the following transforms chemical energy into mechanical energy?

1- Bone
2- Stomach
3- Small intestine
4- Skeletal muscle

Question 2

Which of the following is the appropriate label for "7" in this schematic figure of the cross section of the spinal cord (see image)?

1- Grey matter
2- Ventral root
3- Spinal nerve
4- Dorsal root ganglion

Question 3

A patient is referred to physical therapy with a C6 nerve root injury. Which of the following clinical findings would not be expected with this type of injury?

1- diminished sensation on the anterior arm and index finger.
2- weakness in the biceps and supinator.
3- diminished brachioradialis reflex.
4- paresthesias of the long and ring fingers.

Question 4

Which of the following is the most appropriate position for stretching of radial nerve?

1- Shoulder girdle elevation, full elbow flexion, wrist extension, forearm supination.
2- Shoulder girdle elevation, abduct shoulder, flexion of elbow, lateral rotate the arm, pronate forearm.
3- Shoulder girdle depression, abduct shoulder, elbow extension, laterally rotate the arm and supinate forearm.
4- Shoulder girdle depression, slightly abduct the shoulder, extend the elbow, medially rotate the arm, pronate forearm.

Question 5

A patient employed in a machine shop is referred to physical therapy with a diagnosis of carpal tunnel syndrome. The patient indicates that he is scheduled for a diagnostic test that may help to confirm the diagnosis. Which of the following electrodiagnostic tests would be the most appropriate?

1- electroencephalography
2- evoked potentials
3- nerve conduction velocity
4- electromyography



Question 6

In carpal tunnel syndrome the most affected ms is:

1- flexor degotorum profundus
2- flexor carpi ulnaris
3- adductor pollicis longus

Question 7

With a traction injury to the anterior division of the brachial plexus you would expect to see weakness of the elbow flexors, wrist flexors and forearm pronators. You would also expect to find additional weakness in:

1- forearm supination
2- lateral rotation of the shoulder
3- thumb abduction
4- wrist extension

Question 8

A physical therapist examines a patient diagnosed with post-polio syndrome. Which of the following areas is the least likely to be affected based on the patient's diagnosis?

1- strength
2- sensation
3- endurance
4- functional mobility

Question 9

The contractile elements of intrafusal muscle fibers are supplied by which of the following?

1- Interneurons
2- Ia afferent neurons
3- Alpha motor neurons
4- Gamma motor neurons

Question 10

The problems associated with ankylosing spondylitis in its early stages can best be managed in physical therapy by:

1- postural education
2- stretching of scapular stabilizers
3- Pain management
4- joint mobilization

Question 11

To prepare a patient with an incomplete T12 paraplegia for ambulation with crutches, the upper quadrant muscles that would be MOST important to strengthen include the:

1- deltoid, triceps, and wrist flexors.
2- lower trapezius, latissimusdorsi, and triceps.
3- Middle trapezius, latissimus, dorsi and triceps.
4- Upper trapezius, rhomboids, and levator scapulae.

Question 12

A 42-year old female is admitted to a rehabilitation hospital after sustaining a stroke. During the examination the physical therapist identifies significant sensory deficits in the antero-lateral spinothalamic system. Which sensation would be most affected?

1- barognosis
2- kinesthesia
3- graphesthesia
4- temperature

Question 13

A physical therapist examines the gait of a 62-year-old male with peripheral neuropathy. The therapist observes that the patient's right foot has a tendency to slap the ground during the loading response. The observation can best be explained by weakness of the:

1- iliopsoas
2- tibialis anterior
3- tibialis posterior
4- gastrocnemius

Question 14

A 40-year-old patient complaining of radiating pain in his left leg with a tingling sensation in the anterior part of his leg. Visual examination showed that he is walking with a foot drop. Which of the following nerves is the most likely cause of these symptoms?

1- Sciatic nerve
2- L1 nerve root
3- L5 nerve root
4- Deep peroneal nerve

Question 15

Which of the following nerve injury blocks the nerve conduction without degeneration?

1- Neuropathy
2- Neurapraxia
3- Axonotmesis
4- Neurotmesis

Question 16

A 70-year-old woman complains of a weak bladder. She reports occasional leakage of urine with coughing and sneezing over the past year. She has 6 to 7 episodes of urge incontinence per day. "Physical examination reveals she has a large cystocele and reduced rectal sphincter tone ". Which of the following is the most appropriate management?

1- Surgical
2- Stretching
3- Biofeedback
4- Hormone replacement therapy



Question 17

A therapist evaluating a patient with a neurological disorder assesses that the patient has delayed relaxation of previously contracted muscles associated with muscle stiffness. Which of the following is the most likely diagnosis?

1- Myotonia
2- Myopathies
3- Myasthenia gravis
4- Motor neuron disorders

Question 18

A 60-year-old hemiplegic patient is treated in the outpatient Physical Therapy Clinic, the therapist recommended splint to be used with extended elbow joint. Which of the following is the aim of splint application?

1- Reducing spasticity and pain.
2- Increase strength of elbow flexors.
3- Decreasing flexibility of olbow flexors.
4- Enhancing the proprioception of elbow joint.

Question 19

A therapist starts on the bed training program of a stroke patient. Left upper limb and lower limb power 1/5 and GCS is 8/15. Which of the following positions result in more complications?

1- Sitting in a chair with the head and trunk in the midline.
2- Lying on supine position with arm extended abducted .
3- Lateral side flexion of the head and trunk toward the affected side.
4- Lying on the sound side with straight trunk and arm extended and resting on a pillow.

Question 20

Which of the following positions is associated with the lowest amount of muscle work?

1- Hanging
2- Standing
3- Lying prone
4- Lying supine

Question 21

A physiotherapist receives a referral for a patient. The referring physician has outlined all clinical findings in the referral, one of which is “tremor”. What does this term mean?

1- Involuntary rhythmic oscillation of eyeball .
2- Fine type of repetitive involuntary movement.
3- Inability to perform purposeful voluntary movements .
4- Involuntary repetitive movement characterized by twisting .

Question 22

A gait pattem of narrow base support, short step length bilaterally, and decreased trunk rotation is observed in a patient with pathology in which of the following?

1- Basal ganglia
2- Motor area 4
3- Lumbar spine
4- Cerebral cortex

Question 23

A physical therapist completes a vertebral artery test on a patient diagnosed with a cervical strain. Which component of the vertebral artery test is most likely to assess the patency of the intervertebral foramen?

1- rotation
2- lateral flexion
3- flexion
4- extension

Question 24

A motor unit is the smallest functional unit of the motor system, and it consists of which of the following?

1- Motor neuron, its axon, and all the muscle fibres innervated by the axon.
2- Sensory neuron, its axon, and all the muscle fibres innervated by the axon.
3- Neurons in the white matter of the spinal cord, their axons, and all the muscle fibres innervated by these neurons.
4- Neurons in the posterior horm of the spinal cord, their axons, and all the muscle fibres innervated by these neurons.

Question 25

A physician indicates that a patient rehabilitating from a cerebrovascular accident has significant perceptual deficits. Which anatomical region would most likely be affected by the stroke?

1- primary motor cortex
2- sensory cortex
3- basal ganglia
4- cerebellum

Question 26

A 46-year-old patient with bradykinesia was referred to the Physical Therapy Clinic for treatment. Which type of exercise is required for this case?

1- Walking
2- Stretching
3- Strengthening
4- Range of motion

Question 27

A 22-year-old male status posttraumatic brain injury receives physical therapy services in a rehabilitation hospital. The patient is presently functioning at Rancho Los Amigos Level VI. The patient has progressed well in therapy, however, has been bothered by diplopia. Which treatment strategy would be the most appropriate to address diplopia?

1- provide verbal and non-verbal instructions within the patient's direct line of sight.
2- place a patch over one of the patient's eyes.
3- ask the patient to turn his head to one side when he experiences diplopia.
4- instruct the patient to carefully focus on a single object.

Question 28

A Image(s) A 34-vear-old patient with long-standing low back pain is referred to a Physiotherapy Clinic for treatment and management. Examination reveals skin sensory loss on the lateral aspect of the foot as shown (see Image). The sensory loss is most ikely due to the involvement of which of the following nerve roots?



1- L2
2- L3
3- L4
4- S1

Question 29

30-year-old man injured in a car accident has been diagnosed with spinal cord injury at L2,3 and 4 spinal cord levels. Which of the following tendon jerks is expected to be lost during the physical evaluation?

1- Knee
2- Biceps
3- Triceps
4- Brachioradialis

Question 30

A physiotherapist receives a referral for a patient. The referring physician has outlined all clinical findings in the referral, one of which is "aphasia". What does this term mean?

1- Double vision
2- Difficulty swallowing
3- Slowness of movement
4- Speech related problems

Question 31

A 59 year-old ex-machinist demonstrates significant age-related hearing loss, presbycusis. When trying to communicate with this patient you would NOT suspect:

1- Bilateral hearing loss, at all frequencies since he has had this problem for a number of years.
2- Decreased language comprehension.
3- Poor auditory discrimination
4- unilateral hearing loss.

Question 32

A physiotherapist receives a referal for a patient The referring physician has outlined all clinical findings in the referral, one of which is "agnosia". What does this term mean?

1- Absence of sensation
2- Slowness of movement
3- Loss of ability to perceive objects
4- Loss of coordination in movement

Question 33

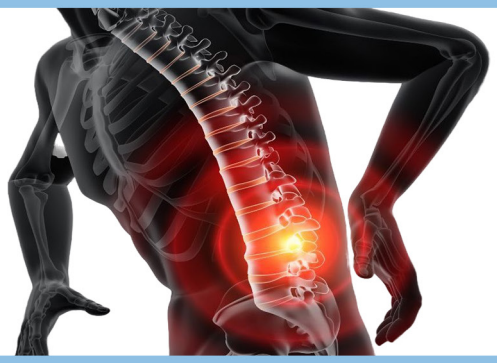
A 13-year-old female diagnosed with cerebral palsy is referred to physical therapy. The patient exhibits slow, involuntary, continuous writhing movements of the upper and lower extremities. This type of motor disturbance is most representative of:

1- spasticity
2- ataxia
3- hypotonia
4- athetosis

Question 34

A 30-year-old patient has a fracture-dislocation of the 4th and 5th lumbar vertebrae (L4 and L5) as a result of a car accident. Which of the following complications easily result from this type of fracture?

1- Spastic diplegia
2- Flaccid paraplegia
3- Spastic paraplegia
4- Flaccid quadriplegia



Chapter

6.B Knowledge



Question 1

What is the body mass index for someone who is overweight?

1. 25 - 30
2. 30 – 35
3. 35 - 40

Question 2

Glasgow coma scale, what is the patient's score?

1. 0
2. 3
3. 5
4. 9

Question 3

A therapist is ordered to evaluate and treat a full-term infant. After reviewing the chart, the therapist discovers that at 1 minute after birth the infant exhibited the following symptoms: bluish color in the body and extremities, heart rate of 85 beats/minute, slow respirations, no response to reflex irritability, and some resistance of the extremities to movement. What was the infant's APGAR score at 1 minute after birth?

1. 1
2. 2
3. 3
4. 4

Question 4

A 72-year-old woman complains of axial skeletal pain. Her T score on DEXA scan is -3.5. Which of the following is the most appropriate Borg scale of perceived exertion for trunk exercise in osteoporosis

1. 12/20
2. 14/20
3. 16/20
4. 18/20

Question 5

What is the T source of a bone mineral density (BMD) scan as per the World Health Organization criteria for the diagnosis of osteoporosis?

1. 1.1 to 2.0
2. 1.0 to - 0.
3. 2.5 or less
4. 1.0 to - 2.4

Question 6

A 23 - year - old football player suffered a knee injury during a During the assessment of the knee, the physical therapist is .game Which of the following is .thinking about anterior cruciate ligament injury? the most appropriate radiological examination

1. X – ray
2.Ultrasound
3. Computed tomography scan
4. Magnetic resonance imaging

Question 7

which is not correct about oxford scale?

1. 0 no contraction
2. flicker contraction
3. 4 can make full ROM against resistance

Question 8

Pulmonary resuscitation?

1. 1 cycle of 2 and 20 compression
2. 12 breath and 10 compressions
3. continue 2 breath and 30 compression

Question 9

it is considered illegal?

1. Burn of patient after hot pack
2. Attitude toward patient and behavior

Question 10

self-actualization mean

1.person frustrated because can not gain target
2.sense of blessing due to age
3. person has gain something and proud
4.no sense

Question 11

During a treatment session in a rehabilitation hospital a patient asks the physical therapist assistant to let him see the physical therapy progress notes in his medical recor4. The physical therapist assistant shoul4.

1. contact the patient's physician and explain the situation.
2. refuse to let the patient see the recor4.
3. allow the patient to see the notes.
4. get the permission of the physical therapist supervisor before giving the patient the chart.

Question 12

All of the following are objective measurements used in evaluation except:

1. time used for performance activity
2. visual analogue scale
3. goniometric ROM
4. disability devices

Question 13

Which of the following should a therapist evaluate first when performing a jo2.site analysis?

1. Job description and duties.
2. Bathroom accessibility
3. Lighting conditions
4. Parking conditions.



Question 14

A therapist is instructing a physical therapy student in writing a SOAP note. The student has misplaced the following phrase: Patient reports a functional goal of returning to playing baseball in 5 weeks. Where should this phrase be placed in a SOAP note?

1.Subjective
2.Objective
3.Assessment
4.Plan

Question 15

How often dose the joint commission on accreditation of healthcare organization (JCAHO) require that all electrical equipment in hospitals be inspected ?

1. Every 3 months
2. Every 6 months
3. Every 12 months
4. Every 3 years

Question 16

A physical therapist is setting up a portable whirlpool unit in the room of a severely immobile patient. What is the most important task of the physical therapist before the patient is placed in the whirlpool?

1. Check for a ground fault circuit interruption outlet.
2. Check to make sure the water temperature is below 110°
3. Make sure the whirlpool agitator is immersed in the water.
4. Obtain the appropriate assistance to perform a transfer.

Question 17

A 37-year-old man fell and struck his left temple area on the corner of a mat table. He begins to bleed profusely but remains conscious and alert. Attempts to stop blood flow with direct pressure to the area of injury are unsuccessful. Of the following, which is an additional area to which pressure should be applied to stop bleeding?

1. Left parietal bone one-inch posterior to the ear.
2. Left temporal bone just anterior to the ear.
3. Zygomatic arch of the frontal bone
4. Zygomatic arch superior to the mastoid process.

Question 18

A therapist is preparing a poster that will clarify some of the data in an in-service presentation. The poster reflects the mode, median, and mean of a set of dat1. The data consist of the numbers 2, 2, 4, 9, and 13. If presented in the above order (mode, median mean), which of the following is the correct list of answers calculated from the data?

1. 4,2,6
2. 2,4,6
3. 6,2,4
4. 6, 4, 2.

Question 19

A physical therapist is beginning the evaluation of a patient with AIDS. The patient was admitted to the acute floor of the hospital on the previous night after receiving a right total hip replacement. The physician has ordered gait training and dressing change of the surgical site. Of the following precautions, which is the least necessary?

1. Mask
2. Gloves
3. Hand washing
4. Gown.

Question 20

A physical therapist is scheduled to evaluate the shoulder of a patient with hepatitis 2. The therapist notices no open wounds or abrasions and also notices that the patient has good hygiene. The physician has ordered passive range of motion to the right shoulder because of adhesive capsulitis. Which of the following precautions is absolutely necessary to prevent the therapist from being infected?

1. The therapist must wear a gown
2. The therapist must wear a mask
3. The therapist must wear gloves.
4. None of the above.

Question 21

PT is assessing a patient's ability to perform basic activities of daily living. The assessment tool measures bathing, toileting, dressing, transfers, continence and feeding. The tool does not assess the patient's ability to maneuver in a wheel chair The therapist is using which of the following tests?

1. Barthel Index
2. Katz Index of Activities of Daily Living
3. Kenny Self-Care Evaluation
4. Functional Status Index

Question 22

A 67-year-old man has some difficulty reaching overhead and grooming his hair. The physical therapist decided to give him some exercises to help him achieve his goal of improving this. Which of the following aspects of the function according to the International Classification of Functioning (ICF) the PT is looking to improve?

1. Personal factors
2. Activity limitation
3. Physical Impairment
4. Participation restriction

Question 23

main aim of ICF (international classification function)

1. Impairment
2. Activity limitation
3. Participation



Question 24

patient can't make hyperextension of shoulder and put his hand in his pocket according to CIF?

1. activity limitation
2. impairment
3. disability

Question 25

physical therapist working with a patient to improve the difficulty to reaching overhead and brushing hair , what is that according to ICF?

1. participation restriction
2. activity limitation
3. impairment

Question 26

patient have problems in ROM and muscle weakness is classified as

1.participation limitation
2.functional impairment
3.personal factors
4.activity limitation

Question 27

Which negative factor you treat if you increase strength of ms and ROM

1. Personal factor
2. Activity limitation
3. Participant
4. Impairment

Question 28

exercise professional autonomy

1.Hard work
2.Patient safety
3. Critical thinking
4. All of the above

Question 29

A part of the musculoskeletal exam may include measuring the end feeling. Which of the following relates to this test?

1. Smoking, physical inactivity, obesity, diet
2 .Hypertension, Smoking, diabetes mellitus, diet
3. Physical inactivity, heart disease, obesity, elevated Low – density lipoprotein cholesterol (LDL) cholesterol
4. Hypertension, heart disease, diabetes mellitus, elevated Low – density lipoprotein cholesterol (LDL) cholesterol

Question 30

A therapist places a pen in front of a patient and asks him to pick it up and hold it as he normally would to write. The patient picks the pen up and holds it between the pad of the thumb and the middle and index fingers. Why type of grasp or prehension is the patient using?

1.Palmar prehension
2.Fingertip prehension
3.Lateral prehension
4.Hook grasp

Question 31

The supervisor of a rehabilitative facility insists on a weekly meeting to discuss drops in productivity. The supervisor is not concerned with the increased demand placed on the already overworked employees. According to the managerial grid, what is the best classification of this manager?

1. 5.5
2. 9.9
3. 1.9
4. 9.1

Question 32

CORF is an abbreviation for which of the following?

1. Certified Owner of a Rehabilitation Facility.
2. Certified Outpatient Rehabilitation Facility
3. Control Organization for Rehabilitation Facilities.
4. Corporation for Organization of Rehabilitation Facilities.

Question 33

In an attempt to establish a home exercise program, the therapist gives a patient written exercise. After 1 week, the patient returns and has not performed any of the exercises. After further questioning, the therapist determines that the patient is illiterate. What is the best course of action?

1. Go over the exercises in a one-on-one review session.
2.Give the patient a picture of the exercises.
3. Give a copy of the exercises to a literate family member.
4. All of the above.

Question 34

In order to determine if an exercise session should be terminated, the patient is asked to assess level of exertion using the Borg Rating of Perceived Exertion Scale (RPE). The patient rates the level of exertion as 9 on the 6-19 scale. A rating of 9 corresponds to which of the following?

1. Very, very light
2. Very light
3. Somewhat hard
4. Hard



Question 35

The therapist is treating a male patient for a secon4.degree acromioclavicular sprain. The patient has just finished the doctor's prescription of 3 sessions/week for 4 weeks. The therapist is treating the patient with iontophoresis (driving dexamethasone), deltoid strengthening exercises, pectoral strengthening exercises, and ice. The patient reports no decline in pain level since the initial evaluation. Which of the following is the best course of action for the therapist?

1. Phone the doctor and request continued physical therapy.
2. Tell the patient to go back to the doctor because he is not making appropriate progress.
3. Discharge the patient because he will improve on his own.
4. Take the problem to the supervisor of the facility

Question 36

A local orthopedic doctor, who is a main referral source for an outpatient clinic, insists on an all open-chain exercise program for all patients who have undergone anterior cruciate ligament reconstruction. The therapist feels a close4.chain and open-chain program is the best course of therapy. What is the best course of action to convince the doctor?

1. Call the physician's nurse and discuss with him or her the new treatment protocol.
1. Phone the doctor and request continued physical therapy.
2. Tell the patient to go back to the doctor because he is not making appropriate progress.
3. Discharge the patient because he will improve on his own.
4. Take the problem to the supervisor of the facility

Question 37

A therapist places a pen in front of a patient and asks him to pick it up and hold it as he normally would to write. The patient picks the pen up and holds it between the pad of the thumb and the middle and index fingers. Why type of grasp or prehension is the patient using?

1. Call the physician's nurse and discuss with him or her the new treatment protocol.
2. Schedule a face-to-face meeting with the doctor to discuss the correct treatment plan.
3. Fax the doctor research supporting use of a combined open- and close4.chain program.
4. Have one of the doctor's colleagues convince him that the new protocol is best

Question 38

Doctors see on patient door word (terminal illness) what do this mean:

1. disease with elderly people
2. with adult people
3. irreversible course
4. neurologic origin

Question 39

when you sit-up a patient from long sitting after prolonged period of bed rest following major back surgery, patient experience an orthostatic -hypotension and suddenly will fall what is your response?

1. immediately return patient to original long sitting
2. call the physician
3. rite down this in patient`s review
4. assure him it`s O.K don't worry it`s normal decrease in Blood Pressure due to a long rest period

Question 40

A therapist is teaching a family to care for a family member at home. The patient is totally bedboun4. To prevent pressure ulcers most effectively, what should be the maximal amount of time between position changes?

1. One hour
2.Two hours
3.Six hours
4.Eight hours

Question 41

A therapist is treating a 35-year-old man diagnosed with lumbar disc degeneration in an outpatient clini3. Through conversation with the patient, the therapist learns that he is also being treated by a chiropractor for cervical dysfunction. What is the best course of action by the therapist?

1. Continue with the current treatment plan and ignore the chiropractor's treatment.
2. Ask the patient what the chiropractor is doing and try the same approach.
3. Stop physical therapy at once and consult with the referring physician.
4. Contact the chiropractor to coordinate his or her plan of care with the physical therapy plan of care.

Question 42

A patient at an outpatient facility experiences the onset of a grand mal seizure. Which of the following is the most appropriate course of action by the therapist?

1. Assist patient to a lying position, move away close furniture, loosen tight clothes, and prop the patient's mouth open.
2. Assist patient to a lying position, move away close furniture, and loosen tight clothes
3. Assist the patient to a seated position, move away close furniture, and loosen tight clothes.
4. Assist the patient to a seated position, move away close furniture, loosen tight clothing, and prop the patient's mouth open.

Question 43

Following a cerebrovascular accident involving the dominant right hemisphere, a patient exhibiting unilateral neglect would generally not

1.eat food only from the right side of the plate.
2.bump a one-arm driven wheelchair into things on the left side.
3.ignore or deny the existence of the left-sided limbs.
4.shave or put make-up only on the left side of the face.

Question 44

The therapist is treating a patient who recently received a below- knee amputation. The therapist notices in the patient's chart that a psychiatrist has stated that the patient is in the second stage of the grieving process. Which stage of the grieving process is this patient most likely exhibiting?

1. Denial
2. Acceptance
3. Depression
4. Anger



Question 45

In yellow and red flags which of the following not considered as red flag in these situations:

1. history of carcinoma
2. **psychosocial factor**
3. bowel problems
4. excess weight loss

Question 46

A 27-year-old man with a diagnosis of incomplete spinal cord injury at the L4 level is being evaluated by a physical therapist. The patient is nearing discharge from the rehabilitation unit. Manual muscle testing reveals the following: right hip flexion = 3+/5, right hip adduction = 5/5, right knee flexion = 2/5, right knee extension = 3+/5, right ankle plantarflexion = 1/5, and right ankle dorsiflexion = 2-/5; left hip flexion = 4+/5, left hip adduction = 4+/5, left knee flexion = 2+/5, left knee extension = 3+/5, left ankle plantarflexion = 2-/5, and left ankle dorsiflexion = 2-/5. What is the appropriate orthotic for this patient? What is his most likely functional outcome?

1. Hip-knee-ankle-foot orthosis (HKAFO) with forearm crutches – household ambulator.
2. Knee-ankle-foot orthosis (KAFO) with forearm crutches – household ambulator.
3. **KAFO with forearm crutches – functional ambulator**
4. HKAFO with forearm crutches – functional ambulator

Question 47

A 60-year-old man sustained a fracture of the hip joint following a fall. He was given skeletal traction for conservative management of the fracture. He received regular physiotherapy to prevent muscle wasting and has started regular cyclic loading exercise plan. How long, in months, will it take him to regain normal muscle strength?

1. 4
2. 5
3. **6**
4. 7

Question 48

Patient have 4/5 muscle power what is the minimum number of sessions per week to achieve full strength?

1. 3
2. 4
3. 5

Question 49

which of the following is the most appropriate duration for normal elderly person to training regular per day?

1. 15
2. **30**
3. 45
4. 60

Question 50

Which of the following is the primary aim of the physical therapy profession?

1. Primary
2. Tertiary
3. Secondary
4. Transitional

Question 51

Which of the following is the most appropriate term to describe prevention with early diagnosis and reduction in the severity of an existing disease?

1. Primary
2. Tertiary
3. Primitive
4. Secondary

Question 52

A 37-year-old man referred to the Physical Therapy Clinic for back pain. The physical therapist needs to implement an intervention according to the prognosis. Which of the following refers to the term "prognosis"?

1. Predictable course of the condition
2. Determining the nature of the condition
3. Tests and measures findings related to the physical exam
4. Tests and measures findings related to medical history

Question 53

A 25-year-old man referred to the Physical Therapy Clinic for hip pain. The physical therapist needs to apply an evidence-based intervention. What is the first step of the evidence-based practice process?

1. Critically analyze the pertinent evidence
2. Integrate the appraised evidence into practice
3. Identify the problem and convert it into a specific question
4. Search the literature and collect clinically relevant scientific studies

Question 54

Which of the following is the main moral duty of a physical therapist?

1- Respect the patient's dignity
2- Disclose the patient's autonomy
3- Promote and protect the hospital interest
4- D Protect the privacy and regards the employer.

Question 55

A physical therapist is searching for evidence to support the use of eccentric exercises to decrease the pain in the patient with tendinopathy. Which of the following study designs should be a priority during the search?

1. Cross-sectional
2. Prospective cohort
3. Quasi-experimental
4. Randomized control trial



Question 56

A 29-year-old woman referred to physical therapy for post-operative rehabilitation. The physical therapist decided to use therapeutic exercises as part of the intervention. What is the ultimate goal of therapeutic exercises in physical therapy?

1. Restore function
2. Improve balance
3. increase strength and ROM
4. Improve cardiopulmonary fitness

Question 57

Patient has depressing; Which cause him to have the following:

1. behavior problems
2. emotional problems
3. memory problems
4. delusion.

Question 58

Perform a blind finger sweep of the throat Which of the following circumstances would normally decrease body temperature in a healthy person?

1. Exercising on a treadmill.
2. Pregnancy
3. Normal ovulation
4. Reaching age of 65 years or older

Question 59

A physical therapist in an outpatient clinic is called into a room to assist an infant who is unconscious and not breathing. The therapist opens the airway of the infant and attempts ventilation. The breaths do not make the chest rise. After the infant's head is repositioned, the breaths still do not cause the chest to move. What should the therapist do next?

1. Give five back blows
2. Look into the throat for a foreign body
3. Have someone call 911

Question 60

A physician is preparing a patient for an upcoming procedure. The physician explains that the procedure will provide a detailed image that appears to be a slice of the brain. This image is obtained with a highly concentrated x-ray beam. What procedure is the patient scheduled to undergo?

1.Angiogram
2.Magnetic resonance imaging (MRI)
3.Positron emission tomography (PET)
4.Computed tomography (CT)

Question 61

Which of the following are tests for peripheral arterial involvement in a patient with complaints of calf musculature pain?

1.Claudication time
2.Homan's sign
3.Percussion test
4.None of the above

Question 62

The therapist is ordered to evaluate a patient in the intensive care unit. The patient appears to be in a coma and is totally unresponsive to noxious, visual and auditory stimuli. What rating on the Rancho Los Amigos Cognitive Functioning Scale is most appropriate?

1.I
2.III
3.IV
4.VI

Question 63

At what age does a human have the greatest amount of fluid in the intervertebral disc?

1.1 year
2.4 years
3.7 years
4.10 years

Question 64

A patient is sitting over the edge of a table and performing active knee extension exercises using an ankle weight as resistance. This exercise demonstrates what class lever?

1. First class
2. Second class
3. Third class
4. Fourth class

Question 65

Which of the following is categorized as illegal or malpractice?

1. Burn caused by prolonged use of hot pack
2. Prolong and detailed physical examination
3. Muscle soreness after high-intensity exercise
4. Strict attitudes and behavior with the patient

Question 66

Which plan of care is typically recommended during the chronic stage of injury?

1. Reduce joint swelling
2. Progress functional activities
3. Control pain, edema and spasm
4. Promote healing of injured tissues

Question 67

When measuring ROM of the wrist put the axis on:

1. trapezoid
2. trapezium
3. lunate
4. capitate.

Question 68

The type exercise where distal part is fixed and proximal part moving over distal part:

1. open pack position
2. closed pack position
3. circuit exercise
4. isometric exercise.



Question 69

Closed pack position of knee is

1. ext. knee
2. flex knee
3. **ext knee with lat tibial rotation**

Question 70

What is the closed packed position of the shoulder?

1. Internal rotation and abduction
2. **External rotation and abduction**
3. Internal rotation and adduction
4. External rotation and adduction

Question 71

A 28-year-old man presents to the clinic a pain in his right hip before one month ago. The physical therapist applies a measurement of the ground reaction force of the hip joint. Where does the ground reaction force pass at heel strike during normal gait?

1. Lateral
2. Central
3. **Anterior**
4. Posterior

Question 72

A 44-year-old woman presents to the clinic with left ankle pain before a week ago. The physical therapist applies a measurement of the ground reaction force of the ankle joint. Where does the ground reaction force pass at heel strike during normal gait

1. Lateral
2. Central
3. Anterior
4. **Posterior**

Question 73

Which of the following is the most likely cause of mechanical joint instability?

1. Mal-alignment
2. Muscle weakness
3. **Ligamentous injury**
4. Loss of proprioception

Question 74

A 66-year-old patient complains of sensory changes due to a narrowed spinal canal as a result of degeneration. Which of the following postures is most likely recommended to increase the canal size?

1. 0-degree Flexion
2. **15-degree Flexion**
3. 0-degree extension
4. 15-degree extension

Question 75

A 70 - year - old man presented with a history of frequent falls. He has been screened for visual acuity issues, which was normal with glasses. His clinical examination was unremarkable, except signs of vestibular dysfunction. His blood tests including plasma glucose levels are well and within normal limits. Which of the following is the most appropriate management to prevent future falls?

1. Electronystagmography
2. Snellen chart assessment
3. **Frenkel coordination exercises**
4. Hip girdle strengthening exercises

Question 76

Muscle strength depend on all except:

1. size of muscle
2. lever arm
3. number of muscle spindle
4. **type of ex**

Question 77

Which of the following is an example of first class lever?

1. Soleus
2. **Trapezius**
3. Hamstring
4. Brachioradialis

Question 78

A therapist is evaluating a wound in a patient with the following signs: the right foot has a toe that is gangrenous, the skin on the dorsum of the foot is shiny in appearance, and no calluses are present. The patient has what type of ulcer?

1.Venous insufficiency ulcer
2.**Arterial insufficiency ulcer**
3.Decubitus ulcer
4.Trophic ulcer

Question 79

A patient who has suffered a recent fracture of the right tibia and fibula has developed foot drop of the right foot during gait. Which nerve is causing this loss of motor function?

1.Posterior tibial
2.Superficial peroneals
3.**Deep peroneals**
4.Anterior tibial

Question 80

A patient presents to therapy with an ankle injury. The therapist has determined that the injury is at the junction of the distal tibia and fibul1. Which of the following functions most in preventing excessive external rotation and posterior displacement of the fibula?

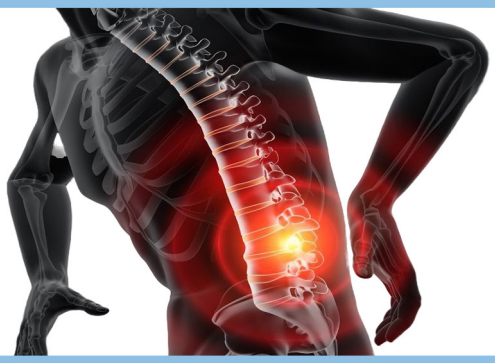
1. **Anterior inferior tibiofibular ligament**
2. Posterior inferior tibiofibular ligament
3. Interosseous membrane
4. None of the above.



Question 81

It is most likely than when treating a patient with Lyme Disease of more than one year's duration, the physical therapy focus will be on management of arthritic changes primarily affecting the:

1.small joints of the hands and feet
2.large joints of the body; especially the knee
3.axial joints, especially the lumbosacral spine
4.axial joints, especially the cervical and thoracic spine



Chapter

6.C Figures

Question 1

Below knee amputation what do you see :

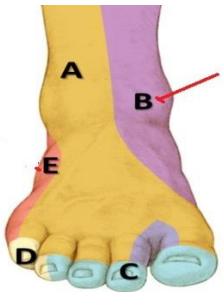
a- there is knee flexion
b- there is no knee flexion
c- there is edematous stump
d- there is deformed stump.



Question 2

Dermatome of letter B

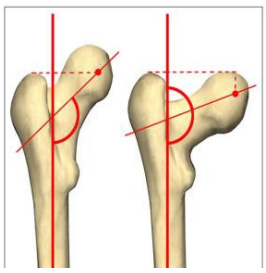
a- L3-4
b- L4-5
c- L4 S1
d- S1-2



Question 3

This angle in the picture means

a- genu valgum
b- coxa valgum
c- coxa vara
d- genu recurvatum



Question 4

To strengthen this muscle, you should

a- maintain dorsiflexion and walking some steps on heel
b- walking on toes
c- sit at edge of plinth, rest foot on ground and fully
d- dorsiflexion the foot.



Question 5

Which nerve supplies the muscle in the picture and its name

a- Axillary nerve and Name deltoid ms
b- suprascapular nerve
c- long thoracic nerve
d- Musculocutaneous nerve



Question 6

What is the deformity shown in x-ray:

a- Hallux valgus deformity
b- hallucis varus
c- hallucis rigidus
d- hammer toe.



Question 7

fixed stiffness of 1st IPJ joint of big toe called:-

a-hallucis varus
b-hallucis valgus
c-hammer toe
d-hallucis rigidus



Question 8

This angle is:

a- Q angle
b- C angle.



Question 9

pt has median Nerve compression assist him by

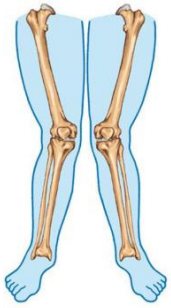
a- Weak of medial 2 fingers & decrease grasping objects
b- Numbness of 3 ½ fingers & weak opponents ms



Question 10

ulnar nerve give supply to

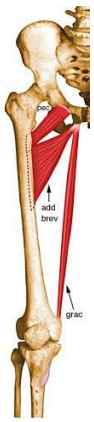
a- whole hand
b- outer3 1/2 fingers
c- inner 1 +1/2 fingers



Question 11

What do u see in this picture

a- bow leg
b- genu valgum
c- coxa valgum
d- genu recurvatum



Question 12

What is the action of these muscle

a- hip Adduction
b- hip abduction
c- hip external rotation
d- hip internal rotation.

Question 13

The least muscle affected by this stretching shown in the figure is

a- vastus lateralis
b- vastus medialis
c- iliopsoas muscle
d- gluteus medius



Question 14



The most muscle affected by this stretching shown in the figure is:

a- lateral rotator of the hip or External rotators

b- medial rotator of the hip

c- iliopsoas muscle

d- adductor muscle

Question 15



The most muscle affected by this stretching shown in the figure is:

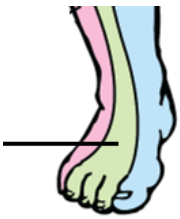
a- lateral rotator of the hip or External rotators

b- medial rotator of the hip

c- iliopsoas muscle

d- adductor muscle

Question 16



Dermatome chosen below is:

a- L3- L4

b- L4

c- L4-L5

d- L5

Question 17



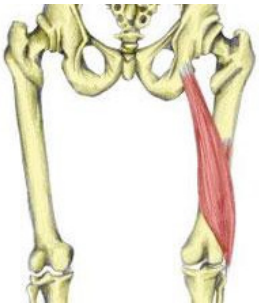
Patient is suffering from pain at the insertion

a- rotator cuff tear

b- infra spinatus tendinitis

c- supraspinatus tendinitis

Question 18



What is true about this muscle and nerve supply?

a- biceps femoris muscle and supplied by tibial nerve

b- adductor brevis muscle

d- sartorius

Question 19



what's the name of this muscle

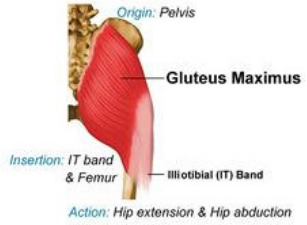
a- Biceps brachii

b- Brachioradialis

Question 20

What is the action of this muscle shape?

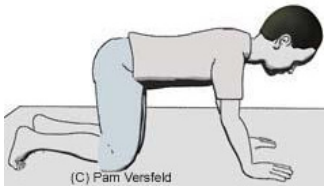
a- hip adduction
b- hip extension
c- back extension



Question 21

What is this position:

a-half kneeling
b-prone kneeling
c-sitting kneeling
d-half kneeling



Question 22

What is the name of this muscle:

a- biceps brachii
b- brachioradialis
c- triceps
d- deltoid.



Question 23

A patient is referred to physical therapy with a ruptured Achilles tendon. The patient reports that he was moving the lawn and was going down a steep incline when he felt a sharp, sudden pain in the left heel region. Which of the following would be proper location for the Achilles tendon insertion?

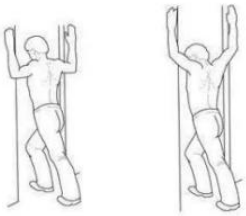
a- Talus
b- Calcaneus
c- Cuboid
d- Navicular



Question 24

A physical therapist instructs a patient to perform a stretch as shown in the image. This type of stretch would most likely be used to stretch the:

a- pectoralis minor
b- triceps
c- upper trapezius
d- middle trapezius



Question 25

A physical therapist instructs a client to perform a standing stretch as shown in the image. This type of stretch would most likely be used to stretch the:

a- pectoralis major
b- inferior capsule
c- horizontal abductors
d- pectoralis minor

